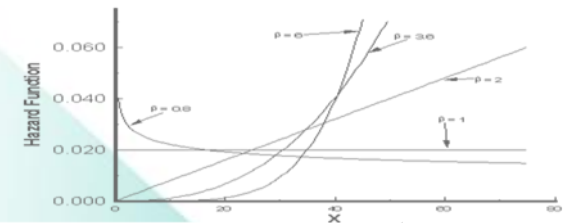


SKG12.3 Reliability and Maintainability Engineering

Level 2

Part 2

Ts Dr Masdi Muhammad CEng (UK), CMRP, CRE
Ts Dr Ainul Akmar Mokhtar
Ts Dr Hilmi Hussin



$$f(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta} \right)^{\beta-1} e^{-\left(\frac{t}{\eta} \right)^{\beta}}$$

Introduction

Name :

Job Function :

What do you do at work (ie. experience with RAM Modeling):

What do you expect from this training :

SKG12.3 TI & R

7-3-13-0003-QR1 v2.0

SKG 12 - Technology Inventory & Ruler
Discipline: 12.3 Reliability Engineering
Subdiscipline: -
Effective Date: 1st January 2020



Technology Inventory			Level 1	Level 2		Level 3			Level 4 Staff			Level 5 Principal			Level 6 Custodian	
				E1		E2			E3			E4			E5	
			20	21	22	E	23	24	E	25	26	E	27	28	E	29
Base	1	Reliability Integrity Management (RIM)	1	2	2	2	2	3	3	3	3	4	4	4	4	5
	2	Data Acquisition, Validation and Decision Analysis (DAVDA)*	1	1	2	2	3	3	3	4	4	4	4	4	4	4
	3	Asset Performance Analysis (APA)*	1	1	2	2	3	3	3	3	3	3	4	4	4	5
	4	Equipment Reliability Strategy (ERS)*	1	2	2	2	3	3	3	4	4	4	4	4	4	4
	5	Root Cause Analysis (RCA)*	1	2	2	2	3	3	3	4	4	4	4	4	4	5
	6	Life Cycle Costing (LCC)	1	1	2	2	2	3	3	3	4	4	4	4	4	4
	7	Benchmarking Process	1	1	1	2	2	2	3	3	3	3	4	4	4	4
Key	1	Risk Assessment (RA)	1	1	1	1	2	2	2	3	3	3	4	4	4	4
	2	System Reliability Modeling	1	1	2	2	2	2	2	3	3	3	4	4	4	5
	3	Advance Analytics	1	1	1	1	2	2	2	2	3	3	3	4	4	4
	4	Predictive Technology	1	1	1	1	2	2	2	3	3	3	3	4	4	4
Pacing	1	Multi State System Reliability Modeling	0	0	0	0	0	1	1	2	2	2	3	3	3	4
	2	Design for Reliability	0	0	0	0	0	0	0	1	1	2	2	3	3	4
Emerging	1	Digital Twins (DT) model in Asset Management	0	0	0	0	0	0	0	1	1	2	2	3	3	3

* indicates compulsory TI

Technology Competence Level

5 Expert

= Have the ultimate knowledge/Skill of Technology Inventory
= Network, originate - advise other's how to do.

4 Advanced

= Have advanced knowledge/Skill of Technology Inventory in question
= Mentor, integrate - teach others how to do

3 Skill

= Fully conversant with Technology Inventory in question
= Applv. demonstrate - do without help

2 Knowledge

= Have basic knowledge, able to perform according to instruction/method available.

1 Awareness

= Contribute, participate - do without help

= Aware of Technology Inventory in question, help needed.

= Recognise, identify - ask when help needed

0 Not Applicable

= Technology Inventory in question is not applicable to Role Level in question.

Course Objectives

At the end of the course, the participants should be able to :

1. Understand the concepts of repairable system and recurrent data analysis (RDA)
2. Perform RDA for repairable system
3. Perform system reliability model and RAM analysis using reliability block diagram (RBD)
4. Propose potential improvements based on RAM analysis
5. Understand Fault Tree Analysis (FTA) and its application
6. Perform Multistate system analysis using Markov modeling

Session schedule

Session	Content	Main Topics	Day	Time
0	Introduction	- Program Agenda and Course Introduction - Pre-test	Day 1	9:00am - 9:30am
1.	Recurrent Data Analysis (RDA)- Dr Masdi (AM) Dr Ainul (PM)	- Repairable versus non-repairable - Trend tests - Analytical Trend Test (Laplace and Mann) - HPP versus NHPP - Crow-AMSAA Model for NHPP - Parameter Estimation - Single system versus multiple system - Cumulative versus instantaneous failure rate - Predicting number of failures - PM and Overhaul - Change of Slope Analysis - Optimum Overhaul	Day 1	9:30am - 10:30am Break (15 Min) 10:45am – 1:00pm Lunch/Prayer (1:00pm – 2:00pm) 2:00pm – 3:30pm Break (15 Min) 3:45pm – 5:00pm
2.	System Reliability Modeling - Dr Hilmi (AM) Dr Ainul (PM)	- Definition of system - Reliability block diagram (RBD) - Monte-Carlo Simulation - Failure model - Repair Model - Repairable system in RBD - System Availability (RAM Modelling) - Case Study	Day 2	9:00am - 10:30am Break (15 Min) 10:45am – 1:00pm Lunch/Prayer (1:00pm – 2:00pm) 2:00pm – 3:30pm Break (15 Min) 3:45pm – 5:00pm

Session Schedule

Session	Content	Main Topics	Day	Time
3.	Advanced System Reliability- (Dr Ainul)	• Case Study Presentation • Maintenance model • Standby Configuration • Load Sharing Configuration • Throughput Analysis • RAM and Life Cycle cost	Day 3 (AM)	9:00am - 10:30am Break (15 Min) 10:45am – 1:00pm Lunch/Prayer (1:00pm – 2:00pm)
4.	Fault Tree Analysis- (Dr Hilmi)	• Introduction to Fault tree Analysis • Fault tree construction • Minimal cut sets • Qualitative analysis • Quantitative analysis	Day 3 (PM) Day 4 (AM)	2:00pm – 3:30pm Break (15 Min) 3:45pm – 5:00pm 9:00am - 10:30am Break (15 Min) 10:45am – 1:00pm Lunch/Prayer (1:00pm – 2:00pm)
5.	Introduction to Multi State system and Markov process in reliability- (Dr Masdi)	• Multi-State system and Markov Model • Discrete Time Markov • Continuous Time Markov • Application of multistate system modelling (IPF) • Post Assessment	Day 4 (PM)	2:00pm – 3:30pm Break (15 Min) 3:45pm – 5:00pm

Methodology

Lectures on basic theories

Application

- Through exercises
- Through case studies

Approach

- Group discussions
- Manual calculations
- Use of spreadsheet/Software

Tools

- Scientific calculator
- Microsoft Excel/Weibull ++/BlockSim

Facilitators' Profile



Dr Masdi bin Muhammad, CEng (UK), CMRP, ASQ-Certified Reliability Engineer is an associate professor in Mechanical Engineering Department and research cluster leader for Facility and Plant Engineering, UTP. He obtained his first and Master degrees, BSc in Mechanical Engineering and MSc in Manufacturing System Engineering, from Lehigh University, USA. His PhD in Mechanical Engineering was obtained from UTP with research on “Reliability Model for Repairable Systems with Multi-State Degradation”. He is actively involved in research collaboration and consultancy with PETRONAS OPU and currently supervising 3 PhDs and 2 MSc in the area of Reliability and Maintenance. He has twelve years of experience working in various positions in one of the leading semiconductor companies, involving process and equipment engineering, product development and material quality prior to joining UTP. He is a chartered engineer and member of BEM, IEM, American Society of Quality and Co-Chair for ASME-Malaysia Section.



Dr Hilmi bin Hussin is a Senior lecturer in Mechanical Engineering Department, UTP. He obtained his BSc in Mechanical Engineering from State University of New York at Buffalo, USA and MSc. in Manufacturing System Engineering from Warwick University, UK. He received his PhD in Mechanical Engineering from UTP, on the topic of RAM analysis at operation phase. He has more than ten years industrial experience and held various positions such as Process Engineer, Senior Quality Engineer and Certified Six Sigma Black Belt in a multi-national manufacturing company.



Dr Ainul Akmar binti Mokhtar is an associate professor in Mechanical Engineering Department and Programme Manager for MSc in Asset Management and Maintenance, UTP. Prior joining UTP, she worked with Seagate Technology (M) Sdn Bhd as a process engineer. She obtained her first degree, BSc in Industrial Engineering from Purdue University, USA. She pursued her postgraduate study in MSc in Manufacturing Systems from Nottingham University, UK and PhD in Mechanical Engineering from UTP. Her PhD research title was “Failure Probability Modelling for Piping Systems subject to Corrosion under Insulation”

Session 1

Reliability Analysis for Repairable System

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Session Objectives

- Understand the concept of recurrent data analysis
- MTBF and RCOF for constant failure rate (HPP)
- Able to predict number of failure and parts required per interval
- Able to perform Crow-AMSAA (Power law) using either Weibull ++ or RGA
- Able to interpret result from the analysis



Types of Systems

Most systems can be categorized into two basic types :-

- One time event or non-repairable systems
 - Sensors, bearings, components of a repairable system

- Reusable or repairable systems
 - Consist of many components
 - Compressors, pumps, turbines
 - Most complex systems fall into repairable category

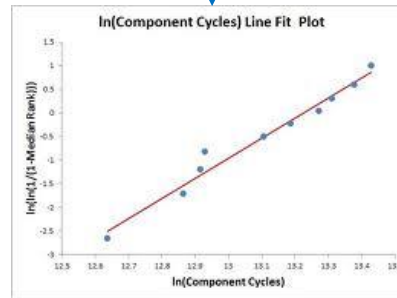
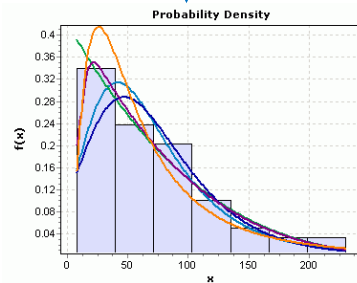
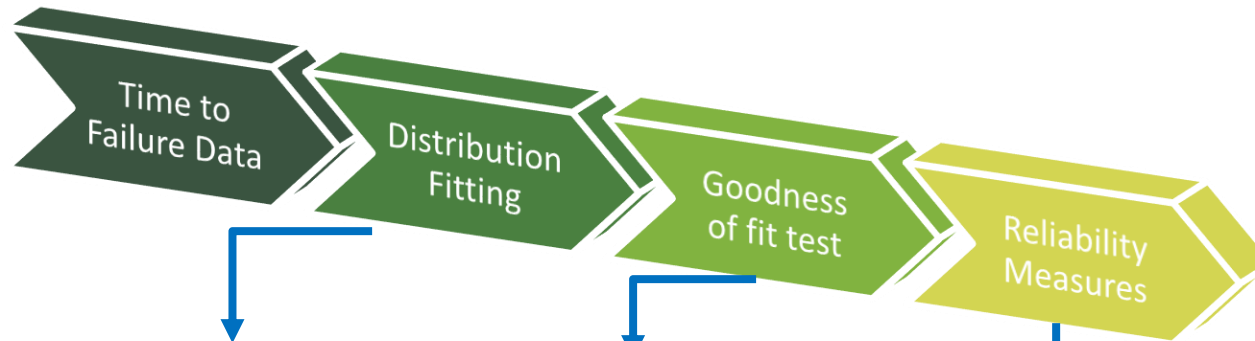


Non-Repairable Component

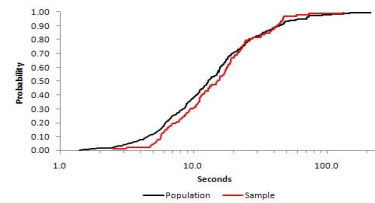
Non-Repairable

- Is defined as equipment after failure can no longer be repaired
- Limited by technical and/or cost of repair
- Discard and replace with 'similar' component after failure
- The component lifetime is a random variable described by a single time to failure
- Non-repairable data analysis or life data analysis is considered as one time event, i.e. failure

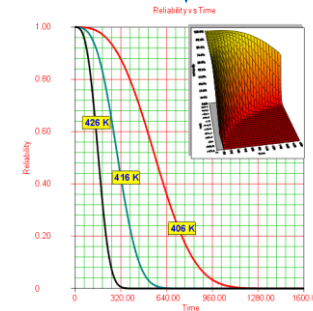
Steps involved in analyzing non-repairable system/component



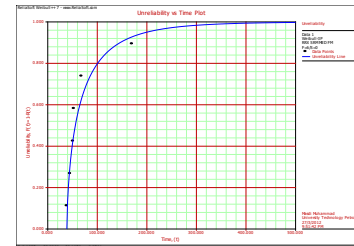
Regression



Kolmogorov-Smirnov



Reliability



Unreliability

Stochastic point process

What is it?

- A stochastic process characterized by isolated events occurring at instants distributed randomly over a time continuum.
- Depending on the repair process, the subsequent time between successive failures may or may not follow the distribution representing the time to first failure.

sto·chas·tic  [stuh·kas·tik]  [Show IPA](#)

—*adjective* *Statistics.*

of or pertaining to a process involving a randomly determined sequence of observations each of which is considered as a sample of one element from a probability distribution.

[dictionary.com](https://www.dictionary.com)

Why use it? (Objective)

- For predictive purpose.
- To model random events (e.g. random variables X_i , time to i_{th} failure)

How to use it?

- First, you need to identify the repair process, whether it's a renewal process or a minimal repair process.
- Then apply the principles governing renewal process or minimal repair process as deemed fit.

Recurrent Data Analysis

For repairable system, the system may accumulate any number of events (failures) over time thus requires different method of analysis

- 2 random variables
 - The number of failure by the age of T
 - The time between successive failures X_i
- The age of the system at the i -th failure, T_i is given by

$$T_i = X_1 + X_2 + X_3 + \dots + X_i$$

- Analysis of repairable system reliability must consider the effects of successive repair actions

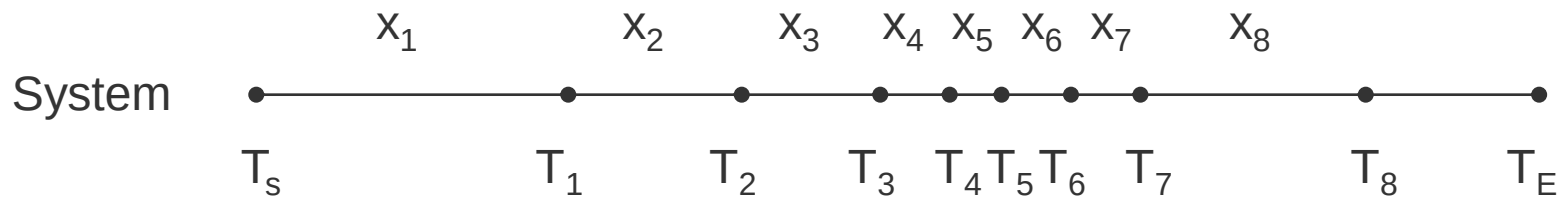
Recurrent Data Analysis

Repairable system

- Definition: a system which, after failing to perform one or more of its functions satisfactorily, can be restored to fully satisfactory performance by any method other than replacement of the entire system

T_i is cumulative time to failure

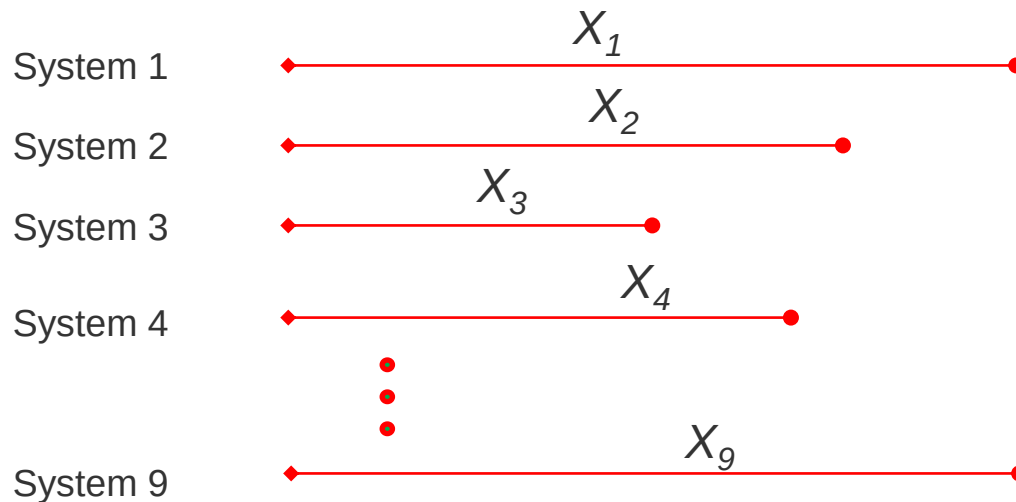
X_i is the inter-arrival time = $T_i - T_{i-1}$



The system can have multiple failures/event

Recurrent Data Analysis

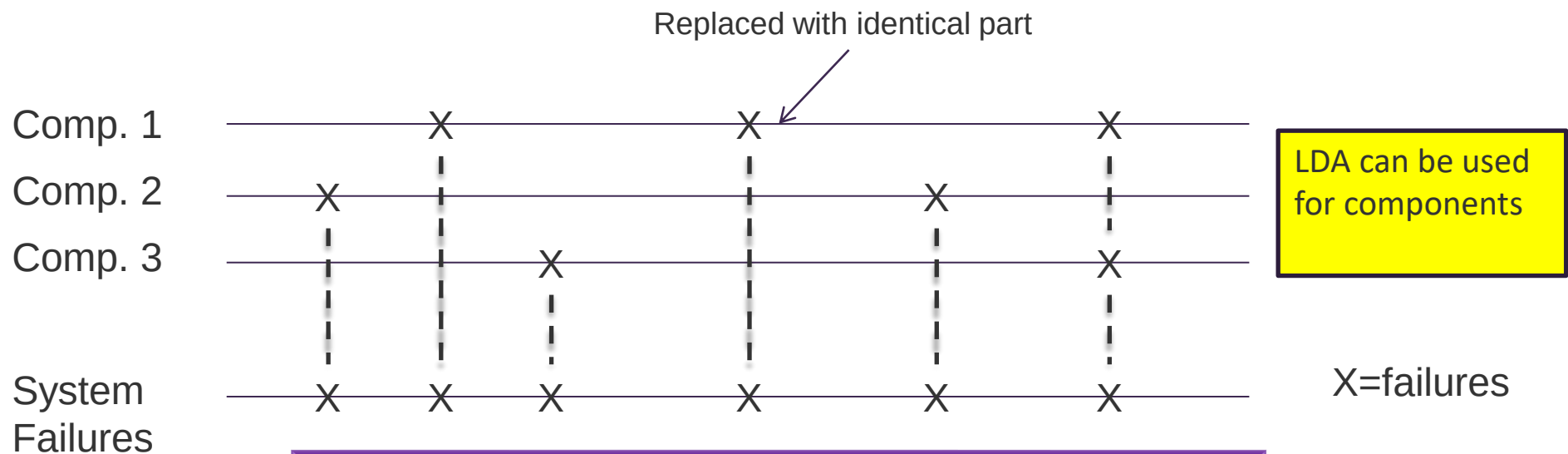
If we use life data analysis on repairable system, it is as though we have 9 different systems where system 1 failed after t_1 , system 2 failed after t_2 and so on...



Recurrent Data Analysis

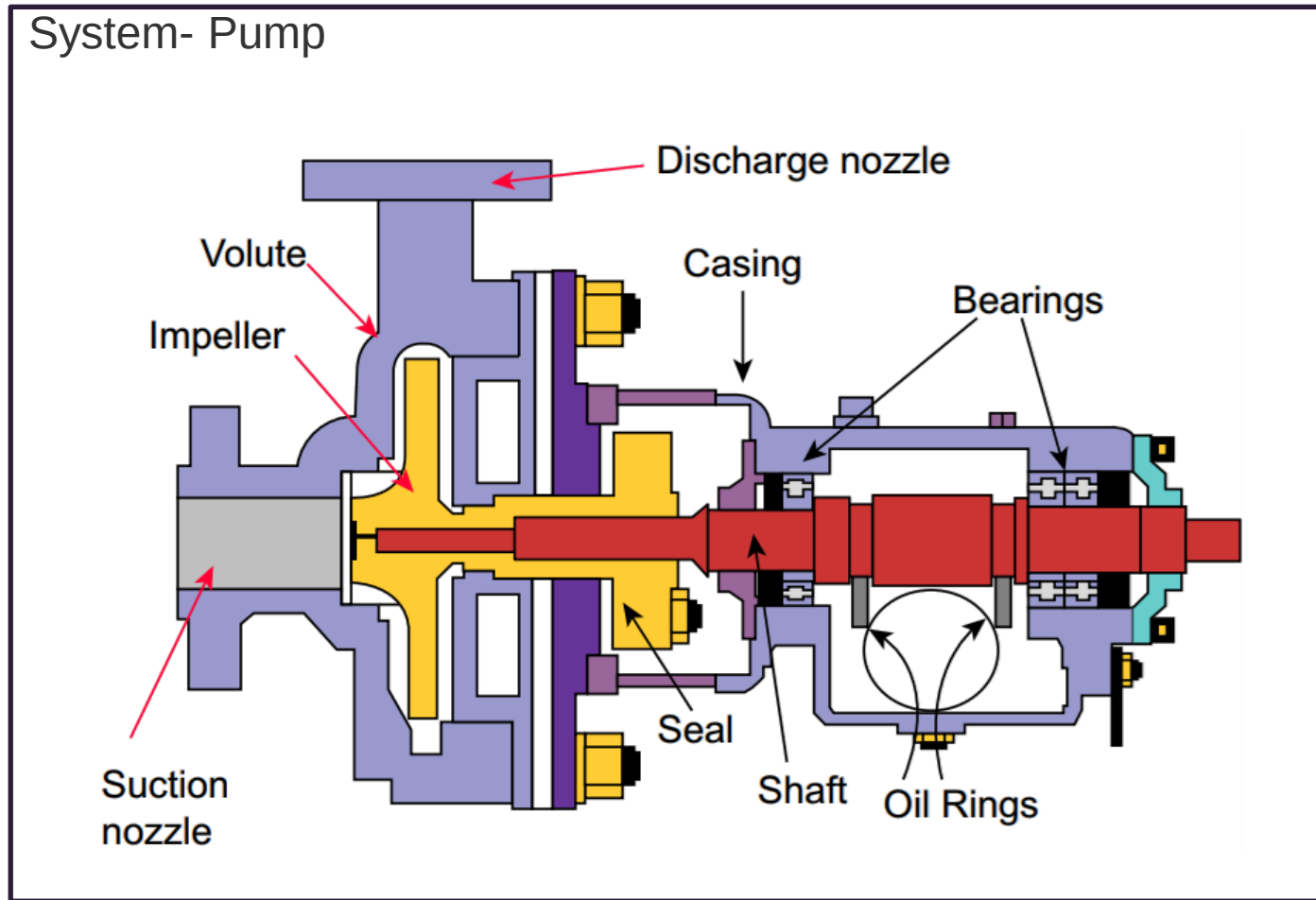
Generally, LDA is more applicable to components that are replaced upon failure.

Systems are normally made up by hundreds of different non-repairable components.



Time to failure for system does not follow the same distribution as components, therefore it is meaningless to fit a distribution.

Example-Pump



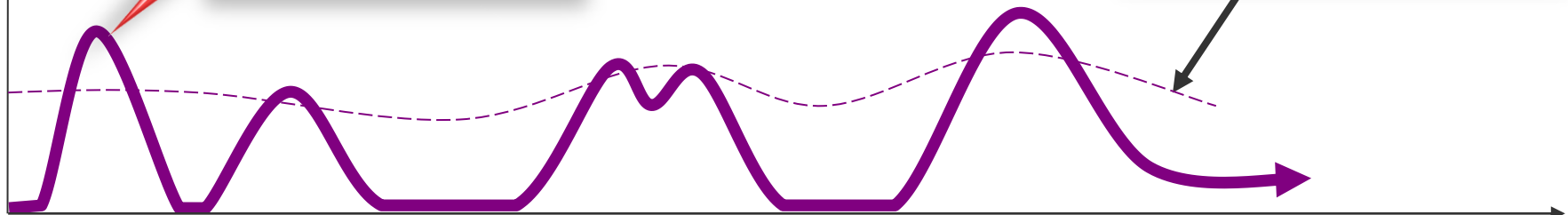
Machine Failure Rate is the Sum of Its Parts Instantaneous Failure Rate

ROCOF

Rate of Occurrence of Failure in a Machine System

A machine's rate of occurrence of failure changes as its parts do, or do not fail.

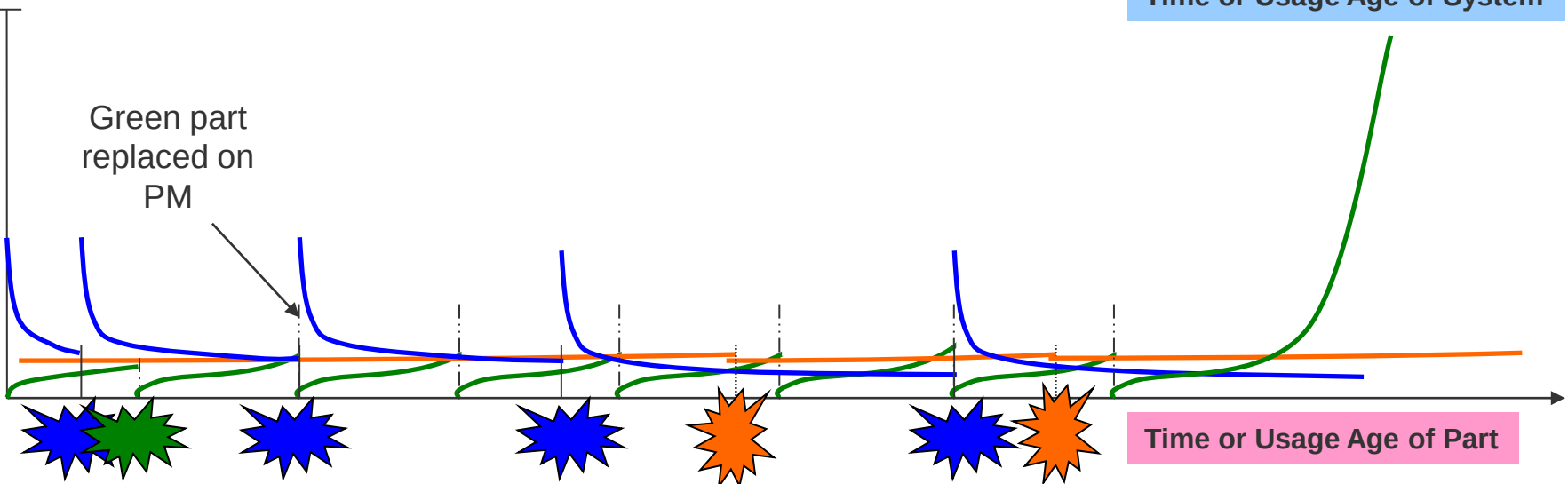
With more parts ROCOF becomes approximately constant (Drenick's Theorem)



Time or Usage Age of System

Hazard Rate, $h(x)$

Green part replaced on PM



Time or Usage Age of Part

Rate of Occurrence of Failure (ROCOF)

- For repairable systems, the 'failure rate' is known as ROCOF
- Given $N(t)$ is the counting of the number of failure occurs in the interval $(0, t]$,

$$ROCOF \equiv \frac{N(t + \Delta t) - N(t)}{\Delta t} \equiv \frac{N(t, t + \Delta t)}{\Delta t}$$

- ROCOF is the probability of failure in a small interval of time divided by the length of the interval

Repair Assumptions

- Unfortunately, as good as new assumption rarely occurs in practice.

- Other Repair Assumptions:



- The system is in the same state after repair just as before it fails or also known as **'as bad as old'**
- The system's state falls between 'as good as new' and 'as bad as old' state
- The system's state is worse than old
- The system state is better than new
 - Through defect elimination

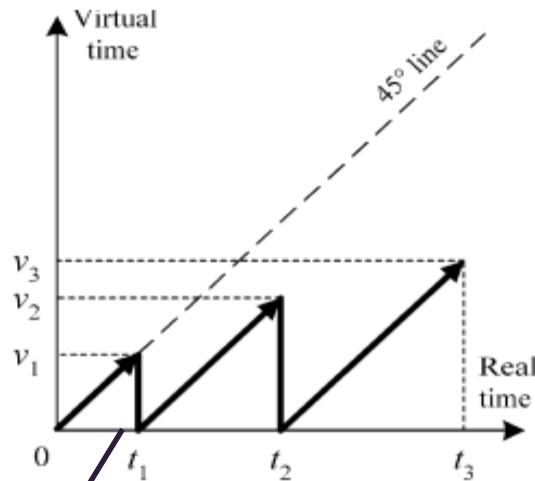
As Good As New
 ...or maybe even better?

Warranty topped up to new
 3 years free servicing
 3 months deferred payments

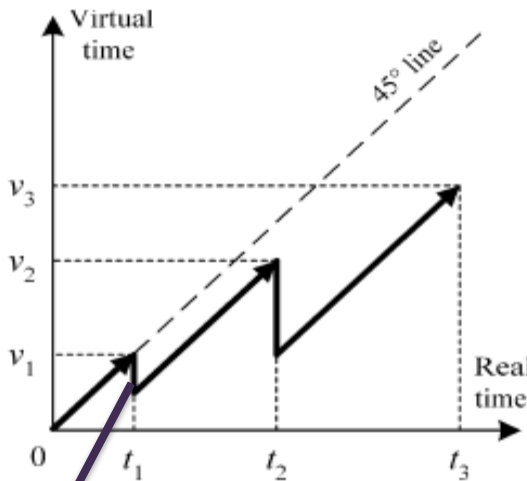
On over 300 selected Ex-Management and Demonstrator Stock



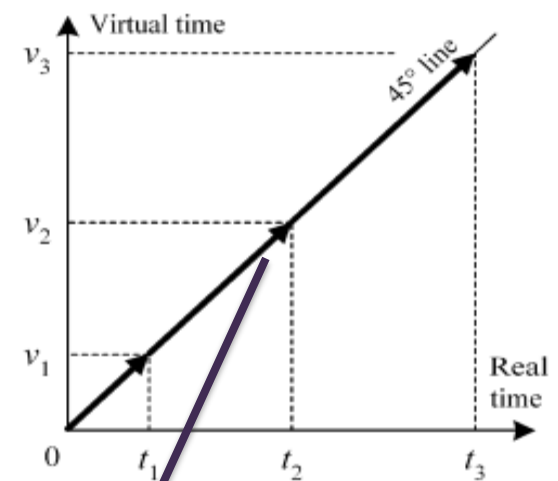
Effect of repairs



System's age is back to zero after each repair

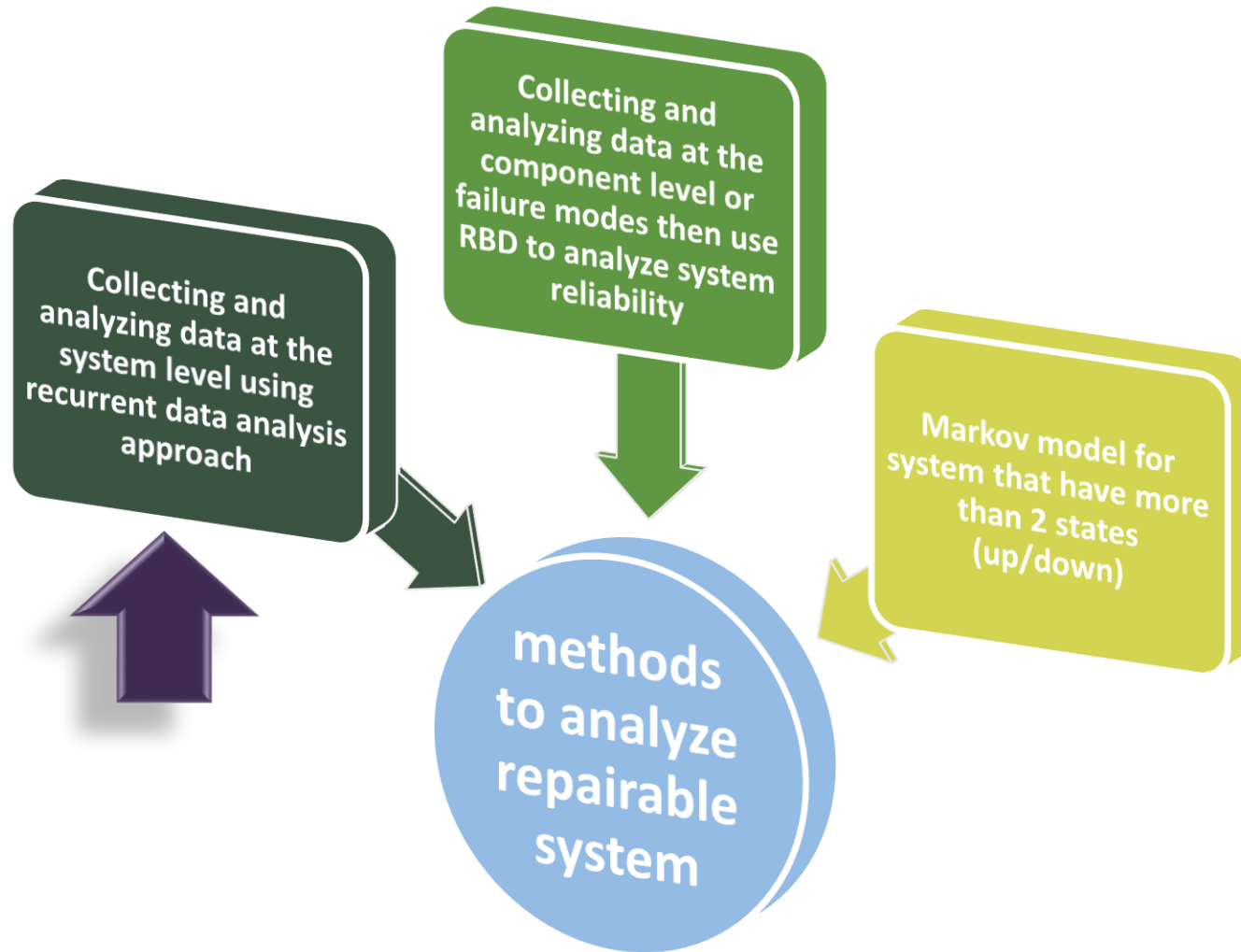


System's age is in between old and new after repair



No change in system's age after repair

Analysis for Repairable System



Recurrent Data Analysis (RDA)

Objectives of Data Analysis :

1. Estimate the reliability and the probability of failure for the system in next time interval
2. Estimate the expected number of failure for a given time or interval
3. Estimate the effectiveness of repair
4. For a given cost of repair (unplanned) and cost of PM (planned), optimum overhaul time can be estimated.

This can be done even with limited number of data

Repairable System Analysis (RBD)

Modelling of the system based on the non-repairable components

The system reliability characteristics are estimated based on the component's failure distribution and the arrangement of each component to make up the system.

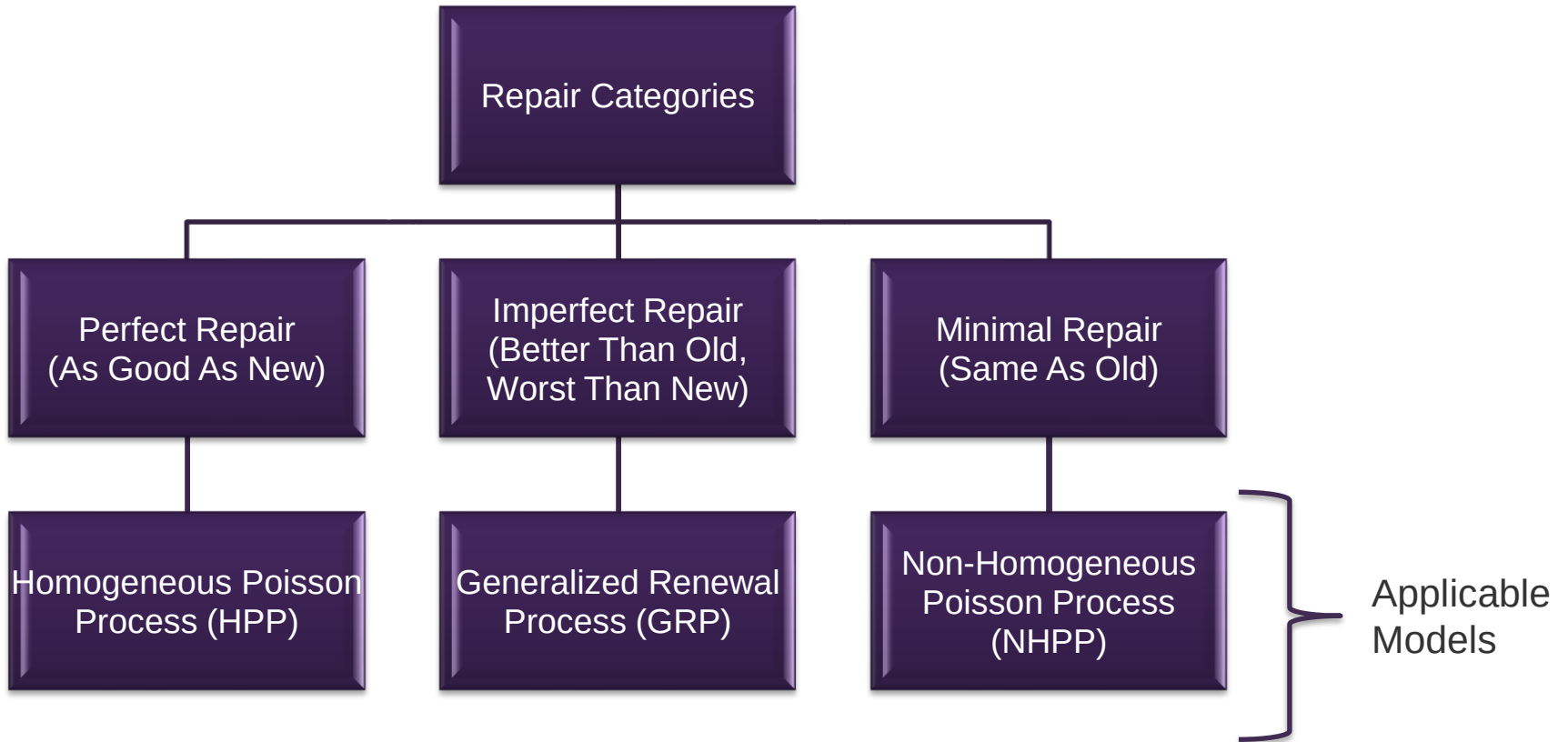
Advantage:

- Can model complex system configuration such as spare parts stocks, maintenance policies, inspection policy, number of crews, throughput etc.
- Able to identify critical system and weak point of the system
- Able to perform sensitivity analysis

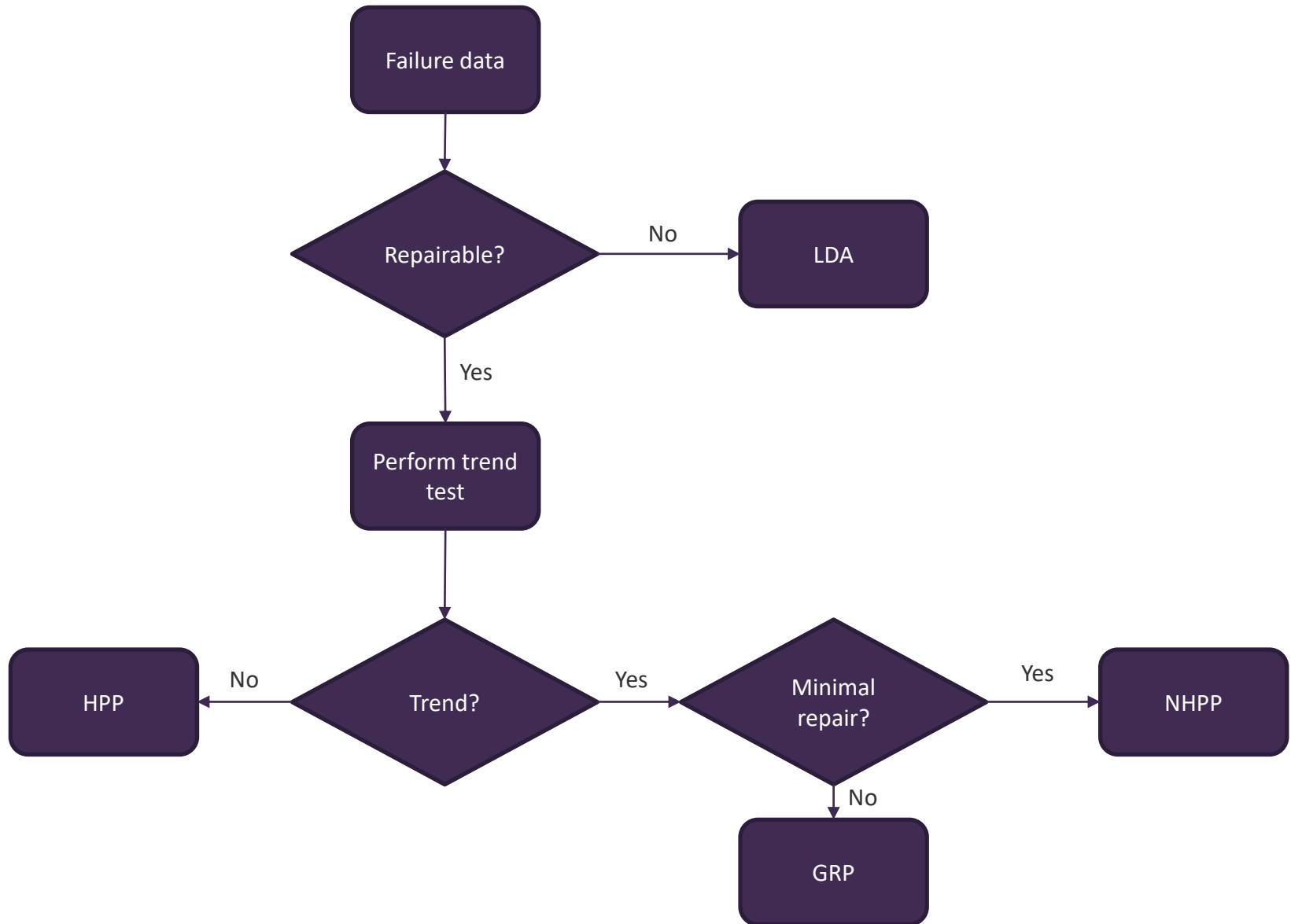
Limitation

- Require a lot of data
- Require deep understanding of system operation and configuration

Repairable System



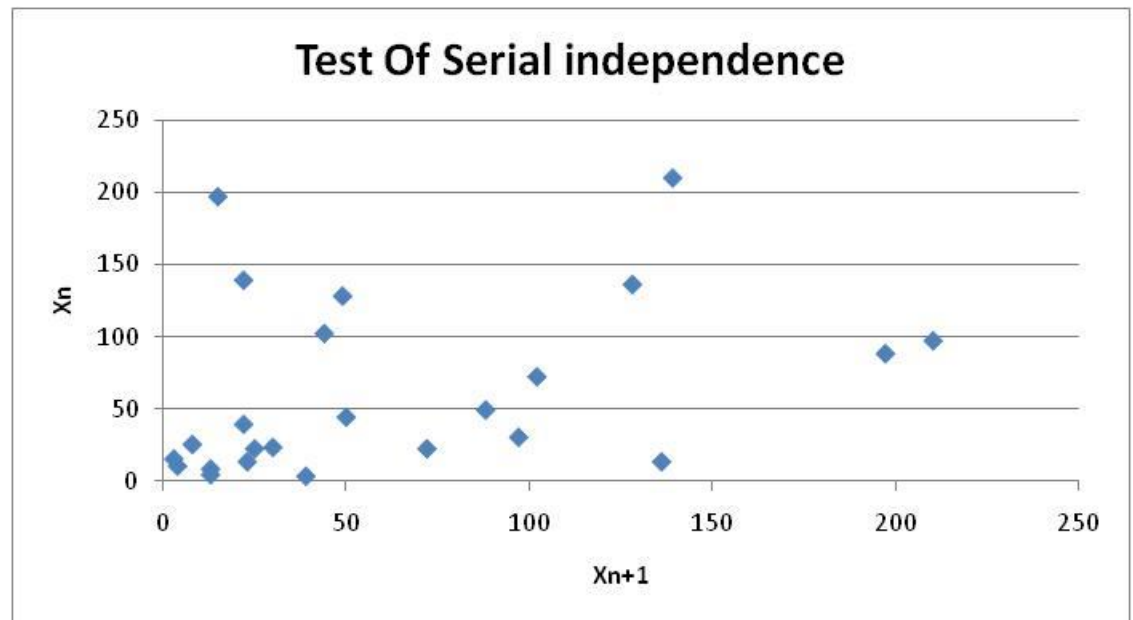
Methodology-Simplified



Test of independence-Serial Correlation Test

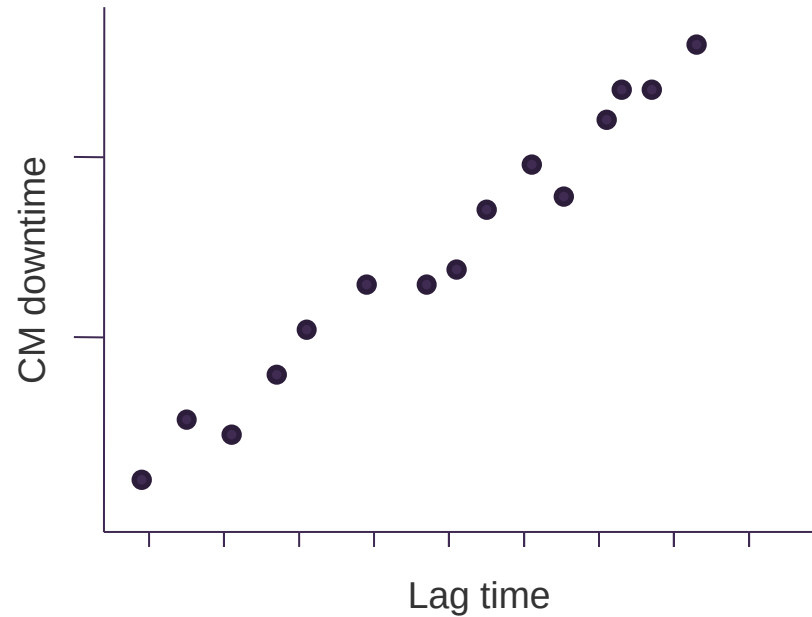
No of Failure	Time to Failure (chronological)
1	50
2	44
3	102
4	72
5	22
6	39
7	3
8	15
9	197
10	88
11	49
12	128
13	136
14	13
15	8
16	25
17	22
18	139
19	210
20	97
21	30
22	23
23	13
24	4
25	10

Are these time to failures independent?



No obvious cluster, independent assumption is valid

Test of independent



Not independent

Trend Test: Graphical Techniques

Arrange time to failure in chronological order

Plot cumulative failure vs. time

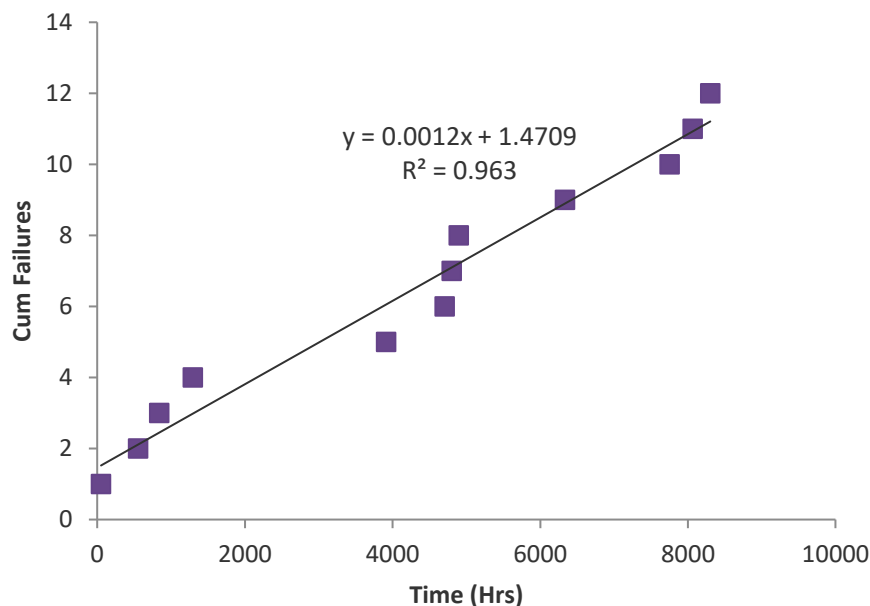
Visually check whether or not trend exists.

- If there is no trend, same distribution for TTF is accepted and LDA can be performed
- Otherwise, LDA is not applicable

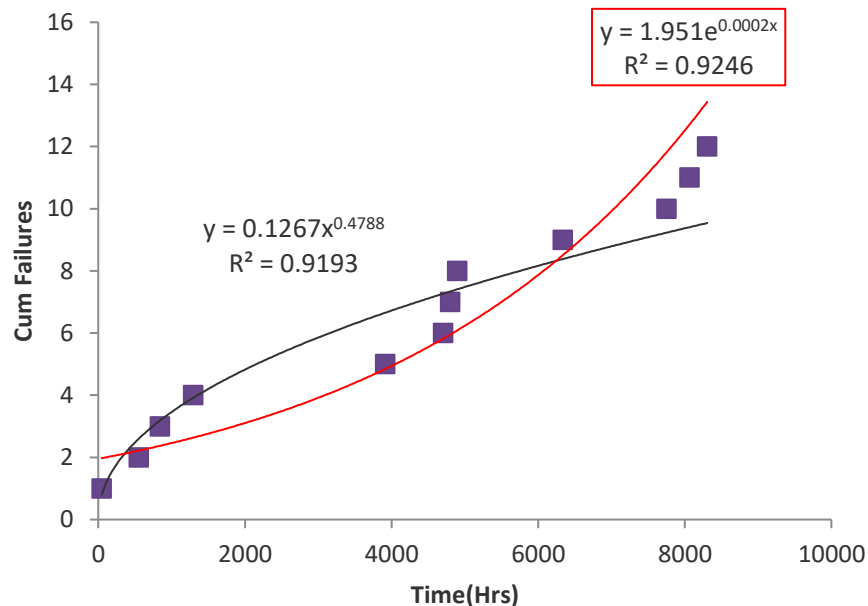
Use R^2 values to check the goodness of fit to straight line

Graphical Trend Test

Cumulative Failure vs Time



Cumulative Failure vs Time



R^2 value show a better fit compared with concave downward (decreasing) or concave upward (increasing)

Result: No trend detected

Mann Trend Test

Null hypothesis: The data exhibit **renewal process (RP)**

If accepted, data can be analyzed by fitting distribution to TBF data

The test statistic is calculated by counting the number of *reverse arrangement*, M , among the times between failure

- If $T_1, T_2, \dots, T_n$ are times between failures, reverse arrangement occurs whenever $T_i < T_j$ for $i < j$
- Example: given TBFs as 21, 17, 48, 37, 64, 13 (days), find M .

M value is compared against expected value

Mann test values

Values of M indicating significant improvement

Number of Repairs	Minimum M for 90%	Minimum M for 95%
4	6	6
5	9	9
6	12	13
7	16	17
8	20	22
9	25	27
10	31	33
11	37	39
12	43	46

www.itl.nist.

Values of M indicating significant degradation

Number of Repairs	Max M for 90%	Max M for 95%
4	0	0
5	1	1
6	3	2
7	5	4
8	8	6
9	11	9
10	14	12
11	18	16
12	23	20

www.itl.nist.

For more than 12 repairs (improvement),

For degradation, M is replaced by $(r(r-1)/2 - M)$

$$Z = \frac{M - \frac{r(r-1)}{4} + 0.5}{\sqrt{\frac{(2r+5)(r-1)r}{72}}}$$

z value can be compared
Against standard normal table

r - number of repair
 M - number of reversal

Detailed calculation of the values can be referred to

<http://www.itl.nist.gov>

Mann Test - Example

Given TBFs as 21, 17, 48, 37, 64, 13 (days), find M .

No of Failure	Time to Failure (chronological)	Cum. Time to Failure	Mann Test
1	21	21	2
2	17	38	1
3	48	86	2
4	37	123	1
5	64	187	1
6	13	200	0

$$M = 2+1+2+1+1=7$$

Compare this M with Mann Test value

Since $M = 7$ does not fall between the values in the Mann test table, M does not indicate significant improvement or degradation → RP assumption is valid.

Laplace Trend Test

Null hypothesis: HPP

- If the test is accepted, we can assume the data is **IID exponentially** distributed

Alternative hypothesis: NHPP

- If rejected, need to use minimal repair assumption

General idea is to compare the mean value of TTF with the mid point of the interval given by the following

$$L = \frac{\frac{\sum_{j=1}^n T_j}{\hat{n}} - \frac{1}{2}(b + a)}{\sqrt{\frac{1}{12\hat{n}}(b - a)^2}}$$

a- start of observation time

b- end of observation time

T_j – cumulative time between failure

n' –number of failure

Use the following example to calculate L values using Excel

- Failures occurred at 5, 40, 43, 175, 389, 712, 747, 795, 1299 and 1478 hours, with the observation ending at 1500 hours.

Summary of trend tests

- Mann Test – No trend detected. RP assumption is valid

$$M = \sum_{i=1}^{n-1} \sum_{j=i+1}^n I(T_i < T_j)$$

T_j - Age at j th failure
 n – Observed number of failures

- Laplace test – No trend, Use HPP

$$L = \frac{\frac{\sum_{j=1}^{\hat{n}} T_j}{\hat{n}} - \frac{1}{2}(b + a)}{\sqrt{\frac{1}{12\hat{n}}(b - a)^2}}$$

T_j - Age at j th failure
 $[a, b]$ – observation interval
 n – Observed number of failures

- Serial correlation test for test of independence

Homogenous Poisson Process (HPP)

HPP implies that system's failure rate is constant (i.e. the system is neither improving nor deteriorating with age)

The probability of having $N(t)$ number of failure at a given t is, where λ is the failure rate

$$P\{N(t) = n\} = \frac{(\lambda t)^n}{n!} e^{-\lambda t}$$

Spare part and Renewal Process

- When to use?
 - Renewal process, **AND**
 - Distribution of time-between-failures is exponential (CFR)
- The **probability distribution** for the **number of failures** by time t, is given by:

$$\Pr\{N(t) = j\} = \frac{(\lambda t)^j e^{-\lambda t}}{j!}$$

j = number of failures
 λ = failure rate

- Expected number of failures by time t, is given by:

$$E[N(0, t)] = \lambda t$$

- The cumulative probability that **k number of failures or fewer** is going to take place by time t, is given by:

$$\Pr_{\text{cum}}\{N(t) = k\} = \sum_{j=0}^k \Pr\{N(t) = j\}$$

Homogeneous Poisson Process -Example

$\Lambda = 0.000050$ per hr

MTTF= 20000 hr

No. of Brgs= 20

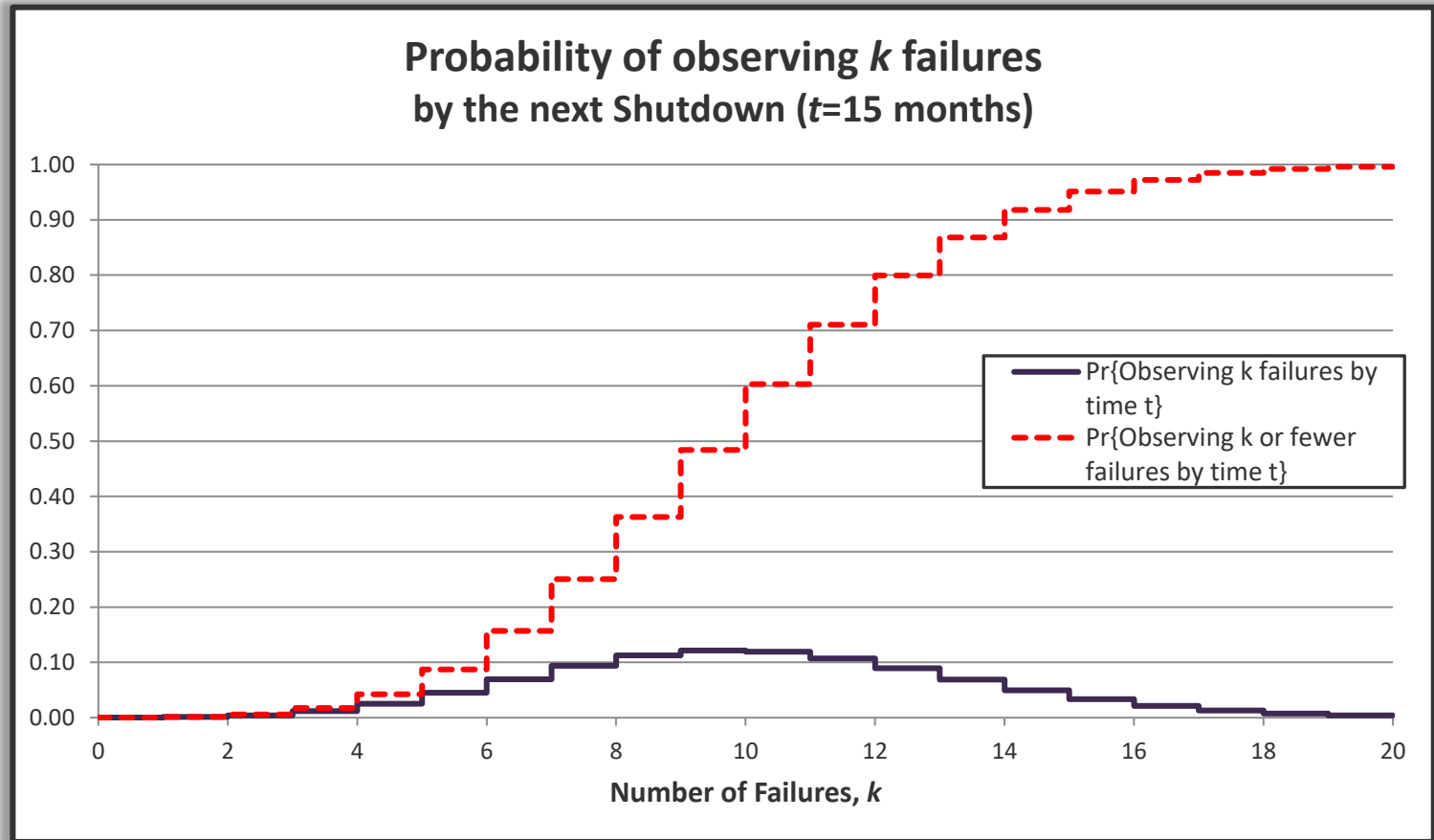
Ops run, $t = 216000$ hr

$(\Lambda) \times (t) = 11$

- Failure records for the past three years for a certain bearing (both DE and NDE) on a group of motors indicate random critical failures.
- There are 20 such bearings with CFR of 0.00005 per operating hour.
- The maintenance team plans to stock **10 units of spare bearings**.
- If the period until the next shutdown for the process unit is 15 months from now, what is the probability that the number of spare bearings is enough?
- Establish the probability distribution and the cumulative probability of failures for the coming 15 months.
- How many bearings would you recommend that maintenance keep?

No. of Failures k	Pr{Observing k <u>failures</u> by time t }	Pr{Observing k or fewer <u>failures</u> by time t }
No. of Spare Bearings k		Pr{ k or fewer bearings is enough by time t }
1	0.00	0.00
2	0.00	0.00
3	0.00	0.01
4	0.01	0.02
5	0.02	0.04
6	0.04	0.09
7	0.07	0.16
8	0.09	0.25
9	0.11	0.36
10	0.12	0.48
11	0.12	0.60
12	0.11	0.71
13	0.09	0.80
14	0.07	0.87
15	0.05	0.92
16	0.03	0.95
17	0.02	0.97
18	0.01	0.99
19	0.01	0.99
20	0.00	1.00

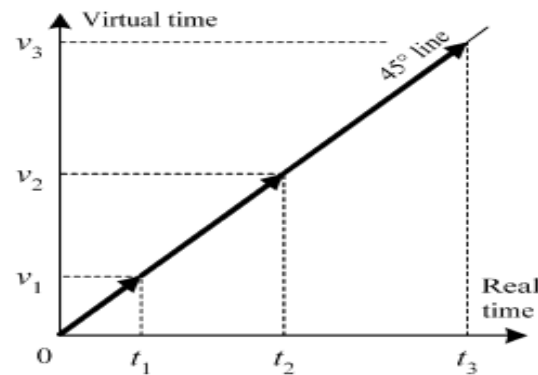
Homogeneous Poisson Process -Example



Non-Homogenous Poisson Process (NHPP)

NHPP assumes that the system is at the same state after repair as just before failure

Recall:



Minimal repair assumption or as bad as old after repair

Assumption is based on the fact that most repairs involve small fraction of system's component, all other components remain at the same age.

Non-Homogenous Poisson Process (NHPP)

The two most popular model for NHPP are:

- Power Law model or also know as Crow/AMSAA
- NHPP Exponential law which is applicable when the Duane plot show obvious curvature (will not be covered)

Power Law

Power law assumes that the distribution for time to first failure (TTFF) is following Weibull distribution

Recall for Weibull Distribution:

$$\lambda(t) = \frac{f(t)}{R(t)} = \frac{\beta}{\eta} \left(\frac{t - \gamma}{\eta} \right)^{\beta-1}$$

$$f(T_i|T_{i-1}) = \frac{\beta}{\eta} \left(\frac{T_i}{\eta} \right)^{\beta-1} \cdot e^{-\frac{1}{\eta^\beta} (T_i^\beta - T_{i-1}^\beta)}$$

The probability density function (pdf) of the i^{th} event given that the $(i-1)^{\text{th}}$ event occurred at T_{i-1}

For power law the expected number of failures, $M(t)$ is given by

$$E[N(t)] = M(t) = \lambda t^\beta$$

The derivative of equation above will give the failure intensity function:

$$u(t) = M'(t) = \lambda \beta t^{\beta-1}$$

$$MTBF_{\text{instantaneous}} = \frac{1}{u(t)}$$

Power Law

Probability of having, n failure is given by Power law Poisson process

$$\Pr[N(t) = n] = \frac{[M(t)]^n e^{-M(t)}}{n!}; n = 1, 2, \dots$$

Power Law-Graphical Method

λ and β can be estimated by plotting cumulative failure versus time on log-log paper which would produce a straight line.

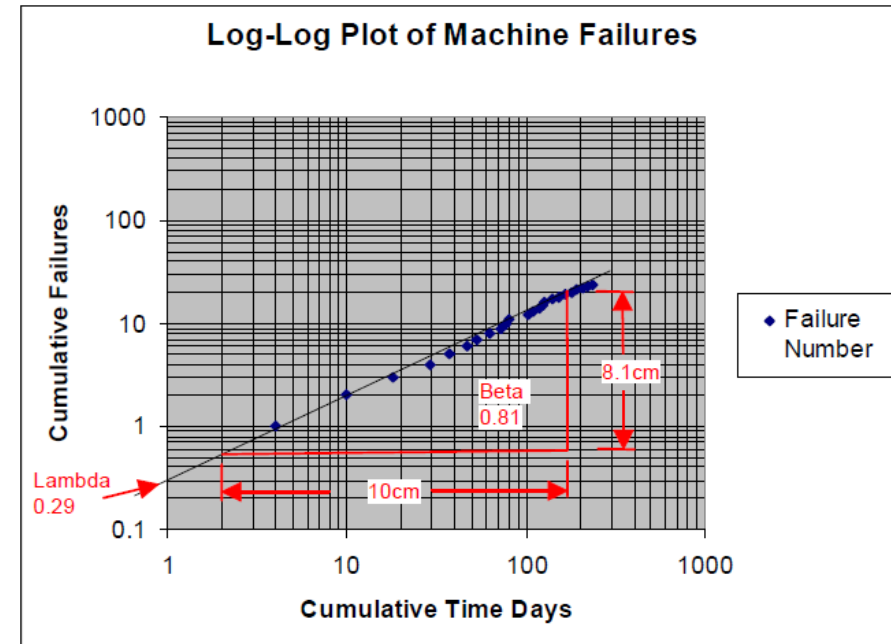
- λ intercept of the line to the x axis
- β the slope of the line

Similar to Weibull plot

$\beta < 1$, decreasing failure rate 😊

$\beta = 1$, constant failure rate, HPP

$\beta > 1$, increasing failure rate 😞



Power Law

Alternatively, λ and β can be estimated using maximum likelihood estimation.

For a fixed observation time, T , and r failures occurring at times $t_1, t_2, t_3, \dots, t_r$

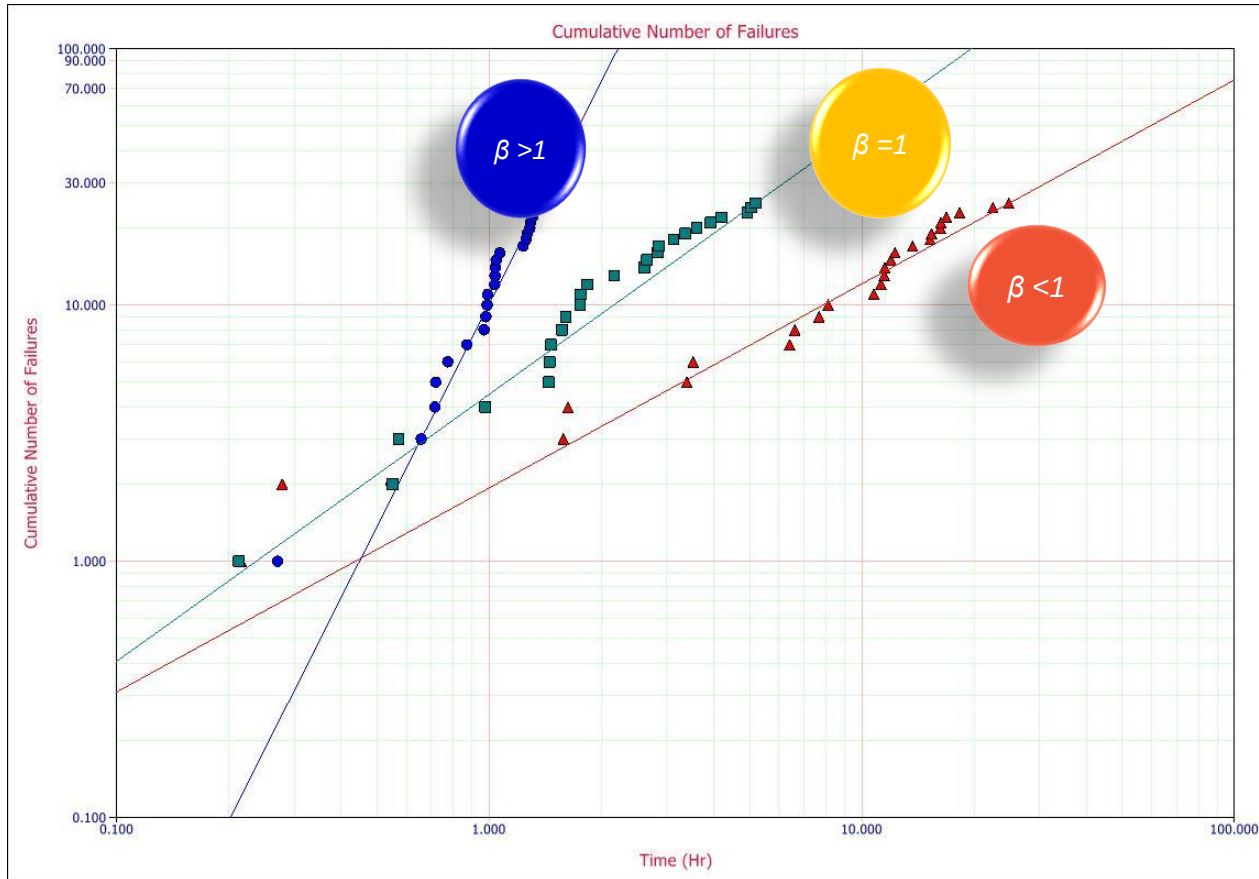
$$\hat{\beta} = \frac{r}{r \ln(T) - \sum_{i=1}^r \ln(T_i)}$$
$$\hat{\lambda} = \frac{r}{T^{\hat{\beta}}}$$

Where

r - number of failure

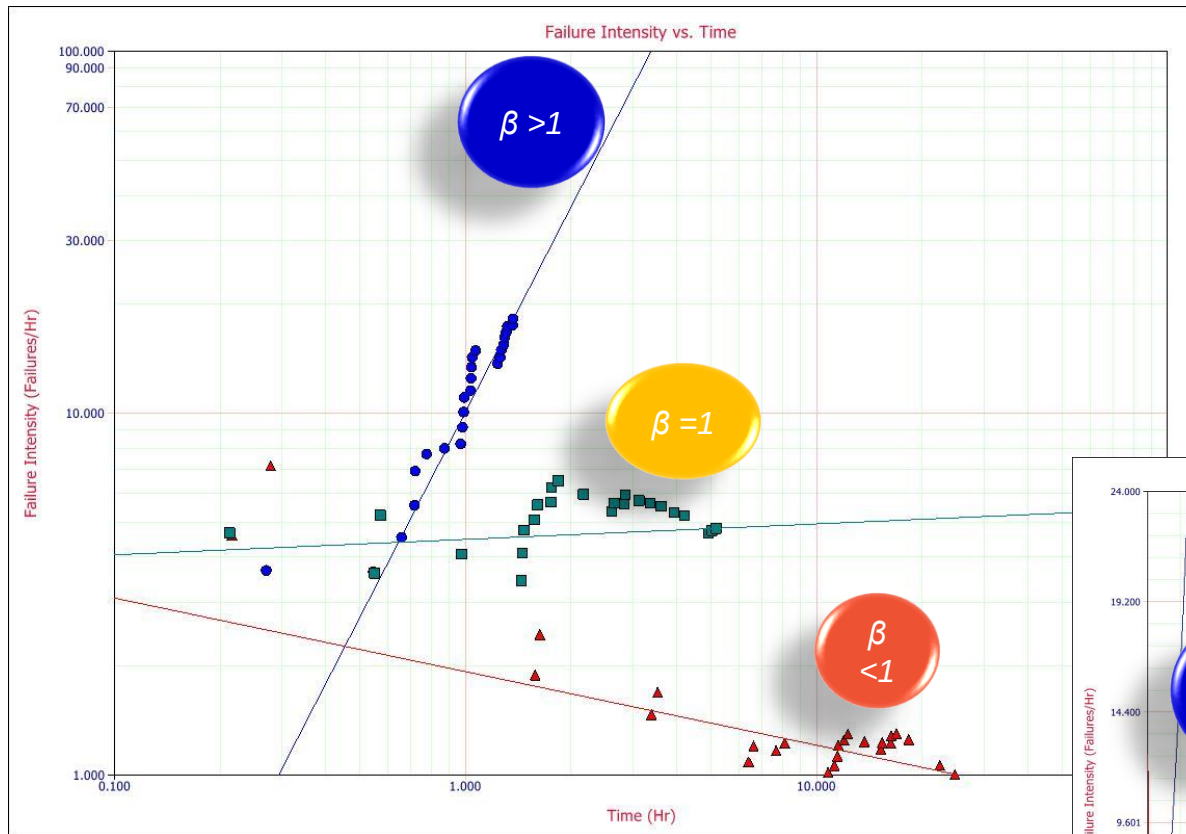
T - observation time

Power Law



$$E[N(t)] = M(t) = \lambda t^{\beta}$$

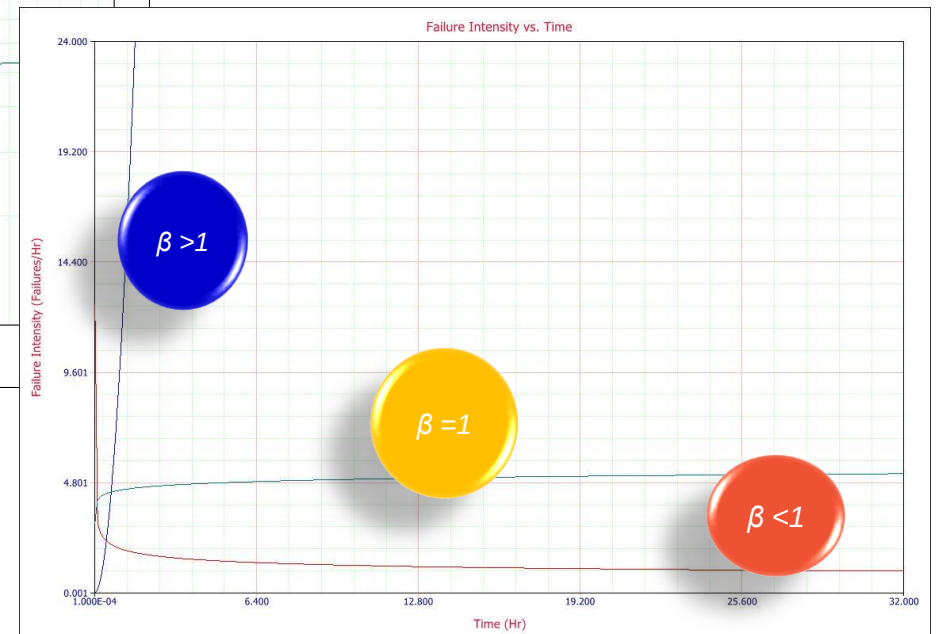
Power Law



Log-Log Scale

Failure intensity function

$$u(t) = M'(t) = \lambda \beta t^{\beta-1}$$



Linear Scale

Power Law-Example

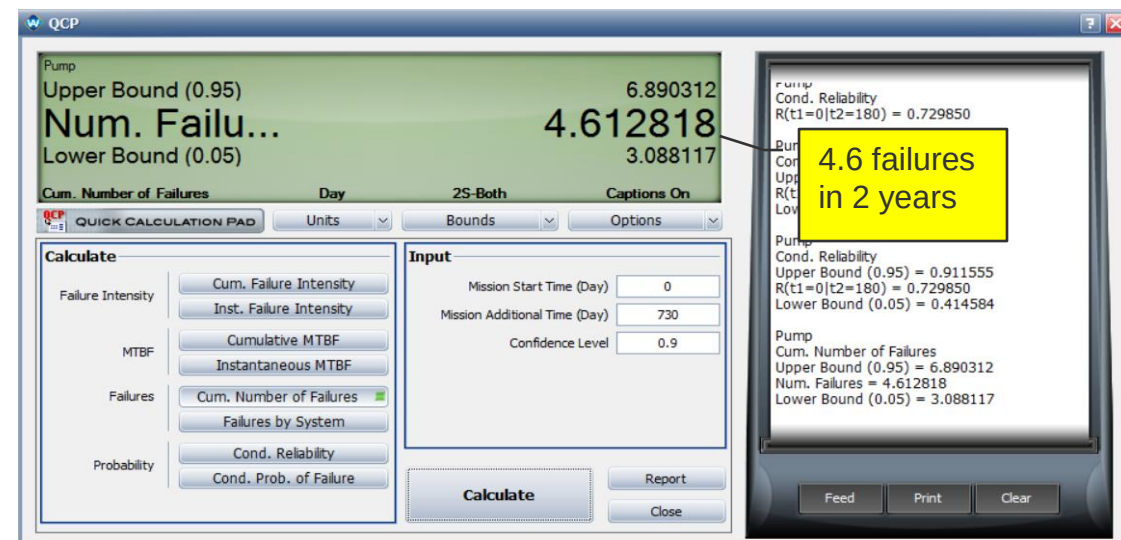
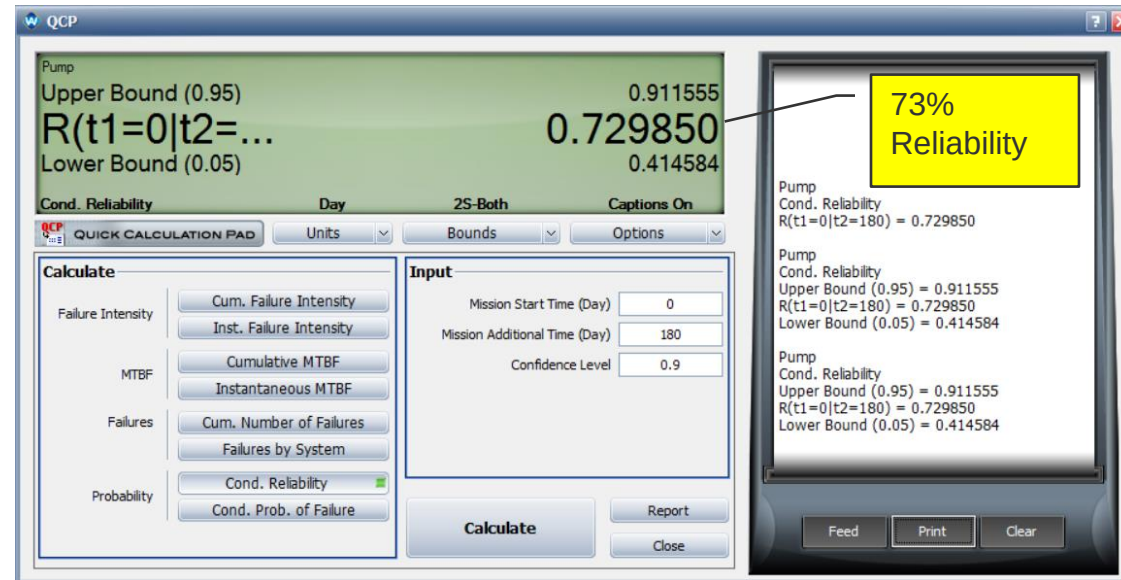
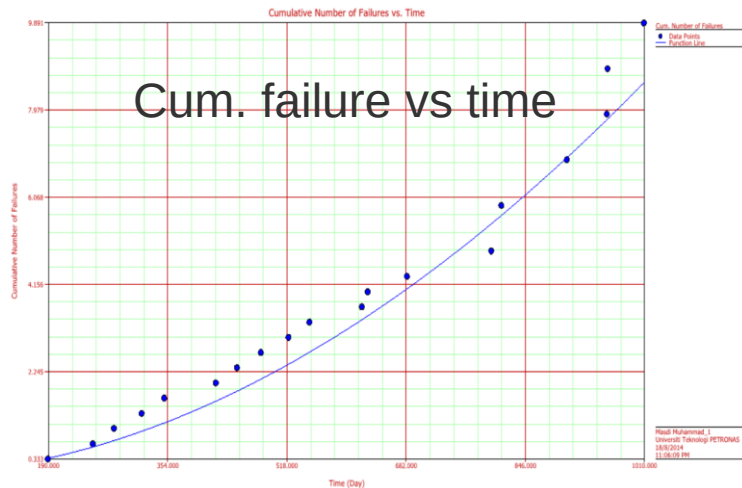
- Calculation can also be performed with Weibull ++
- Failure data for 3 pumps running at similar condition (A,B,C) with the following **cumulative time to failure.**

	System ID	Event F or E	Time to Event
1	A	F	281
2	A	F	421
3	A	F	450
4	A	F	483
5	A	F	550
6	A	F	630
7	A	F	800
8	A	F	814
9	A	F	904
10	A	F	959
11	A	F	960
12	A	F	1010
13	A	E	1080
14	B	F	190
15	B	F	319
16	B	F	521
17	B	F	622
18	B	E	730
19	C	F	252
20	C	F	350
21	C	F	684
22	C	E	730
23			

Power Law

- Apply the model to answer the following questions:
 - What is the probability there is no failure in 6 months time (180 days) operation?
 - How many times that the system will fail within 2 years?

Note that the reliability is conditional



Example using Weibull ++

- To show the error in using LDA for repairable system
 - Similar data used in previous example
 - based on data the best fit distribution is Weibull 2 parameters

	State F or S	Time to F or S	Subset ID
1	F	281	A
2	F	140	A
3	F	29	A
4	F	33	A
5	F	67	A
6	F	80	A
7	F	170	A
8	F	14	A
9	F	90	A
10	F	55	A
11	F	1	A
12	F	50	A
13	S	70	A
14	F	190	B
15	F	129	B
16	F	202	B
17	F	101	B
18	S	108	B
19	F	252	C
20	F	98	C
21	F	334	C
22	S	46	C

Open Excel File

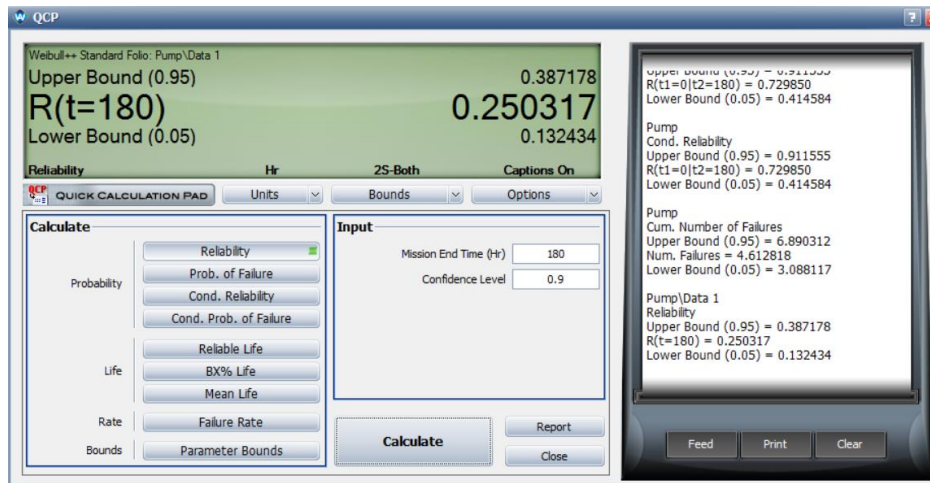
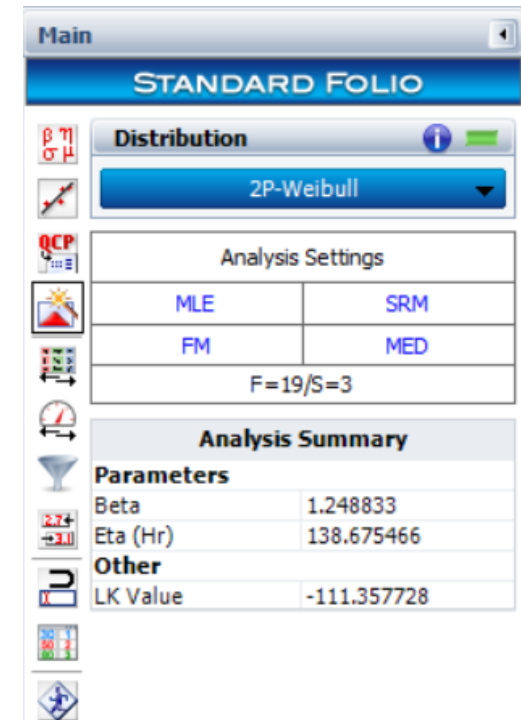
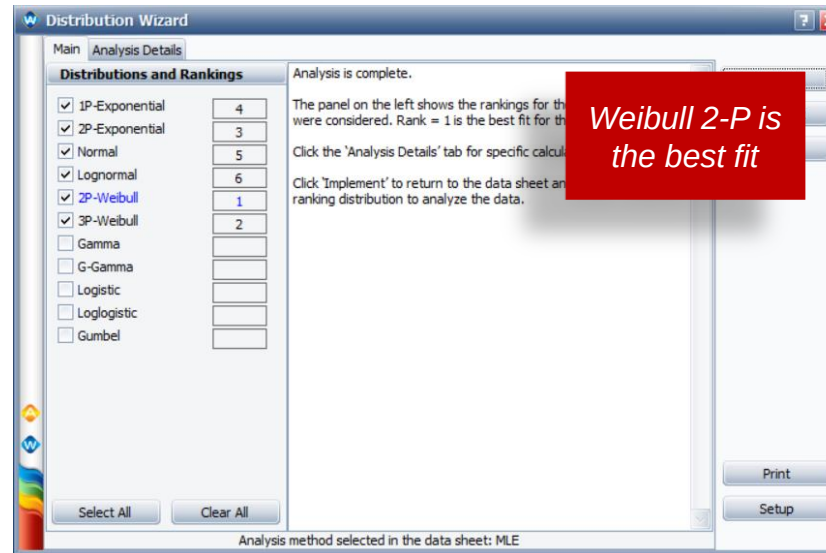
Example using Weibull ++

- Using MLE, the estimated parameters are:

$$\beta = 1.2488$$

$$\eta = 138.6755$$

- What's the reliability at 180 days?



Power Law-Example 1

Failure data for a compressor in an offshore installation is as shown in table below. The observation ends at the last failure. Estimate the number of failure 6 months (180 days) after the observation ended.

Failure No	Time between failures (Days)
1	272
2	435
3	374
4	45
5	9
6	36
7	84
8	40
9	11
10	67

Answer

System 1	
Beta	3.0600
Lambda (Day)	2.5049E-9

Power Law

Special Cases:

- Time truncated data (observation stopped at specific time)
- Occurrence of scheduled maintenance within the observation period
- If there are more than one system being observed.

Power Law-Example 2

Failure data for a compressor in an offshore installation is as shown in table below. The observation ends after 1500 days. Estimate the number of failure 6 months (180 days) after the observation ended.

Failure No	Time between failures (Days)
1	272
2	435
3	374
4	45
5	9
6	36

Answer

Beta	1.6665
Lambda (Day)	2.95E-5

$$\hat{\beta} = \frac{r}{r \ln(T) - \sum_{i=1}^r \ln(T_i)}$$

$$\hat{\lambda} = \frac{r}{T^{\hat{\beta}}}$$

Power Law - Multiple System

For multiple system or fleet, the systems have:

- same configuration
- same maintenance policy
- same operational environment.

With the assumption that the observation starts at $t=0$, same observation period, T , for K number of system

$$\hat{\lambda} = \frac{\sum_{q=1}^K N_q}{KT^{\beta}}$$
$$\hat{\beta} = \frac{\sum_{q=1}^K N_q}{\sum_{q=1}^K \sum_{i=1}^{N_q} \ln(\frac{T}{X_{iq}})}$$

Power Law- Example 3

Failure data for a compressor in an offshore installation is as shown in table 1. The observation ends after 1500 days. Another similar system is also observed for the same period and the failures are given in table 2. Estimate the number of failure 6 months (180 days) after the observation ended.

Table 1

Failure No	Cum Time
1	433
2	680
3	703

Table 2

Failure No	Cum Time
1	272
2	707
3	1081

Answer

Beta	1.08
Lambda (Day)	0.0011

Estimates from Power Law Model

- Expected number of failure between periods (T_1 and T_2)

$$E\{N(T_2) - N(T_1)\} = \lambda (T_2^\beta - T_1^\beta)$$

- Expected Reliability between periods (T_1 and T_2)

$$R(T_1, T_2) = e^{-\lambda(T_2^\beta - T_1^\beta)}$$

- The probability that the system will survive to age ($t + d$) without failure is given by:

$$R(t) = e^{-[\lambda(t+d)^\beta - \lambda t^\beta]}$$

This is the mission reliability for a system of age t and mission length d

- MTBF between periods (T_1 and T_2)

$$\text{MTBF}(T_1, T_2) = \frac{(T_2 - T_1)}{\lambda(T_2^\beta - T_1^\beta)}$$

Optimum Overhaul

In addition to reliability metrics, RDA can also be used to determine the optimum time to overhaul (lowest cost)

Overhaul policy only make sense if $\beta > 1$ (increasing failure rate)

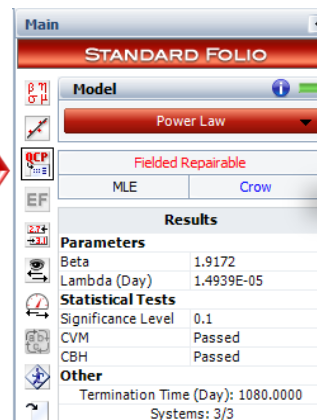
Data needed:

- Cost of overhaul
- Cost of repair (including consequential cost of failure)

Optimum Overhaul

Recall the pump failure data. Assume cost of unscheduled repair is RM200 per failure and cost of overhaul is RM20,000.

F-RS	System ID	Event	Time to Event (Day)
1	A	S	0
2	A	F	281
3	A	F	421
4	A	F	450
5	A	F	483
6	A	F	550
7	A	F	630
8	A	F	800
9	A	F	814
10	A	F	904
11	A	F	959
12	A	F	960
13	A	F	1010
14	A	E	1080
15	B	S	0
16	B	F	190
17	B	F	319
18	B	F	521
19	B	F	622
20	B	E	730
21	C	S	0
22	C	F	252
23	C	F	350
24	C	F	684
25	C	E	730



STANDARD FOLIO

Model: Power Law

Fielded Repairable

MLE Crow

Results

Parameters

Beta 1.9172

Lambda (Day) 1.4939E-05

Statistical Tests

Significance Level 0.1

CVM Passed

CBH Passed

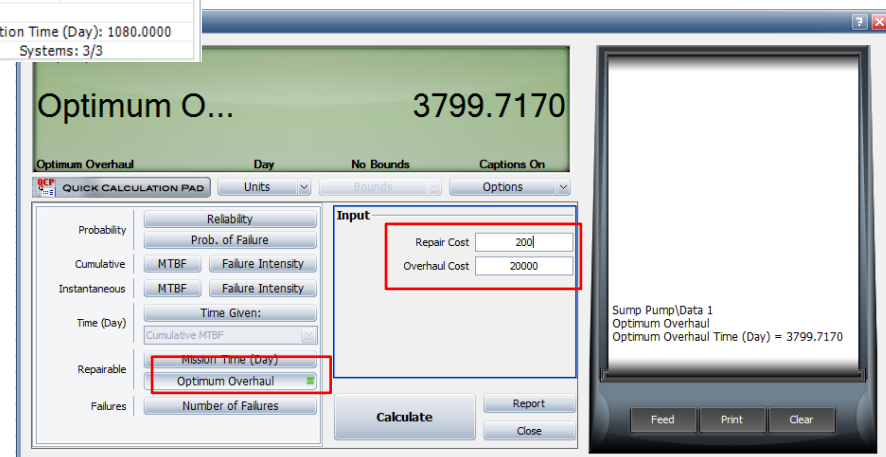
Other

Termination Time (Day): 1080.0000

Systems: 3/3

Equation for optimum overhaul time where C1 is cost for unscheduled repair and C2 is cost of overhaul

$$T_0 = \left[\frac{C_2}{\lambda(\beta - 1)C_1} \right]^{1/\beta}$$



Optimum O... 3799.7170

Optimum Overhaul Day No Bounds Captions On

QUICK CALCULATION PAD Units Bounds Options

Probability Reliability Prob. of Failure

Cumulative MTBF Failure Intensity

Instantaneous MTBF Failure Intensity

Time (Day) Time Given: Cumulative MTBF

Repairable Mission Time (Day) Optimum Overhaul

Failures Number of Failures

Input

Repair Cost 200

Overhaul Cost 20000

Calculate Report Close

Sump Pump(Data 1
Optimum Overhaul
Optimum Overhaul Time (Day) = 3799.7170

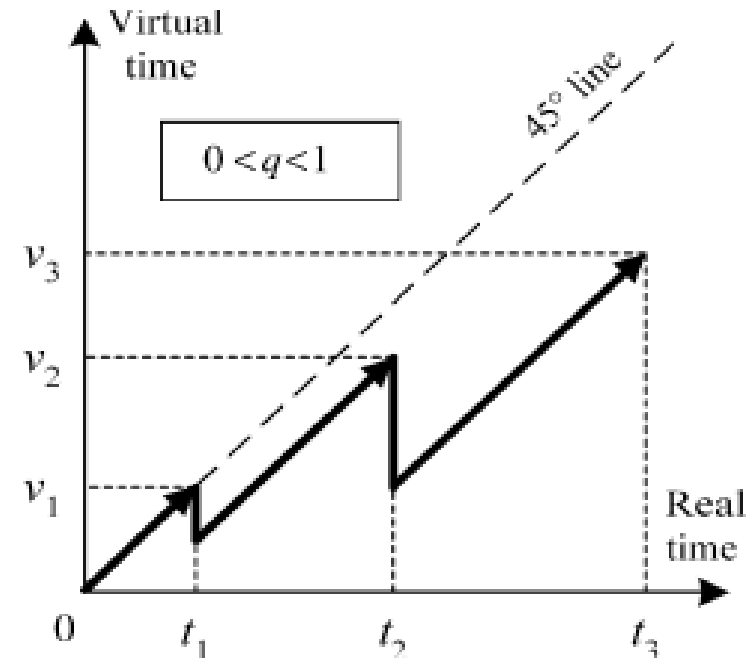
Feed Print Clear

Generalized Renewal Process

Recall that most repairs result in equipment being in between 'AGAN' and 'ABAO'.

Any failure is caused by faulty component. Upon replacement, the whole system's age will be 'newer' than old due to new component being replaced.

The actual age and thus system's reliability is estimated using GRP.



GRP Models

There are two types of GRP model

– Type I:

- Assumes the repair can only remove the damage incurred from last repair

$$v_i = v_{i-1} + qx_i$$

x_i – failure times
 v_i - virtual age of the system right after i repair
 q - repair effectiveness factor

– Type II

- Assumes the repair can remove all cumulative damage

$$v_i = q(v_{i-1} + x_i)$$

x_i – failure times
 v_i - virtual age of the system right after i repair
 q - repair effectiveness factor

GRP Example-Data

Pump data with cumulative time to failure in days. Observation ends at 1539 days.

C30			
	System ID	Event F or E	Time to Event
1	1	F	50
2	1	F	94
3	1	F	196
4	1	F	268
5	1	F	290
6	1	F	329
7	1	F	332
8	1	F	347
9	1	F	544
10	1	F	632
11	1	F	681
12	1	F	809
13	1	F	945
14	1	F	958
15	1	F	966
16	1	F	991
17	1	F	1013
18	1	F	1152
19	1	F	1362
20	1	F	1459
21	1	F	1489
22	1	F	1512
23	1	F	1525
24	1	F	1529
25	1	E	1539
26			
27			
28			
29			
30			
31			

GRP Example

GRP assumes that the times to first failure (TTFF) follow a Weibull distribution

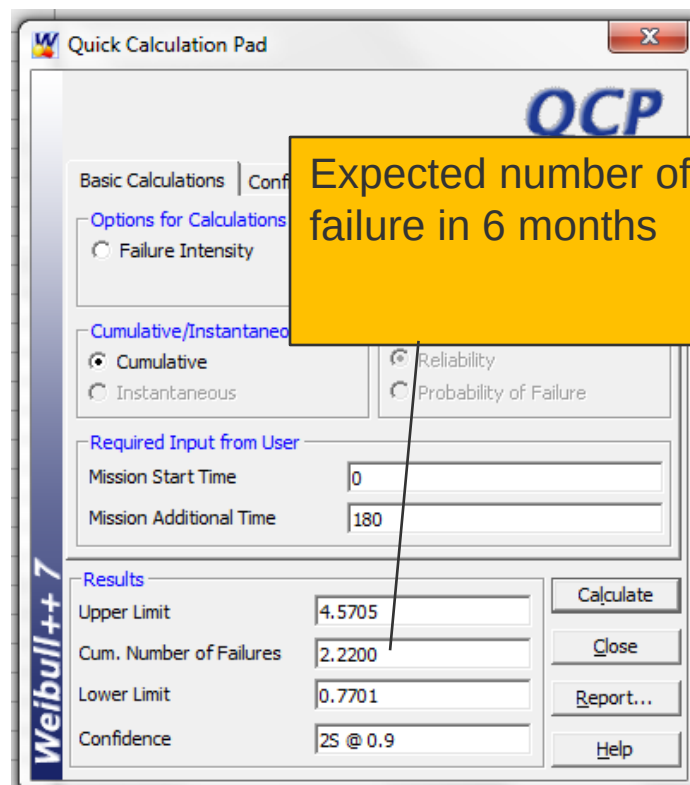
Using Weibull ++ the parameters estimated :

$$\beta = 1.2111$$

$$\lambda = 0.0048$$

$$\eta = \left(\frac{1}{\lambda} \right)^{\frac{1}{\beta}} = 82.168$$

$$q = 0.0920$$



Quick Calculation Pad

Basic Calculations | Conf

Options for Calculations

☐ Failure Intensity

Cumulative/Instantaneous

☒ Cumulative

☐ Instantaneous

Reliability

☐ Reliability

☒ Probability of Failure

Required Input from User

Mission Start Time: 0

Mission Additional Time: 180

Results

Upper Limit: 4.5705

Cum. Number of Failures: 2.2200

Lower Limit: 0.7701

Confidence: 2S @ 0.9

Calculate

Close

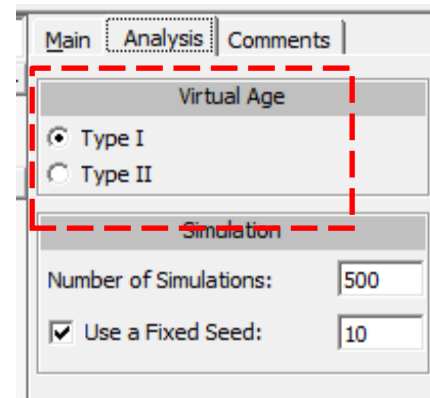
Report...

Help

Weibull++ 7

OCP

Expected number of failure in 6 months



Main | Analysis | Comments

Virtual Age

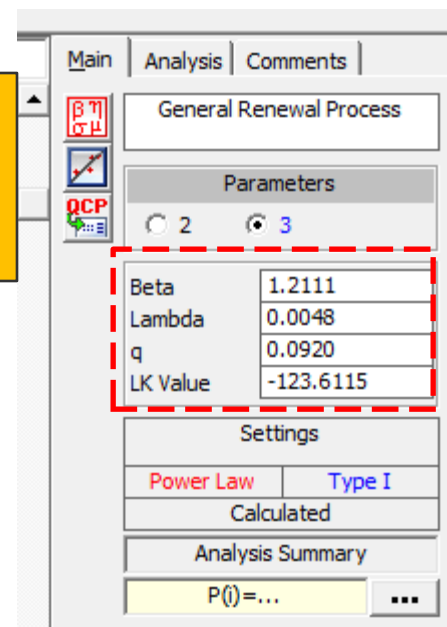
☒ Type I

☐ Type II

Simulation

Number of Simulations: 500

☒ Use a Fixed Seed: 10



Main | Analysis | Comments

General Renewal Process

Parameters

Beta: 1.2111

Lambda: 0.0048

q: 0.0920

LK Value: -123.6115

Settings

Power Law | Type I

Calculated

Analysis Summary

P(i)=...

Compare GRP and Power Law

With different q values, Weibull++ can apply both GRP and Power Law

Results comparison as follows:

GRP

$$\beta = 1.2111$$

$$\lambda = 0.0048$$

$$\eta = 78.26$$

$$q = 0.0920$$

Power Law

$$\beta = 1.0680$$

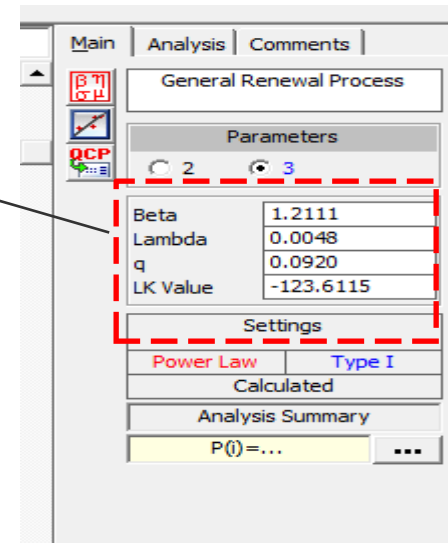
$$\lambda = 0.0095$$

$$\eta = \left(\frac{1}{\lambda} \right)^{\frac{1}{\beta}} = 82.168$$

$$q = 1.00$$

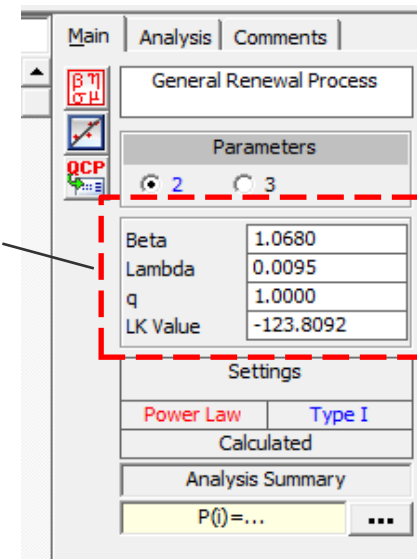
GRP

NHPP



Parameters	
Beta	1.2111
Lambda	0.0048
q	0.0920
LK Value	-123.6115

Settings	
Power Law	Type I
Calculated	
Analysis Summary	
P(i)=...	



Parameters	
Beta	1.0680
Lambda	0.0095
q	1.0000
LK Value	-123.8092

Settings	
Power Law	Type I
Calculated	
Analysis Summary	
P(i)=...	

Compare GRP and Power Law

NHPP

GRP

Quick Calculation Pad

QCP

Basic Calculations | Confidence Bounds

Options for Calculations

☐ Failure Intensity ☒ Number of Failures

☐ Conditional Probability

Cumulative/Instantaneous

☒ Cumulative ☐ Instantaneous

Results Options

☒ Reliability ☐ Probability of Failure

Required Input from User

Mission Start Time: 0

Mission Additional Time: 180

Parameters

Beta: 1.0680
Lambda: 0.0095
q: 1.0000
LK Value: -123.8092

Settings

Power Law | Type I

Calculated

Analysis Summary

P(t)=...

Results

Upper Limit: 5.6169
Cum. Number of Failures: 2.4259
Lower Limit: 1.0478
Confidence: 2S @ 0.9

Calculate
Close
Report...
Help

Weibull++ 7

No of failure in the next 6 months is 2.4259

Quick Calculation Pad

QCP

Basic Calculations | Confidence Bounds

Options for Calculations

☐ Failure Intensity ☒ Number of Failures

☐ Conditional Probability

Cumulative/Instantaneous

☒ Cumulative ☐ Instantaneous

Results Options

☒ Reliability ☐ Probability of Failure

Required Input from User

Mission Start Time: 0

Mission Additional Time: 180

Parameters

Beta: 1.2111
Lambda: 0.0048
q: 0.0920
LK Value: -123.6115

Settings

Power Law | Type I

Calculated

Analysis Summary

P(t)=...

Results

Upper Limit: 4.5705
Cum. Number of Failures: 2.2200
Lower Limit: 0.7701
Confidence: 2S @ 0.9

Calculate
Close
Report...
Help

Weibull++ 7

No of failure in the next 6 months is 2.2200

Summary

- For reliability analysis results to be useful, proper methods discussed earlier need to be followed.
- Wrong assumptions could lead to costly conclusions.
- Data integrity is still the prerequisite for useful results.

References

1. Feingold, H., and Ascher H. Repairable systems reliability: Modelling, inference, misconceptions and their causes. New York: Marcel Dekker, 1984
2. Louit, D.M., Paccual, R. and Jardine, A.K.S., A practical procedure for the selection of time-to-failure models based on assessment of trends in maintenance data, Reliability Engineering & System Safety, 2009: 1618-1628
3. Mettas, A. and Zhao, W. "Modeling and analysis of repairable system with general repair." Annual Reliability and Maintainability Symposium. Alexandria, Virginia: IEEE, 2005
4. Wolstenholme, L.C. Reliability modeling: A statistical approach. Chapman & Hall, 1999
5. Yanez, M., Joglar, F. and Modarres, M. "Generalized renewal process for analysis of repairable system with limited failure experience." Reliability Engineering and System Safety, 2002: 167-180

Session 2

System reliability Modelling

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Session schedule

Session	Content	Main Topics	Day	Time
0	Introduction	- Program Agenda and Course Introduction - Pre-test	Day 1	9:00am - 9:30am
1.	Recurrent Data Analysis (RDA)- Dr Masdi (AM) Dr Ainul (PM)	- Repairable versus non-repairable - Trend tests - Analytical Trend Test (Laplace and Mann) - HPP versus NHPP - Crow-AMSAA Model for NHPP - Parameter Estimation - Single system versus multiple system - Cumulative versus instantaneous failure rate - Predicting number of failures - PM and Overhaul - Change of Slope Analysis - Optimum Overhaul	Day 1	9:30am - 10:30am Break (15 Min) 10:45am – 1:00pm Lunch/Prayer (1:00pm – 2:00pm) 2:00pm – 3:30pm Break (15 Min) 3:45pm – 5:00pm
2.	System Reliability Modeling - Dr Hilmi (AM) Dr Ainul (PM)	- Definition of system - Reliability block diagram (RBD) - Monte-Carlo Simulation - Failure model - Repair Model - Repairable system in RBD - System Availability (RAM Modelling) - Case Study	Day 2	9:00am - 10:30am Break (15 Min) 10:45am – 1:00pm Lunch/Prayer (1:00pm – 2:00pm) 2:00pm – 3:30pm Break (15 Min) 3:45pm – 5:00pm

Session Objectives

- Understand the concept and methods of system reliability modelling
- Able to create RBD model for system (including standby and load sharing)
- Able to estimate system availability using simulation
- Able to estimate system throughput and operating costs



System Reliability

A system is defined as a collection of subsystems, assemblies and/or components arranged in a specific design to achieve the desired functionality.

Your definition of a “system” can be the level that is of interest to you.

Given a system made up of several components, our interest is to determine the reliability of the system.

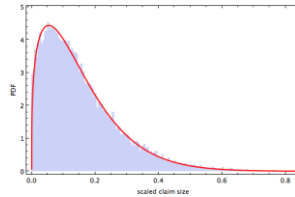
The reliabilities of the components can be estimated using:

- Accelerated life testing
- Available database (OREDA, CPSS, US MILITARY HANDBOOK 217, OEM etc)
- Field data
- Engineering knowledge (expert elicitation)

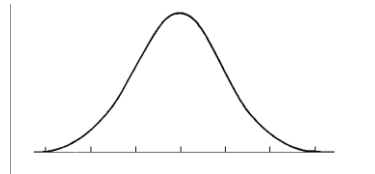
System Reliability

Component reliability is quantified using a probability density function (pdf).

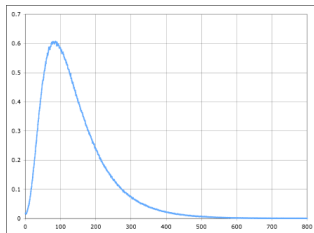
Use the component models to create a system model.



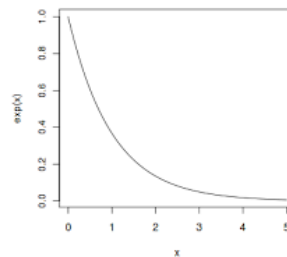
Bearing- Weibull Distribution



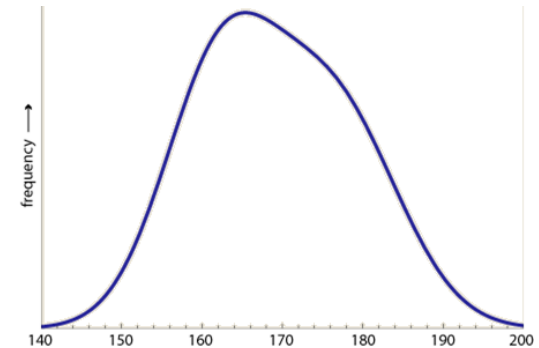
Shaft- Normal Distribution



Mech Seal- Lognormal Distribution



Impeller- Normal Distribution



System Distribution

Reliability Block Diagram

A reliability block diagram (RBD) is a graphical representation of the components/subsystems of the system and how they are connected “reliability-wise”.

It can be thought as a reliability logic diagram.

It consists of

- Blocks representing the components of the system



- Lines connecting the blocks
 - The structure of these connections define the reliability configuration of the system



RBD Block

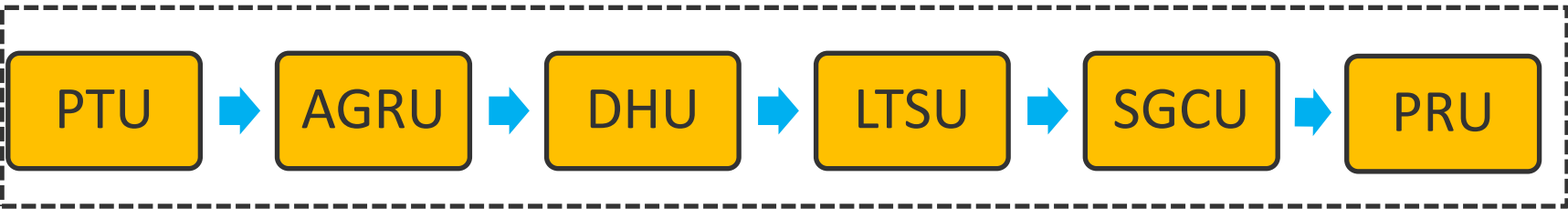
A block can be:

- A component
- An assembly (a system or a subsystem)

Reliability blocks have properties including

- Time-to-failure distribution (minimum requirement)
- Time-to-repair distribution
- Other maintenance and inspection requirements
- Throughput properties
- Repair costs

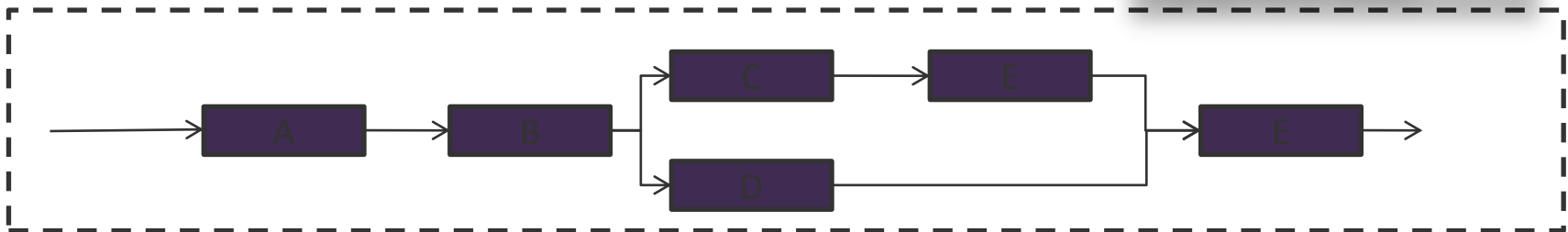
Example of RBD for Acid Gas Removal Unit (AGRU)



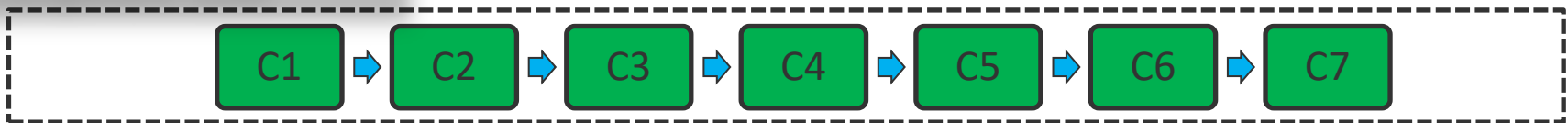
Sub-system Level



Equipment Level



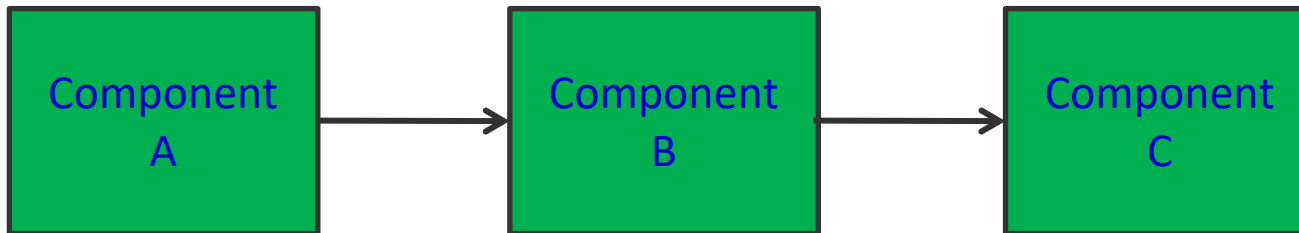
Component Level



Creating RBD

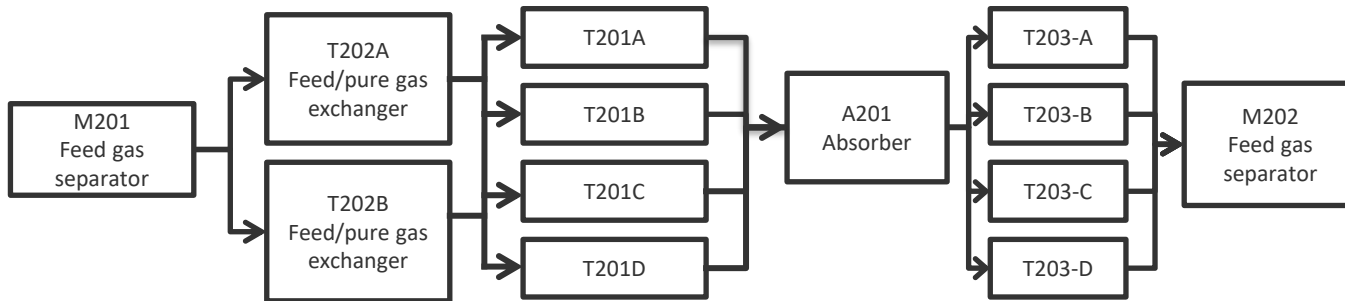
Once the block characteristics are determined, they can then be connected in a reliability-wise manner to create a RBD for the system.

Once the RBD of the system has been created, it can be analyzed to estimate the system reliability and availability



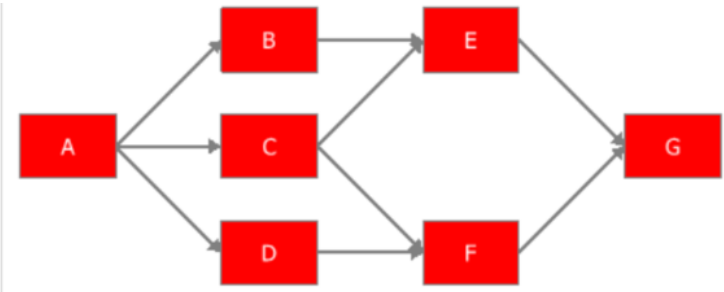
Types of Configurations

- Series
- Parallel
- Combination



Example RBD for Acid Gas Removal Unit (Equipment Level)

- Complex
System can not be Broken down into groups of series or parallel system



Series Configuration

- Blocks are connected in a series.
- It can be thought of as an “OR” relationship (i.e. The system fails if A OR B fails).
- It implies no redundancy in the components.
- If units are in series, then all units must for the system to work. If any unit in the series fails, then the system fails.
- The reliability of the system is given by:

$$R_s = R_1 \times R_2 \times \dots \times R_n$$



Reliability Calculation for Series System

Calculate system reliability given $R_1 = 0.90$, $R_2 = 0.95$ and $R_3 = 0.98$.



$$\begin{aligned}
 R_s &= R_1 \times R_2 \times R_3 \\
 &= (0.90)(0.95)(0.98) \\
 &= 0.8379
 \end{aligned}$$



Reliability Calculation for Series System

What is the system reliability and failure rate?

Assuming that the components are having a **constant failure rate**.

Then, the system reliability is

$$\begin{aligned}
 R_s(t) &= R_1(t) \times R_2(t) \times R_3(t) \\
 &= e^{-\lambda_1 t} \times e^{-\lambda_2 t} \times e^{-\lambda_3 t} \\
 &= e^{-(\lambda_1 + \lambda_2 + \lambda_3)t}
 \end{aligned}$$



So, the failure rate for the system is

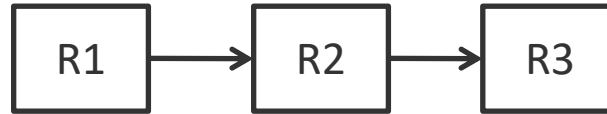
$$\lambda_s = \lambda_1 + \lambda_2 + \lambda_3$$

Exercise for Series System

- Consider a system with three components in series. All the components are mutually independent. From the records, it showed that all the components follow random failure pattern.
 - You are required to achieve a system reliability of 0.98 over a 800-hours non-stop operation.
1. What would be the target failure rate for the system?
 2. What would be the system MTBF?
 3. Assuming the component failures are identically distributed,
 - a) What should be the component failure rate?
 - b) What would be the component MTBF?
 - c) What should be the component reliability?

Exercise for Series System

Consider a system with three components in series.

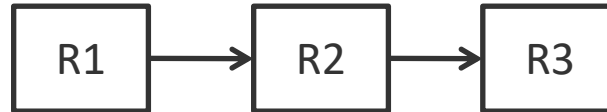


You are required to achieve a system reliability of 0.98 over a 800-hours non-stop operation.

1. What would be the target failure rate for the system?

Exercise for Series System

Consider a system with three components in series.



You are required to achieve a system reliability of 0.98 over a 800-hours non-stop operation.

2. What would be the system MTBF be?

Exercise for Series System

3. Assuming the component failures are identically distributed,

a) What should be the component failure rate?



b) What would be the component MTBF?

c) What should be the component reliability?

Parallel Configuration

A system will fail when all the units fail. Also known as active redundancy.

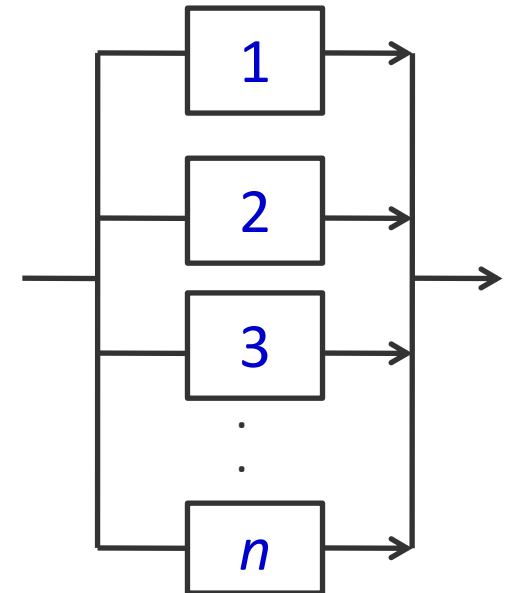
Assumption: probabilities of failure for the various subsystems are totally independent for the entire mission time

It can be thought of as an “AND” relationship (i.e. the system fails if 1 and 2 and ... and n fail)

At least one unit must succeed for a successful mission.

The reliability of the system is given by:

$$R_s = 1 - [(1-R_1) \times (1-R_2) \times \dots \times (1-R_n)]$$

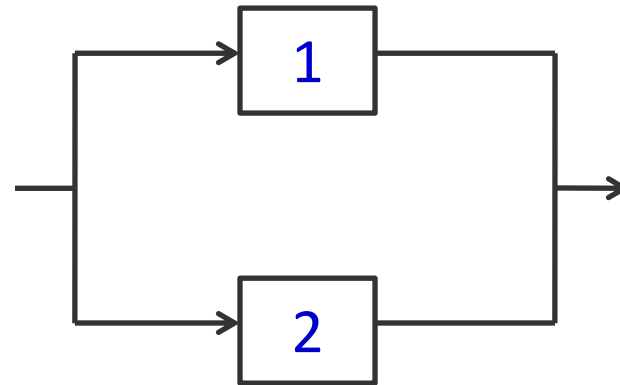




Reliability Calculation for Parallel System

Calculate system reliability given $R_1 = 0.90$ and $R_2 = 0.98$.

$$\begin{aligned} R_S &= 1 - [(1 - R_1)(1 - R_2)] \\ &= 1 - [(1 - 0.90)(1 - 0.98)] \\ &= 1 - (0.10)(0.02) \\ &= 1 - 0.002 \\ &= 0.998 \end{aligned}$$

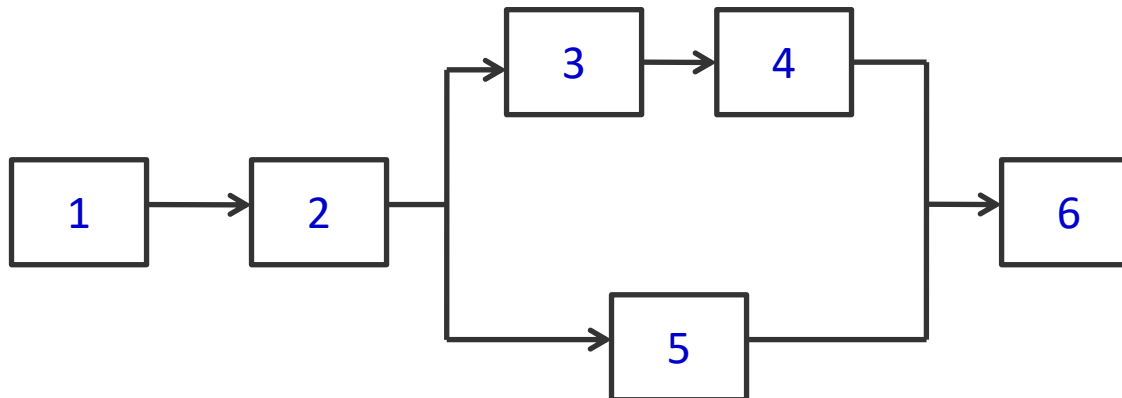


Combination of Basic Configurations

Also known as series-parallel model

The configuration types can be used simultaneously in one diagram.

Consider a system having subsystems.



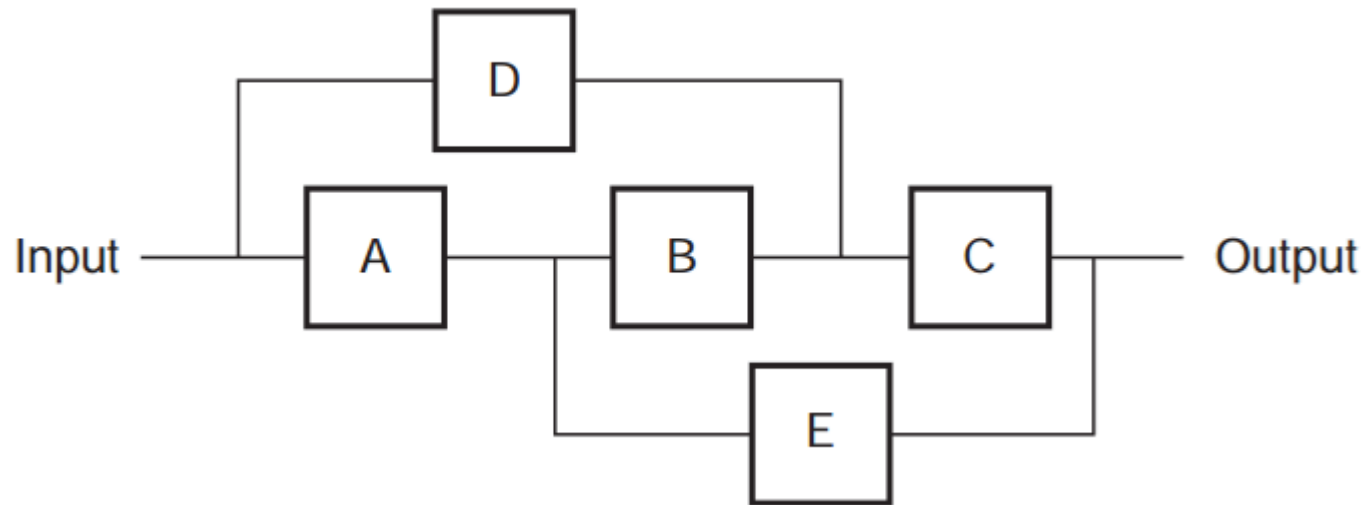
Steps to calculate system reliability for combined series-parallel configuration

1. Break the system into smaller series and parallel arrangements.
2. Calculate reliability of each arrangement identified in step 1.
3. Finally, calculate R_S using the reliability obtained in step 2.



Analysis For Complex System

Complex system is a system that can not be reduced to simple system.



Can be solved using Bayes' Theorem and truth table method

Analysis For Complex System-Bayes' Theorem

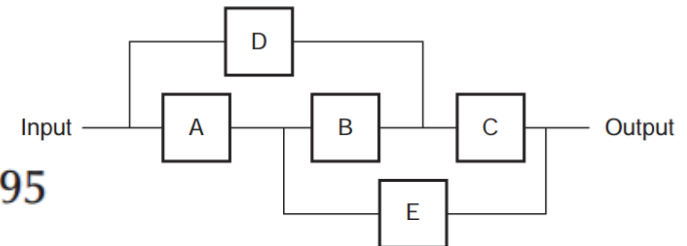
- Choose one subsystem. With that subsystem assumed to be first in a success state,
- Reduce the remaining system into a series– parallel arrangement
- Find the reliability of the remaining system and multiply by the probability that the subsystem has not failed.
- Let subsystem assumed to be in a failed state, find the reliability of the remaining system and multiply by the probability that the subsystem has failed.
- The sum of these two values is the system reliability.

Bayes' Theorem-Example

The subsystem reliability values are:

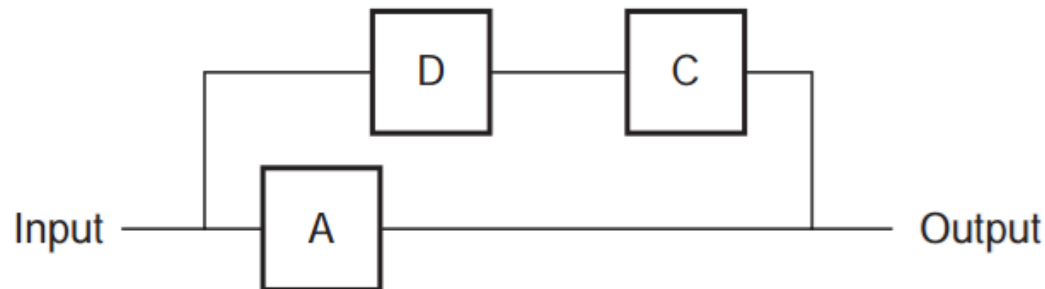
$$R_A = R_B = R_C = 0.95$$

$$R_D = R_E = 0.99$$



Choose subsystem E.

Assume E is in a success state. The system model will reduce to

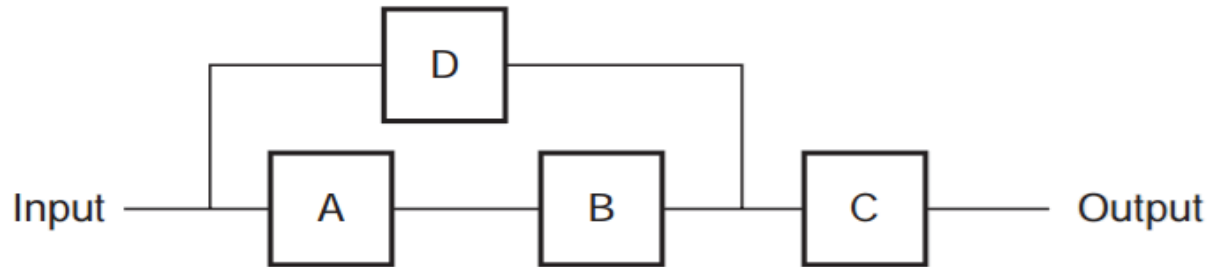


and the system reliability with E good is

$$R_{(E \text{ success})} = 1 - [1 - (.99)(.95)][1 - .95] = 0.9970.$$

Bayes' Theorem-Example

Assume E is in a failed state. The system model will reduce to



and the system reliability with E failed is

$$R_{(E \text{ failed})} = [1 - (1 - .99)(1 - (.95)(.95))] [.95] = 0.9491.$$

Bayes' Theorem-Example

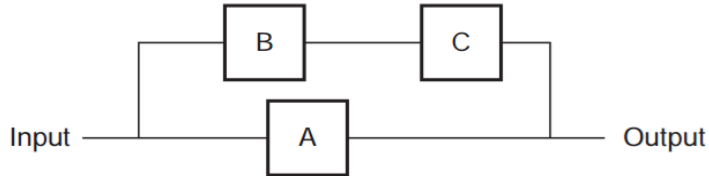
The system reliability is

$$R_{\text{System}} = R_{(\text{E success})} \times P(\text{E success}) + R_{(\text{E failed})} \times P(\text{E failed})$$

$$R_{\text{System}} = (0.9970)(0.99) + (0.9491)(1 - 0.99)$$

$$R_{\text{System}} = 0.9965.$$

Analysis using truth table



$$R_A = 0.85$$

$$R_B = 0.90$$

$$R_C = 0.92$$

The total number of events is $2^3 = 8$.

Construct a truth table using S for success and F for failure.

Event #	Subsystem			System	Event probability
	A	B	C		
1	S	S	S	S	$(.85)(.90)(.92) = 0.7038$
2	S	S	F	S	$(.85)(.90)(.08) = 0.0612$
3	S	F	S	S	$(.85)(.10)(.92) = 0.0782$
4	S	F	F	S	$(.85)(.10)(.08) = 0.0068$
5	F	S	S	S	$(.15)(.90)(.92) = 0.1242$
6	F	S	F	F	
7	F	F	S	F	
8	F	F	F	F	

The system reliability is the sum of the probabilities of the events that result in system success:

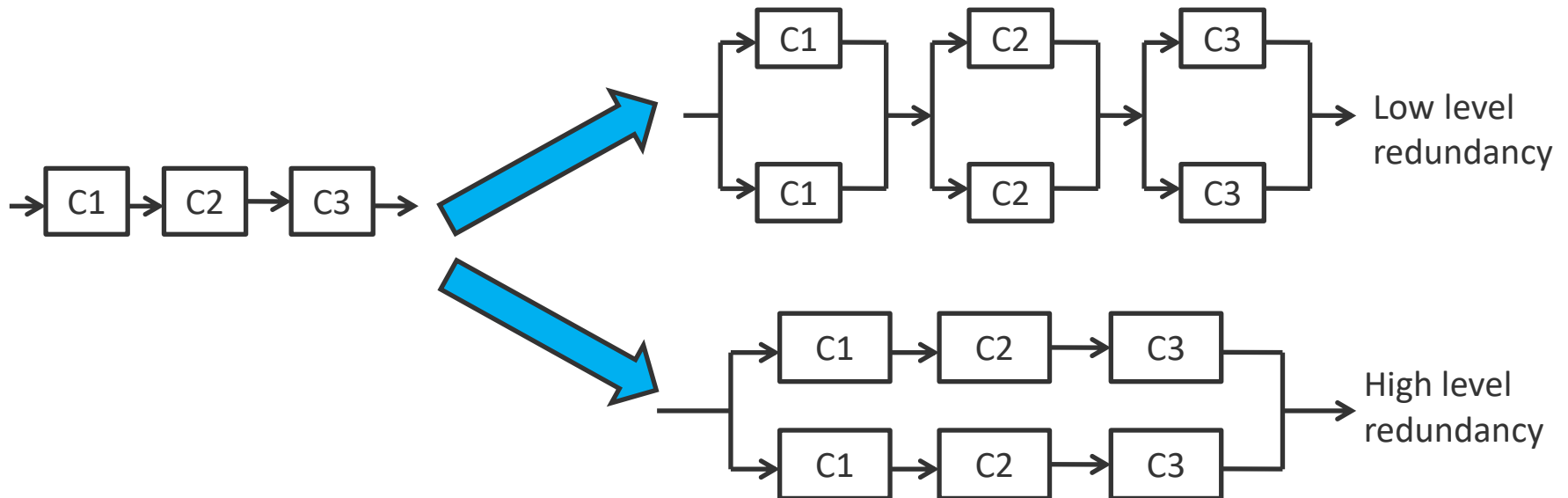
$$R_{\text{System}} = 0.7038 + 0.0612 + 0.0782 + 0.0068 + 0.1242 = 0.9742$$

Redundancy

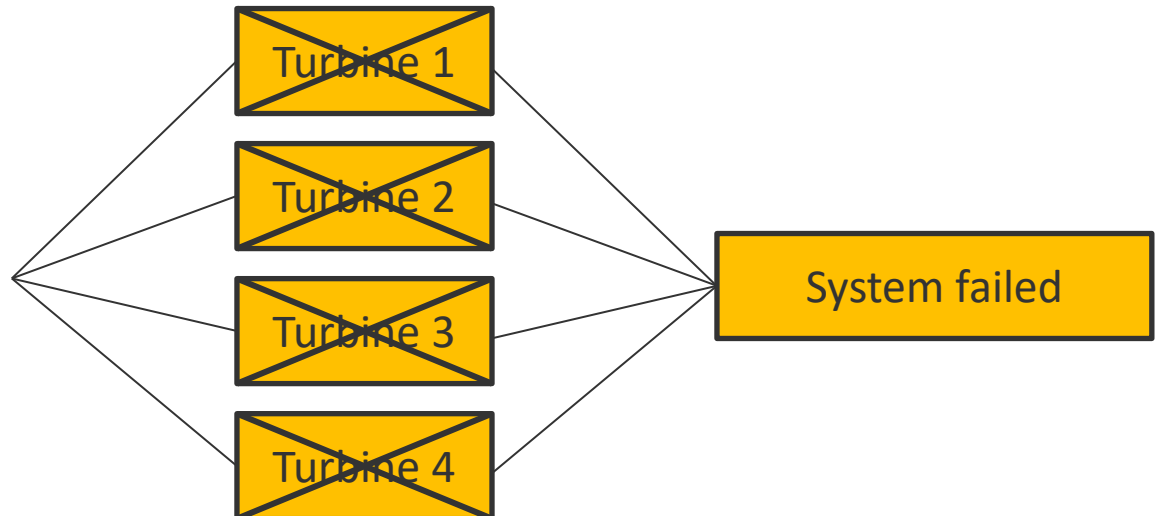
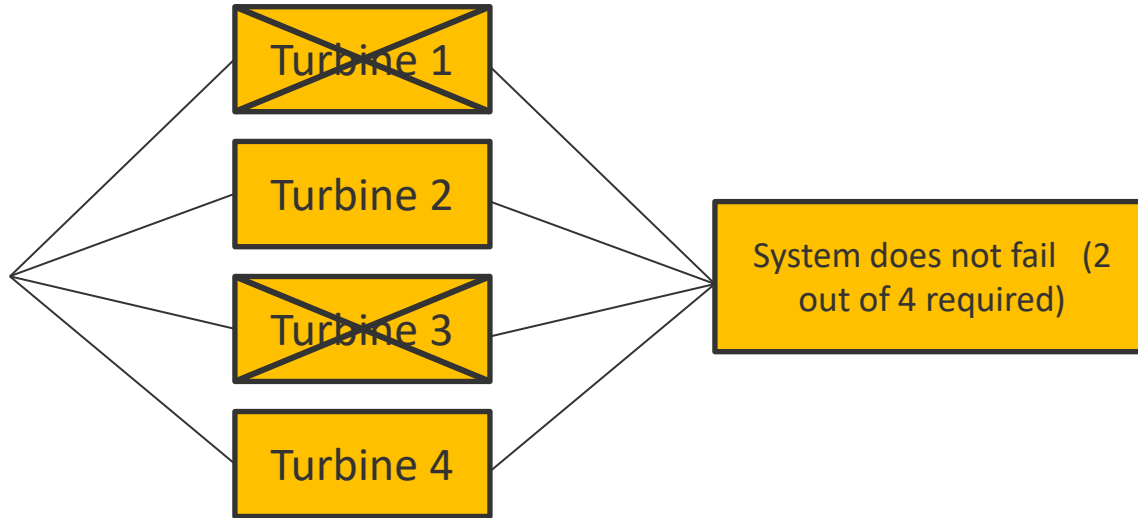
System redundancy may be obtained in 2 ways

1. Low level redundancy
 - At component level
2. High level redundancy
 - For the entire system

Consider this simple system...

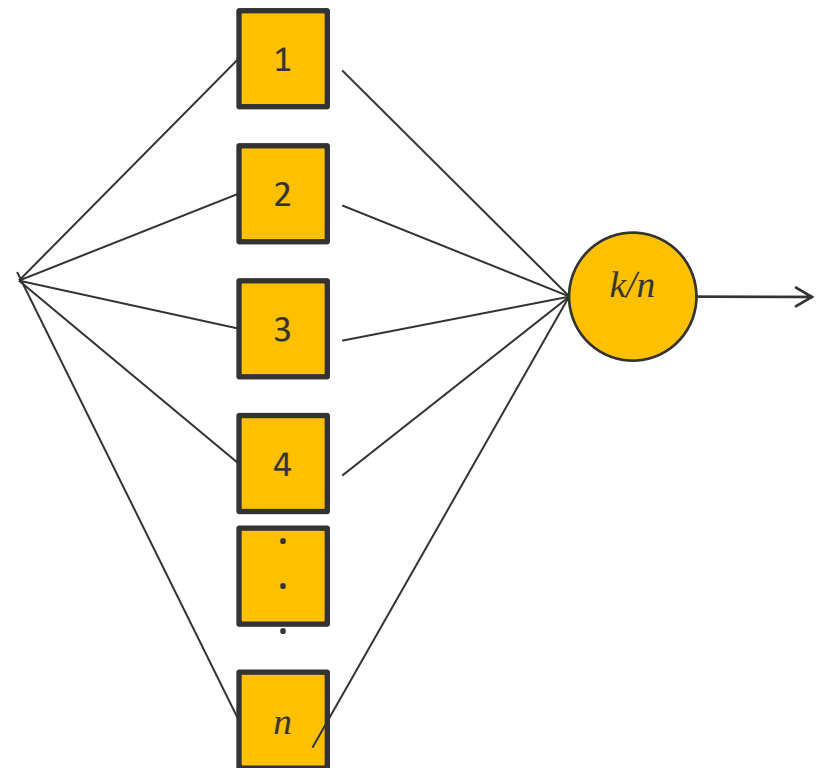
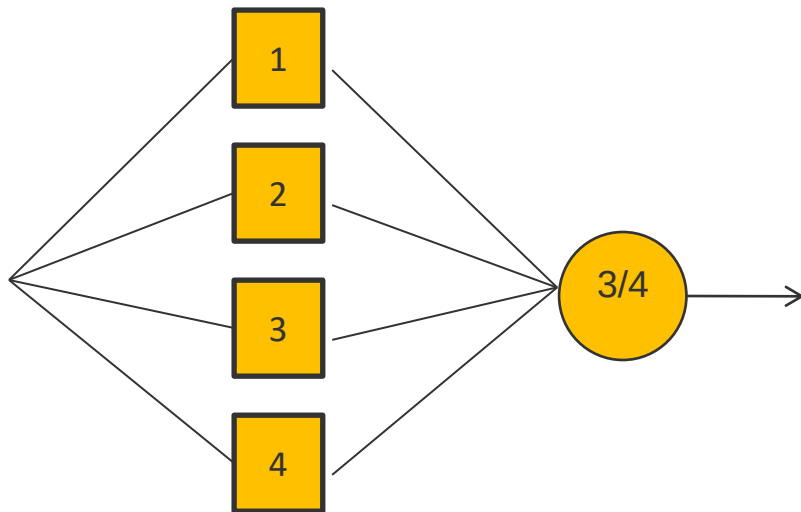


k -out-of- n Redundancy



k-out-of-*n* Redundancy

At times, a system function is such that *k*-out-of-*n* of its components need to be working for the system to function.



k -out-of- n Redundancy

A node is used to signify k -out-of- n redundancy.

The basic property of the node is to define the number of incoming paths that must be “Good” for the system to be “Good”.

k-out-of-n Redundancy

For n identical components (i.e. same reliability values), the system reliability is calculated as

$R_s = \text{Prob}(\text{at least } k \text{ components are working})$

$$= \sum_{x=k}^n P(x)$$

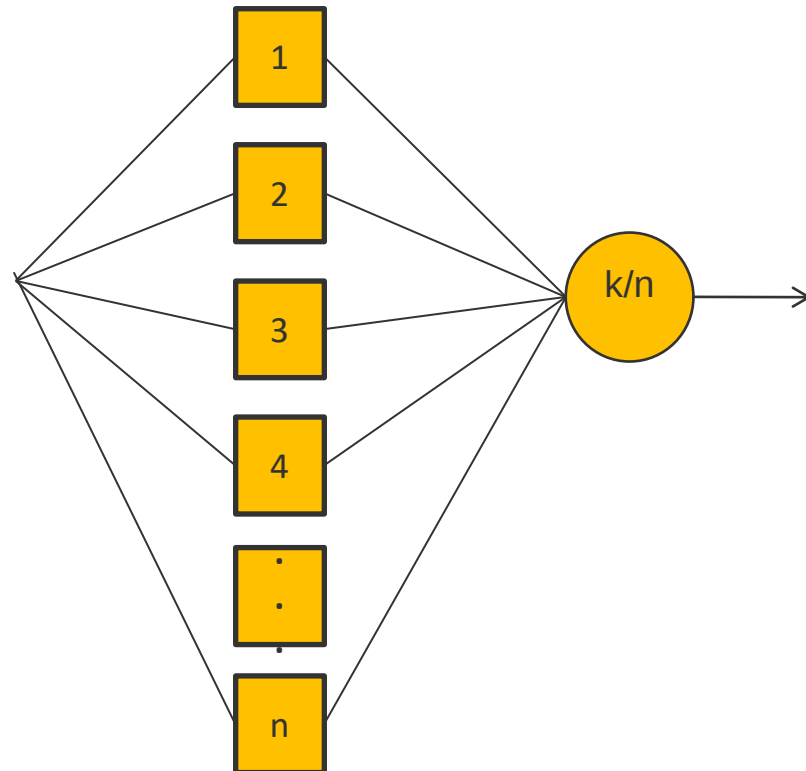
where

$$P(x) = \binom{n}{x} R^x (1-R)^{n-x}$$

and

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

Binomial
distribution



Practically, never happens. Normally, each equipment/component follows different distribution and as such, can only be solved through simulation

Example: k -out-of- n Redundancy

A high pressure boiler is mounted with 5 identical pressure relief valves. Pressure inside the boiler is successfully controlled by any three of these valves. If the failure probability of a relief valve is 0.05, compute the reliability of pressure relief valve system.

Solution: This is 3-out-of-5 system where $n = 5$, $R = 1 - 0.05 = 0.95$.

$$\begin{aligned} R_s &= \sum_{x=k}^n \binom{n}{x} R^x (1-R)^{n-x} \\ &= \frac{5!}{3!(5-3)!} \times 0.95^3 \times (1-0.95)^{5-3} + \frac{5!}{4!(5-4)!} \times 0.95^4 \times (1-0.95)^{5-4} \\ &\quad + \frac{5!}{5!(5-5)!} \times 0.95^5 \times (1-0.95)^{5-5} \\ &= 0.99884 \end{aligned}$$

RAM Modelling and Analysis using BlockSim

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Introduction

In recent years, RAM have expanded their influence in various industries and fields, thus serve as integral quality elements in the organization system and manufacturing process.

It has been established as a useful tool for risk analysis, production availability studies and design of systems.

System performance depends on reliability and availability of the system/components, operating environment, maintenance efficiency, operation process and technical expertise of operators, etc.

Availability of the system can be improved by improvement in its reliability and maintainability.

Introduction

Availability has been considered as an important measure of performance for many industrial systems which are generally considered as repairable ones.

Maintainability is also to be a key index to enhance the performance of these systems.

Maintenance is any activity carried out on an asset in order to ensure that the asset continues to perform its intended function(s).

RAM as an engineering tool evaluates the equipment performance at different stages in design process and can be used to improve and predict the performance of the system.

Introduction

RAM addresses both operation and safety issues and aims to identify areas within the system or process where improvement actions can be initiated.

RAM model provides a quantitative basis for judging the acceptability of the design, and the tools needed to facilitate justification and prioritization activities for all future reliability and availability decision making.

Factors that affect RAM of a repairable industrial system

- Machinery
- Operating conditions
- Maintenance conditions
- Infra-structural facilities

Plant life cycle and RAM

	Conceptual design	Basic design	Detailed design	Commissioning & startup	Operations
Main deliverables	Simple PFD Process screening using simple shortcut models	Detail PFS Simplified engineering design Equipment specifications Control strategy	Develop P & ID diagram Safety systems Detailed drawings, material selection	Startup procedures As built drawings Installation procedures	Operate and maintain
Variables impacting RAM	• Process configuration • Initial inherent failure and repair rates • assumed maintenance policy	Plant layout Equipment rated capacity	Installed spare components Detail of maintenance plans Vendor selection	Installation errors Operator training programs Equipment procurement	Maintenance plan Size of maintenance crew Spares in warehouse Logistical and administrative delays.... Human errors
Data	Generic and benchmark data		Data from vendors	Testing data	"Actual" or operational data

Reported Benefits of RAM

Company	Benefits
Marathon Ashland Petroleum (Matusheski and Andrews, 2002)	Saved \$3 million lost opportunity costs in one year by avoiding heat exchanger failures at a total cost of about \$500,000
ExxonMobil ¹	The reliability and maintenance system program, since its introduction in 1994, has reduced maintenance costs (about \$1 billion) by about 30% while improving mechanical availability by about 2%
Shell's Pulau Bukom refinery ²	The design and operational modifications made during 1996 turnaround results in a four year run of its long-residue catalytic cracking unit (LRCCU) with only 21 hours of downtime
Toa refinery, Japan ³	With the help of Shell Global Solutions International BV's maintenance and reliability (Merit) program saved \$10 million in its first year and \$17 million in the second year
Lima refinery (Paul, 1997)	Over \$1.4 million dollars per year were saved in pump repairs by increasing the MTBF (Mean time between failure) of the pumps
Conoco Refinery ⁴	Maintenance costs dropped by 21% and unscheduled lost profit opportunities were down 47% (\$34 million) due to improved equipment reliability and streamlined maintenance practices

¹ ExxonMobil financial and operating review, 2001

² *Impact Magazine* (internal magazine of Shell Global Solutions), Issue 2, 2001

³ *Impact Magazine* (internal magazine of Shell Global Solutions), Issue 4, 2001

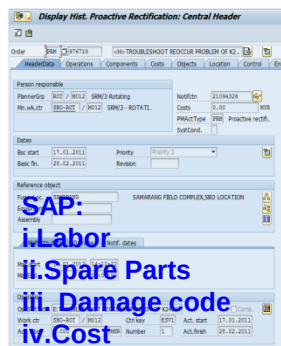
⁴ Dupont consulting website (<http://www.dupont.com/consulting/solutions/conoco.html>)

Methodology

Input



Eqp. Running Hrs Record



Display Hist. Proactive Rectification: Central Header

Person responsible: SPM's Rotating

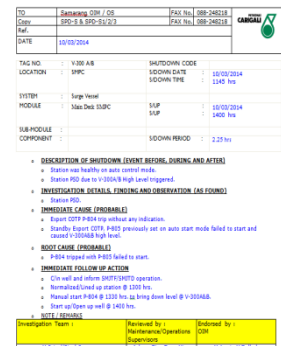
Reference object: SARANG PULU COMPLEXED LOCATION

SAP: i.Labor

ii. Spare Parts

iii. Damage code

iv. Cost



DESCRIPTION OF SHUTDOWN EVENT BEFORE, DURING AND AFTER

- Station was healthy on auto control mode.
- Station PSD due to V304A/B High Level triggered.

IMMEDIATE CAUSE (PROBABLE)

- Station PSD
- Station PSD High Level triggered without any indication.
- Station Report CTR, PSD previously set on auto start mode failed to start and caused V304A/B High Level.

IMMEDIATE ACTION

- Station PSD
- Station PSD High Level triggered without any indication.
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ROOT CAUSE (PROBABLE)

- PSD High Level triggered without any indication.
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IMMEDIATE ACTION

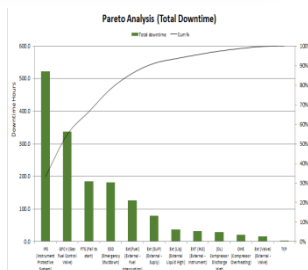
- Station PSD
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ROOT CAUSE (PROBABLE)

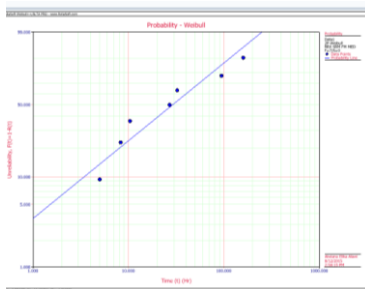
- PSD High Level triggered without any indication.
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- Station PSD High Level triggered without any indication.
- Station Report CTR, PSD previously set on auto start mode failed to start and caused V304A/B High Level.

EFR

Categorization & Analysis

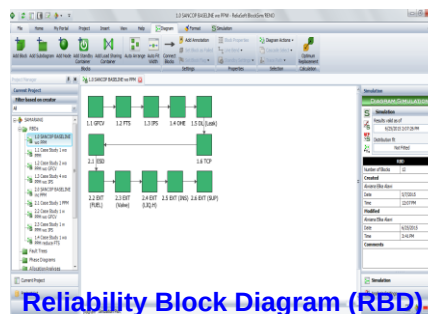


Pareto Analysis:
i. Failure Category
ii. Total Downtime

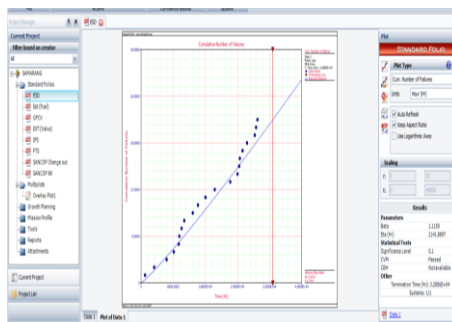


Identify Failure Distribution, MTBF and MTTR

Modelling

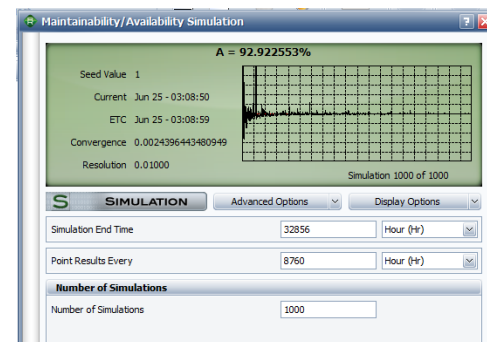


Modelling



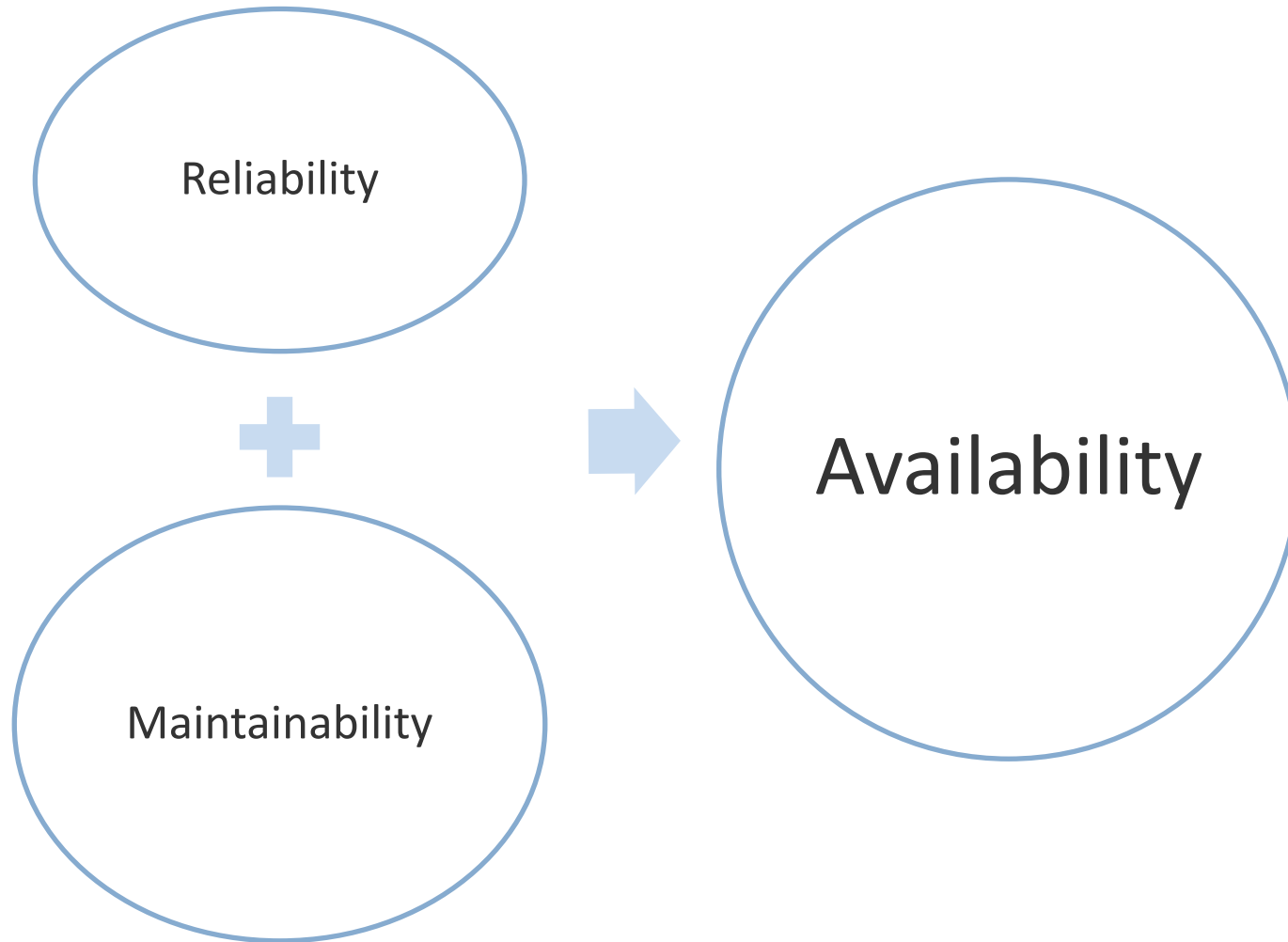
Reliability Growth Analysis (RGA)
Modelling

Output



System Availability/Cost/Throughput

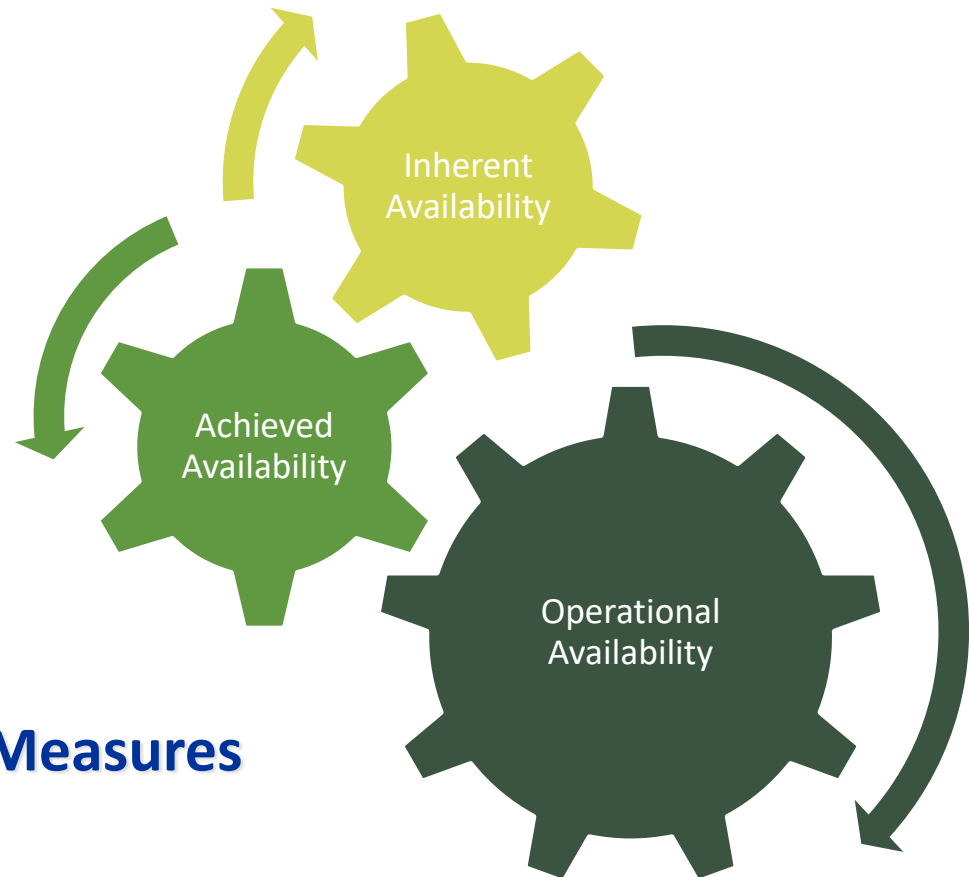
Mathematical equations for RAM analysis



Availability

“The probability that a system or component is performing its required function at a given point in time or over a stated period of time when operated and maintained in prescribed manner”

(Ebeling, 1997)



Availability Measures

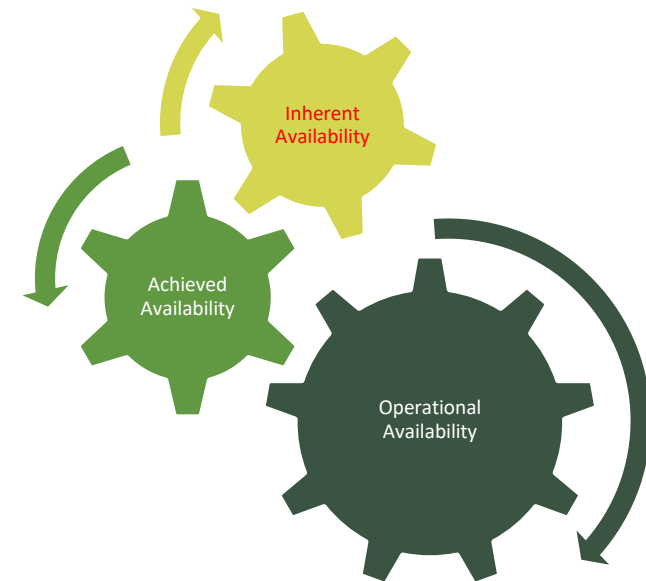
Inherent Availability

● Inherent Availability

- The steady state availability which considers only the corrective maintenance (CM).
- Assuming that CM actions occur at a constant rate, it can be estimated as:

$$A_I = \frac{MTBF}{MTBF + MTTR}$$

where *MTBF* is the mean time between failure and *MTTR* is the mean time to repair.



Achieved Availability

- Similar to inherent availability with the exception that PM downtimes are also included.
- Sometimes referred to as the availability seen by the maintenance department (i.e. do not include logistic delays, supply delays or administrative delays).
- It can be estimated as:

$$A_A = \frac{MTBMA}{MTBMA + MMT}$$

where *MTBMA* is the mean time between maintenance actions (i.e. preventive and corrective) and *MMT* is the mean maintenance time.

- Assuming constant failure rate *MTBMA* can be calculated as:

$$MTBMA = \frac{1}{\lambda + f_{PM}}$$

where λ is the failure rate (assuming all failures are repaired) and f_{PM} is the frequency of preventive maintenance, the reciprocal of PM cycle.



Achieved Availability

- It can be estimated as:

$$A_A = \frac{MTBMA}{MTBMA + MMT}$$

where $MTBMA$ is the mean time between maintenance actions (i.e. preventive and corrective) and MMT is the mean maintenance time.

- The mean maintenance time MMT can be decomposed into the effects of preventive and corrective maintenance as:

$$MMT = \frac{\lambda MTTR + f_{PM} MPMT}{\lambda + f_{PM}}$$

where $MTTR$ is the mean CM time and $MPMT$ is the mean PM time.

Operational Availability

- A measure of the “real” average availability over a period of time in an actual operational environment.
- Includes all experienced sources of downtime (i.e. administrative downtime, logistic downtime etc.)
- Can be calculated using

$$A_{op} = \frac{MTBMA}{MTBMA + MDT}$$

where *MDT* is the mean maintenance downtime and *MDT* is *MMT* + *logistic delay time* + *administrative delay time*



Example of Availability Calculation

Calculate the **achieved availability** for a system with the demonstrated failure rate of 1 failure per 1000 hours of operation. Preventive maintenance is scheduled every 500 hours or operation and average takes 6 hours. To repair a failed system takes on average 16 hours.

The frequency of preventive maintenance is $f_{PM} = \frac{1}{500} = 0.002h^{-1}$ and the failure rate is $\lambda = \frac{1}{1000} = 0.001h^{-1}$.

$$MTBMA = \frac{1}{\lambda + f_{PM}} = \frac{1}{0.001 + 0.002} = 333.33 \text{ h}$$

$$MMT = \frac{\lambda MTTR + f_{PM} MPMT}{\lambda + f_{PM}} = \frac{0.001(16) + 0.002(6)}{0.001 + 0.002} = 9.33 \text{ h}$$

$$A_A = \frac{MTBMA}{MTBMA + MMT} = \frac{333.33}{333.33 + 9.33} = \mathbf{0.973}$$

Maintainability

The probability that a failed component or system will be restored or repaired to a specified condition within a period of time when maintenance is performed in accordance with prescribed procedures.

A key maintainability figure of merit is the mean time to repair (MTTR) and the other is a limit for the maximum repair time.

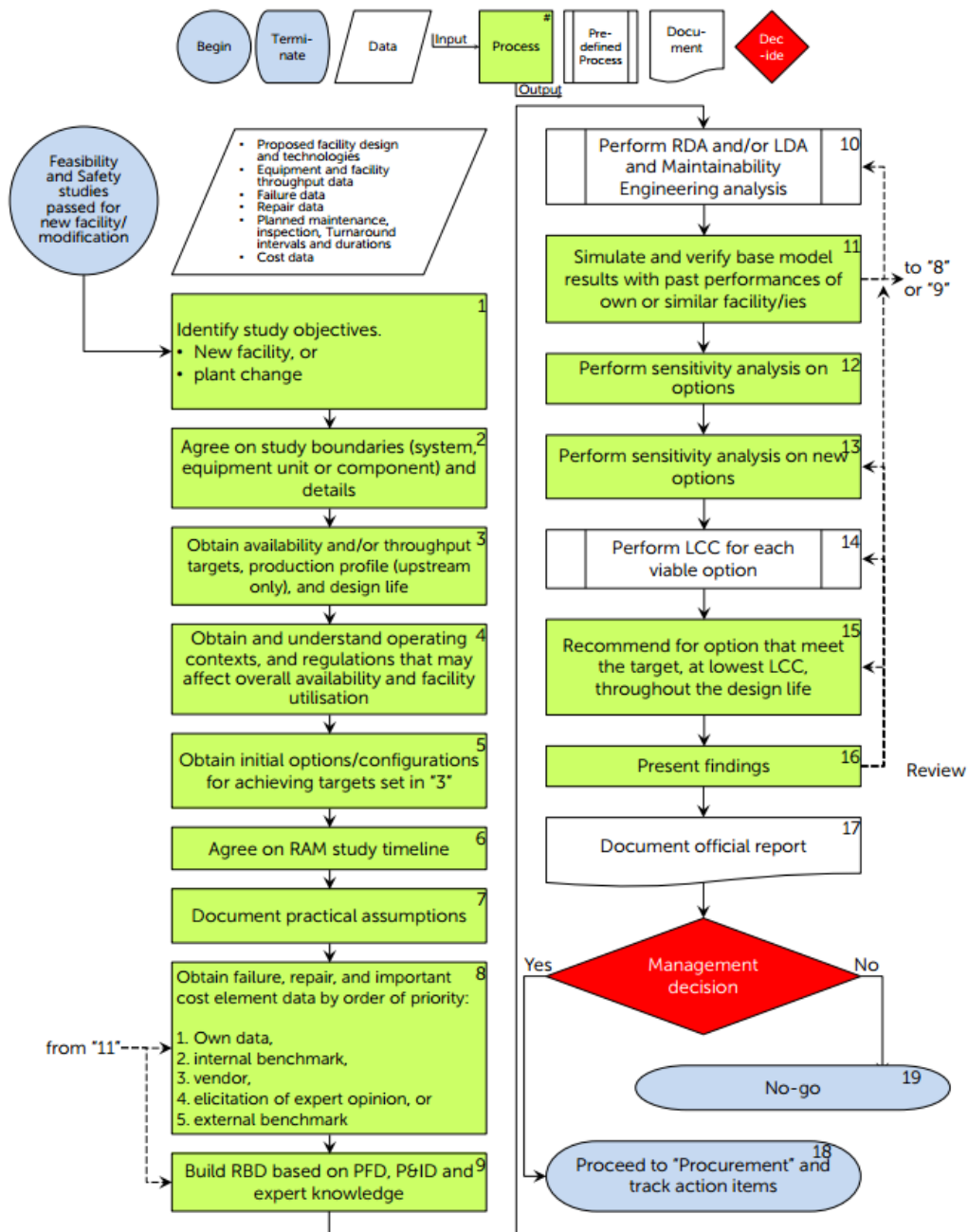
The maintainability of a component is calculated as

$$M(t) = 1 - e^{-\frac{t}{MTTR}} = 1 - e^{-\mu t}$$

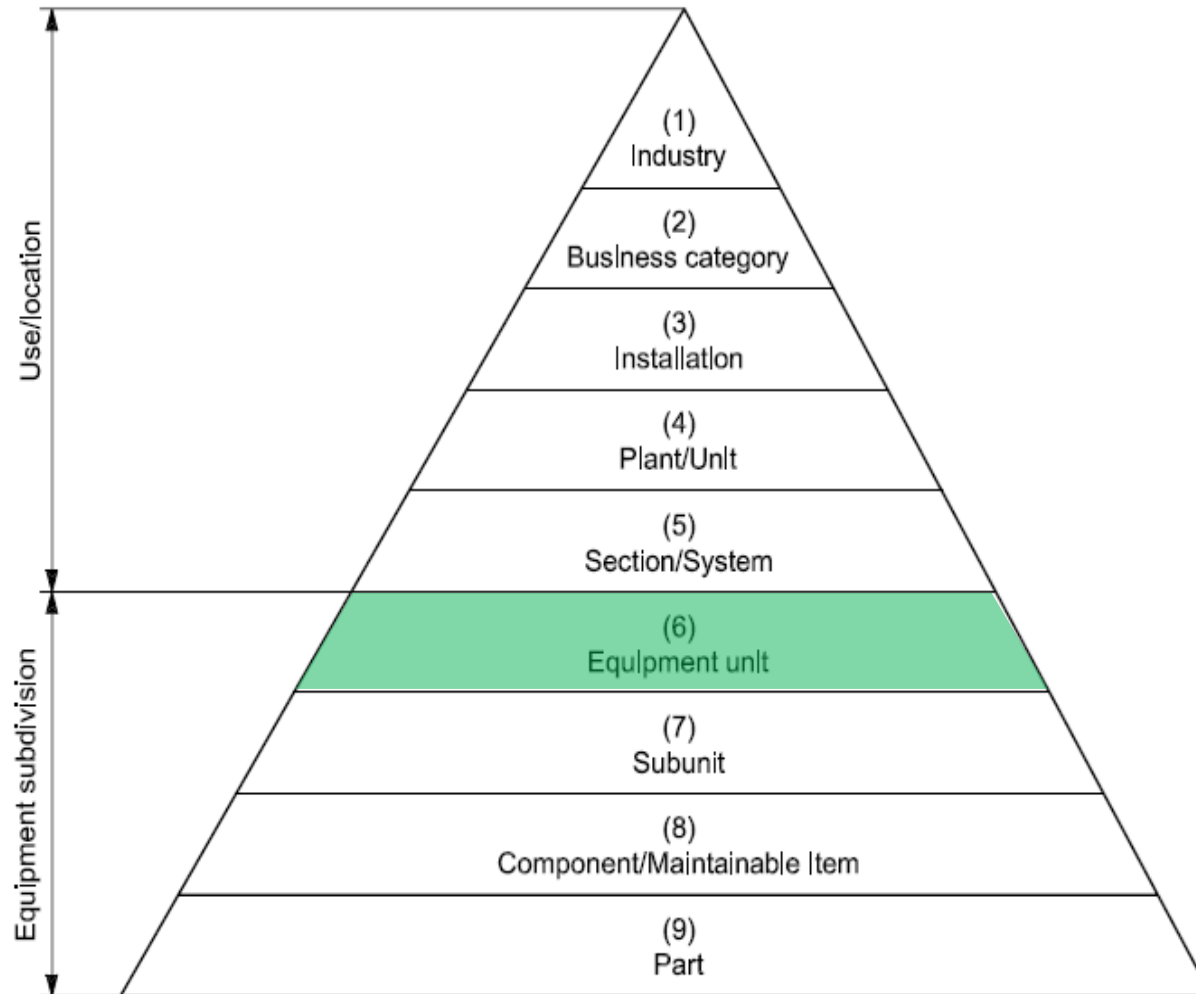
Maintenance times tend to be lognormally distributed.

RAM Workflow (RIM 150522 V5)

RAM work flow



ISO 14224 Taxonomy



RAM Software Packages Comparison

		BlockSim ¹	AvSim ²	TITAN ³	MAROS ⁴	SPAR ⁵	SPARC ⁶
Type	Simulation/Analytical	Hybrid	Simulation	Simulation	Simulation	Simulation	Analytical
Model	RBD/FT/Markov/Petri Net	RBD	RBD/FT	RBD	RBD	RBD	RBD
GUI		+ ⁷	+	+	+	+	+
Input data	Failure rate	Constant	+	+	+	+	+
		Time varying	+	+	+	+	
		Different modes	+	+	+	+	+
	Maintenance data	Corrective	+	+	+	+	+
		Preventive	+	+	+	+	+
Modeling capability	redundancy	Active parallel	+	+	+	+	+
		Standby	+	+	+	+	
	Intermediate storage		+	+	+	+	
	Spares		+	+	+	+	
	Cost analysis		+	+	+	+	+
	User defined logical restrictions			+		+	
Output	Average and point reliability/availability		+	+	+	+	+
	Spare and stocks optimization		+	+	+	+	
	Reliability optimization		+				

¹ ReliaSoft Corporation, 115 S. Sherwood Village Drive, Suite 103, Tucson, AZ 85710

² Item Software, Inc., 2030 Main Street, Suite 1130, Irvine, CA 92614

³ Fidelis Group, 4545 Post Oak Place STE 347 Houston, TX 77027

⁴ Jardine and Associates Ltd, Nine Holyrood Street, London SE1 2EL, United Kingdom

⁵ Clockwork Designs, Inc., 3432 Greystone Drive, Suite 202, Austin, TX 78731

⁶ IES Products 2811 NV Reeuwijk Reeuwijkse Poort 301 The Netherlands

⁷ '+' denotes YES

INTRODUCTION TO MONTE CARLO

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Session Objectives

- To understand the basic concept of Monte Carlo simulation technique
- To explain the application of Monte Carlo in deterministic and stochastic models
- To describe the application of Monte Carlo in reliability

Definition of Monte Carlo

Simulation method that uses random sampling to study properties of systems with component that behave in random and deterministic fashion

- Example of deterministic: approximation of π or finding solution for integration
- Example of random: Predicting system reliability that consist of multiple components

The Monte Carlo method provides approximate solutions to a variety of mathematical problems by performing statistical sampling experiments on a computer.

The Monte Carlo method of reliability prediction is useful
when system complexity makes the formulation of exact
models essentially impossible.

History of Monte Carlo

The name *Monte Carlo* refers to the famous casino in Monaco, where randomness is also used in a repetitive way.

The development of the Monte Carlo method as a statistical computing tool goes back to the mid-1940s, when the first electronic computers were built.

John von Neumann and Stanislaw Ulam who first worked on the idea of using random numbers generated by a computer in order to solve problems encountered in the development of the atomic bomb



John von Neumann (28.12.1903-8.2.1957)

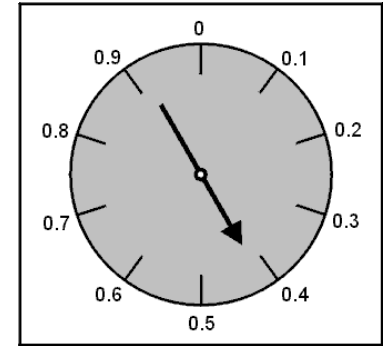
Random number generation

Manual generation using coin toss, dice rolling or roulette wheels – impractical.

Good random number is characterized by its uniformly distributed and statistically independent.

Modern techniques uses computer to generate the so called pseudorandom numbers (i.e. based on deterministic formulation) such as in Microsoft Excel.

In Excel, the formula is `RAND()` to generate uniformly distributed pseudorandom number between 1 and 0.



Right number of samples

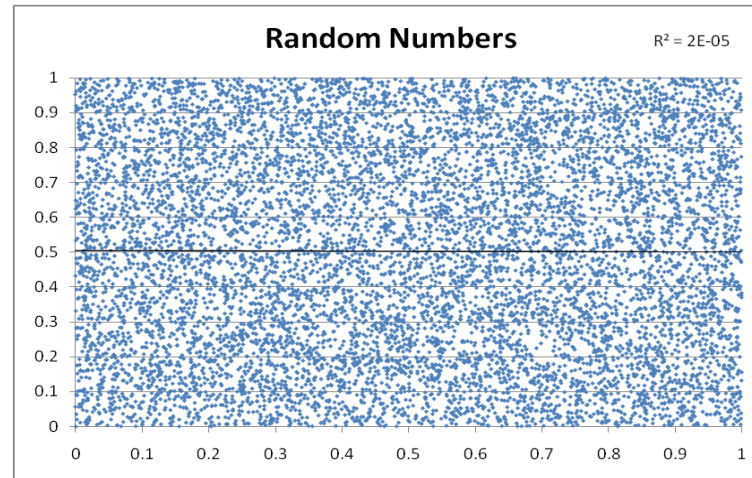
Theoretically, correct solution is achieved only when we sample the whole population which is impossible.

A trade off between practically 'good enough' solution versus the cost of getting more samples (in terms of time and resources).

A rule of thumb - Enough samples when the result reaches a convergence (not much changes in solution as the number of sample increases).

Test of randomness

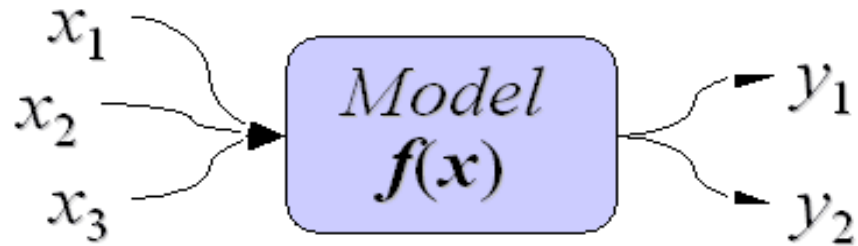
Scatter plot of 1k of random numbers (x_i, y_i) shows uniform pattern (use of Excel *RAND()* function)



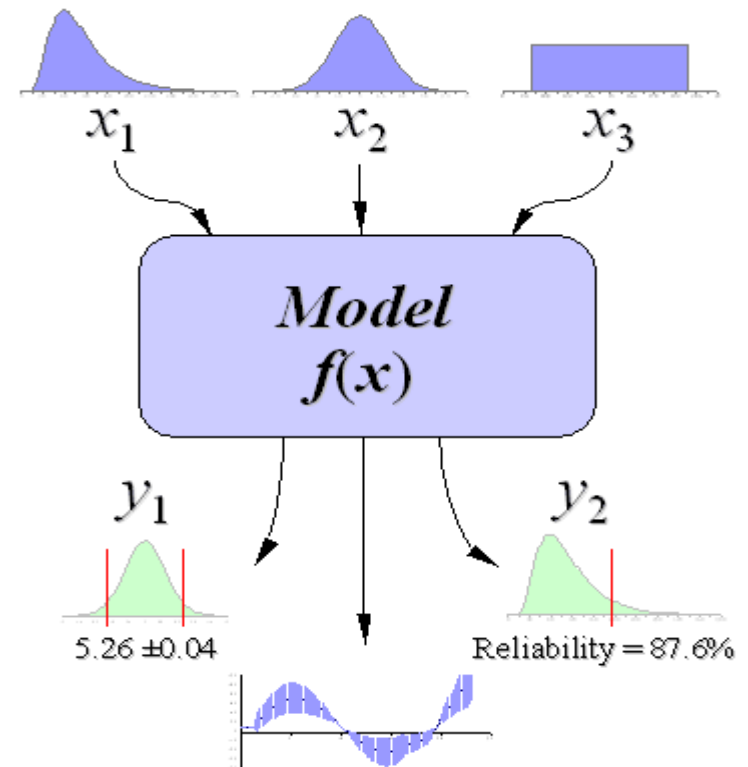
No correlation between two variables, $R^2 \sim 0$

Samples should have average of 0.5 and standard deviation of 0.28

Deterministic vs. Stochastic



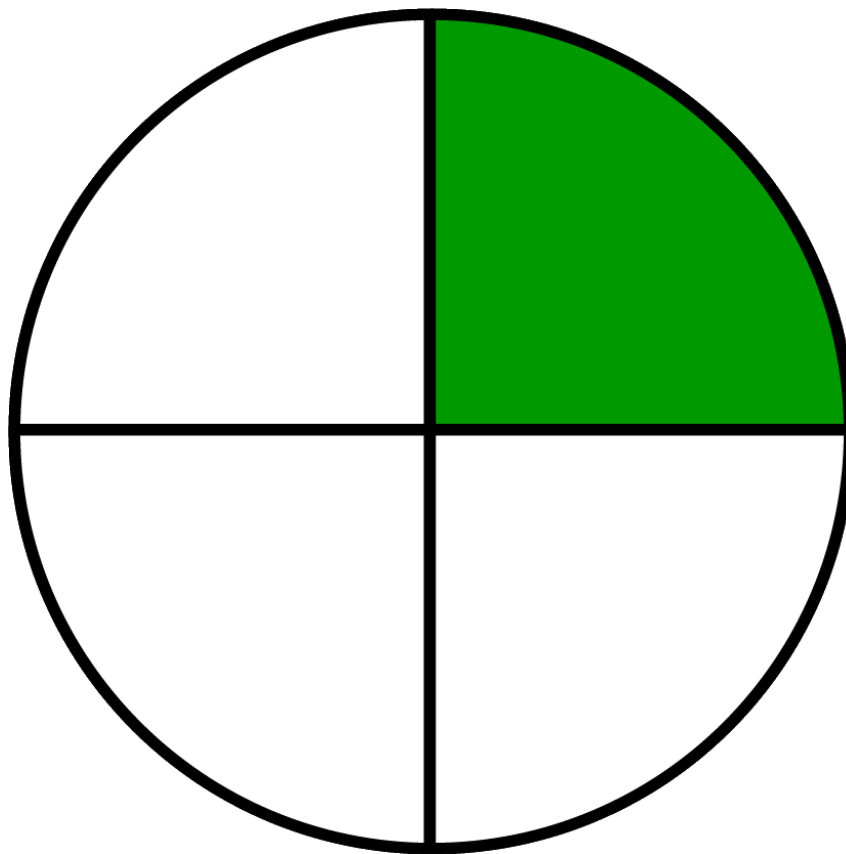
Deterministic Model
Same answer every time you run



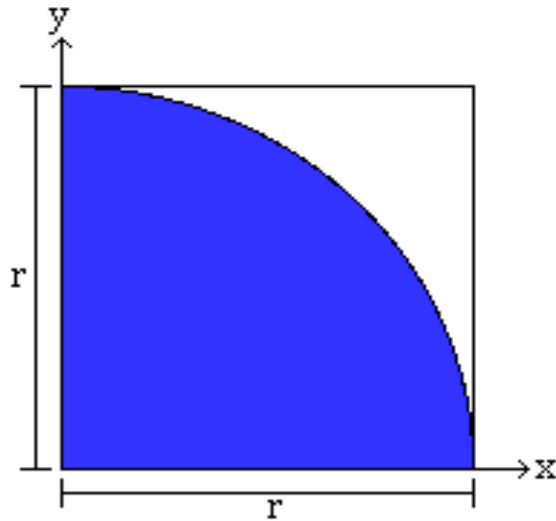
Stochastic Model-
different answer for every run

Deterministic MC Example

Estimating π



Monte Carlo Example



A dart player throwing darts randomly at the above figure, it should be apparent that of the total number of darts that hit within the square, the number of darts that hit the shaded part (circle quadrant) is proportional to the area of that part. In other words,

$$\frac{\# \text{ darts hitting shaded area}}{\# \text{ darts hitting inside square}} = \frac{\text{area of shaded area}}{\text{area of square}}$$

Monte Carlo Example

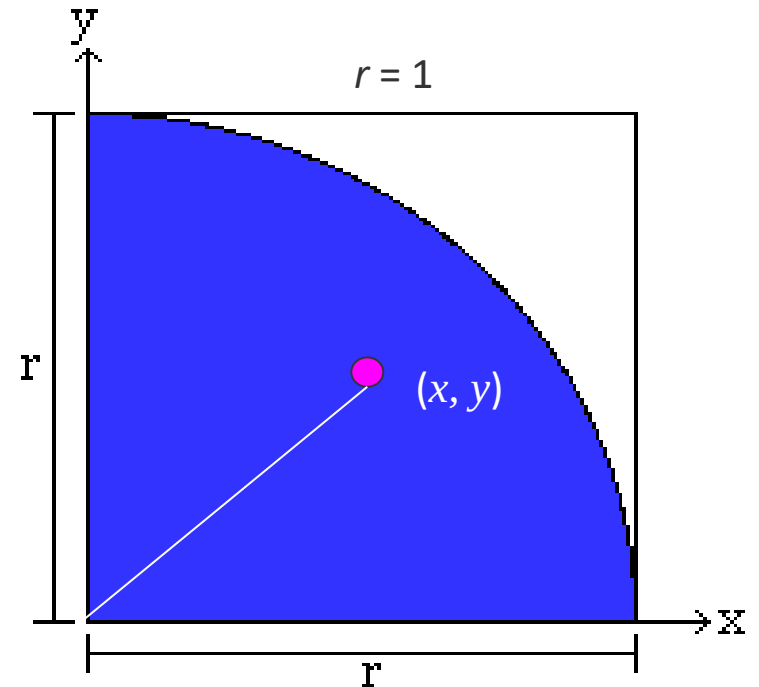
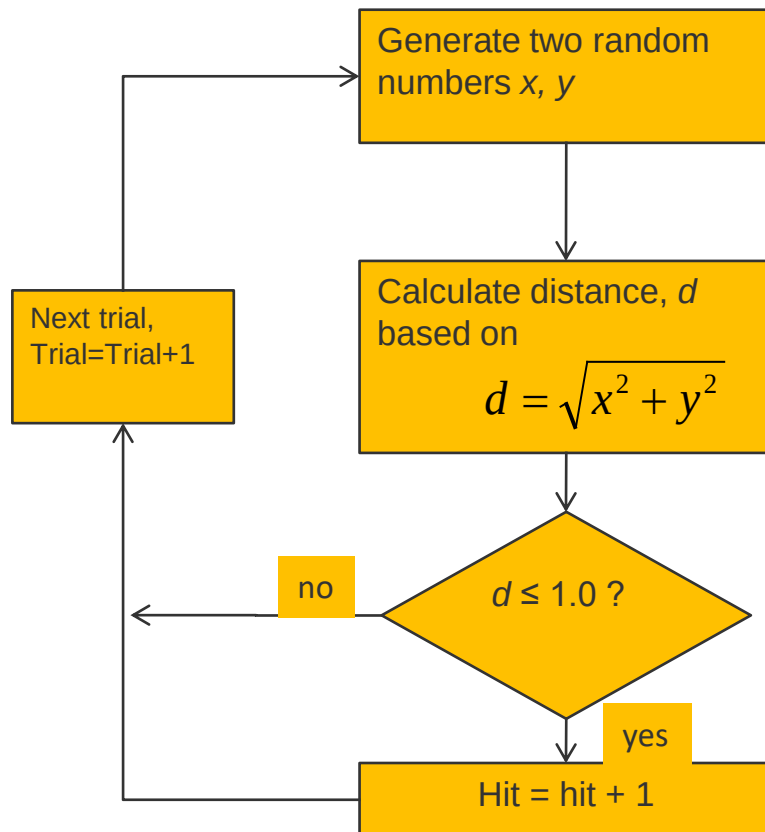
$$\frac{\# \text{ darts hitting shaded area}}{\# \text{ darts hitting inside square}} = \frac{\text{area of shaded area}}{\text{area of square}}$$

$$\frac{\text{area of shaded area}}{\text{area of square}} = \frac{\frac{1}{4}\pi r^2}{r^2} = \frac{1}{4}\pi$$

$$\frac{\# \text{ darts hitting shaded area}}{\# \text{ darts hitting inside square}} = \frac{1}{4}\pi$$

$$\pi = \frac{4 \times \# \text{ darts hitting shaded area}}{\# \text{ darts hitting inside square}}$$

Monte Carlo Example



$$\pi = 4 \times \frac{Hit}{Trial}$$

Estimating π in Excel (Class activity)

Use of RAND() function to generate 1000 of random number (x,y) in two columns

Remember to select the option of manual calculation(in Excel option)

Enter the formula for distance

Enter the accept/reject criteria

Estimate π based on the formula

Analytical vs. Monte Carlo

Estimating integration using Monte Carlo and analytical solution. Compare result

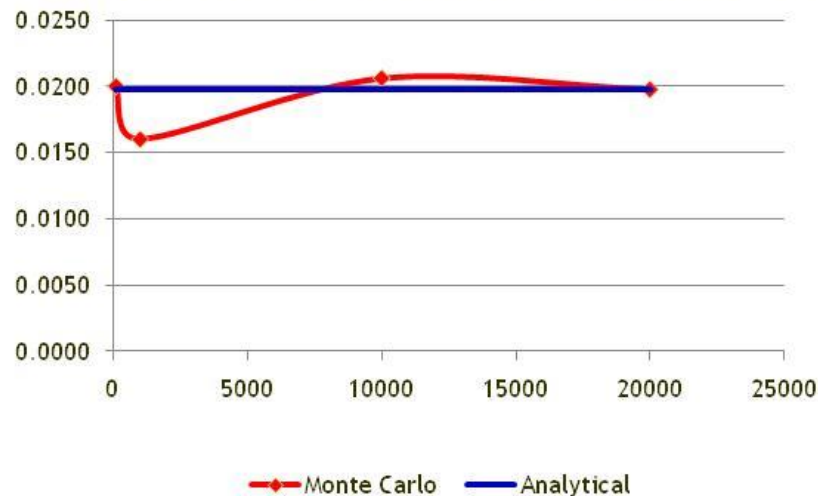
- Use exponential function

$$g(x) = \lambda e^{-\lambda x}$$

Run Monte Carlo with different number of sample

- 100, 1000, 10,000, 20,000
- Does result approaching analytical solution with more sample?
- Plot graph

Comparison between MC and analytical



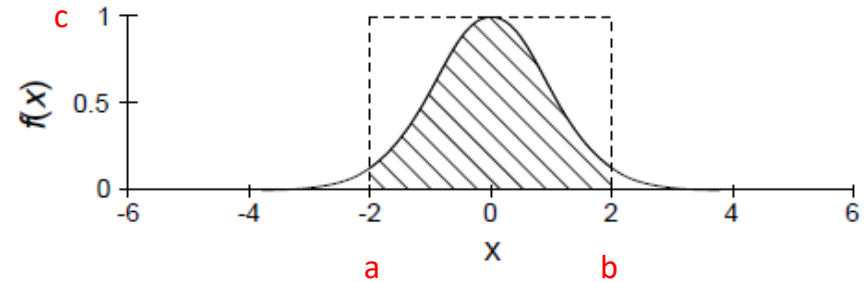
Analytical vs. Monte Carlo: Example

Function

$$g(x) = \lambda e^{-\lambda x}$$

$\lambda =$ 0.02
 $a =$ 0
 $b =$ 1
 $c =$ 1

} Limits



	U _i	U' _i	X _i	g(x _i)	check
1	0.938680	0.284515	0.938680	0.019628	0
2	0.423587	0.604533	0.423587	0.019831	0
3	0.528722	0.606402	0.528722	0.019790	0
4	0.884874	0.226273	0.884874	0.019649	0
5	0.139981	0.756931	0.139981	0.019944	0
6	0.868036	0.841022	0.868036	0.019656	0
7	0.891575	0.664271	0.891575	0.019647	0
8	0.691136	0.795935	0.691136	0.019725	0
9	0.147848	0.489732	0.147848	0.019941	0
10	0.104544	0.866998	0.104544	0.019958	0
11	0.697974	0.567184	0.697974	0.019723	0
12	0.244295	0.220843	0.244295	0.019903	0
13	0.806606	0.514958	0.806606	0.019680	0

	Result	
Sample	Monte Carlo	Analytical
100	0.0200	0.0198
1000	0.0260	0.0198
10000	0.0196	0.0198
20000	0.0197	0.0198

Generating Random Number from continuous distribution

Exponential distribution

- assuming U = uniformly distributed random number between 0 and 1, TTF, can be generated.

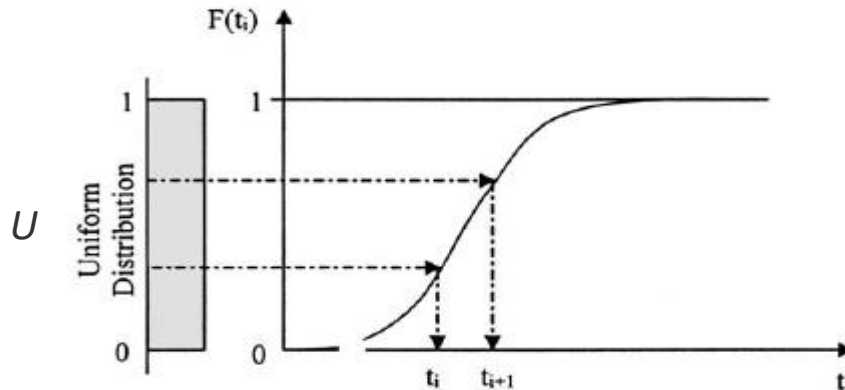
$$U = F(t) = 1 - e^{-\lambda t}$$

Inverse probability transformation

$$1 - U = e^{-\lambda t}$$

$$\ln(1 - U) = -\lambda t$$

$$t = \frac{1}{\lambda} \ln \left(\frac{1}{1 - U} \right)$$



where λ = failure rate
 t = time to failure (TTF)
 U = random number

Generating Random Number for various distributions in Excel

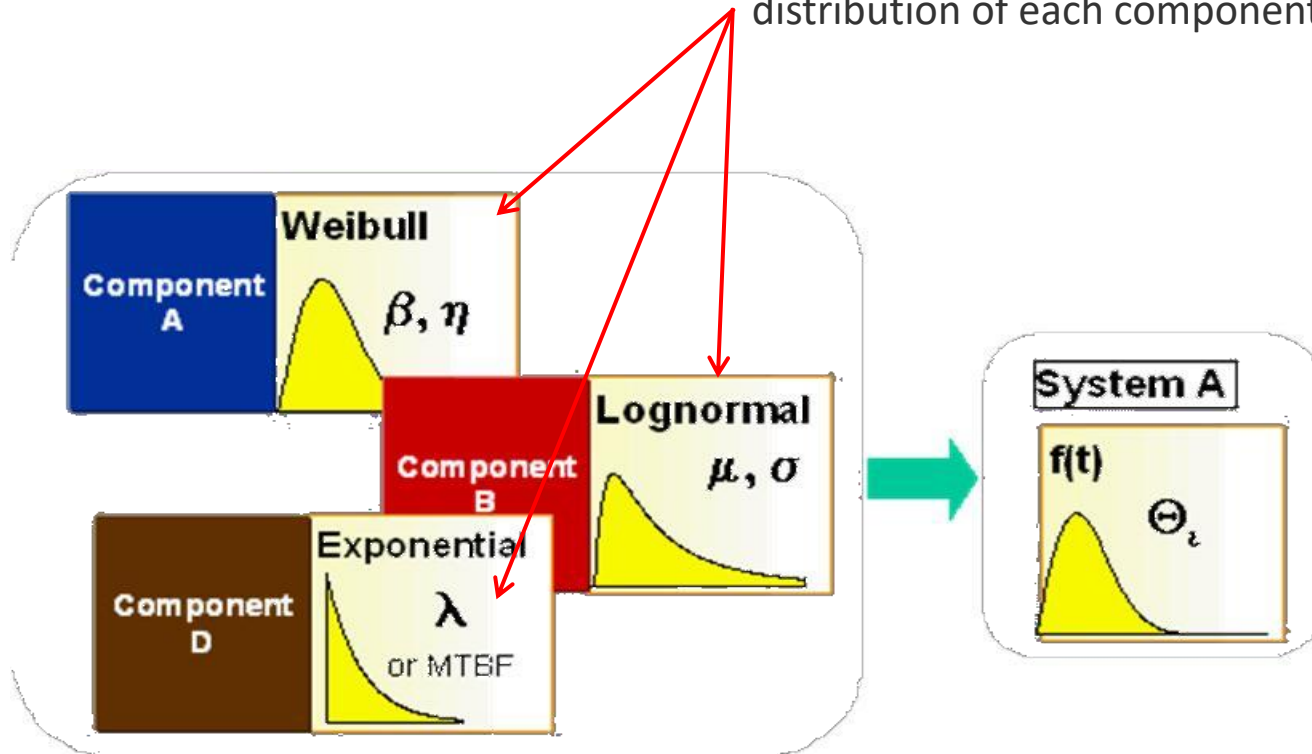
Distribution	PDF or f(t)	Excel Expression for time to failure
Exponential	$\lambda \exp(-\lambda t)$	$-\text{LN}(1-\text{RAND()})/\lambda$ $\lambda = \text{failure rate}$
Weibull	$\frac{\beta}{\alpha} \left(\frac{t}{\alpha}\right)^{\beta-1} \exp\left\{-\left(\frac{t}{\alpha}\right)^{\beta}\right\}$	$\alpha \{-\text{LN}(1-\text{RAND()})\}^{(1/\beta)}$ $\alpha = \text{scale parameter,}$ $\beta = \text{shape parameter}$
Normal	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2\right\}$	$\text{NORMINV}(\text{RAND()}, \mu, \sigma)$ $\mu = \text{mean, } \sigma = \text{variance}$
Uniform	$1/T$	$T \times \text{RAND()}$ $T = \text{max time}$
Lognormal	$\frac{1}{\sigma t \sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(\frac{\ln t - \mu}{\sigma}\right)^2\right\}$	$\text{LOGINV}(\text{RAND()}, \mu, \sigma)$ $\mu = \text{mean, } \sigma = \text{variance}$

Application of MC in RBD

Parallel system

Series system

Sample random number from the distribution of each components

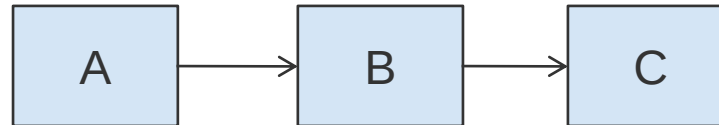


MC Example: Series/parallel system

RBD in Blocksim

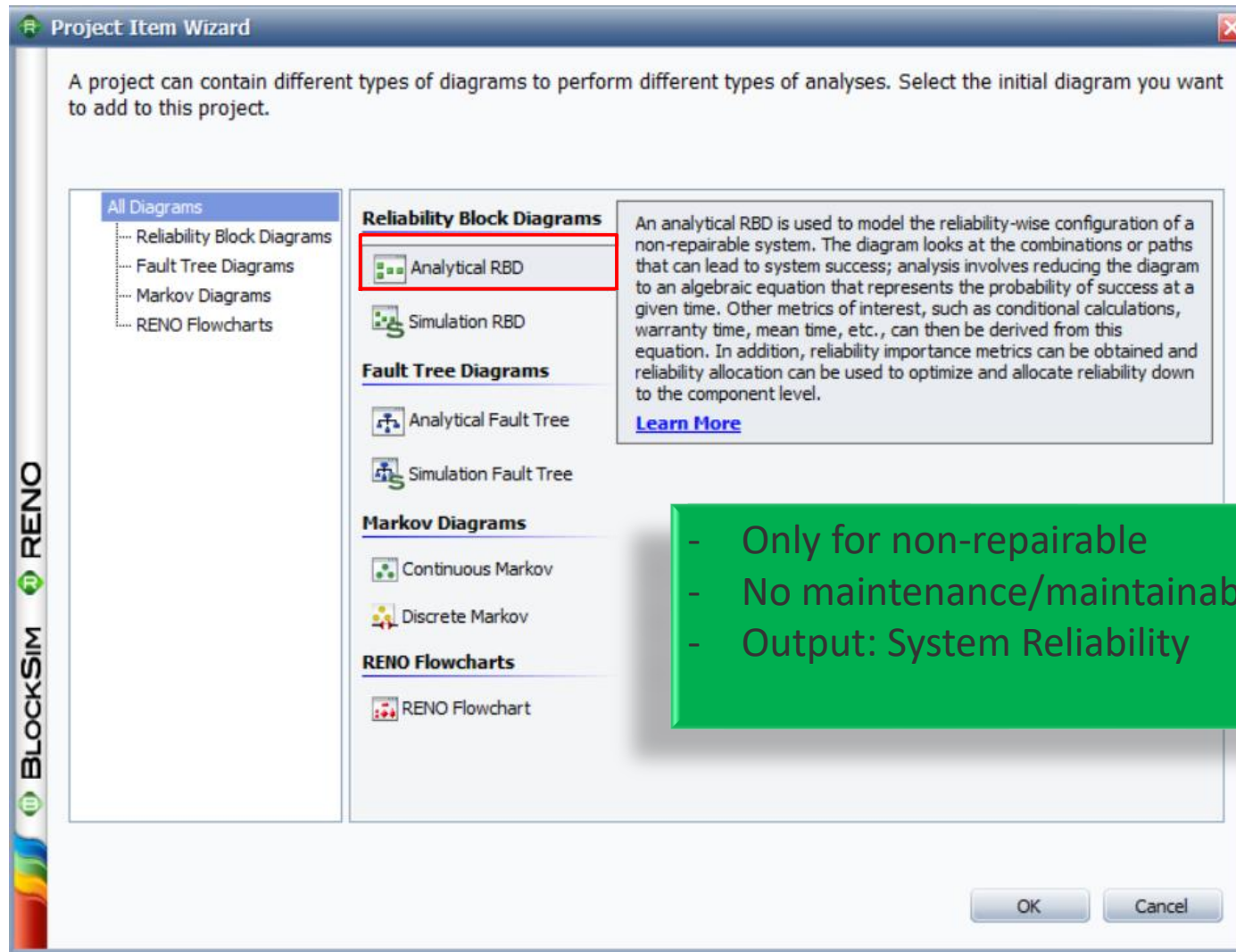
Calculate the reliability and failure rate of the system at

- i. 1000 hrs
- ii. 4320 hrs
- iii. 8640 hrs



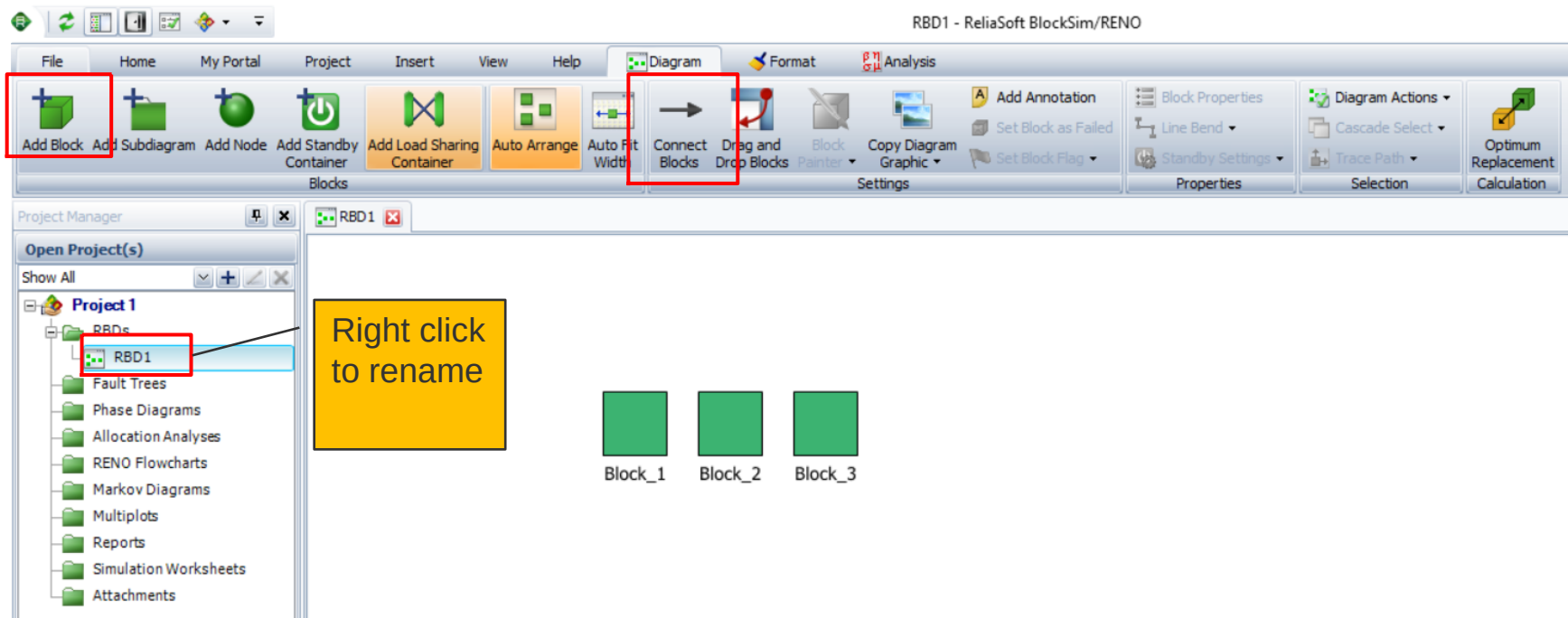
	A	B	C
Dist.	Exponential	Exponential	Exponential
Parameters	$\lambda = .00005$	$\lambda = .00003$	$\lambda = .00008$

Blocksim :Step 1 – Select Analytical RBD



- Only for non-repairable
- No maintenance/maintainability data
- Output: System Reliability

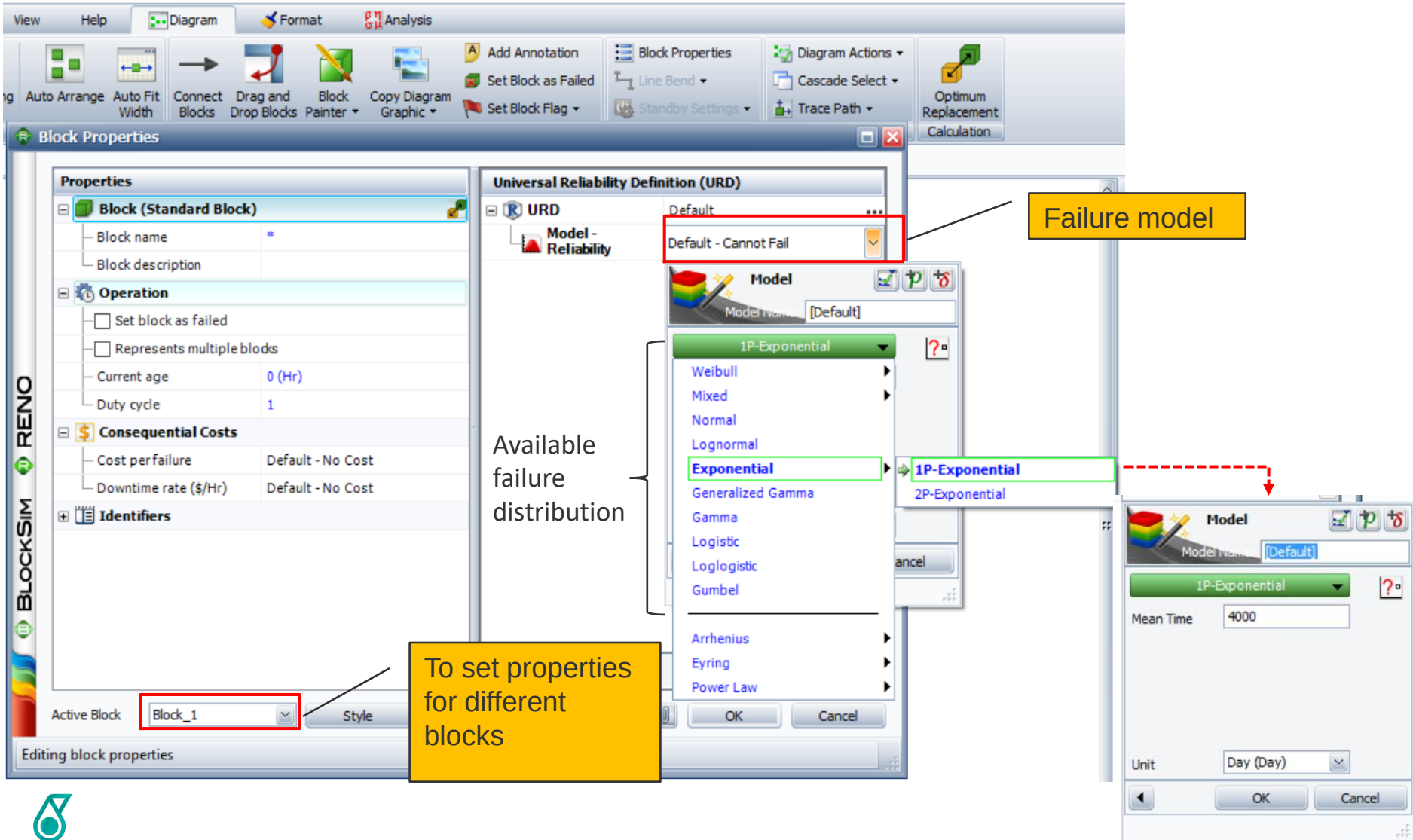
Blocksim :Step 2– create block configuration



Click arrow key to connect the block in series configuration

Blocksim : Step 3 – input block properties

RBD1 - ReliaSoft BlockSim/RENO



Block Properties

Properties

Block (Standard Block)

Block name *

Block description

Operation

☐ Set block as failed

☐ Represents multiple blocks

Current age 0 (Hr)

Duty cycle 1

Consequential Costs

Cost per failure Default - No Cost

Downtime rate (\$/Hr) Default - No Cost

Identifiers

Active Block Block_1

Style

Editing block properties

Universal Reliability Definition (URD)

URD

Default

Model - Reliability

Default - Cannot Fail

Model

Model Name [Default]

1P-Exponential

Weibull

Mixed

Normal

Lognormal

Exponential

Generalized Gamma

Gamma

Logistic

Loglogistic

Gumbel

Arrhenius

Eyring

Power Law

OK

Cancel

Model

Model Name [Default]

1P-Exponential

Mean Time 4000

Unit Day (Day)

OK

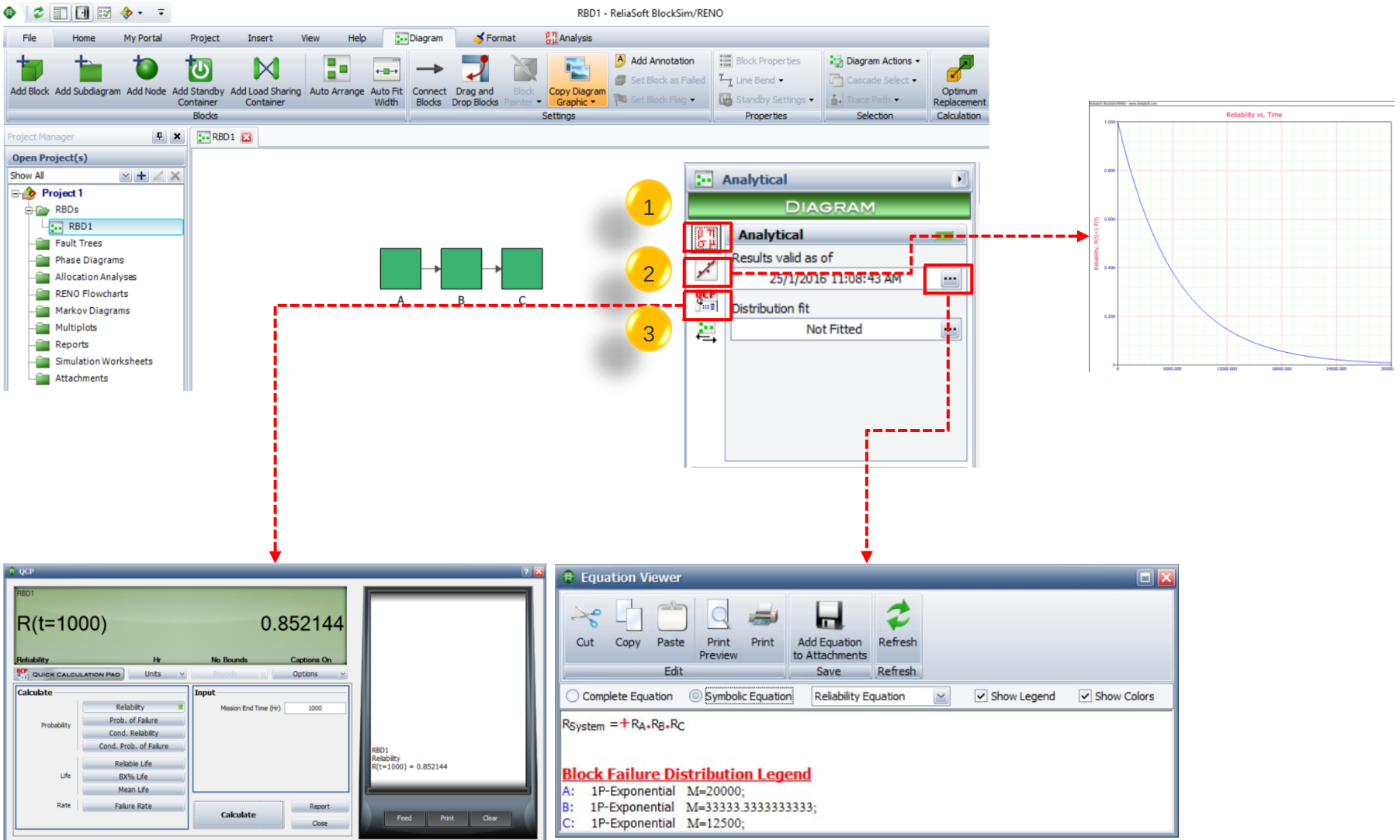
Cancel

Failure model

Available failure distribution

To set properties for different blocks

Blocksim : Step 4 – System Reliability



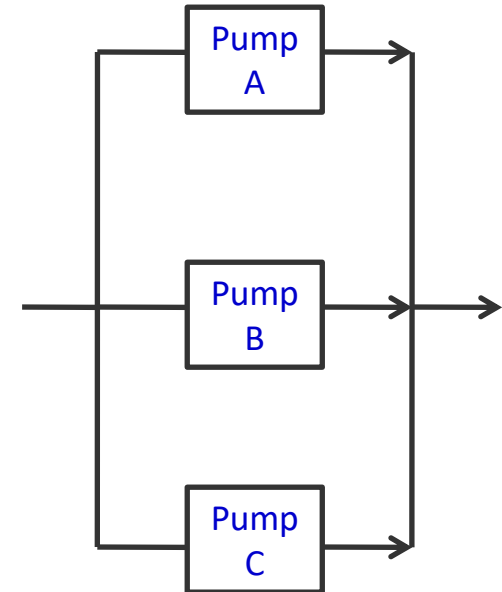
Blocksim – k out of n system

Example 2:

Calculate the reliability of the system at 8640 hrs for these *k-out-of-n* configurations:

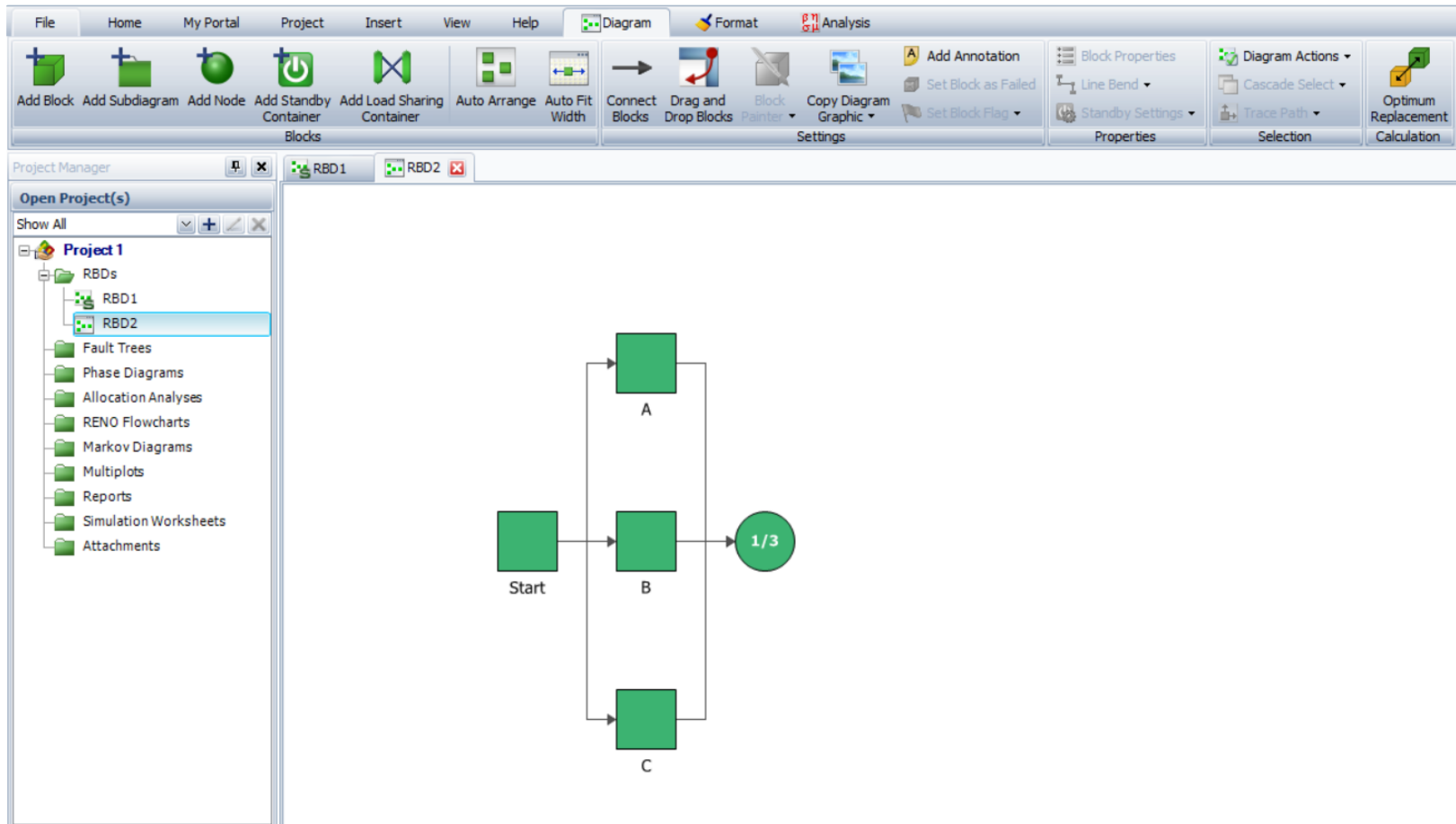
- i. 1 out of 3
- ii. 2 out of 3

Assume they are **non-repairable** items

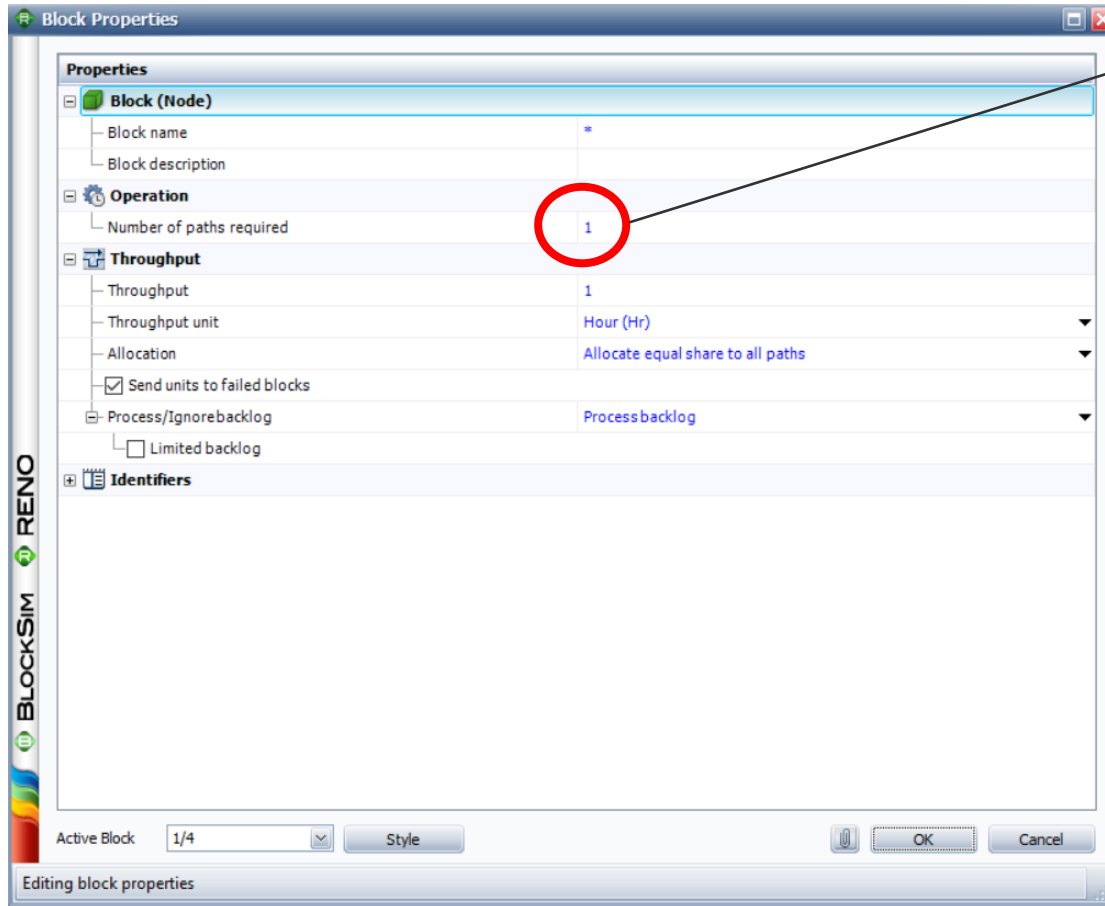


	Pump A	Pump B	Pump C	
Dist.	Exponential	Weibull	Weibull	
Parameters	$\lambda = .00005$	$\beta = 1.15$ $\eta = 20000 \text{ hrs}$	$\beta = 0.8475$	$\eta = 6581 \text{ hrs}$

Blocksim – k out of n system



Blocksim : Node (k-out-of-n)

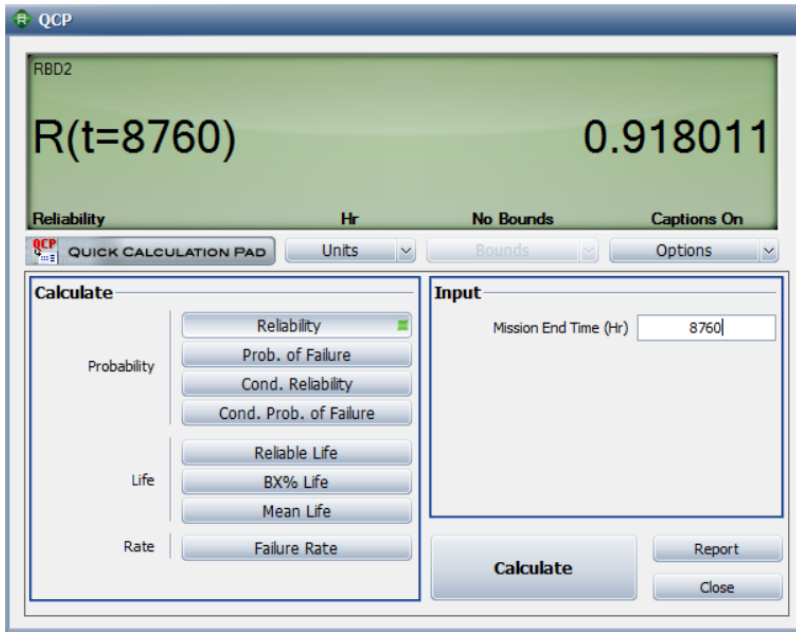


The image shows the 'Block Properties' dialog box in the Blocksim software. The 'Block (Node)' section is expanded, showing the 'Block name' as '*'. The 'Operation' section is expanded, showing the 'Number of paths required' as '1'. The 'Throughput' section is expanded, showing the 'Throughput' as '1', 'Throughput unit' as 'Hour (Hr)', and 'Allocation' as 'Allocate equal share to all paths'. The 'Identifiers' section is collapsed. The 'Active Block' dropdown at the bottom shows '1/4'. The 'Style' button is visible. The status bar at the bottom indicates 'Editing block properties'.

Section	Property	Value
Block (Node)	Block name	*
	Block description	
Operation	Number of paths required	1
Throughput	Throughput	1
	Throughput unit	Hour (Hr)
	Allocation	Allocate equal share to all paths
	☒ Send units to failed blocks	
	Process/Ignorebacklog	Processbacklog
Identifiers	☐ Limited backlog	

Change the number to change configuration

Blocksim : Node (k-out-of-n)



QCP

RBD2

R(t=8760) 0.918011

Reliability Hr No Bounds Captions On

QUICK CALCULATION PAD Units Bounds Options

Calculate

Probability

Reliability ☒

Prob. of Failure

Cond. Reliability

Cond. Prob. of Failure

Life

Reliable Life

BX% Life

Mean Life

Rate

Failure Rate

Input

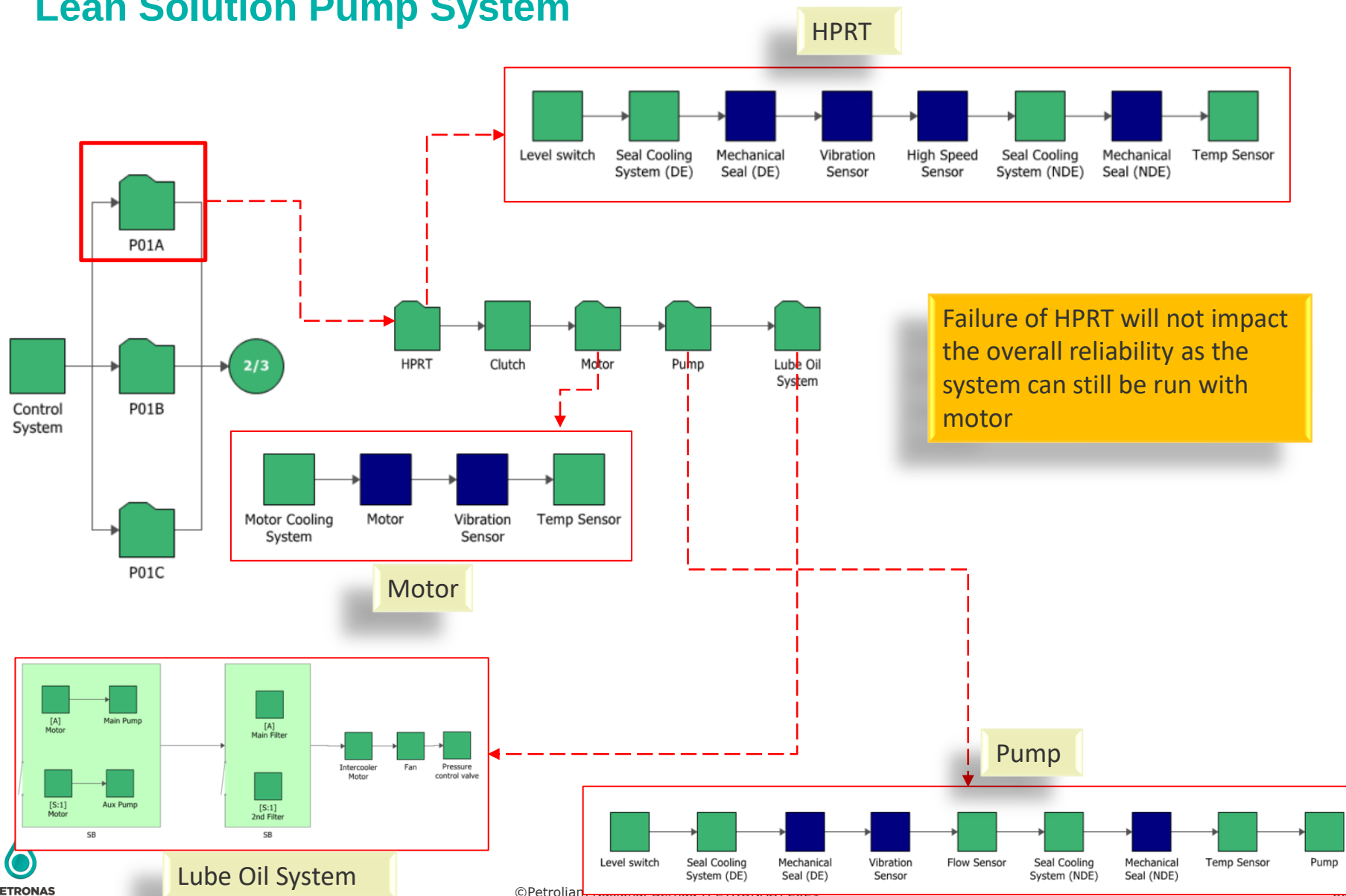
Mission End Time (Hr) 8760

Calculate **Report** **Close**

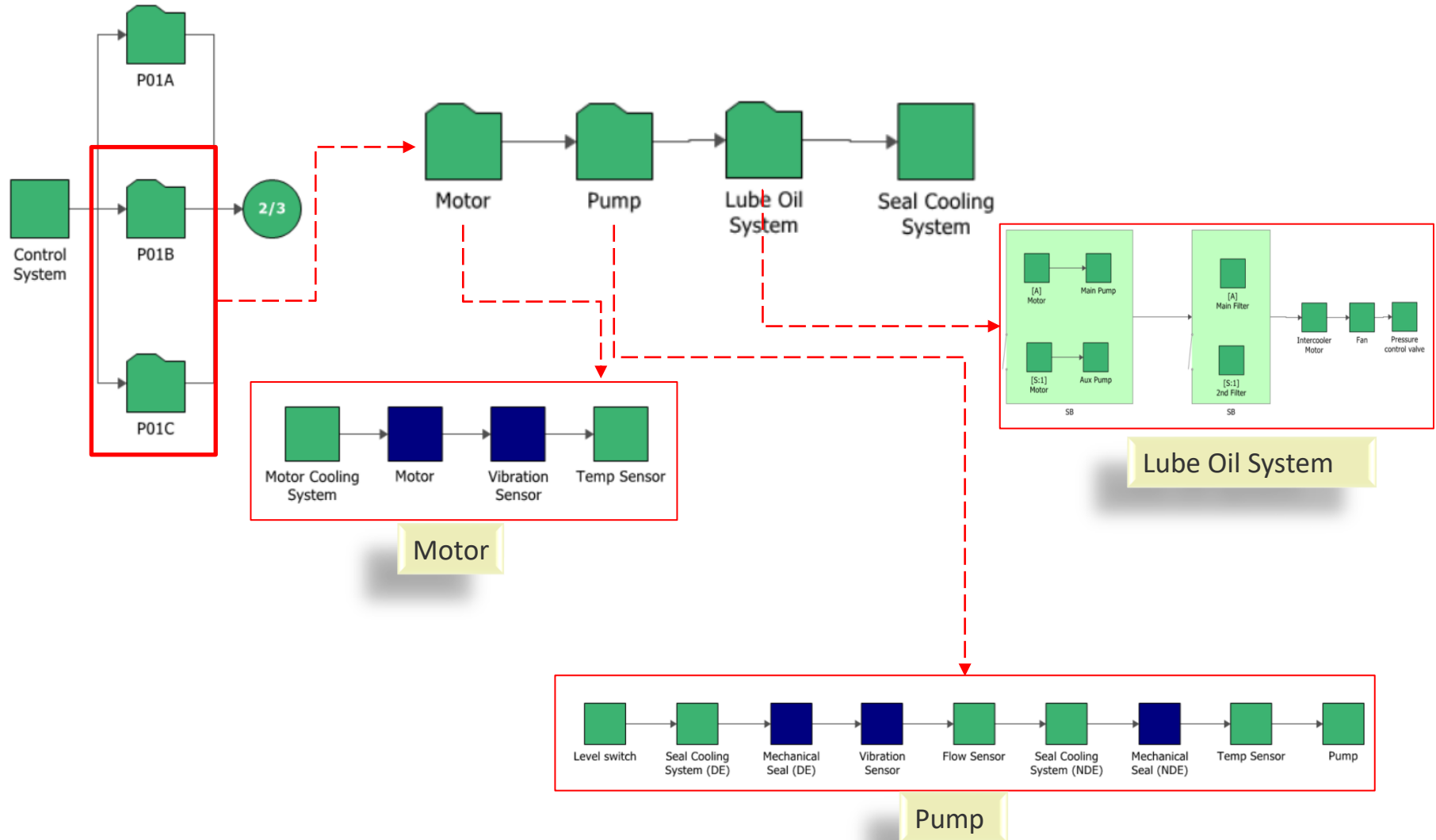
Reliability of the system at 8760 hrs for
1 out of 3 configurations
= 91.8%

Reliability of the system at 8640 hrs for
2 out of 3 configurations
=

Block Diagram-Main Configuration Lean Solution Pump System

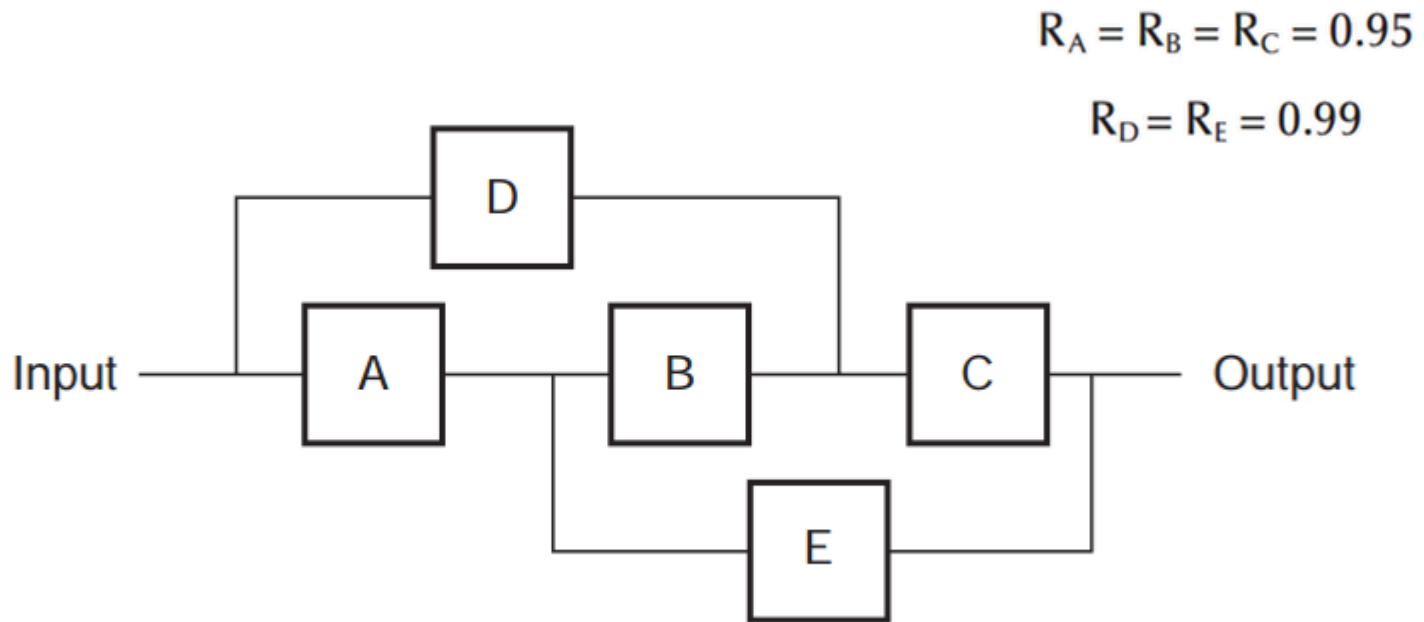


Block Diagram (P01B/C)



Exercise : Analysis For Complex System using BlockSim

Complex system is a system that can not be reduced to simple system.



Use BlockSim to estimate the system reliability

Recall: Availability Measures

1. Inherent, A_i

$$A_i = \frac{MTBF}{(MTBF + MTTR)}$$

2. Achieved, A_a

$$A_a = \frac{MTBM}{(MTBM + MMT)}$$

3. Operational, A_o

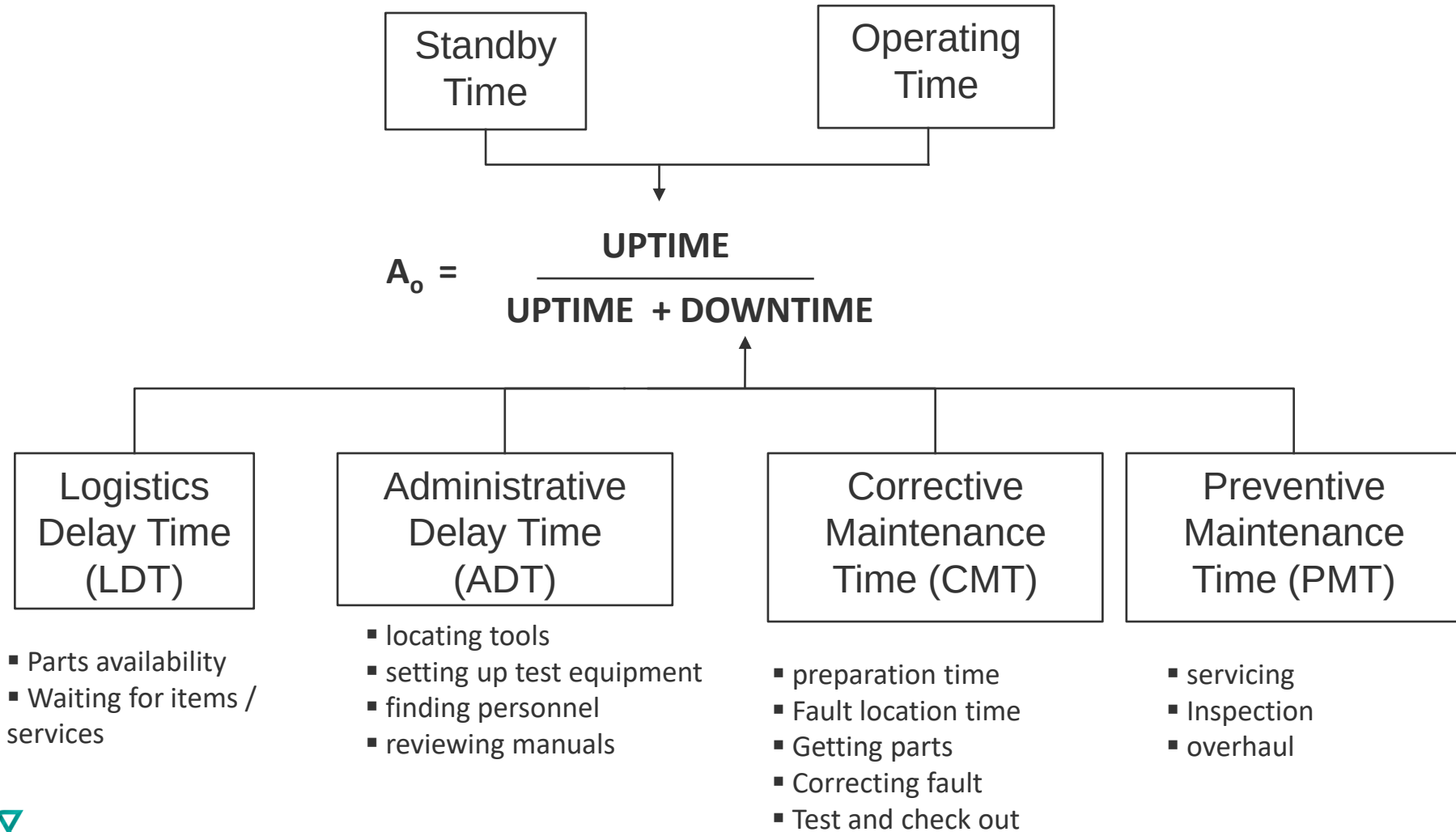
$$A_o = \frac{MTBM}{(MTBM + MMT + \overset{\nearrow}{MLDT})}$$

(LDT + ADT)

$$A_o = \frac{\text{Uptime}}{(\text{Uptime} + \text{Downtime})}$$

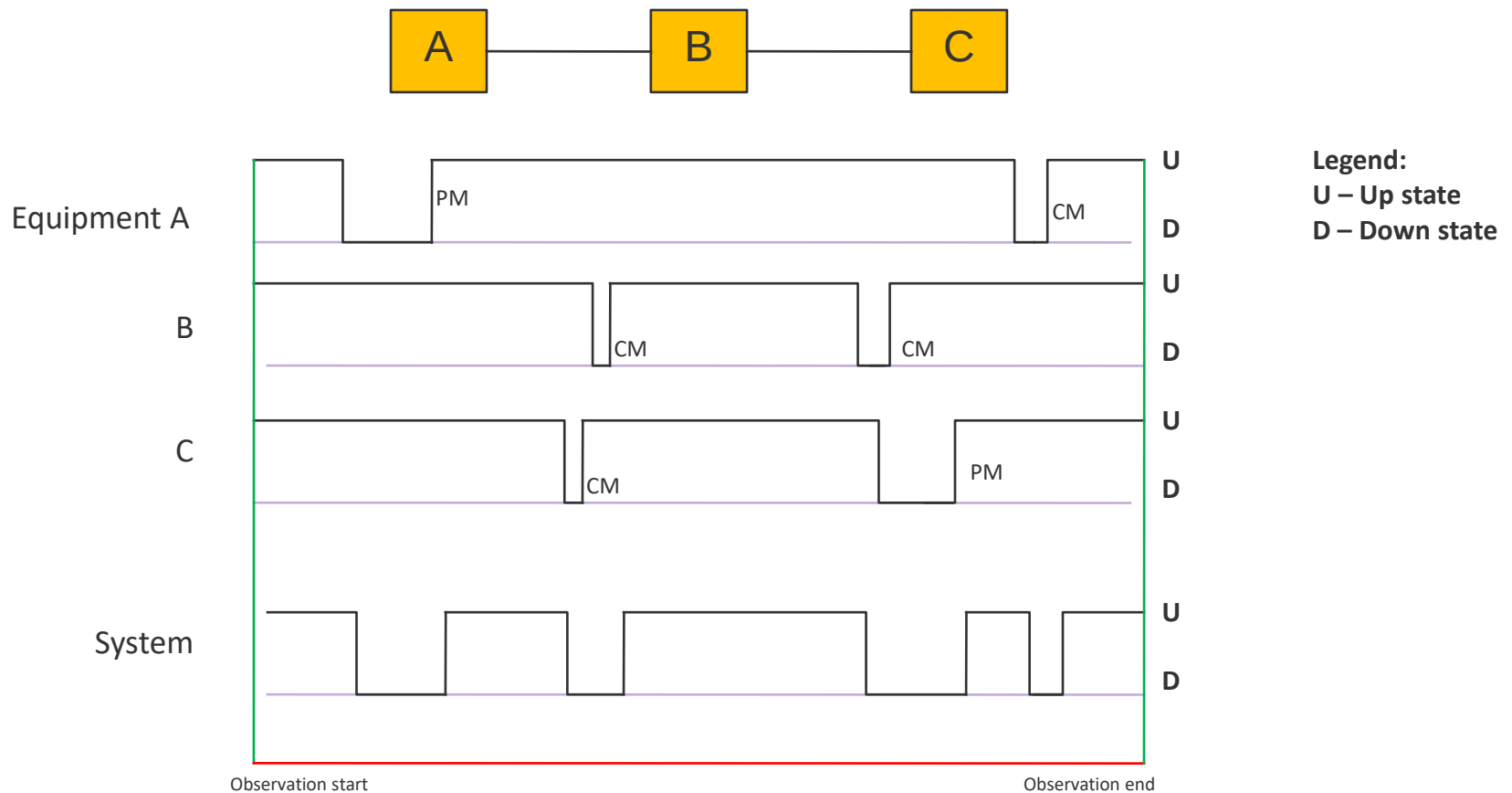
MTBF = mean time between failure
MTTR = mean time to repair
MTBM = mean time between maintenance
MMT = mean maintenance time
MLDT = mean logistics down time
LDT = logistics delay time
ADT = administrative delay time

Operational Availability



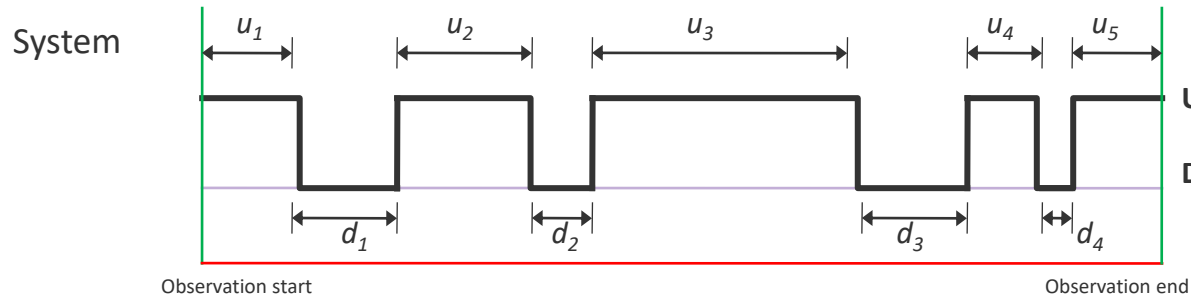
System Availability

Consider a system having 3 equipment in **series**. A down state of any equipment will bring the system down. Equipment can be in down state due to CM and PM actions



System Availability

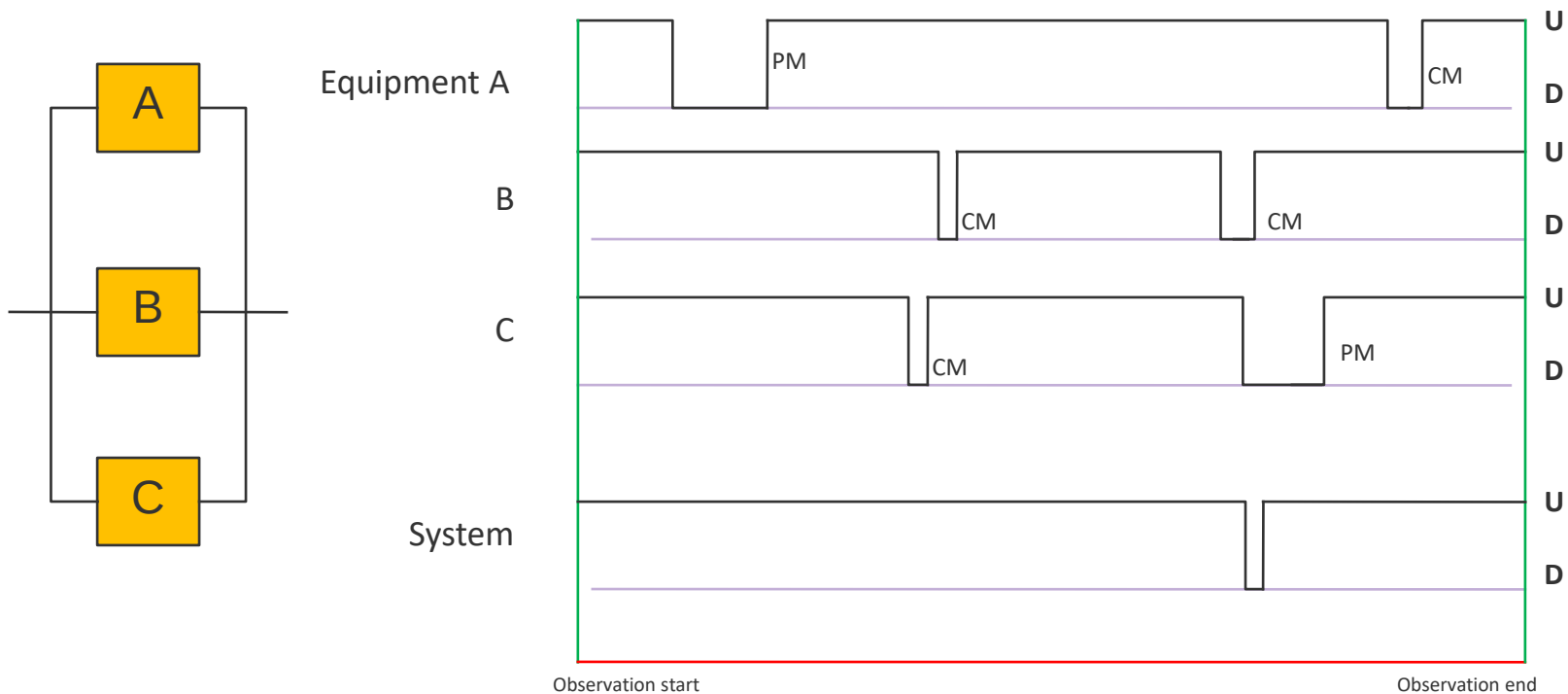
$$\text{System Availability} = \frac{\text{UPTIME}}{\text{UPTIME} + \text{DOWNTIME}}$$



$$\text{System Availability} = \frac{\sum_{i=1}^n u_i}{\sum_{i=1}^n u_i + \sum_{i=1}^n d_i}$$

System Availability

Now consider a system having 3 equipment in **parallel** configuration. The system is down when 2 equipment are in down state (2-out-of-3)



Legend:
U – Up state
D – Down state

Reliability & Maintainability vs Availability

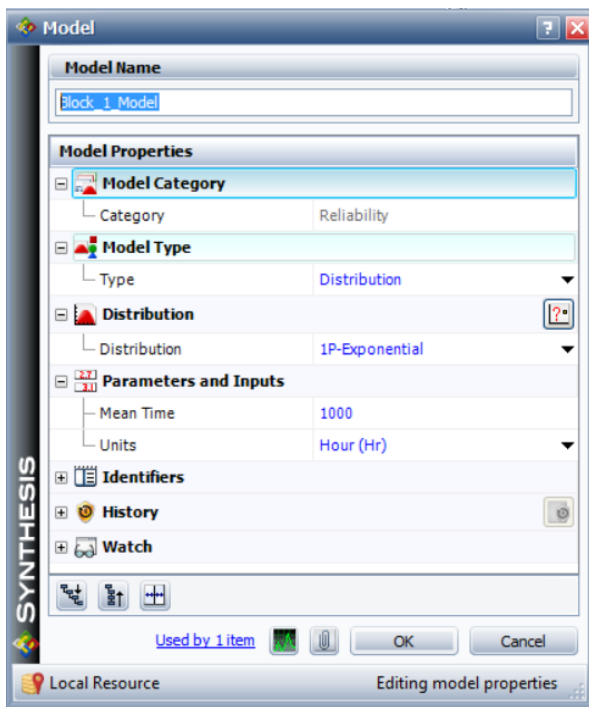
Availability is a function of reliability and maintainability

Reliability	Maintainability	Availability
↑	↑	↑
↑	→	↑
→	↑	↑
↓	→	↓
→	↓	↓
↓	↓	↓

The highest improvement in availability can be achieved by improving both reliability and maintainability !

Availability Calculation in BlockSim

Reliability



Model

Model Name: Block_1_Model

Model Properties

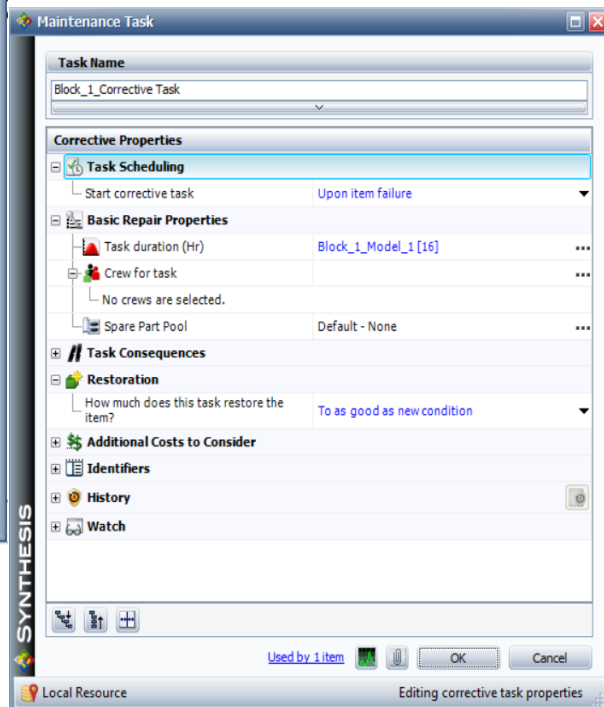
- Model Category: Reliability
- Model Type: Distribution
- Distribution: 1P-Exponential
- Parameters and Inputs:
 - Mean Time: 1000
 - Units: Hour (Hr)
- Identifiers
- History
- Watch

Used by 1 item

OK Cancel

Local Resource Editing model properties

Repair



Maintenance Task

Task Name: Block_1_Corrective Task

Corrective Properties

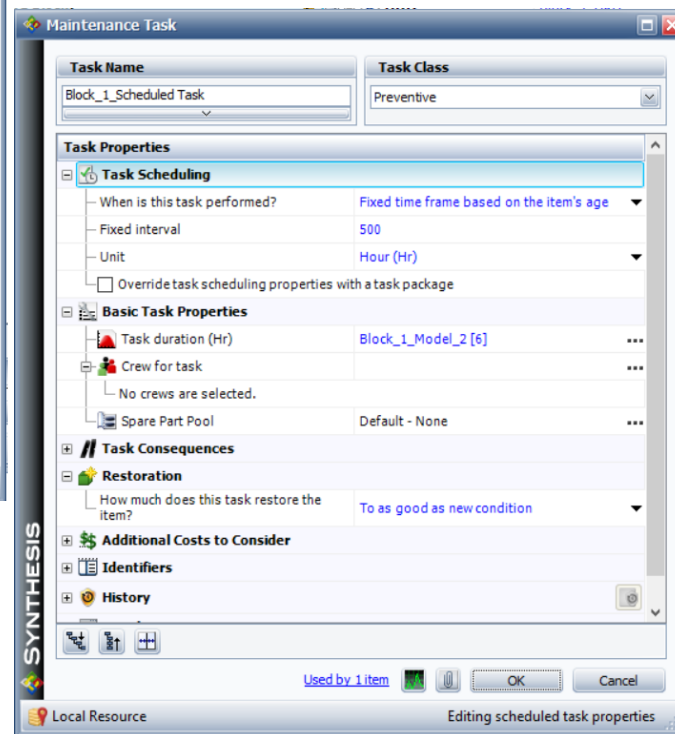
- Task Scheduling:
 - Start corrective task: Upon item failure
- Basic Repair Properties:
 - Task duration (Hr): Block_1_Model_1[16]
 - Crew for task: No crews are selected.
 - Spare Part Pool: Default - None
- Task Consequences
- Restoration:
 - How much does this task restore the item?: To as good as new condition
- Additional Costs to Consider
- Identifiers
- History
- Watch

Used by 1 item

OK Cancel

Local Resource Editing corrective task properties

PM



Maintenance Task

Task Name: Block_1_Scheduled Task

Task Class: Preventive

Task Properties

- Task Scheduling:
 - When is this task performed?: Fixed time frame based on the item's age
 - Fixed interval: 500
 - Unit: Hour (Hr)
 - Override task scheduling properties with a task package: ☐
- Basic Task Properties:
 - Task duration (Hr): Block_1_Model_2[6]
 - Crew for task: No crews are selected.
 - Spare Part Pool: Default - None
- Task Consequences
- Restoration:
 - How much does this task restore the item?: To as good as new condition
- Additional Costs to Consider
- Identifiers
- History

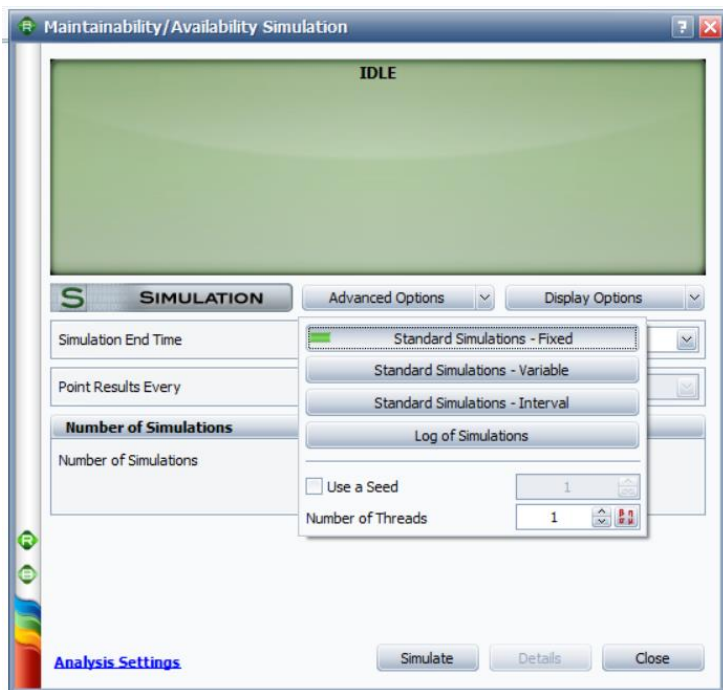
Used by 1 item

OK Cancel

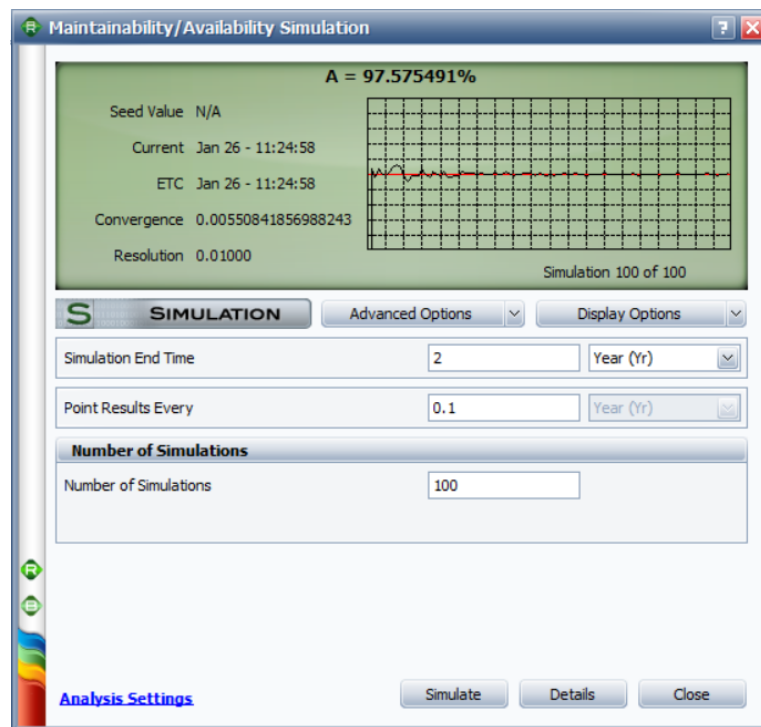
Local Resource Editing scheduled task properties

Availability Calculation in BlockSim

Simulation Setup



Simulation Result



References

Wolstenholme, L.C. (1999). *Reliability Modeling: A statistical approach*. Boca Raton, Florida: Chapman & Hall/CRC.

Verma, A.K., Srividya, A. and Gaonkar, R.S.P. (2007). *Fuzzy-reliability engineering: Concepts and application*. New Delhi, India: Narosa Publishing House.

<http://www.weibull.com/basics/lifedata.htm>

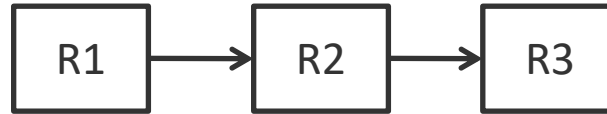
<http://www.reliabilityeducation.com/rbd.pdf>

<http://www.win.tue.nl/~mchaudro/.../Reliability%20Block%20Diagrams.pdf>

<http://www.chinareel.com/hotwire/issue27/hottopics27.htm>

Exercise for Series System (Solution)

Consider a system with three components in series.



You are required to achieve a system reliability of 0.98 over a 800-hours non-stop operation.

1. What would be the target failure rate for the system?

$$R_s(t) = e^{-\lambda_s t}$$

$$0.98 = e^{-\lambda_s (800)}$$

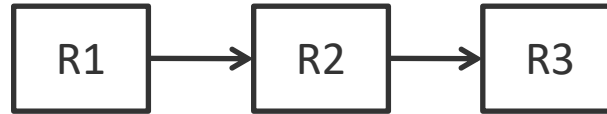
$$\ln(0.98) = -\lambda_s (800)$$

$$\lambda_s = -\frac{\ln(0.98)}{800}$$

$$\lambda_s = 2.53 \times 10^{-5} \text{ per hour}$$

Exercise for Series System (Solution)

Consider a system with three components in series.



You are required to achieve a system reliability of 0.98 over a 800-hours non-stop operation.

2. What would be the system MTBF be?

$$\begin{aligned} MTBF_s &= \frac{1}{\lambda_s} \\ &= \frac{1}{2.53 \times 10^{-5}} \\ &= 39599 \text{ hours} \\ &= 1650 \text{ days} \end{aligned}$$

Exercise for Series System (Solution)

3. Assuming the component failures are identically distributed,

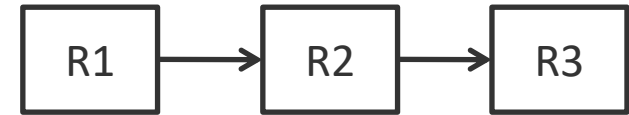
a) What should be the component failure rate?

$$\lambda_s = \lambda_1 + \lambda_2 + \lambda_3$$

$$2.53 \times 10^{-5} = 3\lambda$$

$$\lambda = \frac{2.53 \times 10^{-5}}{3}$$

$$\lambda = 8.42 \times 10^{-6} \text{ per hour}$$



b) What would be the component MTBF?

$$\begin{aligned}
 MTBF &= \frac{1}{\lambda} = \frac{1}{8.42 \times 10^{-6}} \\
 &= 118,796 \text{ hours} \\
 &= 4950 \text{ days}
 \end{aligned}$$

c) What should be the component reliability?

$$\begin{aligned}
 R(t) &= e^{-\lambda t} \\
 &= e^{-(8.42 \times 10^{-6})(800)} \\
 &= 0.993
 \end{aligned}$$

Session 3

Advanced System Modelling

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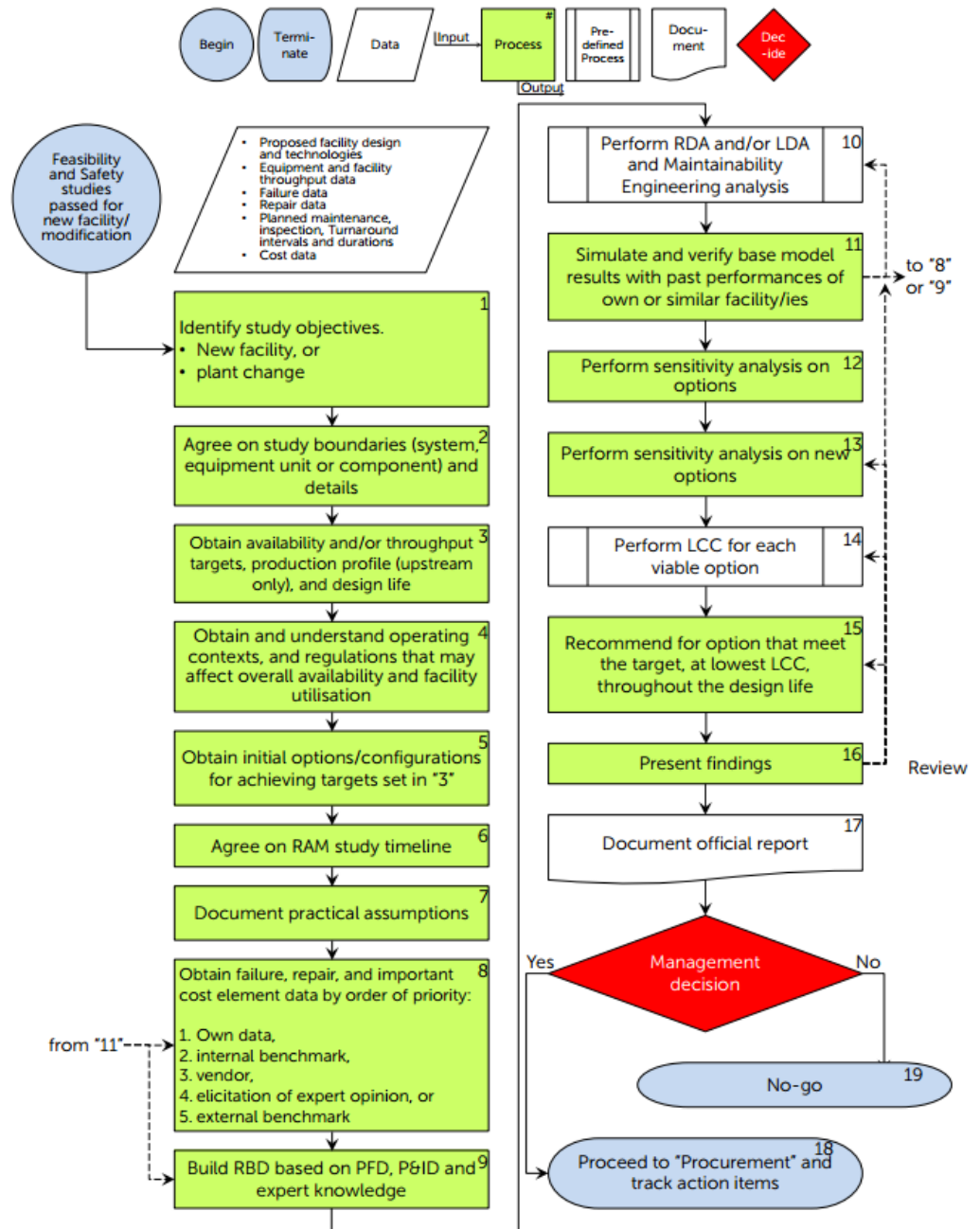
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Session Schedule

Session	Content	Main Topics	Day	Time
3.	Advanced System Reliability- (Dr Ainul)	- Case Study Presentation - Maintenance model - Standby Configuration - Load Sharing Configuration - Throughput Analysis - RAM and Life Cycle cost	Day 3 (AM)	9:00am - 10:30am Break (15 Min) 10:45am – 1:00pm Lunch/Prayer (1:00pm – 2:00pm)
4.	Fault Tree Analysis- (Dr Hilmi)	- Introduction to Fault tree Analysis - Fault tree construction - Minimal cut sets - Qualitative analysis - Quantitative analysis	Day 3 (PM) Day 4 (AM)	2:00pm – 3:30pm Break (15 Min) 3:45pm – 5:00pm 9:00am - 10:30am Break (15 Min) 10:45am – 1:00pm Lunch/Prayer (1:00pm – 2:00pm)
5.	Introduction to Multi State system and Markov process in reliability- (Dr Masdi)	- Multi-State system and Markov Model - Discrete Time Markov - Continuous Time Markov - Application of multistate system modelling (IPF) - Post Assessment	Day 4 (PM)	2:00pm – 3:30pm Break (15 Min) 3:45pm – 5:00pm

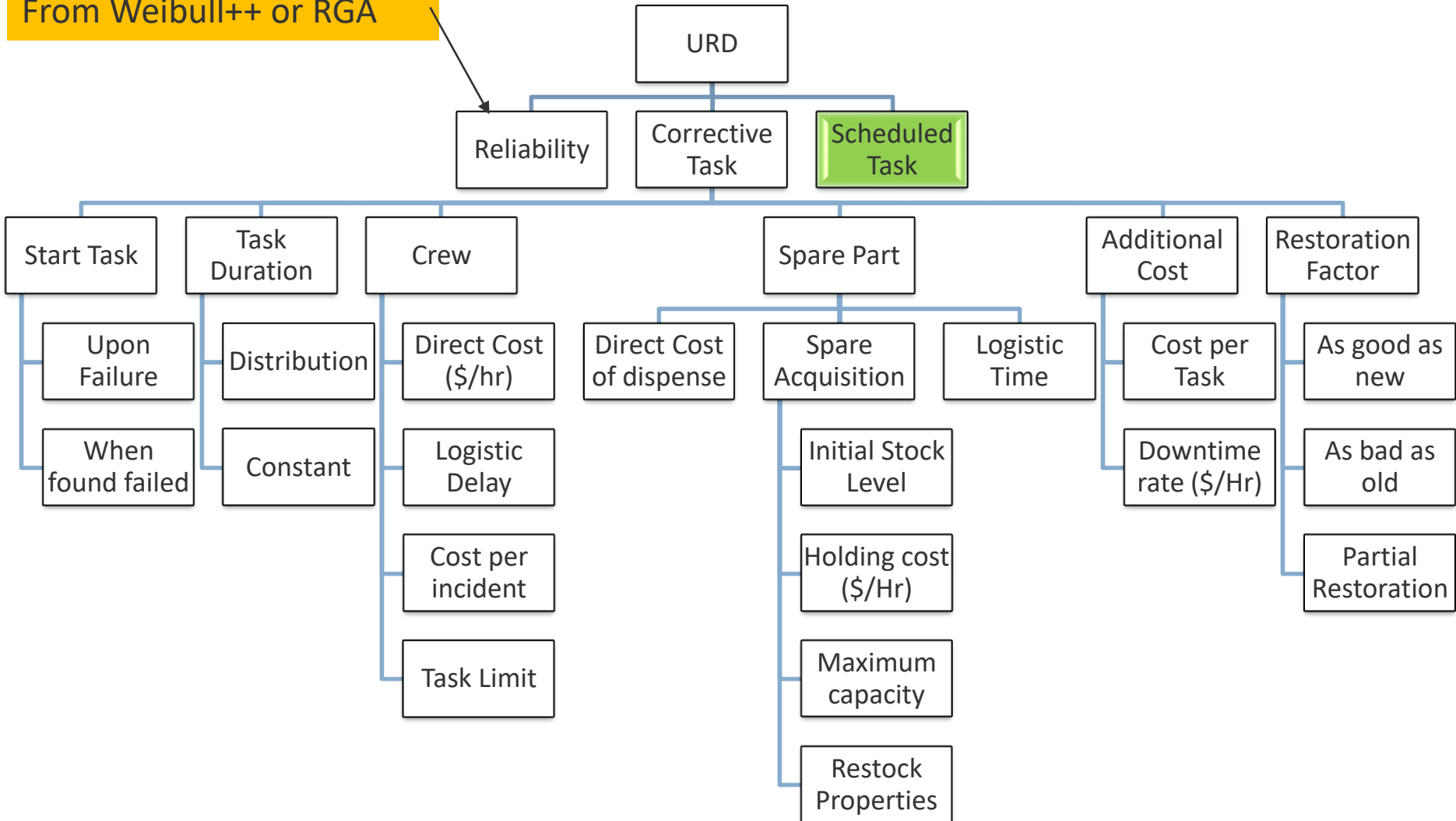
RAM Workflow (RIM 150522 V5)

RAM work flow

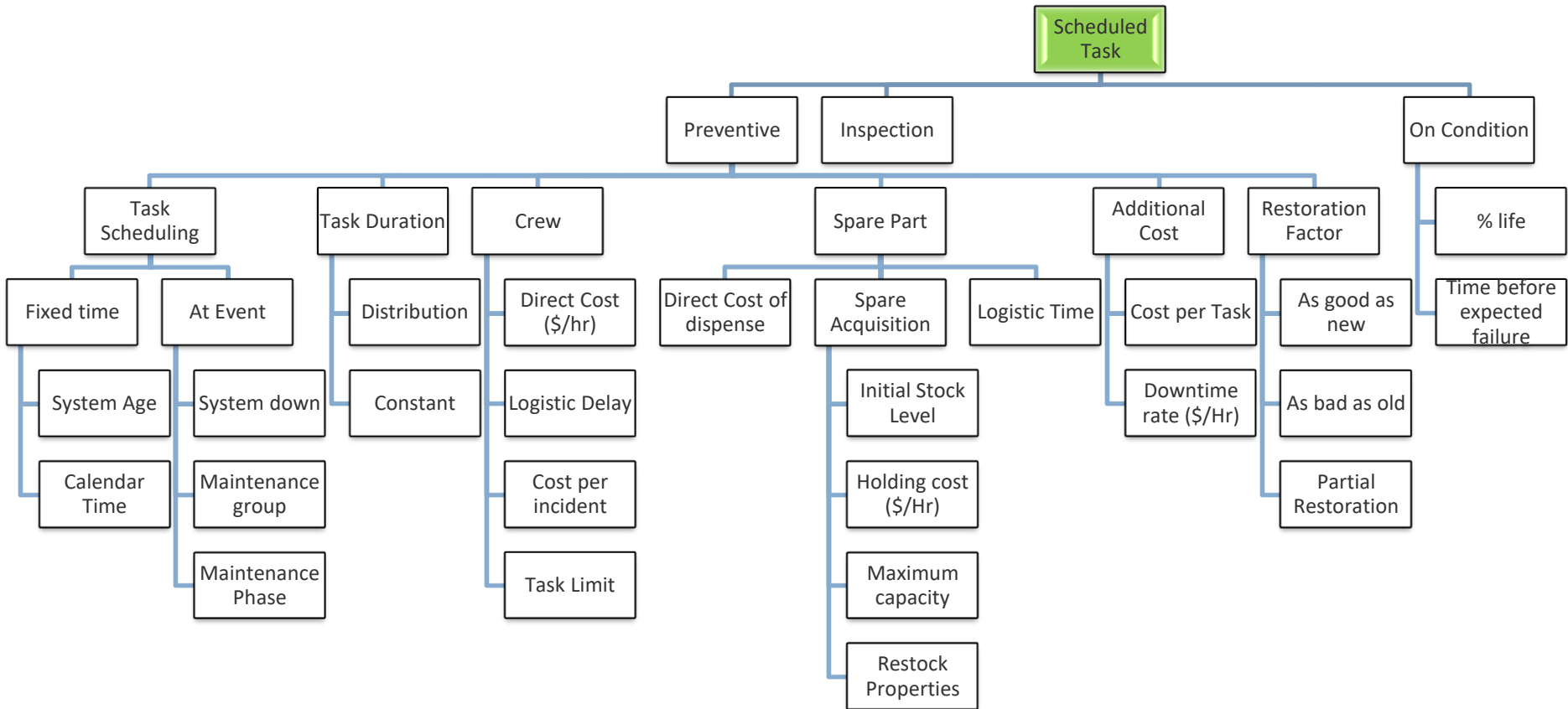


BlockSim Simulation: URD

From Weibull++ or RGA



Scheduled Task

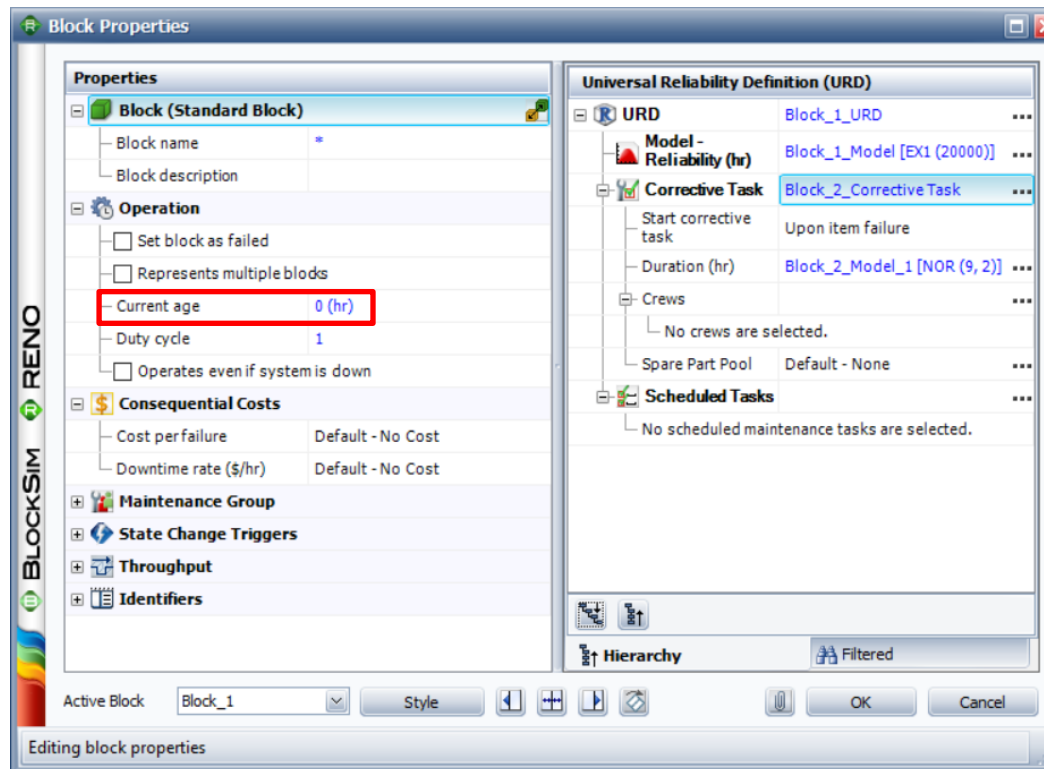


Block with used component

- Some equipment or system can be at certain age at the start of the simulation which would affect its reliability
- The conditional reliability is given as

$$R(t|T) = \frac{R(t + T)}{R(T)}$$

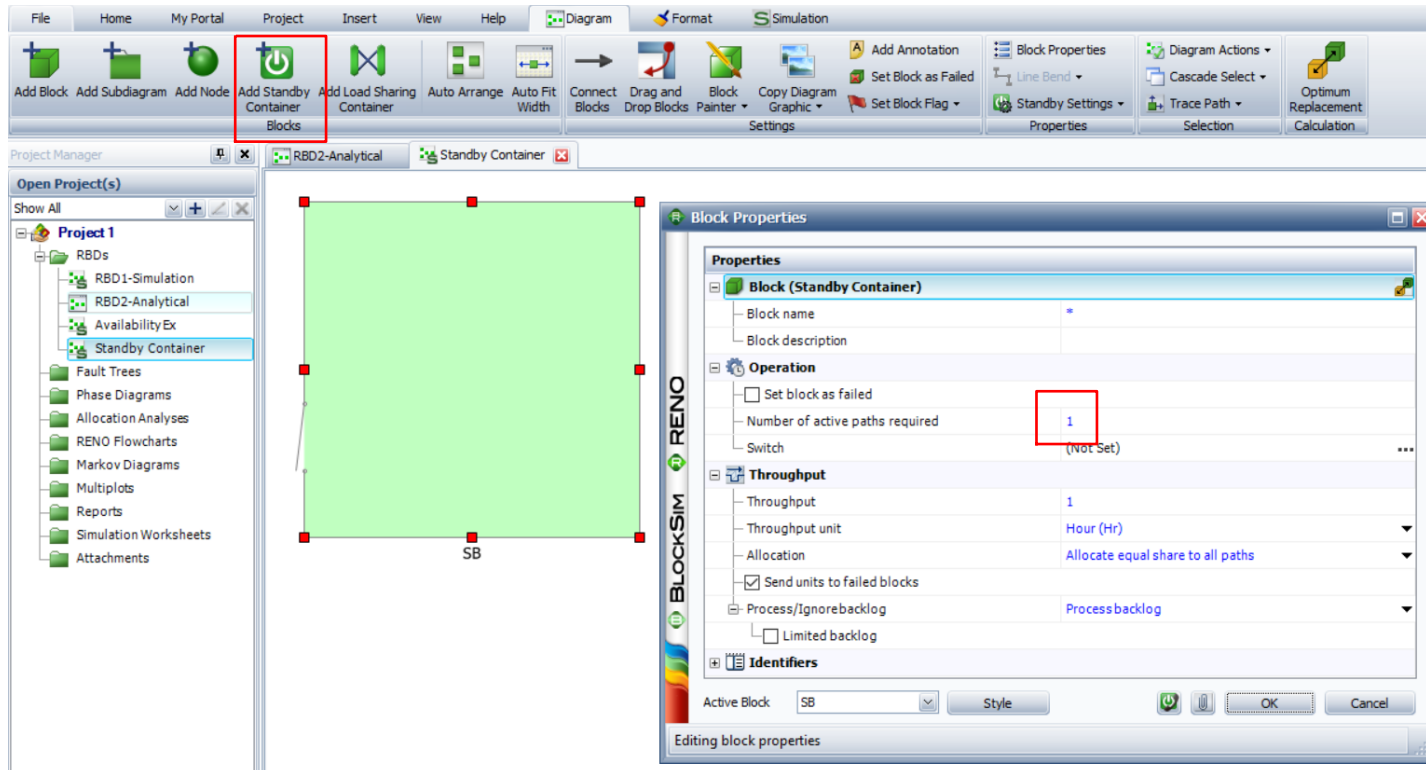
Where : T = Current Age, t = mission time



Block with standby component

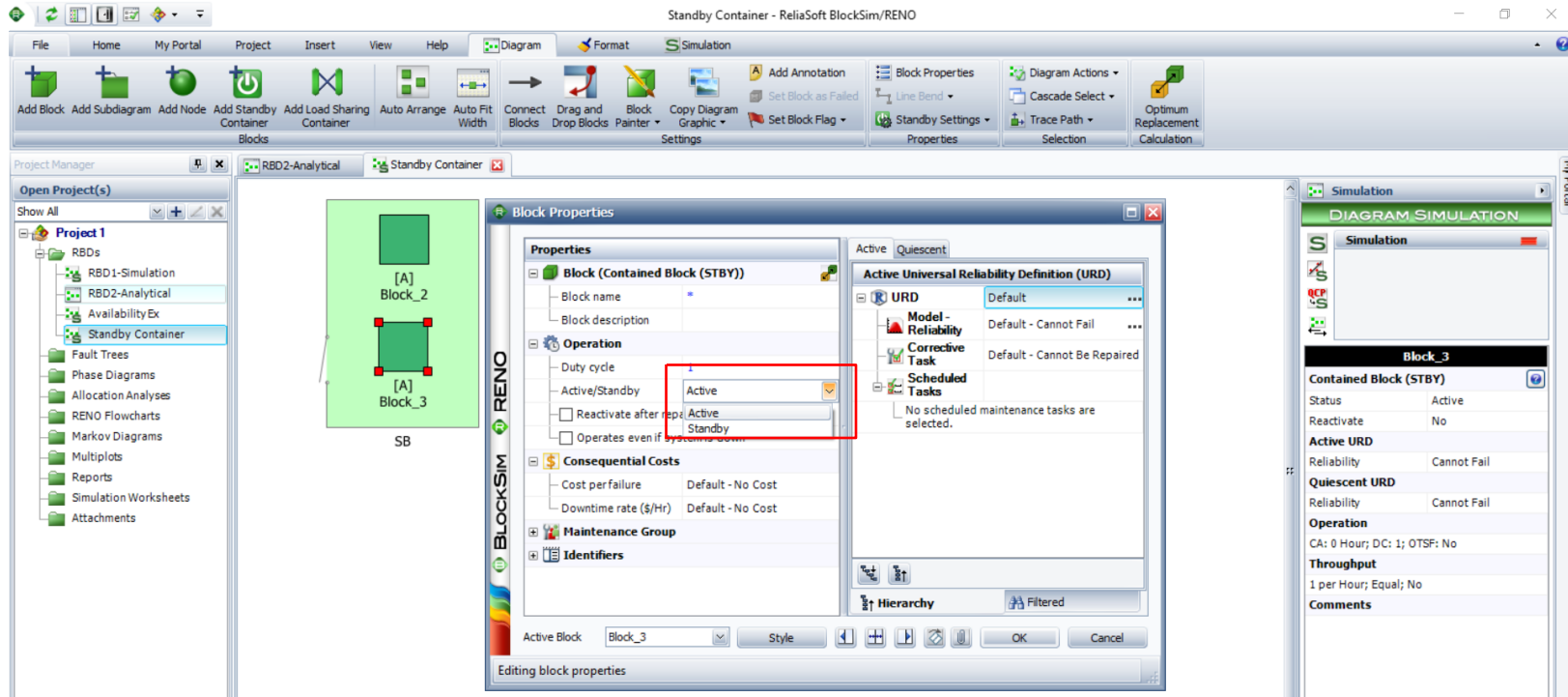
- **The role of the standby container.**
 - Units in a standby container are considered to be in a standby redundancy.
 - The container also acts as the switching mechanism (i.e., is responsible for turning active units on/off).
 - A standby container can fail and be repaired. A failed container does not cause the system to fail. However, a failed standby container will not activate units if needed.
 - The container is also used to define the number of required active units, k/n .
 - There is no single starting point requirement. Each new starting point defines a new component in standby.

BlockSim-Standby Container



The screenshot displays the BlockSim software interface. The top menu bar includes File, Home, My Portal, Project, Insert, View, Help, Diagram, Format, and Simulation. The 'Diagram' tab is active, showing a toolbar with icons for adding blocks, subdiagrams, nodes, and containers. The 'Add Standby Container' icon is highlighted with a red box. Below the toolbar, the 'Project Manager' pane on the left shows a tree view of the project structure, with 'Standby Container' selected. The main workspace shows a green rectangular block labeled 'SB'. The 'Block Properties' dialog box is open on the right, showing the properties for the 'Block (Standby Container)'. The 'Operation' section is expanded, and the 'Number of active paths required' is set to 1, which is also highlighted with a red box. Other settings include 'Throughput' set to 1, 'Throughput unit' set to Hour (Hr), and 'Allocation' set to Allocate equal share to all paths. The 'Identifiers' section shows the 'Active Block' as SB.

BlockSim-Standby Container

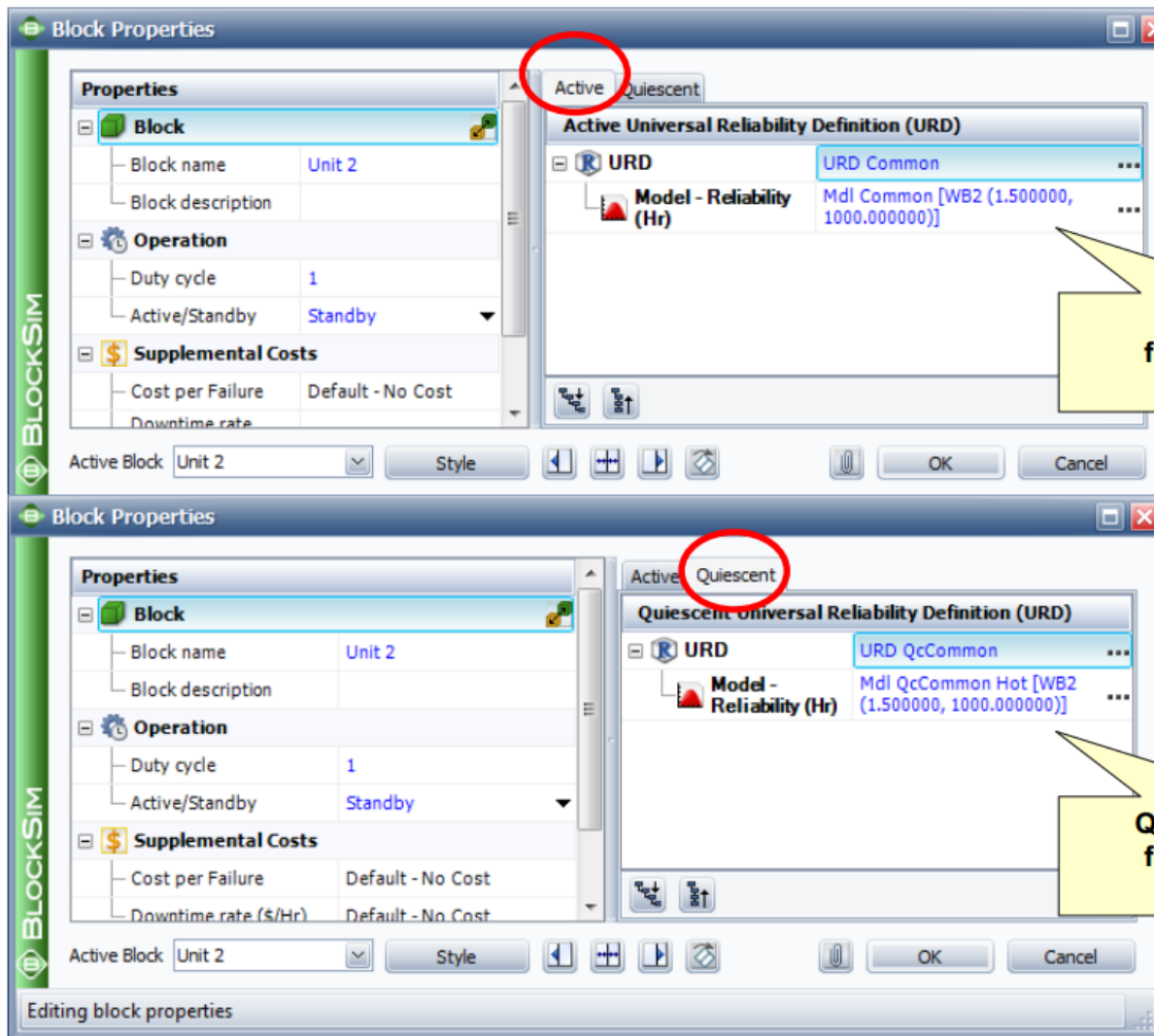


The screenshot displays the ReliaSoft BlockSim/RENO software interface. The main window is titled "Standby Container - ReliaSoft BlockSim/RENO". The interface includes a menu bar (File, Home, My Portal, Project, Insert, View, Help), a toolbar with various diagram and simulation tools, and a Project Manager on the left. The Project Manager shows a tree structure under "Project 1" with items like "RBDs", "RBD1-Simulation", "RBD2-Analytical", "AvailabilityEx", "Standby Container", "Fault Trees", "Phase Diagrams", "Allocation Analyses", "RENO Flowcharts", "Markov Diagrams", "Multiplots", "Reports", "Simulation Worksheets", and "Attachments".

The central workspace shows a diagram with two blocks: "[A] Block_2" and "[A] Block_3". Block_3 is highlighted with a red box, and its properties are displayed in the "Block Properties" dialog. The "Block Properties" dialog has tabs for "Properties", "Active", and "Quiescent". The "Properties" tab is active, showing fields for "Block name", "Block description", "Operation", "Consequential Costs", "Maintenance Group", and "Identifiers". The "Operation" section includes a "Duty cycle" dropdown menu, which is currently set to "Active". The "Active" tab is also visible, showing the "Active Universal Reliability Definition (URD)" settings.

On the right side, the "Simulation" panel is open, displaying "DIAGRAM SIMULATION" and "Simulation" results. It includes a table for "Block_3" with columns for "Status", "Reactive", "Active URD", "Quiescent URD", "Operation", "Throughput", and "Comments".

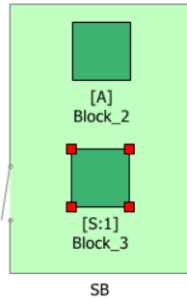
BlockSim-Standby Container



Active (energized)
failure distribution for a
Contained Block

Quiescent (de-energized)
failure distribution for a
Contained Block

BlockSim-Standby Container



Maintainability/Availability Simulation

A = 99.991361%

Seed Value: N/A
Current: Jan 27 - 10:52:37
ETC: Jan 27 - 10:52:37
Convergence: 8.72684329585871E-05
Resolution: 0.01000

Simulation 100 of 100

SIMULATION Advanced Options Display Options

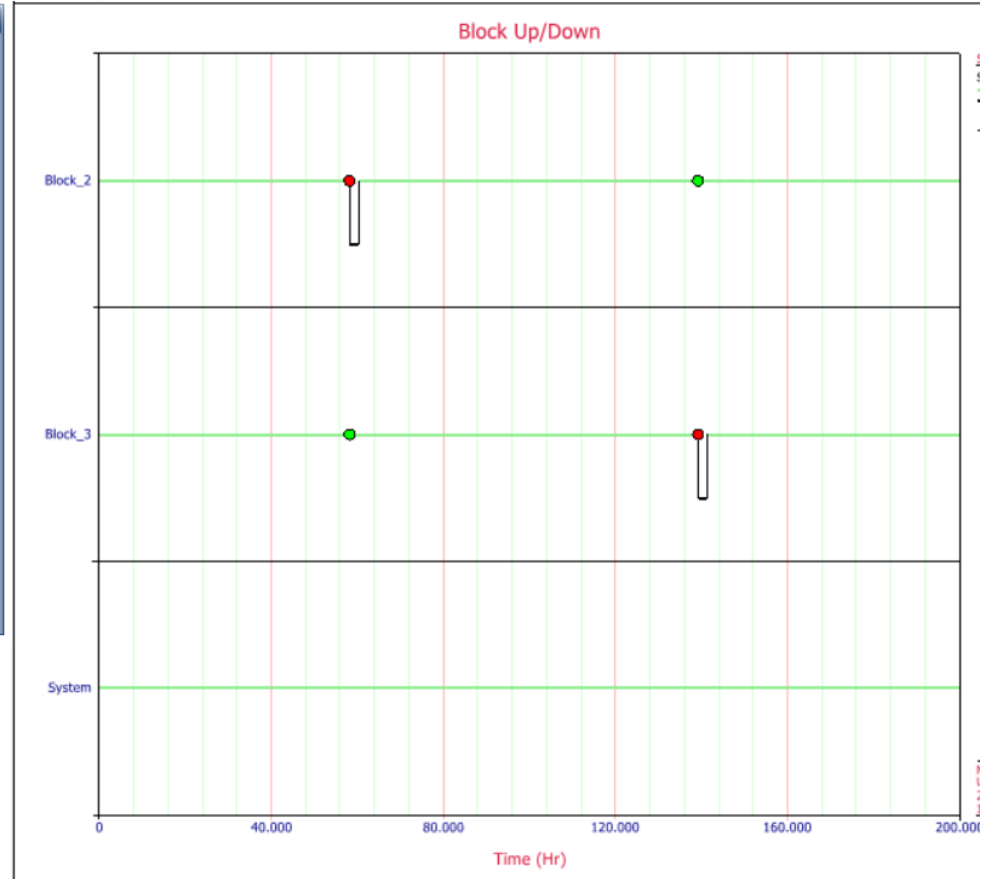
Simulation End Time: 1000 Hour (Hr)

Point Results Every: 0.1 Hour (Hr)

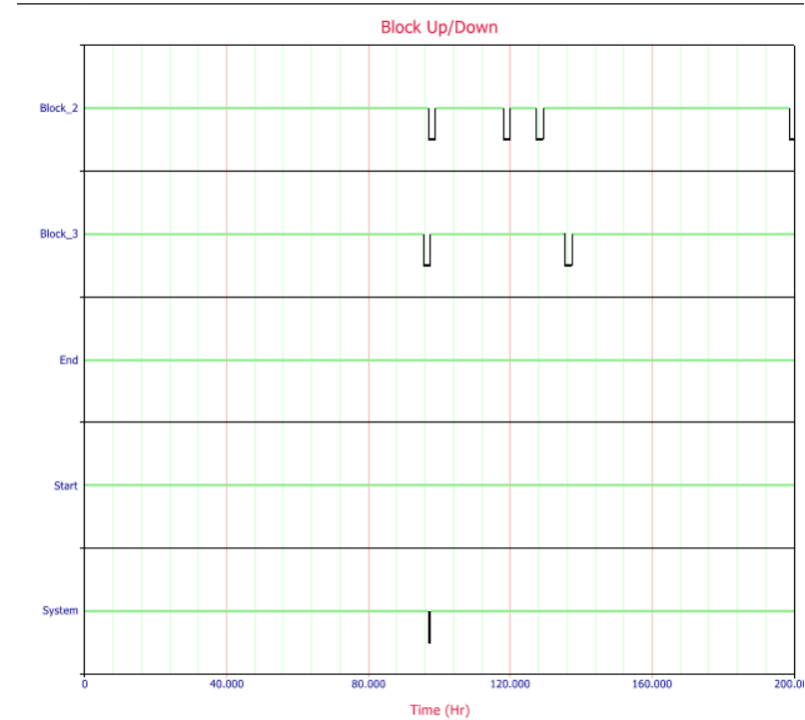
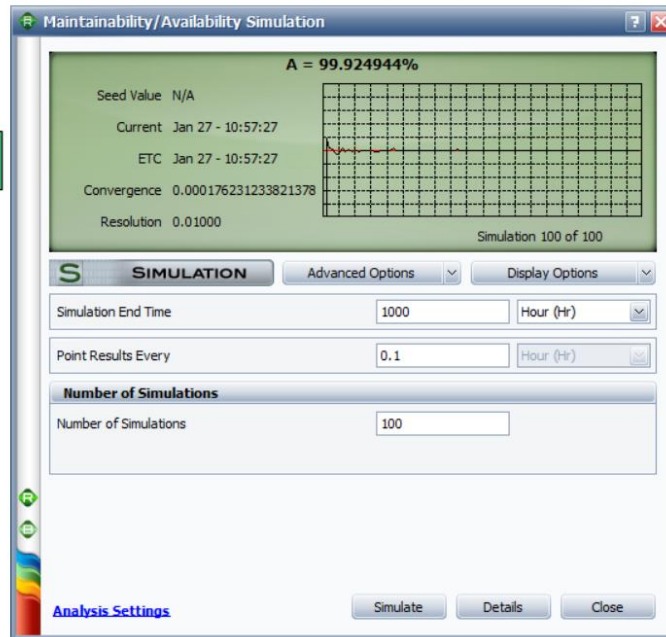
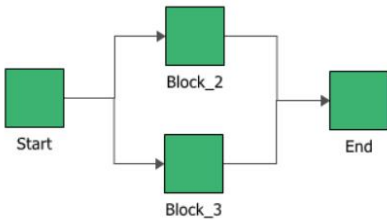
Number of Simulations

Number of Simulations: 100

Analysis Settings Simulate Details Close



Simple Parallel



Identifying Critical component

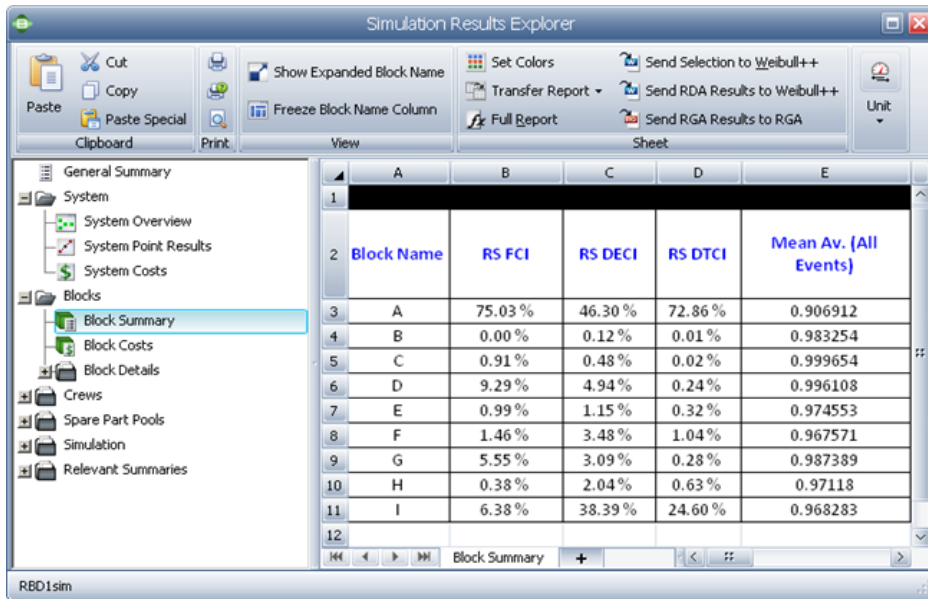
- **RS FCI (Failure Criticality Index)**
 - The index is a relative measure of the number of system failures caused by this component.
 - Percentage of times that failure of this component caused the system to fail.

$$RS\ FCI = \frac{\textit{Number of System Downing Failures (Component)}}{\textit{Number of System Failures}}$$

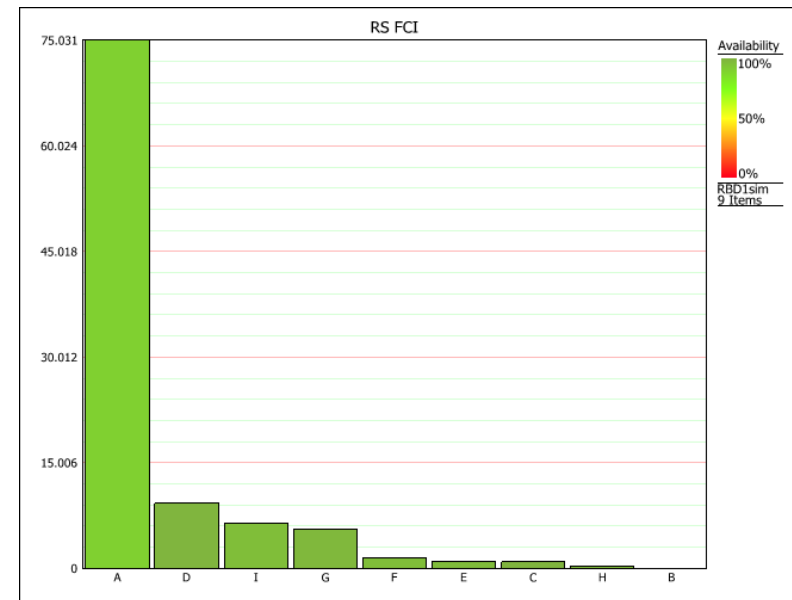
- It only includes failure events (it excludes PM and inspection downing events).

Example: RS FCI

If we simulate the system's operation for 5,000 hours in BlockSim, we obtain the following Block Summary report.



Block Name	RS FCI	RS DECI	RS DTCT	Mean Av. (All Events)
A	75.03 %	46.30 %	72.86 %	0.906912
B	0.00 %	0.12 %	0.01 %	0.983254
C	0.91 %	0.48 %	0.02 %	0.999654
D	9.29 %	4.94 %	0.24 %	0.996108
E	0.99 %	1.15 %	0.32 %	0.974553
F	1.46 %	3.48 %	1.04 %	0.967571
G	5.55 %	3.09 %	0.28 %	0.987389
H	0.38 %	2.04 %	0.63 %	0.97118
I	6.38 %	38.39 %	24.60 %	0.968283



For component A, RS FCI = 75.03%. This implies that 75.03% of the times that the system failed, a component A failure was responsible.

Identifying Critical component

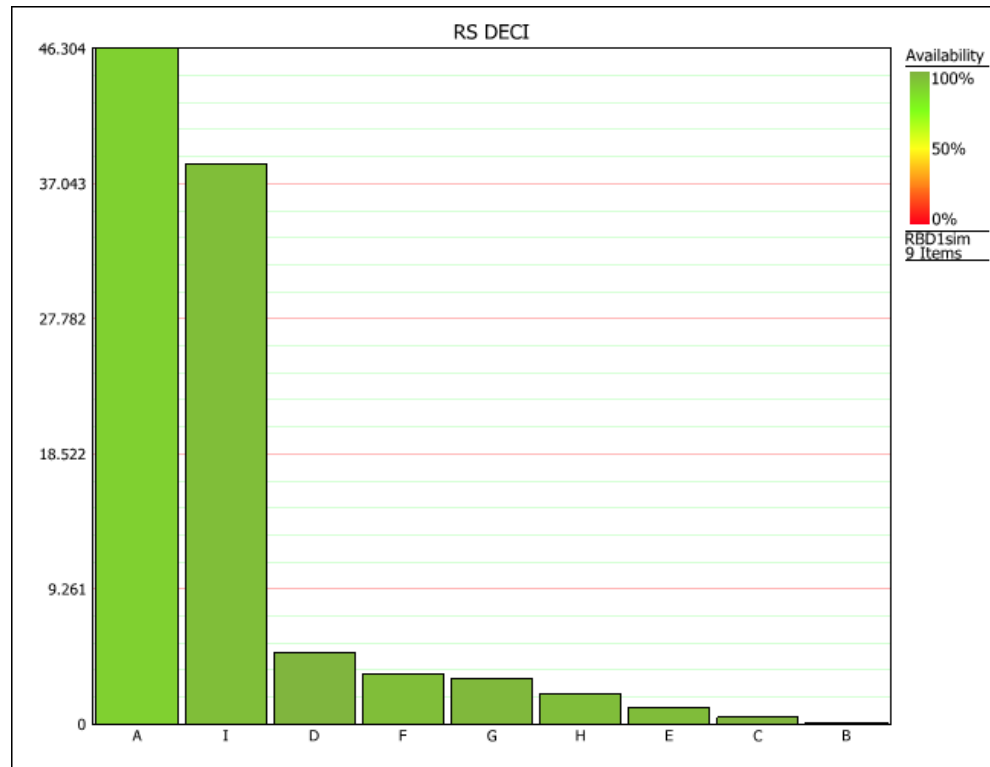
- **RS DECI (Downing Event Criticality Index)**
 - The index is a relative measure of the number of system downing events caused by this component.
 - Percentage of times that any downing event of this component caused the system to go down.

$$RS\ DECI = \frac{\text{Number of System Downing Events (Component)}}{\text{Number of System Downing Events}}$$

- It includes all downing events.

Example: RS DECI

In the simulation results, we see that for component A, RS DECI = 46.30%. This implies that 46.30% of the times that the system was down were due to component A being down.



Identifying Critical component

Other measures:

1. Block Failure Ranking-Ranking based on number of failure
2. Block Downtime Ranking- Ranking based on total downtime
3. Block Availability Ranking-Ranking based on the block availability

Case Study

Modelling Periodical Inspection and Early Failure Detection

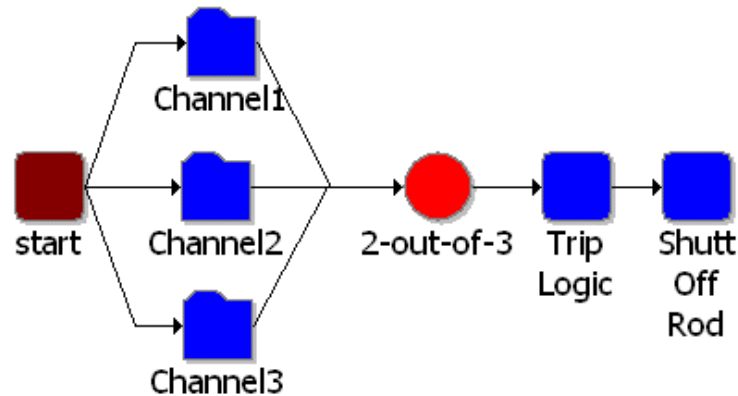
To improve availability, integrity, productivity and safety, many organizations implement regular monitoring of critical parts of the system, enabling regular feedback about the status of the equipment and the degradation over time before reaching failure. This allows quick response to disturbances and helps avoid unplanned shutdowns or safety problems.

There is an increased trend toward introducing modern censoring and early failure detection techniques into the design of advanced control centres in various applications such as manufacturing, navy ships, process industry and nuclear plants. The role of these computerized systems is to detect faults and degraded performances and alert the operators to anomalies in the system to help them manage the problems as quickly as possible with minimal adverse consequences.

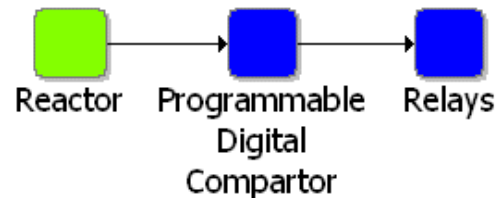
Case Study

Modelling Periodical Inspection and Early Failure Detection

Let us consider the example of a Shutdown System in a nuclear power plants control system. The simplified Reliability Block Diagram (RBD) of the system is shown below.



The channels are represented by the following RBD.



Source: <http://www.weibull.com/hotwire/issue62/hottopics62.htm>

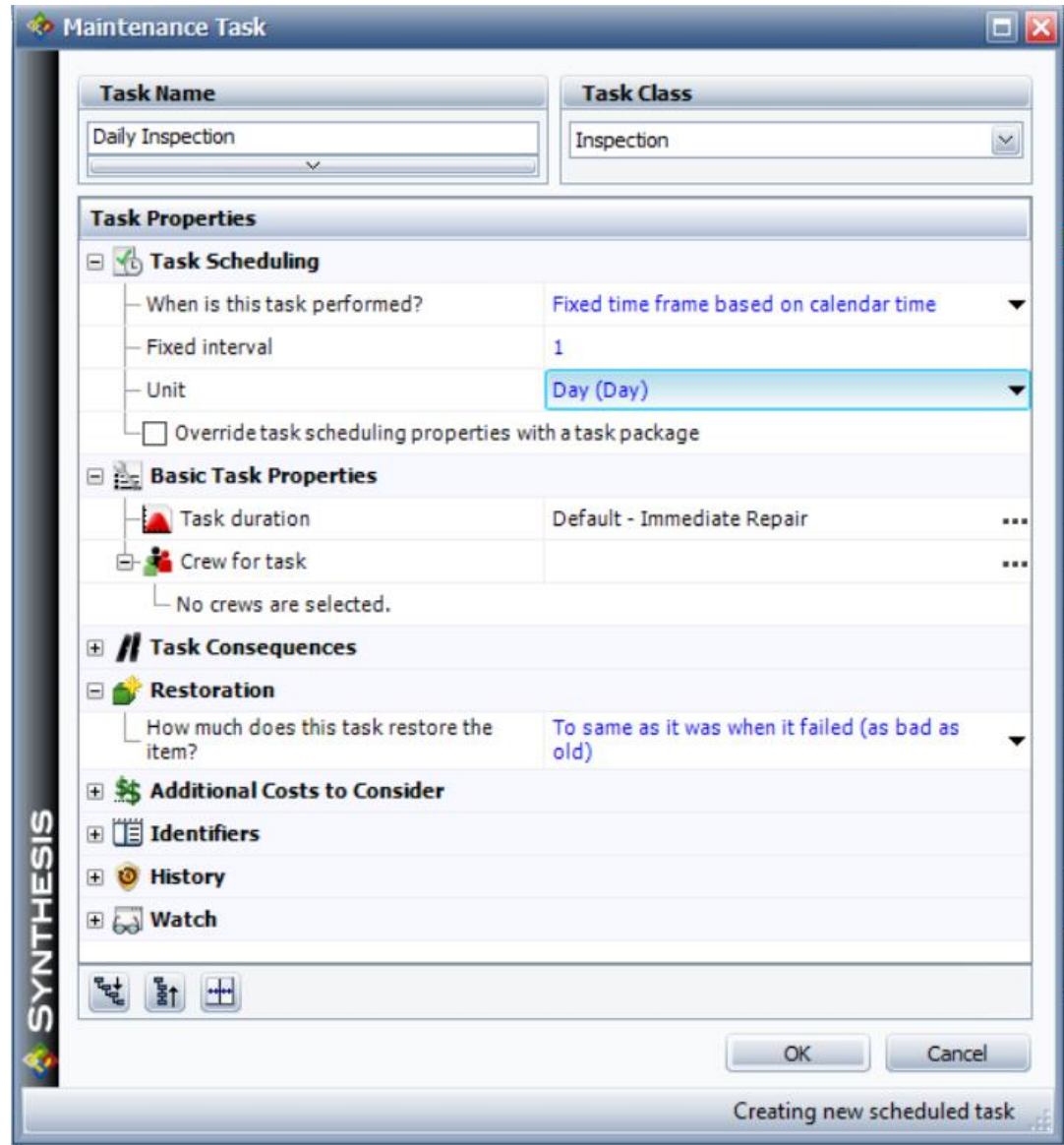
Case Study

The Reactor component of the system is considered to be a critical component that needs regular monitoring and improved visibility to allow early detection of degradation that indicates an imminent failure. The following table summarizes the failure distributions and the repair distributions and cost of repair for the blocks in the above two RBDs.

	Failure Distribution	Repair Distribution	Cost of Repair
Reactor	WBL(Beta=3, Eta=10,000)	NOR (Mean=400, Std=10)	200,000
Programmable Digital Comparator	WBL(Beta=2.4, Eta=80,000)	NOR (Mean=7, Std=2)	5,000
Relays	WBL(Beta=3, Eta=70,000)	NOR (Mean=5, Std=2)	3,000
Trip Logic	WBL(Beta=2.4, Eta=100,000)	NOR (Mean=4, Std=1)	1,000
Shut Off Rod	WBL(Beta=2.6, Eta=50,000)	NOR (Mean=16, Std=2)	5,000
start	Does not fail		

Case Study

Sensors are used in the Reactors to detect degradation in their parameters. The sensors monitor the Reactor daily. In BlockSim, this can be modelled using the Inspection Policy by clicking the Create New Inspection Policy button, in the Inspection tab of the Reactor's Block Properties, selecting Upon fixed time interval based on: and System Age and entering the value that represents the frequency (every 24h) of the sensors.



Maintenance Task

Task Name
Daily Inspection

Task Class
Inspection

Task Properties

- ☒ **Task Scheduling**
 - When is this task performed? Fixed time frame based on calendar time
 - Fixed interval 1
 - Unit Day (Day)
 - ☐ Override task scheduling properties with a task package
- ☒ **Basic Task Properties**
 - Task duration Default - Immediate Repair
 - Crew for task No crews are selected.
- ☒ **Task Consequences**
- ☒ **Restoration**
 - How much does this task restore the item? To same as it was when it failed (as bad as old)
- ☒ **Additional Costs to Consider**
- ☒ **Identifiers**
- ☒ **History**
- ☒ **Watch**

OK Cancel

Creating new scheduled task

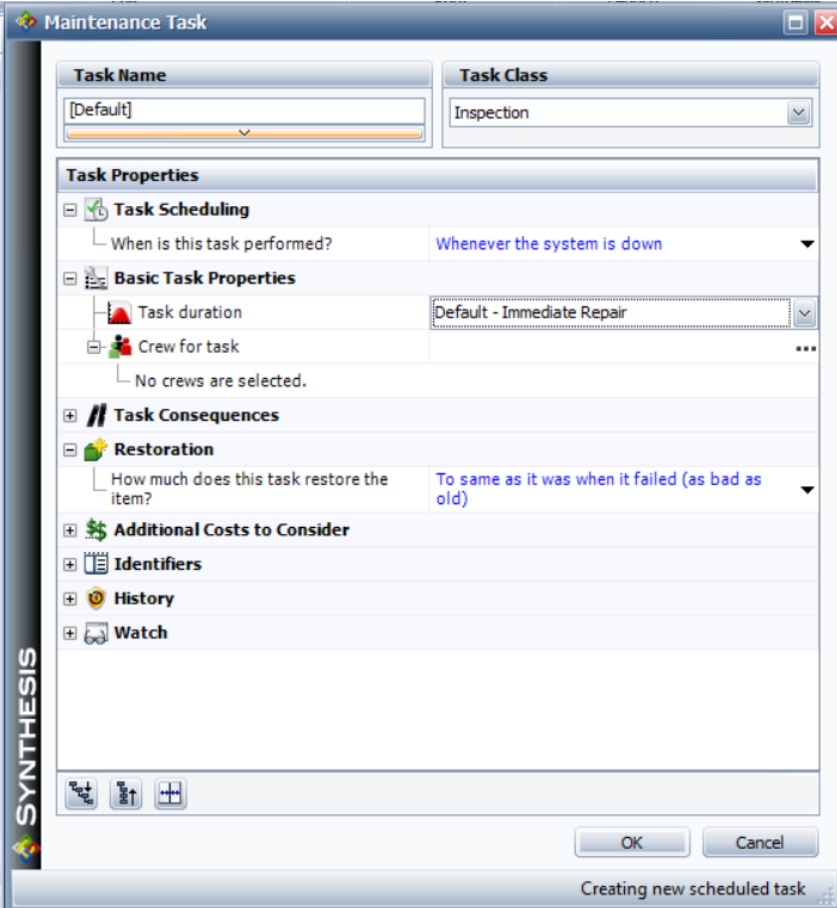
Case Study

The implemented sensor system can detect degradation in the Reactor and predict when a failure is about to happen. The sensitivity of the sensor is estimated at 90% from failure time, i.e. at about 90% of the failure time, the impending failure is detected.

This is quantified in BlockSim using the Failure Detection Threshold (FDT), which is a number from 0 to 1 that indicates the percentage of an item's life that must elapse before an approaching failure can be detected. For example, if the FDT is 0.9 and the item will fail at 1000 days, the approaching failure can begin to be detected at 900 days.

Note that even though the sensor will be measuring degradation, the threshold to trigger a preventive maintenance action in BlockSim is based on simulated failure times.

From the relationship between time and degradation, the degradation threshold can be transferred to a triggering time threshold. The Failure Detection Threshold (FDT) field is found in the Inspection tab of the Reactor's Block



Maintenance Task

Task Name: [Default]

Task Class: Inspection

Task Properties

- Task Scheduling**
 - When is this task performed? Whenever the system is down
- Basic Task Properties**
 - Task duration: Default - Immediate Repair
 - Crew for task: No crews are selected.
- Task Consequences**
- Restoration**
 - How much does this task restore the item? To same as it was when it failed (as bad as old)
- Additional Costs to Consider**
- Identifiers**
- History**
- Watch**

OK Cancel

Creating new scheduled task

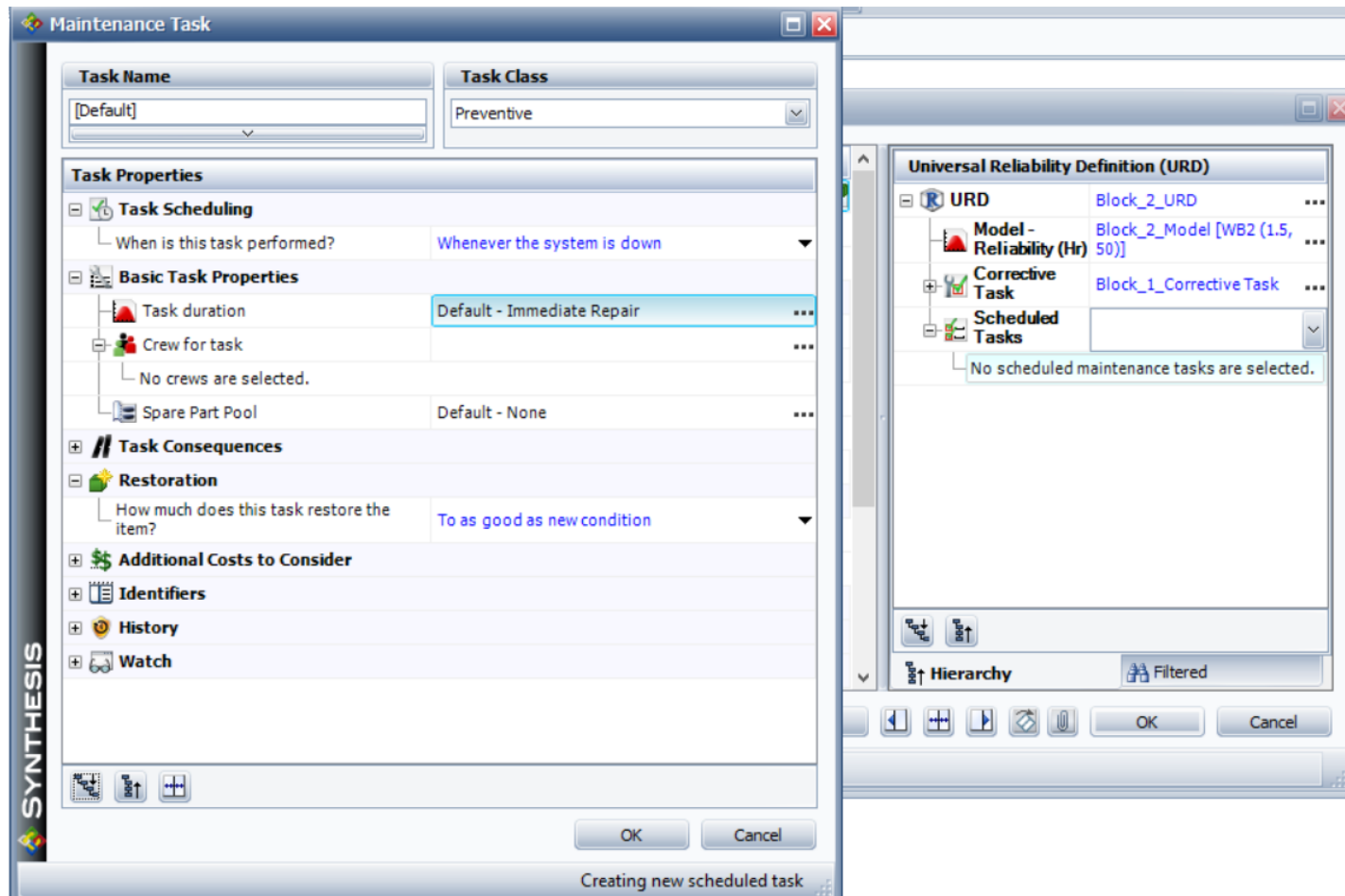
Case Study

When a failure is anticipated, the controller issues an order to the on-site operators to perform a preventive action. The cost and preventive repair time are described in the following table.

Preventive Repair Distribution	Cost of Preventive Repair	Inspection Cost	Cost of Preventive Repair
NOR (Mean=9, Std=2)	8,000	\$10/inspection	\$8,000
Sensor Equipment and Computerized System Cost	\$500,000		

Case Study

This preventive inspection plan is defined in BlockSim by clicking the Create New Preventive Action Policy button, and setting the appropriate Preventive Action Policy.



Case Study

Usually, preventive maintenances are performed on a regular basis (either based on item age or system age) or when the system is down. In this example, neither of these preventive maintenance types is performed. In the above figure, the Upon Maintenance of another Group Item was selected; however, since no groups of blocks were created in this RBD, this selection will not have an influence on the analysis. The goal of this preventive maintenance policy is to associate a preventive repair action to the time when the Failure Detection Threshold is reached.

Case Study

Results & Discussions

If the inspection plan is not implemented (*i.e.* no settings are made under the Preventive tab or the Inspection tab in the *Reactor's* Block Properties), then for 5 years (43,800h) of operation, the average availability is estimated to be 0.9951 and the cost of maintaining the Shutdown System is \$2,581,671 as shown in the simulation's results summary.

System Overview	
<u>General</u>	
Mean Availability (All Events):	0.9951
Std Deviation:	0.0048
Mean Availability (w/o PM & Inspection):	0.9951
Point Availability (All Events) at 43800:	0.994
Reliability at 43800:	0.1
Expected Number of Failures:	1.883
MTTFF:	26077.815
<u>System Uptime/Downtime</u>	
Uptime:	43583.5686
CM Downtime:	216.4314
Inspection Downtime:	0
PM Downtime:	0
Total Downtime:	216.4314
<u>System Downing Events</u>	
Number of Failures:	1.883
Number of CMs:	1.883
Number of Inspections:	0
Number of PMs:	0
Total Events:	1.883
<u>Costs</u>	
Total Costs:	2581671
<u>Throughput</u>	
Total Throughput:	0

Case Study

Results & Discussions

Before simulating the entire Shutdown System with the Reactor's preventive and inspection settings, the Reactor (one Reactor) is simulated by itself for 5 years of operation. The following is an overview summary of the Reactor's simulation results if the regular inspection plan is implemented.

The above table indicates that about 5 preventive maintenances are expected to be initiated (based on the early detection of failures).

The system is simulated for 5 years of operation.

System Overview	
<u>General</u>	
Mean Availability (All Events):	0.999
Std Deviation:	0.0002
Mean Availability (w/o PM & Inspection):	0.999
Point Availability (All Events) at 43800:	1
Reliability at 43800:	1
Expected Number of Failures:	0
MTTFF:	63190042.79
<u>System Uptime/Downtime</u>	
Uptime:	43754.7436
CM Downtime:	0
Inspection Downtime:	0
PM Downtime:	45.2564
Total Downtime:	45.2564
<u>System Downing Events</u>	
Number of Failures:	0
Number of CMs:	0
Number of Inspections:	0
Number of PMs:	5.028
Total Events:	5.028
<u>Costs</u>	
Total Costs:	58464
<u>Throughput</u>	
Total Throughput:	0

Case Study

Results & Discussions

The following is a summarized overview of the system's performance with the Reactor's preventive and inspection settings.

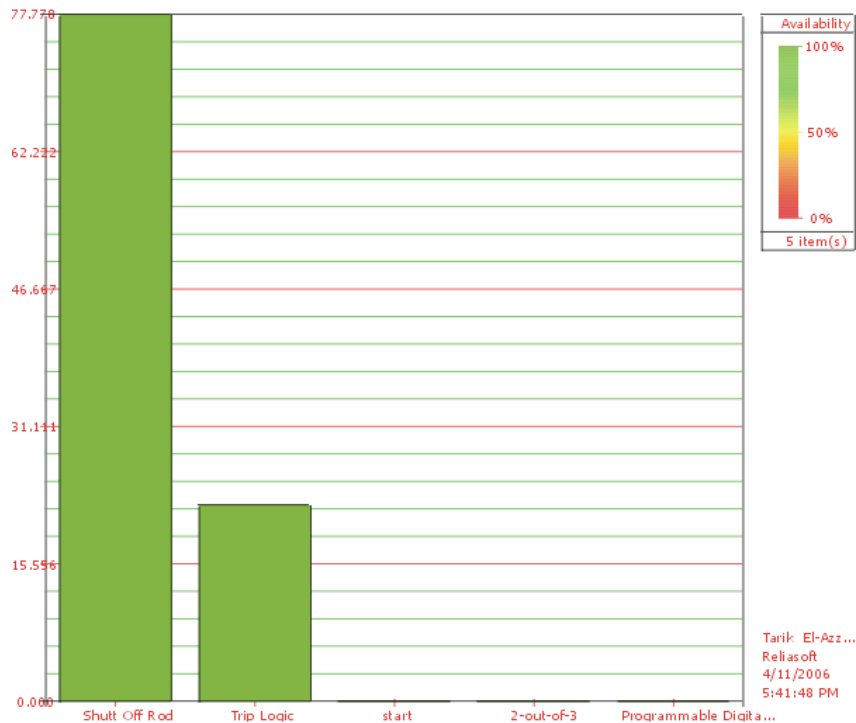
The table shows that for 5 years of operation, the average availability is estimated to be 0.9997 and the cost of maintaining the Shutdown System is \$183,153 + \$500,000 = \$683,153 (after adding the sensor equipment and the computerized system costs) which is a major improvement compared to the results obtained if no inspection and preventive maintenance is implemented at each Reactor. The table summarizes the comparison analysis between implementing the inspection plan and not implementing it.

	Average Availability	Cost
No Inspection	0.9951	\$2,581,671
With Inspection	0.9997	\$683,153

System Overview	
<u>General</u>	
Mean Availability (All Events):	0.9997
Std Deviation:	0.0003
Mean Availability (w/o PM & Inspection):	0.9997
Point Availability (All Events) at 43800:	0.999
Reliability at 43800:	0.387
Expected Number of Failures:	0.756
MTTFF:	56539.7681
<u>System Uptime/Downtime</u>	
Uptime:	43786.7192
CM Downtime:	12.8758
Inspection Downtime:	0
PM Downtime:	0.4049
Total Downtime:	13.2808
<u>System Downing Events</u>	
Number of Failures:	0.756
Number of CMs:	0.756
Number of Inspections:	0
Number of PMs:	0.052
Total Events:	0.808
<u>Costs</u>	
Total Costs:	183153
<u>Throughput</u>	
Total Throughput:	0

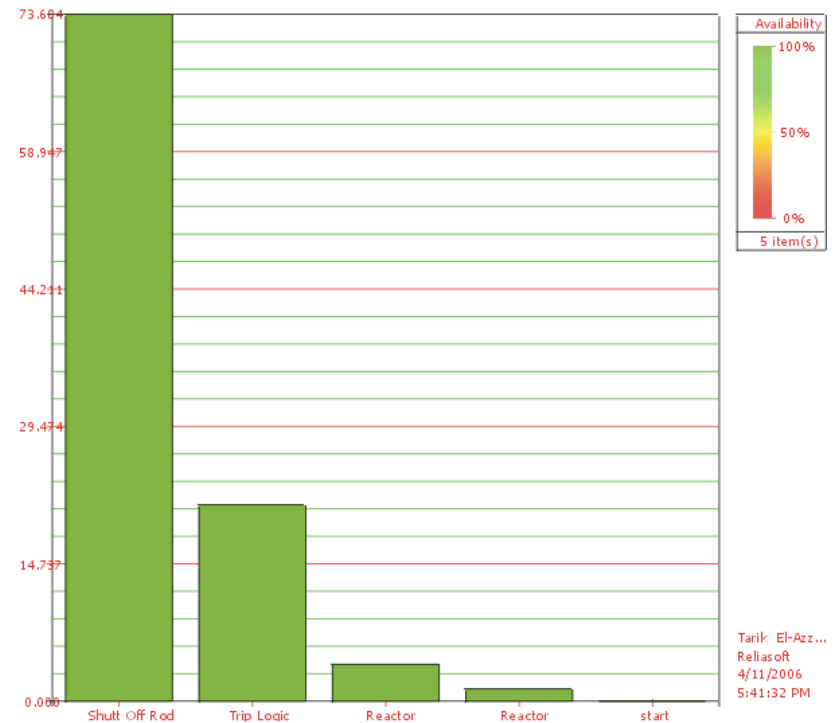
Case Study

RS FCI



A plot of the components' FCI (Failure Criticality Index) values, which is obtained from the number of system downing failures divided by the number of system failures.

RS DECI



A plot of the components' DECI (Downing Event Criticality Index) values, which is obtained from the number of system downing events divided by the number of system failures.

Case Study

These two plots show that even though the Reactor component is less reliable in comparison with the other components in the system, its contribution to the overall system unreliability is not significant, which is due mainly to the preventive maintenances that are implemented once a Reactor failure is anticipated. This analysis illustrates the benefit of implementing regular inspection policies and the great advantages gained when failures can be anticipated and dealt with in a timely fashion.

Throughput Analysis

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Throughput Analysis

In many different industries, there is a need to model and analyse systems that are designed to process or make something so that analysts can estimate production capacity, identify potential bottlenecks, etc.

When using the Reliability Block Diagram (RBD) approach for system modelling and analysis, it is possible to add information about the capacity of each component and then use simulation to perform a variety of throughput calculations for the system.

This includes System Throughput, Component Throughput, Excess Capacity, Backlog and a variety of other useful results.

Throughput Analysis

In this case, the system configuration is not only the system's reliability-wise configuration but also its production/processing sequence.

For example, the first component processes/produces 10 items per unit time (ipu) and the second component can process/produce up to 20 ipu. However, a block can only process/produce items it receives from the blocks before it. Therefore, the second component in this case is only receiving 10 ipu from the block before it. If we assume that neither component can fail, then the maximum items processed from this configuration would be 10 ipu. If the system were to operate for 100 time units, then the throughput of this system would be 1000 items (i.e. 1000×10).



Throughput Metrics and Terminology

System Throughput: System throughput is the total amount of items processed/produced by the system over the defined period of time. In the two-component example, this is 1000 items over 100 time units.

Component Throughput: The total amount of items processed or produced by each component (block). In the two-component example, this is 1000 items each.

Component Maximum Capacity: The maximum number of items that the component (block) could have processed/produced. This is simply the block's throughput rate multiplied by the run time. In the two-component example, this is $100 \times 10 = 1000$ for the first component and $100 \times 20 = 2000$ for the second component.

Throughput Metrics and Terminology

Component Uptime Capacity: The maximum number of items the component could have processed/produced while it was up and running. In the two-component example, this is 1000 for the first component and 2000 for the second component, since we are assuming that the components cannot fail. If the components could fail, this number would be the component's uptime multiplied by its throughput rate.

Component Excess Capacity: The additional amount a component could have processed/produced while up and running. In the two-component example, this is 0 for the first component and 1000 for the second component.

Throughput Metrics and Terminology

Component Actual Utilization: The ratio of the component throughput and the component maximum capacity. In the two-component example, this is 100% for the first component and 50% for the second component.

Component Uptime Utilization: The ratio of the component throughput and the component uptime capacity. In the two-component example, this is 100% for the first component and 50% for the second component. Note that if the components had failed and experienced downtime, this number would be different for each component.

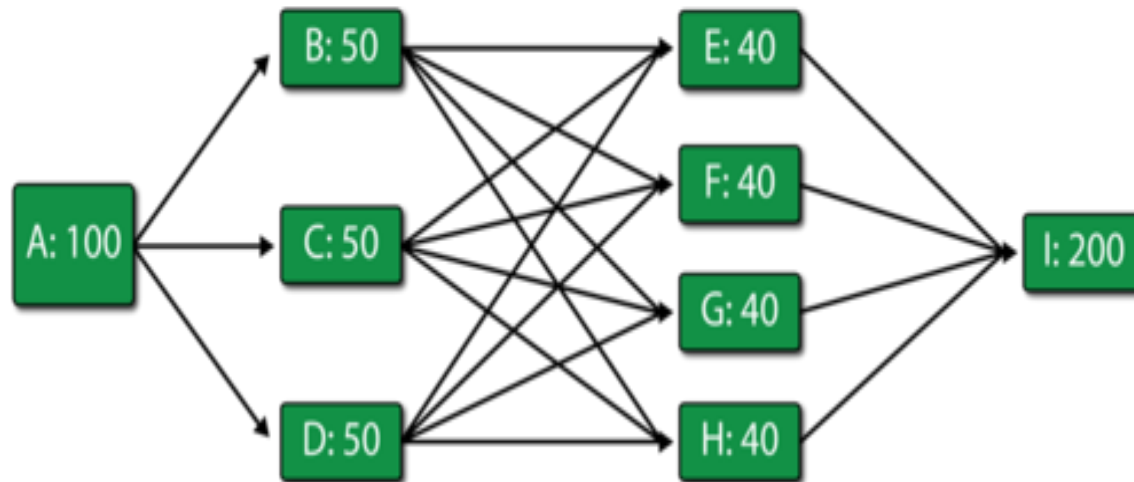
Throughput Metrics and Terminology

Backlog: Items that the component could not process are kept in a backlog. Depending on the settings, a backlog may or may not be processed when an opportunity arises. The available backlog metrics include:

- **Component Backlog:** The amount of a backlog present at the component at the end of the run (simulation).
- **Component Processed Backlog:** The amount of backlog processed by the component.
- **Excess Backlog:** Under specific settings in BlockSim, components can accept only a limited backlog. In these cases, a backlog that was rejected is stored in the Excess Backlog category.

Scenario 1

Consider the system shown in the figure below.



Blocks *A* through *I* produce a number of items per unit time as identified next to each letter (e.g., *A*: 100 implies 100 items per time unit for *A*). The connections shown in the RBD show the physical path of the items through the process (or production line). For the sake of simplicity, also assume that the blocks can never fail and that items are routed equally to each path.

Scenario 1

This then implies that the following occurs over a single time unit:

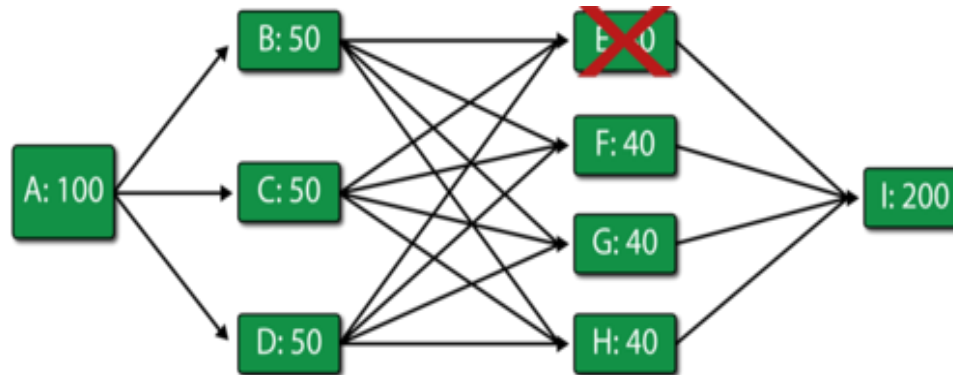
- Unit **A** makes 100 items and routes 33.33 to **B**, 33.33 to **C** and 33.33 to **D**.
- In turn, **B**, **C** and **D** route their 33.33 to **E**, **F**, **G** and **H** (8.33 to each path).
- **E**, **F**, **G** and **H** route 25 each to **I**.
- **I** processes all 100.
- The system produces 100 items.

Thus, the following table would represent the throughput and excess capacity of each block after one time unit.

Block Throughput Summary		
Block Name (Diagram)	Throughput	Excess Cap
A: 100	100	0
B: 50	33.3333	16.6667
C: 50	33.3333	16.6667
D: 50	33.3333	16.6667
E: 40	25	15
F: 40	25	15
G: 40	25	15
H: 40	25	15
I: 200	100	100

Scenario 2

Now consider the figure below where it is assumed that block **E** has failed.



Then:

- Unit **A** makes 100 items and routes 33.33 to **B**, 33.33 to **C** and 33.33 to **D**.
- In turn, **B**, **C** and **D** route their 33.33 to **F**, **G** and **H** (11.11 to each path).
- **F**, **G** and **H** route 33.33 each to **I**.
- **I** processes all 100.
- The system produces 100 items.

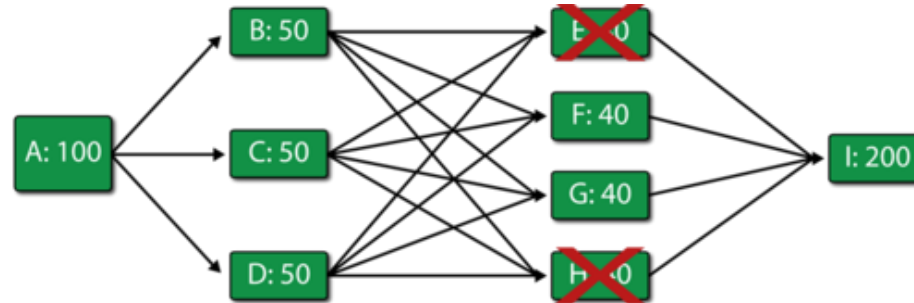
Scenario 2

A summary result table is shown next:

Block Throughput Summary		
Block Name (Diagram)	Throughput	Excess Cap
A: 100	100	0
B: 50	33.3333	16.6667
C: 50	33.3333	16.6667
D: 50	33.3333	16.6667
E: 40	0	0
F: 40	33.3333	6.6667
G: 40	33.3333	6.6667
H: 40	33.3333	6.6667
I: 200	100	100

Scenario 3

Finally, consider the figure below where both *E* and *H* have failed.



Then:

- Unit *A* makes 100 items and routes 33.33 to *B*, 33.33 to *C* and 33.33 to *D*
- In turn, *B*, *C* and *D* route their 33.33 to *F* and *G* (16.66 to each path that has an operating block at the end).
- *F* and *G* get 50 items each.
- *F* and *G* process and route 40 each (their maximum processing capacity) to *I* . Both have a backlog of 10 since they could not process all 50 items they received.
- *I* processes all 80.
- The system produces 80 items.

Scenario 3

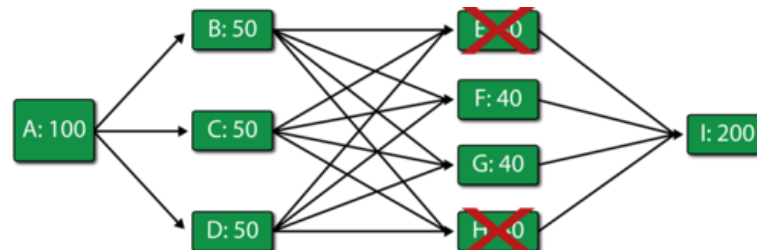
It can be easily seen that the bottlenecks in the system are the blocks **F** and **G**.

Run summary for Scenario 3.

Block Throughput Summary			
Block Name (Diagram)	Throughput	Excess Cap	Backlog
A: 100	100	0	0
B: 50	33.3333	16.6667	0
C: 50	33.3333	16.6667	0
D: 50	33.3333	16.6667	0
E: 40	0	0	0
F: 40	40	0	10
G: 40	40	0	10
H: 40	0	0	0
I: 200	80	120	0

Utilization summary for Scenario 3.

Actual Utilization	
Block Name (Diagram)	Actual Utilization
A: 100	100%
B: 50	67%
C: 50	67%
D: 50	67%
E: 40	0%
F: 40	100%
G: 40	100%
H: 40	0%
I: 200	40%

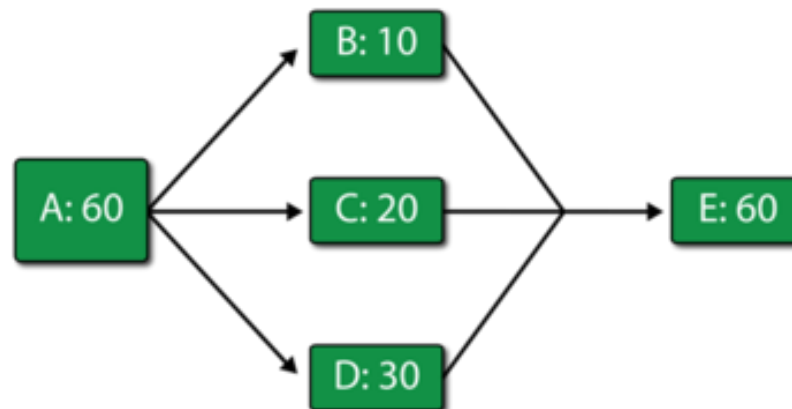


Throughput Analysis Options

In BlockSim, specific throughput properties can be set for the blocks in the diagram.

Throughput: The number of items that the block can process per unit time.

Allocation: Specify the allocation scheme across multiple paths (i.e., equal or weighted). This option is shown in the figure below.



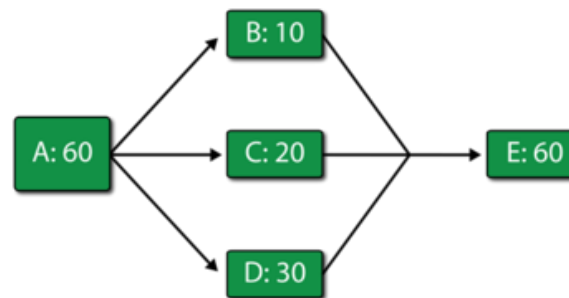
Throughput Analysis Options

To explain these settings, consider the example shown in the figure below, which uses the same notation as before.

If the **Weighted allocation across paths** option is chosen, then the 60 items made by **A** will be allocated to **B**, **C** and **D** based on their throughput capabilities. Specifically, the portion that each block will receive P_i is:

$$P_i = \frac{\text{Throughput}_i}{\sum_{j=1}^N \text{Throughput}_j}$$

The actual amount is then the (portion × available units). In this case, the portion allocated to **B** is $\frac{10}{60}$, the portion allocated to **C** is $\frac{20}{60}$ and the portion allocated to **D** is $\frac{30}{60}$. When a total of 60 units is processed through **A**, **B** will get 10, **C** will get 20 and **D** will get 30.



Throughput Analysis Options

The results would then be as shown in the table below.

Block Throughput Summary					
Block Name (Diagram)	Throughput	Excess Cap	Backlog	Backlog Processed	Excess Backlog
A: 60	60	0	0	0	0
B: 10	10	0	0	0	0
C: 20	20	0	0	0	0
D: 30	30	0	0	0	0
E: 60	60	0	0	0	0

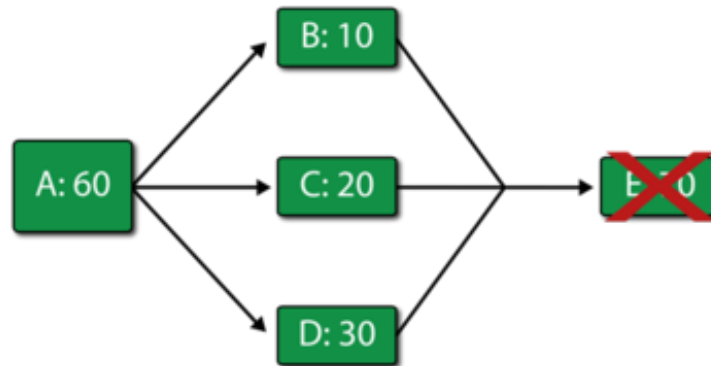
If the **Allocate equal share to all paths** option is chosen, then 20 units will be sent to **B**, **C** and **D** regardless of their processing capacity, yielding the results shown in the table below.

Block Throughput Summary					
Block Name (Diagram)	Throughput	Excess Cap	Backlog	Backlog Processed	Excess Backlog
A: 60	60	0	0	0	0
B: 10	10	0	10	0	0
C: 20	20	0	0	0	0
D: 30	20	10	0	0	0
E: 60	50	10	0	0	0

Throughput Analysis Options

Send units to failed blocks: Decide whether items should be sent to failed parts. If this option is not selected, the throughput units are allocated only to operational units. Otherwise, if this option is selected, units are also allocated to failed blocks and they become part of the failed block's backlog.

In the special case in which one or more blocks fail, causing a disruption in the path, and the **Send units to failed blocks** option is not selected, then the blocks that have a path to the failed block(s) will not be able to process any items, given the fact that they cannot be sent to the forward block. In this case, these blocks will keep all items received in their backlog. As an example, and using the figure below, if **E** is failed (and **E** cannot accept items in its backlog while failed), then **B**, **C** and **D** cannot forward any items to it. Thus, they will not process any items sent to them from **A**. Items sent from **A** will be placed in the backlogs of items **B**, **C** and **D**.



Throughput Analysis Options

Process/Ignore backlog: Identify how a block handles backlog. A block can ignore or process backlog. Items that cannot be processed are kept in a backlog bin and are processed as needed.

Additionally, you can set the maximum number of items that can be stored in the backlog. When you choose to ignore the backlog, BlockSim will still report items that cannot be processed in the backlog column. However, it will let the backlog accumulate and never process it. In the case of a limited backlog, BlockSim will not accumulate more backlog than the maximum allowed and will discard all items sent to the block if they exceed its backlog capacity. It will keep count of the units that did not make it in the backlog in a category called Excess Backlog.

Throughput Analysis Options

To illustrate this, reconsider Scenario 3, but with both **F** and **G** having a limited backlog of 5. After a single time unit of operation, the results would be as shown in the tables below.

Scenario 3 summary with *F* and *G* having a limited backlog and after one time unit.

Block Throughput Summary					
Block Name (Diagram)	Throughput	Excess Cap	Backlog	Backlog Processed	Excess Backlog
A: 100	100	0	0	0	0
B: 50	33.3333	16.6667	10	0	0
C: 50	33.3333	16.6667	0	0	0
D: 50	33.3333	16.6667	0	0	0
E: 40	0	0	0	0	0
F: 40	40	0	5	0	5
G: 40	40	0	5	0	5
H: 40	0	0	0	0	0
I: 200	80	120	0	0	0

Scenario 3 summary with *F* and *G* having a limited backlog and after two time units.

Block Throughput Summary					
Block Name (Diagram)	Throughput	Excess Cap	Backlog	Backlog Processed	Excess Backlog
A: 100	200	0	0	0	0
B: 50	66.6667	33.3333	10	0	0
C: 50	66.6667	33.3333	0	0	0
D: 50	66.6667	33.3333	0	0	0
E: 40	0	0	0	0	0
F: 40	80	0	5	0	15
G: 40	80	0	5	0	15
H: 40	0	0	0	0	0
I: 200	160	240	0	0	0

Throughput and Mirrored Blocks

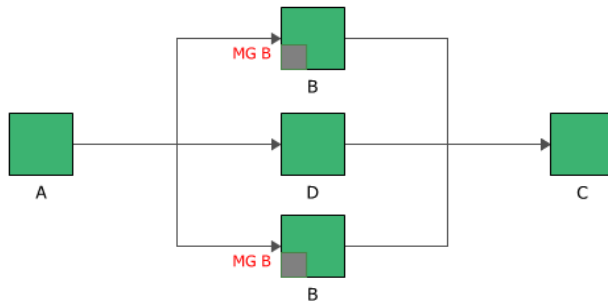
Simulations that contain both mirrored blocks and throughput analysis behave in the following way:

System availability: All blocks within a mirror group will be considered the same block for system uptime/downtime analysis. This means that if the block fails, it will be considered a downed block everywhere that the mirrored block is found.

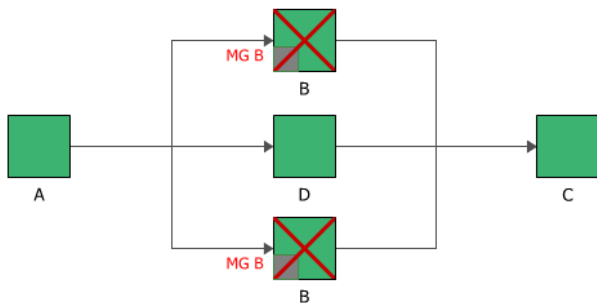
System throughput: The mirrored block throughput is NOT divided between each of the locations within the system diagram. Each instance of the mirrored block can process the amount of throughput assigned to the block.

Throughput and Mirrored Blocks

Example: Imagine a system where block A and block C each have a throughput of 300 units/hour, while block B (mirrored block) and block D each have a throughput of 100 units/hour.



If all the blocks are operating, then the system throughput is 300 units/hour, as each instance of block B is able to handle 100 units/hour.

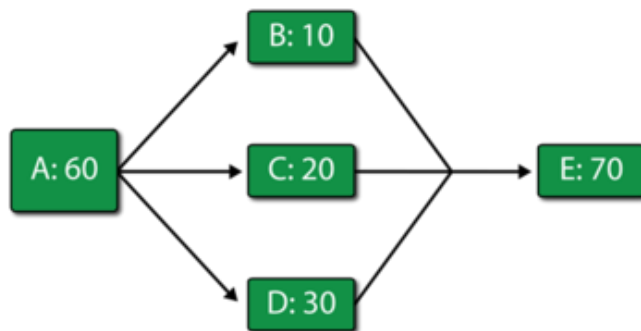


If block B is down, then the system throughput becomes 100 units/hour, as block B is considered down at both locations.

Case Study: A Simple Throughput Analysis

The prior sections illustrated the basics concepts in throughput analysis. However, they did not take into account the reliability and maintenance properties of the blocks and the system. In a complete analysis, these would also need to be incorporated. The following simple example incorporates failures and repairs. Even though the example is trivial, the concepts presented here form the basis of throughput analysis in BlockSim. The principles remain the same no matter how complex the system.

Consider the simple system shown in the figure below. In addition, consider the following deterministic failure and repair characteristics:



Block Name (Diagram)	Fail Every	Repair Duration
A: 60	65	4
B: 10	50	4
C: 20	55	4
D: 30	60	4
E: 70	70	4

Case Study: A Simple Throughput Analysis

Also:

- Set all units to operate through system failure.
- Do not add spare part pools or crews (use defaults).
- Do not send items to failed units.
- Use a weighted allocation scheme.

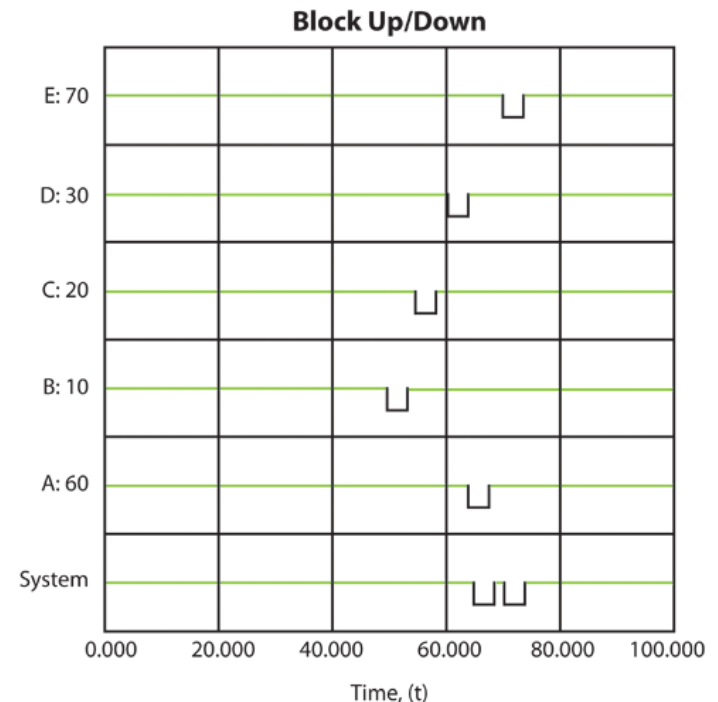
Case Study: A Simple Throughput Analysis

Event History: 0 to 100 Time Units

Then the system behaviour from 0 to 100 time units is given in the table below. The system event history is as follows:

Block	Fail Time	Repaired By
A: 60	65	69
B: 10	50	54
C: 20	55	59
D: 30	60	64
E: 70	70	74

Once the system history has been established, the throughput behaviour of this system from 0 to 100 can be examined by observing the sequence of events and their subsequent effect on system throughput.



Case Study: A Simple Throughput Analysis

Event 1: B Fails at 50

At 50, B fails.

From 0 to 50, A processes $50 \times 60 = 3000$ items.

500 are sent to B , 1000 to C and 1500 to D . There is no excess capacity at B, C or D.

B, C and D process and send 3000 items to E. Because the capacity of E is 3500, E now has an excess capacity of 500.

The next table summarizes these results:

Event 1: B Fails at 50								
		A	B	C	D	E		
Start Time	End Time	60	10	20	30	70		
0	50	3000	500	1000	1500	3000	Processed	Current
		0	0	0	0	500	Excess Capacity	
		0	0	0	0	0	Backlog	
		0	0	0	0	0	Backlog Processed	
		3000	500	1000	1500	3000	Processed	Cumulative
		0	0	0	0	500	Excess Capacity	
		0	0	0	0	0	Backlog	
		0	0	0	0	0	Backlog Processed	

Case Study: A Simple Throughput Analysis

Event 2: B is Down 50 to 54

From 50 to 54, B is down.

A processes 240 items and sends 96 to C and 144 to D.

D and C can only process 80 and 120, respectively during this time. Thus, they get backlogs of 16 and 24, respectively.

The 200 processed are sent to E. E has an excess capacity of 80 during this time period.

The next table summarizes these results:

Event 2: B is Down 50 to 54								
Start Time	End Time	A	B	C	D	E		
		60	10	20	30	70		
50	54	240	0	80	120	200	Processed	Current
		0	0	0	0	80	Excess Capacity	
		0	0	16	24	0	Backlog	
		0	0	0	0	0	Backlog Processed	
		3240	500	1080	1620	3200	Processed	Cumulative
		0	0	0	0	580	Excess Capacity	
		0	0	16	24	0	Backlog	
		0	0	0	0	0	Backlog Processed	

Case Study: A Simple Throughput Analysis

Event 3: All Up 54 to 55

Event 3: All Up - 54 to 55								
		A	B	C	D	E		
Start Time	End Time	60	10	20	30	70		
54	55	60	10	20	30	60	Processed	Current
		0	0	0	0	10	Excess Capacity	
		0	0	0	0	0	Backlog	
		0	0	0	0	0	Backlog Processed	
		3300	510	1100	1650	3260	Processed	Cumulative
		0	0	0	0	590	Excess Capacity	
		0	0	16	24	0	Backlog	
		0	0	0	0	0	Backlog Processed	

Event 4: C is Down 55 to 59

Event 4: C Down - 55 to 59								
		A	B	C	D	E		
Start Time	End Time	60	10	20	30	70		
55	59	240	40	0	120	160	Processed	Current
		0	0	0	0	120	Excess Capacity	
		0	20	0	60	0	Backlog	
		0	0	0	0	0	Backlog Processed	
		3540	550	1100	1770	3420	Processed	Cumulative
		0	0	0	0	710	Excess Capacity	
		0	20	16	84	0	Backlog	
		0	0	0	0	0	Backlog Processed	

Case Study: A Simple Throughput Analysis

Event 5: All Up 59 to 60

Event 5: All Up - 59 to 60								
Start Time	End Time	A	B	C	D	E		
		60	10	20	30	70		
59	60	60	10	20	30	60	Processed	Current
		0	0	0	0	10	Excess Capacity	
		0	0	0	0	0	Backlog	
		0	0	0	0	0	Backlog Processed	
		3600	560	1120	1800	3480	Processed	Cumulative
		0	0	0	0	720	Excess Capacity	
		0	20	16	84	0	Backlog	
		0	0	0	0	0	Backlog Processed	

Event 6: D is Down 60 to 64

Event 6: D Down - 60 to 64								
		A	B	C	D	E		
Start Time	End Time	60	10	20	30	70		
60	64	240	40	80	0	120	Processed	Current
		0	0	0	0	160	Excess Capacity	
		0	40	80	0	0	Backlog	
		0	0	0	0	0	Backlog Processed	
		3840	600	1200	1800	3600	Processed	Cumulative
		0	0	0	0	880	Excess Capacity	
		0	60	96	84	0	Backlog	
		0	0	0	0	0	Backlog Processed	

Case Study: A Simple Throughput Analysis

Event 7: All Up 64 to 65

Event 7: All Up - 64 to 65								
		A	B	C	D	E		
Start Time	End Time	60	10	20	30	70		
64	65	60	10	20	30	60	Processed	Current
		0	0	0	0	10	Excess Capacity	
		0	0	0	0	0	Backlog	
		0	0	0	0	0	Backlog Processed	
		3900	610	1220	1830	3660	Processed	Cumulative
		0	0	0	0	890	Excess Capacity	
		0	60	96	84	0	Backlog	
		0	0	0	0	0	Backlog Processed	

Event 8: A is Down 65 to 69

- Between 65 and 69, A fails. This stops the flow of items in the system and provides an opportunity for the other blocks to process their backlogs. As an example, B processes 40 items from the 60 item in its backlog.

Event 8: A Down - 65 to 69								
		A	B	C	D	E		
Start Time	End Time	60	10	20	30	70		
65	69	0	40	80	84	204	Processed	Current
		0	0	0	36	76	Excess Capacity	
		0	-40	-80	-84	0	Backlog	
		0	40	80	84	0	Backlog Processed	
		3900	650	1300	1914	3864	Processed	Cumulative
		0	0	0	36	966	Excess Capacity	
		0	20	16	0	0	Backlog	
		0	40	80	84	0	Backlog Processed	

Case Study: A Simple Throughput Analysis

Event 9: All Up 69 to 70

Event 9: All Up - 69 to 70								
		A	B	C	D	E		
Start Time	End Time	60	10	20	30	70		
69	70	60	10	20	30	60	Processed	Current
		0	0	0	0	10	Excess Capacity	
		0	0	0	0	0	Backlog	
		0	0	0	0	0	Backlog Processed	
		3960	660	1320	1944	3924	Processed	Cumulative
		0	0	0	36	976	Excess Capacity	
		0	20	16	0	0	Backlog	
		0	40	80	84	0	Backlog Processed	

Case Study: A Simple Throughput Analysis

Event 10: E is Down 60 to 64

- From 70 to 74, E is down. Because we specified that we will not send items to failed units, B, C and D receive items from A but they do not process them, since processing would require that items be sent to E. The items received by B, C and D are added to their respective backlogs. Furthermore, since they could have processed them if E had been up, all three blocks have an excess capacity for this period. Specifically:
- It should be noted that if we had allowed items to be sent to failed blocks, B, C and D would have processed the items received and the backlog would have been at E. The rest of the time, all units are up.

Event 10: E Down - 70 to 74							
Start Time	End Time	A	B	C	D	E	
70	74	60	10	20	30	70	
		240	0	0	0	0	Processed
		0	40	80	120	0	Excess Capacity
		0	40	80	120	0	Backlog
		0	0	0	0	0	Backlog Processed
		4200	660	1320	1944	3924	Processed
		0	40	80	156	976	Excess Capacity
		0	60	96	120	0	Backlog
		0	40	80	84	0	Backlog Processed

Case Study: A Simple Throughput Analysis

Event 11: All Up 74 to 100

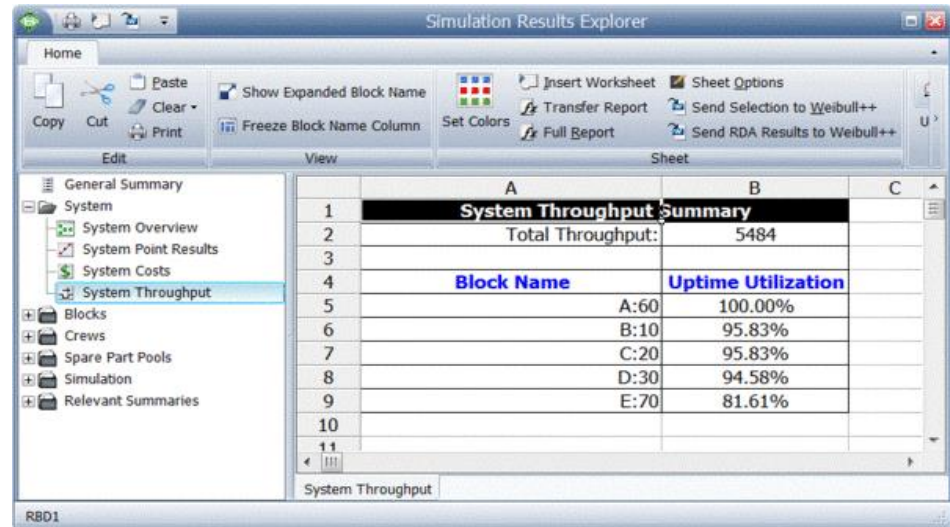
Event 10: E Down - 70 to 74								
		A	B	C	D	E		
Start Time	End Time	60	10	20	30	70		
74	100	1560	260	520	780	1560	Processed	Current Cumulative
		0	0	0	0	260	Excess Capacity	
		0	0	0	0	0	Backlog	
		0	0	0	0	0	Backlog Processed	
		5760	920	1840	2724	5484	Processed	
		0	40	80	156	1236	Excess Capacity	
		0	60	96	120	0	Backlog	
		0	40	80	84	0	Backlog Processed	

Case Study: A Simple Throughput Analysis

Exploring the Results

BlockSim provides all of these results via the Simulation Results Explorer. The figure shows the system throughput summary.

System level results present the total system throughput, which is 5484 items in this example. Additionally, the results include the uptime utilization of each component. The block level result summary, shown next, provides additional results for each item.



Simulation Results Explorer

Home | Edit | View | Sheet

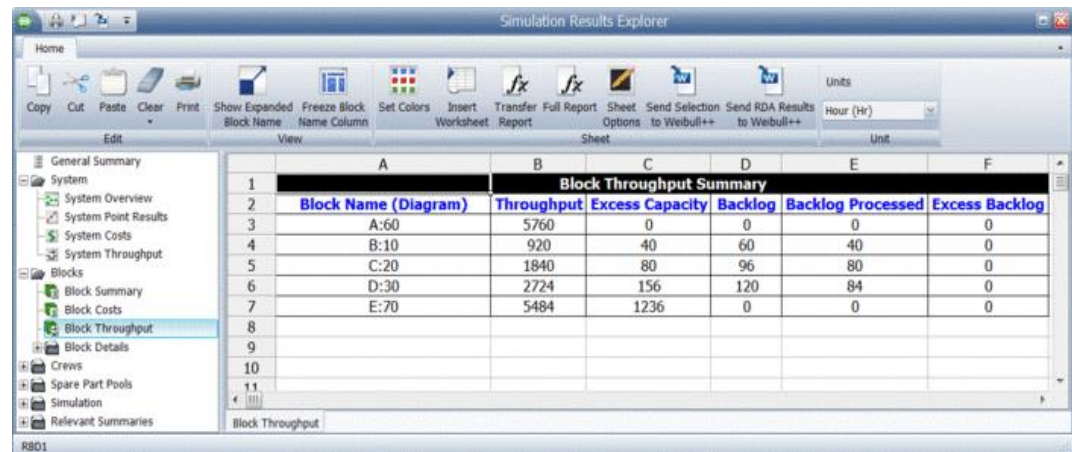
General Summary

- System
 - System Overview
 - System Point Results
 - System Costs
 - System Throughput
- Blocks
- Crews
- Spare Part Pools
- Simulation
- Relevant Summaries

	A	B	C
1	System Throughput Summary		
2	Total Throughput:	5484	
3			
4	Block Name	Uptime Utilization	
5	A:60	100.00%	
6	B:10	95.83%	
7	C:20	95.83%	
8	D:30	94.58%	
9	E:70	81.61%	
10			
11			

System Throughput

RBD1



Simulation Results Explorer

Home | Edit | View | Sheet

General Summary

- System
 - System Overview
 - System Point Results
 - System Costs
 - System Throughput
- Blocks
 - Block Summary
 - Block Costs
 - Block Throughput
 - Block Details
- Crews
- Spare Part Pools
- Simulation
- Relevant Summaries

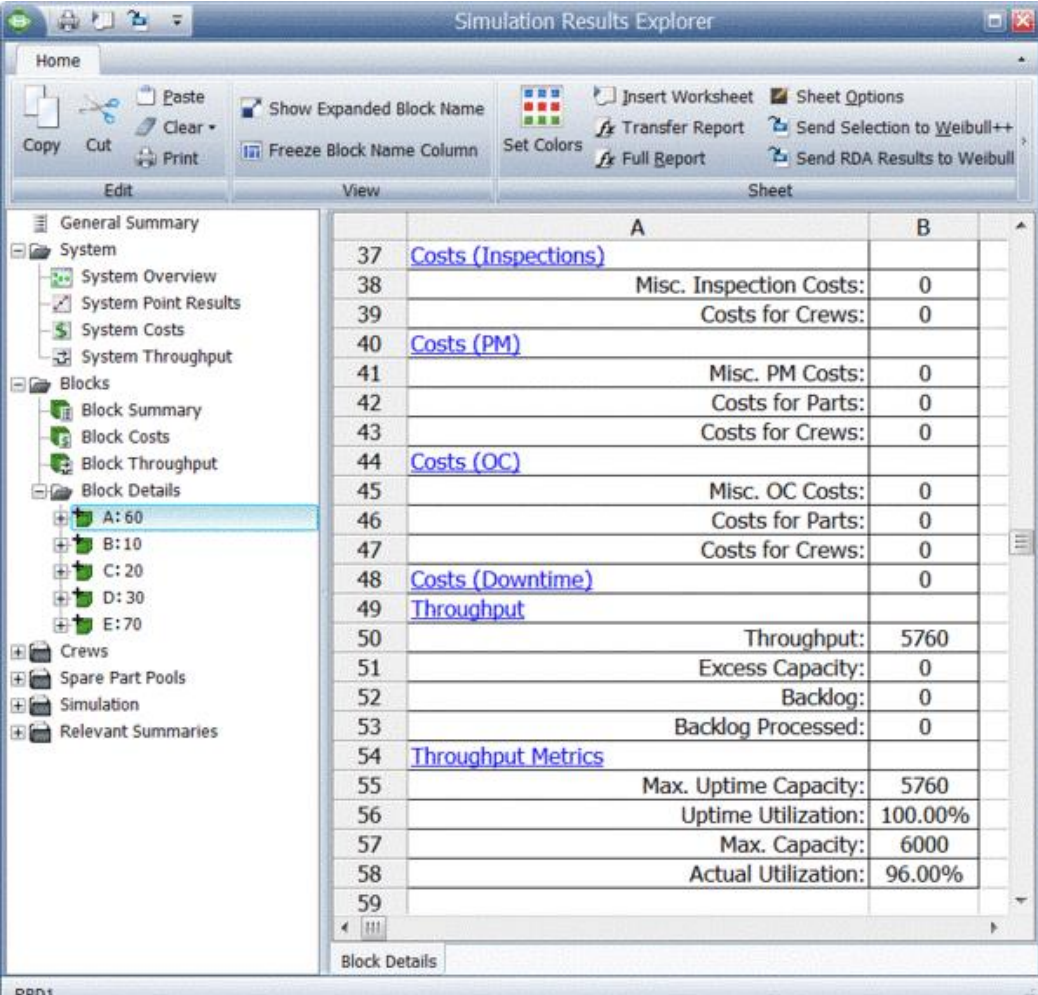
	A	B	C	D	E	F
1	Block Throughput Summary					
2	Block Name (Diagram)	Throughput	Excess Capacity	Backlog	Backlog Processed	Excess Backlog
3	A:60	5760	0	0	0	0
4	B:10	920	40	60	40	0
5	C:20	1840	80	96	80	0
6	D:30	2724	156	120	84	0
7	E:70	5484	1236	0	0	0
8						
9						
10						
11						

Block Throughput

RBD1

Case Study: A Simple Throughput Analysis

Finally, specific throughput results and metrics for each block are provided.



The screenshot shows the 'Simulation Results Explorer' window. The left pane displays a tree view with 'Block Details' expanded, showing 'A: 60' selected. The right pane displays a table of simulation results for Block A: 60.

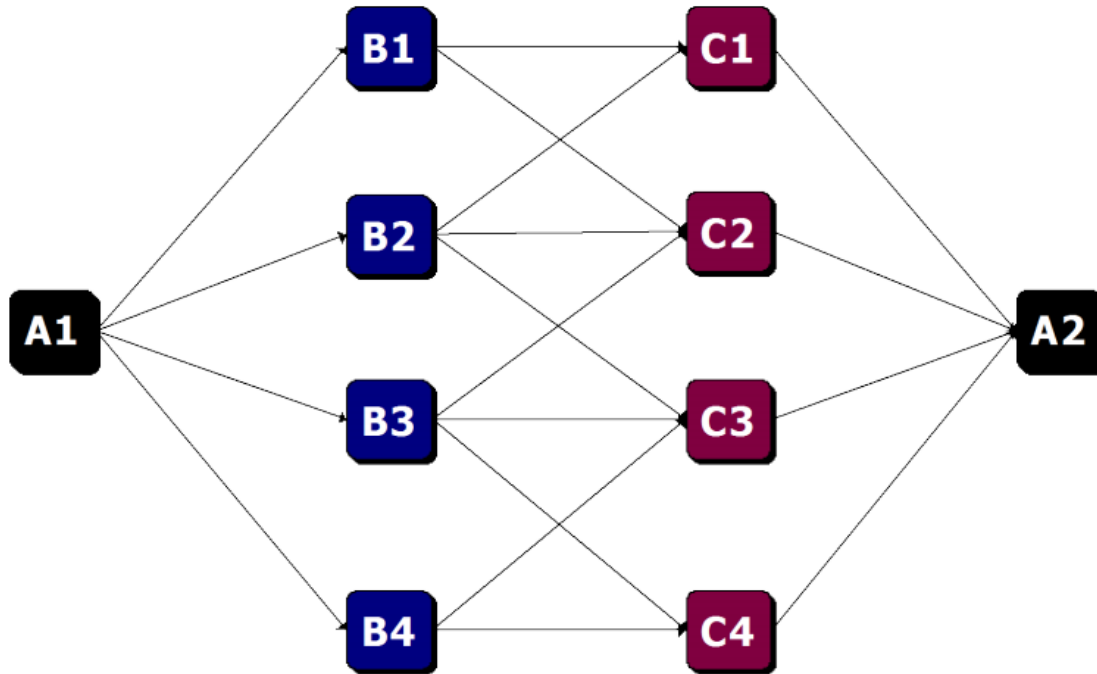
	A	B
37	Costs (Inspections)	
38	Misc. Inspection Costs:	0
39	Costs for Crews:	0
40	Costs (PM)	
41	Misc. PM Costs:	0
42	Costs for Parts:	0
43	Costs for Crews:	0
44	Costs (OC)	
45	Misc. OC Costs:	0
46	Costs for Parts:	0
47	Costs for Crews:	0
48	Costs (Downtime)	0
49	Throughput	
50	Throughput:	5760
51	Excess Capacity:	0
52	Backlog:	0
53	Backlog Processed:	0
54	Throughput Metrics	
55	Max. Uptime Capacity:	5760
56	Uptime Utilization:	100.00%
57	Max. Capacity:	6000
58	Actual Utilization:	96.00%
59		

Block Details

RBD1

BlockSim Simulation-Cost and throughput

Consider the following process line:



BlockSim Simulation-Cost and throughput

With the following properties for each block.

Blocks A		
Failure		
Weibull	β	η
	2	20,000
OTSF	Yes	
Repair		
Exponential	μ	
	100	
Crews Pool & Policies		
Repair Crews	Crew 1	
	Crew 2	
Policy	Upon Failure	
Spare Pool	A Pool	
Throughput		
Parts/hour	4	
Process Backlog	Yes	
Limited Backlog	No	
Other Costs		
Operating Cost /hour	\$10.00	

Blocks B		
Failure		
Weibull	β	η
	2	10,000
OTSF	Yes	
Repair		
Exponential		μ
		50
Crews Pool & Policies		
Repair Crews		Crew 1
		Crew 2
Policy		Upon Failure
Spare Pool		B Pool
Throughput		
Parts/hour		1
Process Backlog		Yes
Limited Backlog		No
Other Costs		
Operating Cost /hour		\$2.00

Blocks C		
Failure		
Weibull	β	η
	3	8,000
OTSF	Yes	
Repair		
Exponential	μ	
	70	
Crews Pool & Policies		
Repair Crews	Crew 1	
	Crew 2	
Policy	Upon Failure	
Spare Pool	C Pool	
Throughput		
Parts/hour	1	
Process Backlog	Yes	
Limited Backlog	No	
Other Costs		
Operating Cost /hour	\$2.00	

BlockSim Simulation-Cost and throughput

And resources.

Pool A	
Cost Per Item	\$5,000.00
Indirect Cost Per Item	1
Initial Stock	1
Logistic Time to Dispense	
Exponential	μ 20
On-Condition Restock	
Restock at	0
Restock items	1
Time for Stock Arrival	
Exponential	μ 30
Emergency	
Can Obtain	Yes
Cost Per Item	\$10,000.00
Time for Emergency Arrival	
Exponential	μ 15

Pool B	
Cost Per Item	\$3,000.00
Indirect Cost Per Item	1
Initial Stock	1
Logistic Time to Dispense	
Exponential	μ 20
On-Condition Restock	
Restock at	0
Restock items	1
Time for Stock Arrival	
Exponential	μ 30
Emergency	
Can Obtain	Yes
Cost Per Item	\$6,000.00
Time for Emergency Arrival	
Exponential	μ 15

Pool C	
Cost Per Item	\$2,000.00
Indirect Cost Per Item	1
Initial Stock	1
Logistic Time to Dispense	
Exponential	μ 20
On-Condition Restock	
Restock at	0
Restock items	1
Time for Stock Arrival	
Exponential	μ 30
Emergency	
Can Obtain	Yes
Cost Per Item	\$4,000.00
Time for Emergency Arrival	
Exponential	μ 15

Crew 1	
Direct Cost per hour	100
Cost per incident	0
Num. Sim. Tasks	1
Logistic Delay	
Exponential	μ 30

Crew 2	
Direct Cost per hour	100
Cost per incident	500
Num. Sim. Tasks	1
Logistic Delay	
Exponential	μ 50

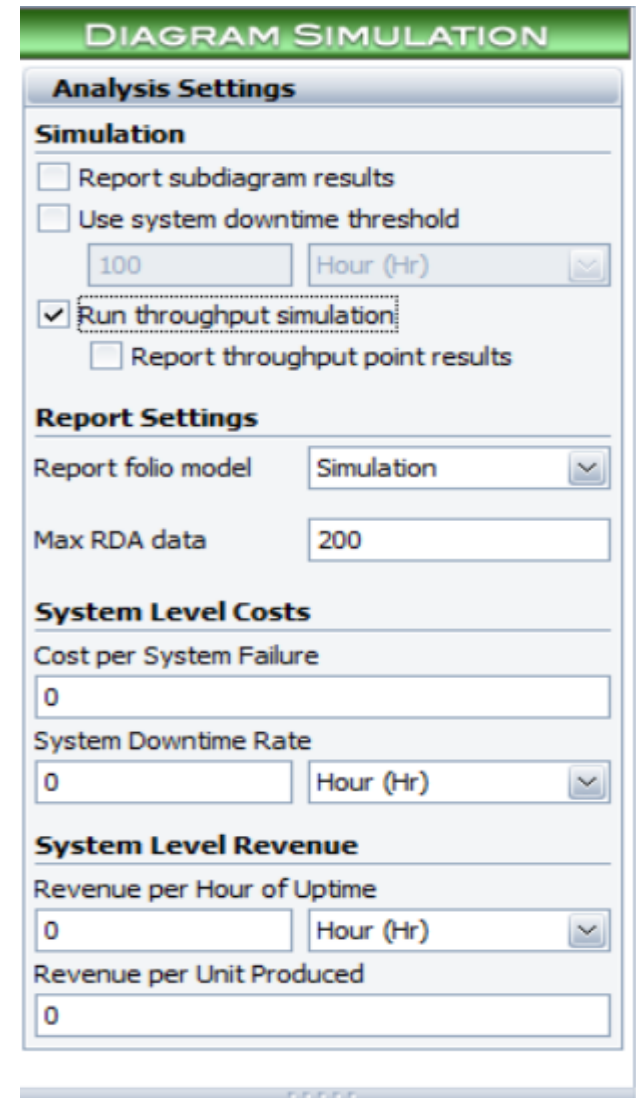
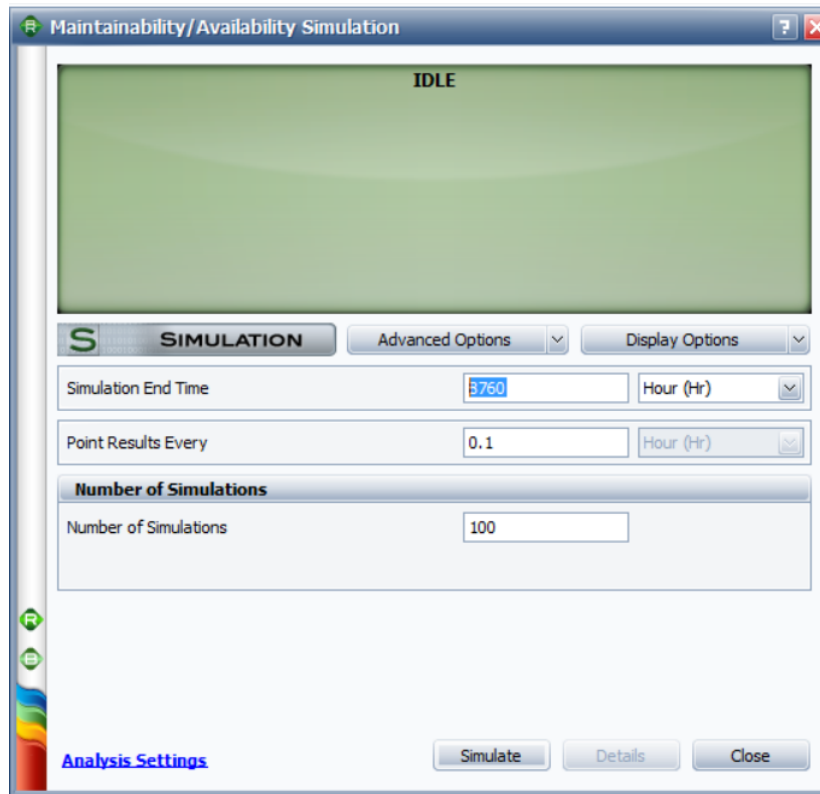
BlockSim Simulation-Cost and throughput

Evaluate the following for 1 year of operation:

1. Mean Availability
2. Total Cost
3. Total Throughput

Example: LCC in BlockSim

This system is then analysed in BlockSim for a period of operation of 8,760 hours, or one year.



Example: LCC in BlockSim

The figure shows the system overview. Most of the variable costs of interest were obtained directly from BlockSim.

System Overview		
<u>General</u>		
Mean Availability (All Events):		0.9941
Std Deviation:		0.0005
Mean Availability (w/o PM & Inspection):		0.9941
Point Availability (All Events) at 8760:		0.9872
Reliability at 8760:		0.6819
Expected Number of Failures:		0.3785
MTTFF:		24409.88
<u>System Uptime/Downtime</u>		
Uptime:		8708.6329
CM Downtime:		51.3671
Inspection Downtime:		0
PM Downtime:		0
Total Downtime:		51.3671
<u>System Downing Events</u>		
Number of Failures:		0.3785
Number of CMs:		0.3785
Number of Inspections:		0
Number of PMs:		0
Total Events:		0.3785
<u>Costs</u>		
Total Costs:		121338.41
<u>Throughput</u>		
Total Throughput:		31651.168

Example: LCC in BlockSim

The figure shows the overall system costs. (Note: If further detail were needed, you could look at additional results. For the purposes of this example, the high level results of Figure 8 are sufficient.)

System Cost Summary		
Misc. Corrective Costs:	0	
Costs for Parts (CM):	21414.9	
Costs for Crews (CM):	84778.705	
<u>Total CM Costs:</u>		<u>106193.605</u>
Misc. Preventive Costs:	0	
Costs for Parts (PM):	0	
Costs for Crews (PM):	0	
<u>Total PM Costs:</u>		<u>0</u>
Misc. Inspection Costs:	0	
Costs for Crews (IN):	0	
<u>Total Inspection Costs:</u>		<u>0</u>
<u>Indirect Pool Costs:</u>		<u>15144.8019</u>
<u>Total Costs:</u>		<u>121338.407</u>

Example: LCC in BlockSim

The total costs from the summary are \$121,338. Note that an additional cost was defined in the problem statement that is not included in the summary. This cost, the operating cost per item per hour of operation, can be obtained by looking at the uptime of each block and then multiplying this by the cost per hour. The results are shown next.

Operating Costs			
Block Name (Diagram)	Block Uptime	Cost per hr	Cost
A1	8734.8373	\$10.00	\$87,348.37
C2	8625.6408	\$2.00	\$17,251.28
C1	8629.5853	\$2.00	\$17,259.17
B1	8694.1258	\$2.00	\$17,388.25
B2	8693.2242	\$2.00	\$17,386.45
C3	8626.9844	\$2.00	\$17,253.97
B3	8693.9473	\$2.00	\$17,387.89
C4	8627.1132	\$2.00	\$17,254.23
B4	8694.1577	\$2.00	\$17,388.32
A2	8733.6418	\$10.00	\$87,336.42
			\$313,254.35

If you also assume a revenue of \$100 per unit produced, then the total revenue is the throughput times the per unit revenue, or $31,651 \times \$100 = \$3,165,100$. The total costs are $121,388 + 313,254 = \$434,642$.

Reporting Results

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Reporting Results

Methodology

- Assumptions
 - Data is assumed verified and correct
 - The deferment amount is correlated to downtime
 - All analysis follow recurrent data unless otherwise noted
 - Only equipment that encountered failures are considered critical and included in the analysis
 - Confidence Level (ie 95% confidence level was used)
- Steps followed in the analysis
 - Identify study duration (start and end date of observation)
 - Identify critical subsystem/equipment (based on P&ID, PFD, DOR)
 - Gather failure data for each equipment or subsystem
 - Perform RDA analysis for all relevant system
 - Create RBD for the subsystem of interest with all the reliability, maintenance and operational data
 - Simulate the model for the study duration
 - Validate model with existing data (actual system availability)
 - Project future failure based on the model through simulation

Reporting Results

Results and discussion

- Overall RBD
- Analysis for each equipment
 - Assumption and summary of results
 - eg. Only 2 failures encountered, assume exponential distribution with MTBF of ____
 - Present downtime table and data sources
 - Exploratory analysis (Pareto, Cum failure vs time etc)
 - RDA Analysis
 - Maintainability analysis (distribution fitting)
- If there are too many equipment, present only summary table and attach the detailed analysis in appendices.

Export Compressor Failure intensity trend (unit in days)				
Subsystem Name				
	Beta	Eta	Current Age	MTBF at Age
Interstage Conditioning	1.13	346.73	1185	263
Interstage Conditioning_NAF	1	395	1185	395
Power Transmission	1	2370	1185	2370
Power Transmission_NAF	1.9	227.56	1185	27
Inletgas Conditioning	N/A	N/A	1185	N/A
Inletgas Conditioning_NAF	1	1185	1185	1185
GG	2.58	691.79	1185	115
GG_NAF	1.17	117.26	1185	67
Lubrication GT	3.37	735.11	1185	70
Lubrication GT_NAF	1	296.25	1185	296.25
Relief valve	1	592.5	1185	592.5
Control & Monitoring	1	395	1185	395
Control & Monitoring_NAF	1.76	428	1185	1.7
Shaft Seal System	1	2370	1185	2370
Compressor Unit_NAF	1.46	147.85	1185	39
Miscellaneous	1	1185	1185	1185
Pigging Activity	1.18	302	1185	201
Unplanned Events	1.19	34.2	1185	15

Reporting Results

Results and discussion

- Validation report
 - Comparing simulation results with actual performance (eg operational availability)
- Identification critical equipment or subsystem and their contribution to overall system performance
- Sensitivity Analysis
 - Changing system configuration to quantify change in performance. eg add redundancy, improve MTBF, reduce MTTR, adding crew etc.

Reporting Results

Conclusion and Recommendations

- Summary of findings and recommendations for improvements
- Highlight any gaps found during the study ie improvement in failure and downtime tracking

Q&A session



Answer for Case Study

System Overview	
<u>General</u>	
Mean Availability (All Events):	0.9948
Std Deviation (Mean Availability):	0.0002
Mean Availability (w/o PM, OC & Inspection):	0.9948
Point Availability (All Events) at 8760:	0.9886
Reliability(8760):	0.6826
Expected Number of Failures:	0.3557
Std Deviation (Number of Failures):	0.0101
MTTFF (Hr):	24421.7695
<u>System Uptime/Downtime</u>	
Uptime (Hr):	8714.4887
CM Downtime (Hr):	45.5113
Inspection Downtime (Hr):	0
PM Downtime (Hr):	0
OC Downtime (Hr):	0
Total Downtime (Hr):	45.5113
<u>System Downing Events</u>	
Number of Failures:	0.3557
Number of CMs:	0.3557
Number of Inspections:	0
Number of PMs:	0
Number of OCs:	0
Number of OFF Events by Trigger:	0
Total Events:	0.3557
<u>Costs</u>	
Total Costs:	92197.64
<u>Throughput</u>	
Total Throughput:	31685.8792

System Cost Summary		
Misc. Corrective Costs:	0	
Costs for Parts (CM):	15575.8	
Costs for Crews (CM):	59034.2533	
<u>Total CM Costs:</u>		<u>74610.0533</u>
Misc. Preventive Costs:	0	
Costs for Parts (PM):	0	
Costs for Crews (PM):	0	
<u>Total PM Costs:</u>		<u>0</u>
Misc. On Condition Costs:	0	
Costs for Parts (OC):	0	
Costs for Crews (OC):	0	
<u>Total OC Costs:</u>		<u>0</u>
Misc. Inspection Costs:	0	
Costs for Crews (IN):	0	
<u>Total Inspection Costs:</u>		<u>0</u>
<u>Downtime Costs:</u>		<u>0</u>
<u>Indirect Pool Costs:</u>		<u>17587.5867</u>
<u>Total Costs:</u>		<u>92197.64</u>



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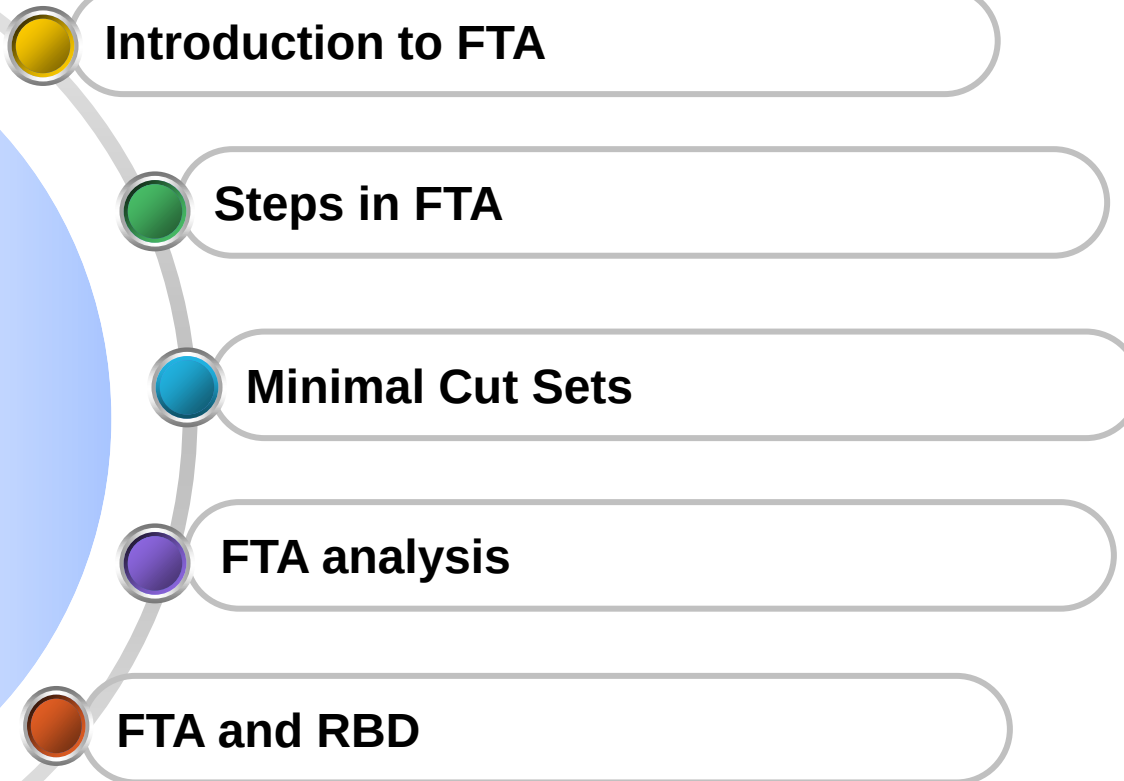
Fault Tree Analysis

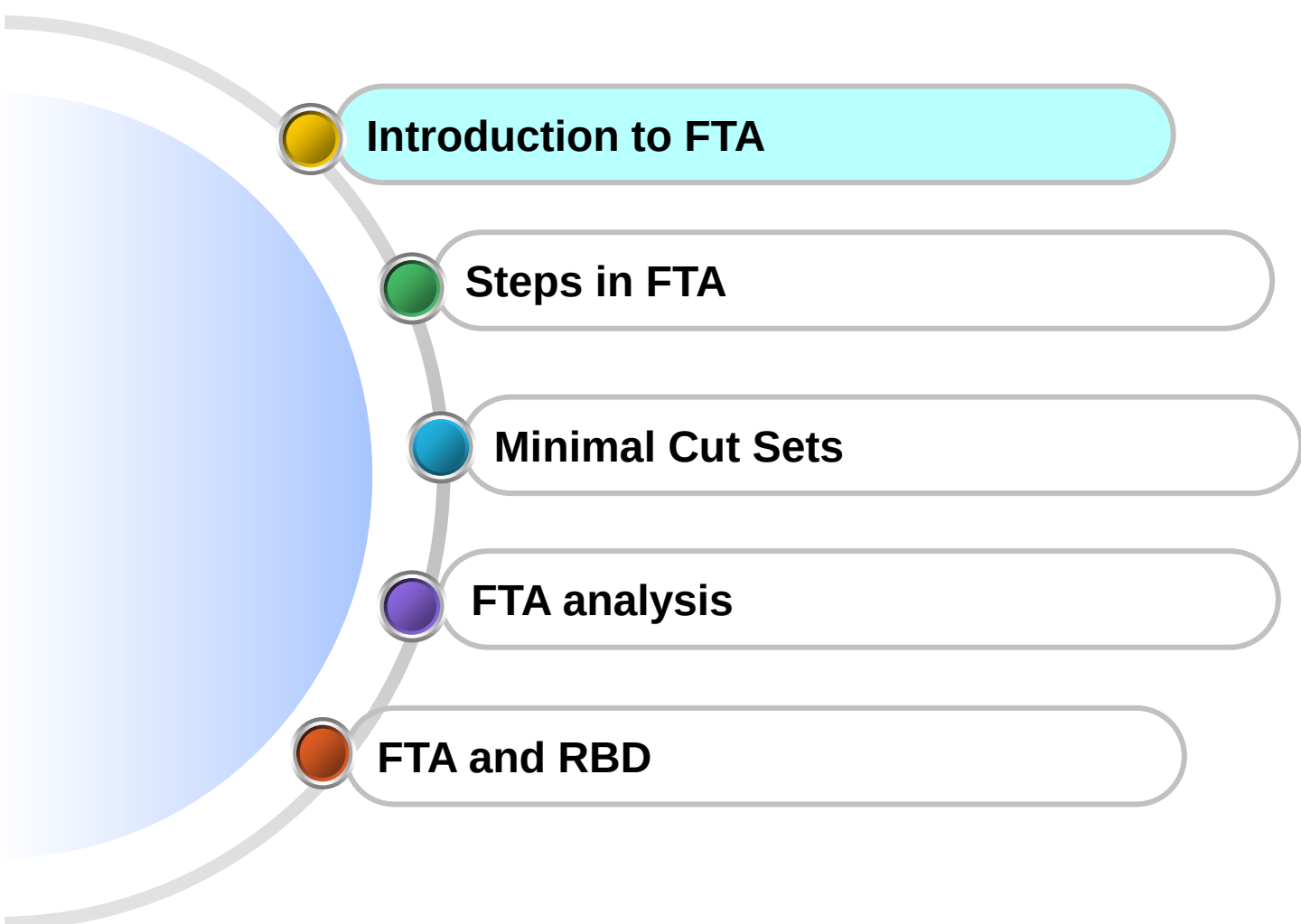
SKG 12.3 Reliability and Maintainability Engineering Part 2 Level 2

Session Schedule

Session	Content	Main Topics	Day	Time
3.	Advanced System Reliability- (Dr Ainul)	- Case Study Presentation - Maintenance model - Standby Configuration - Load Sharing Configuration - Throughput Analysis - RAM and Life Cycle cost	Day 3 (AM)	9:00am - 10:30am Break (15 Min) 10:45am – 1:00pm Lunch/Prayer (1:00pm – 2:00pm)
4.	Fault Tree Analysis- (Dr Hilmi)	- Introduction to Fault tree Analysis - Fault tree construction - Minimal cut sets - Qualitative analysis - Quantitative analysis	Day 3 (PM) Day 4 (AM)	2:00pm – 3:30pm Break (15 Min) 3:45pm – 5:00pm 9:00am - 10:30am Break (15 Min) 10:45am – 1:00pm Lunch/Prayer (1:00pm – 2:00pm)
5.	Introduction to Multi State system and Markov process in reliability- (Dr Masdi)	- Multi-State system and Markov Model - Discrete Time Markov - Continuous Time Markov - Application of multistate system modelling (IPF) - Post Assessment	Day 4 (PM)	2:00pm – 3:30pm Break (15 Min) 3:45pm – 5:00pm

Contents





Introduction to FTA

Steps in FTA

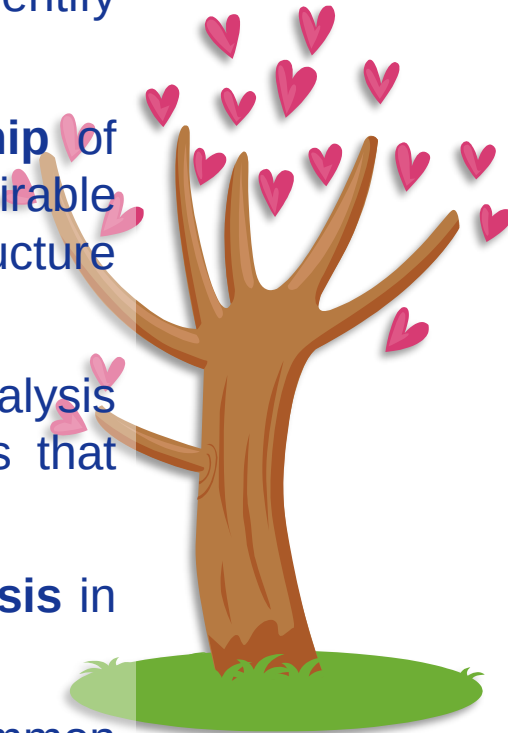
Minimal Cut Sets

FTA analysis

FTA and RBD

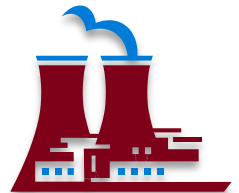
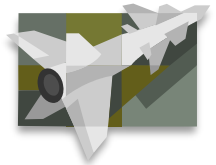
What is Fault Tree Analysis?

- ❖ Fault Tree Analysis (FTA) is an **analytical method** whereby an **undesired event** of the system is specified and the system is then analyzed in the context of its environment and operation to identify all **plausible ways** in which the undesired event can **occur**
- ❖ A fault tree is a **logical** representation of the **relationship** of primary events that cause the occurrence of a specific undesirable event called “the top event” and is described using a tree structure with OR, AND, etc. logic gates
- ❖ FTA is basically a **deductive (top-down) method** of analysis aimed at pinpointing the causes or combination of causes that lead to the defined top event
- ❖ FTA is the most commonly used technique for **causal analysis** in risk, safety and reliability studies
- ❖ FTA is used to identify **possible failures** in a system. A common engineering tool used in the **design stages** of a project



History of FTA

- First introduced by H.A. Watson of Bell Telephone Laboratories in early 1960s to conduct analysis on the Air Force Minuteman Missile Launch Control System
- The technique is further developed by Dave Haasl of Boeing company and used in the design and evaluation of commercial aircraft
- In 1970s, Nuclear Power Energy adopted it and further enhanced FTA's codes and algorithm
- By mid 1980s it was recognized globally and applied in various industries including Petrochemical and Computer Software



FTA Objectives

- Identifying **critical part** of the system with respect to the failure of interest
- Understanding the **functional relationship** of system failures
- Providing **input** to test, maintenance and operational **policies against failures**
- Understanding the **level of protection** that the design concept provides against failures
- Providing an **integrated picture** of some aspects of system operation
- Confirming the **ability of the system** to meet its imposed **safety and jurisdictional requirements**
- Providing a means for **qualitative and quantitative** reliability analysis
- Providing a mean to identify possible **root cause** of problem in **RCFA**



When to perform FTA?

- ❑ Assessing criticality, importance, probability and risk of a system
- ❑ Identifying weaknesses in the a system
- ❑ Investigating the effect of safety barriers
- ❑ As required by customer or for certification
- ❑ Necessitated by the risk involved with the operation or product (risk is high)
- ❑ Evaluating corrective actions or design options
- ❑ Investigating accident/incident/anomaly
- ❑ Finding out the root cause of failure





Introduction to FTA



Steps in FTA



Minimal Cut Sets



FTA analysis



FTA and RBD

What cause Challenger to explode?



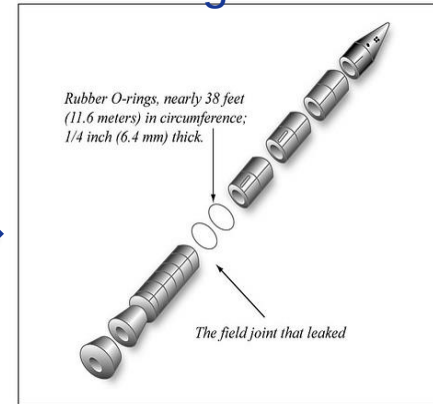
How to perform FTA to find root causes of the accident?



Extreme cold temperature



O-ring failure



What cause Challenger to explode?

VIDEO



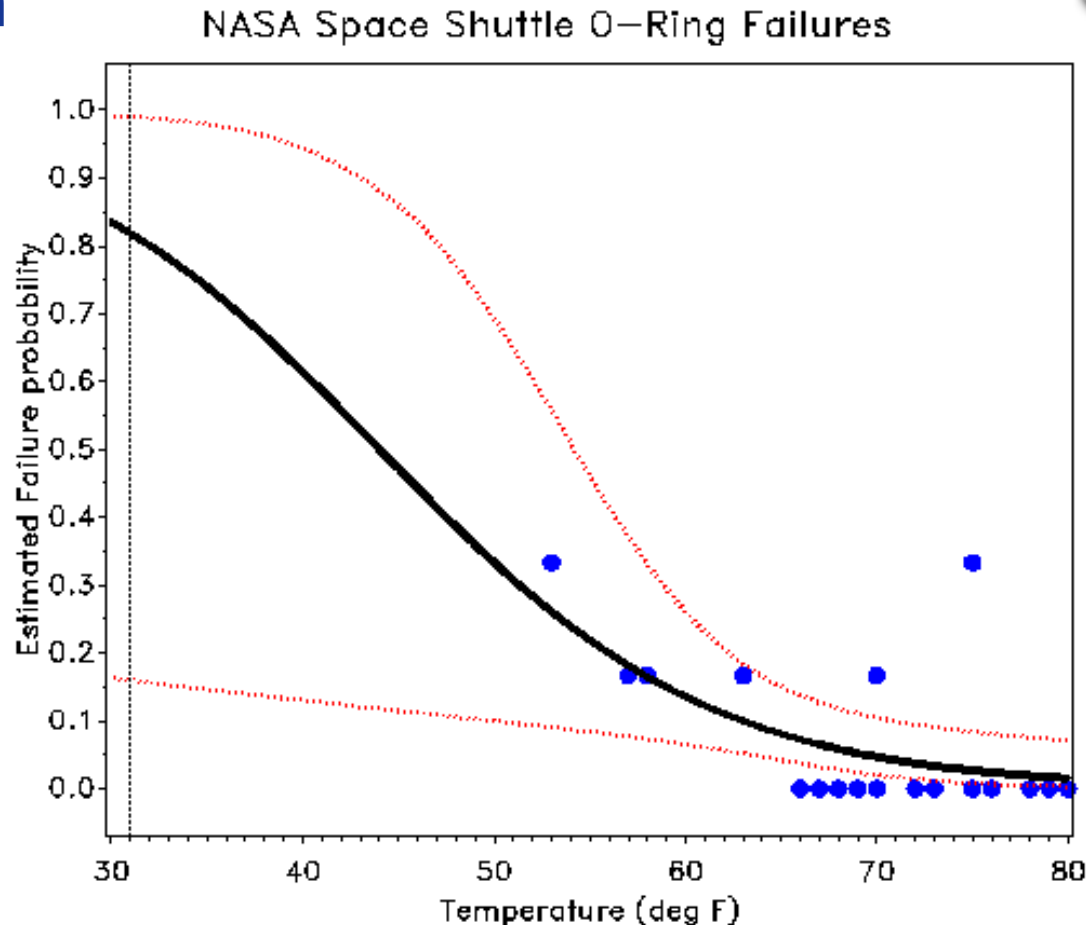
<http://www.youtube.com/watch?v=nspP1YFftAY>



http://www.youtube.com/watch?v=8qAi_9quzUY

What cause Challenger to explode?

- ❖ Insufficient experimental data and wrong conclusion & decision



Preparations for FTA

- ❖ Thorough understanding of design, operation, and maintenance aspect of system under studied
- ❖ Clear definitions on
 - What constitutes system failure : the undesirable event
 - Scope and objective of analysis
 - System physical bounds and boundary
- ❖ Well-defined level of analysis resolution
- ❖ Clear identification of associated assumptions
- ❖ Use the existing FMECA (Failure mode effect criticality analysis), system block diagram or P&ID (process and instrumentation diagram) as a starting point

Steps in FTA

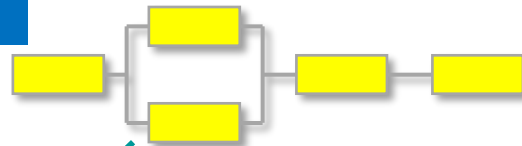
1. Define the system, TOP (top of event) and system boundaries
2. Construct the Fault tree
3. Identify the minimal cut sets
4. Perform Qualitative analysis
5. Conduct Quantitative analysis
6. Report the results



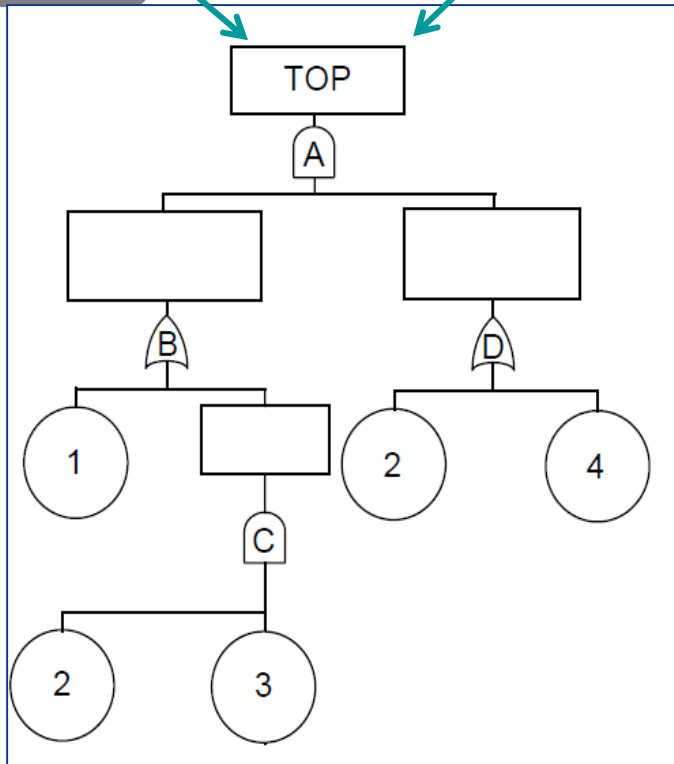
FTA steps - Illustration

① Define system, boundaries and TOP

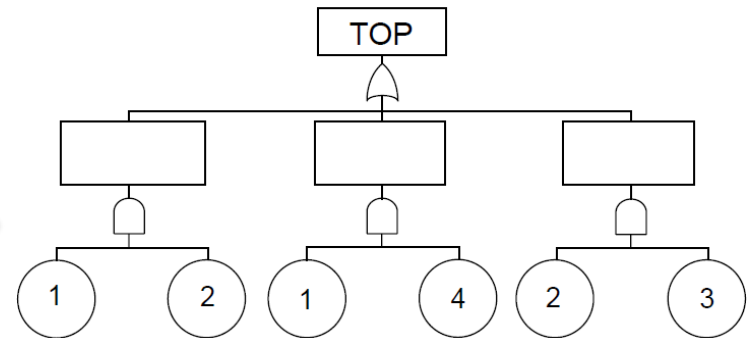
② Construct Fault Tree



System block diagram



③ Find minimal cut sets



④ Qualitative Analysis

⑤ Quantitative Analysis

⑥ Report

Define system and boundaries

- Define the boundary of the system (the scope of the FTA)
 - Which parts of the system are included in the analysis, and which parts are not?
 - Boundary conditions with respect to external stresses (What type of external stresses should be included in the analysis –war, sabotage, earthquake, lightning, etc?)
- Define the basic causal events to be considered (the resolution of the FTA)
 - How detailed should the analysis be?
- Define the initial state of the system
 - What is the system operational status when the TOP event occurs?

Identify Top event

- Define the undesired TOP event to be analyzed (the focus of the FTA)
- Use FMECA, P&ID, system block diagram to define it in a clear and unambiguous way
- Top event normally represents potential high risk event, either due to the severity of the outcome or frequency of occurrence
- State precisely what the fault is and when it occurs

Examples:

The door bell fails to sound when the button is pressed

Car fails to start when ignition key is turned

Fault tree construction

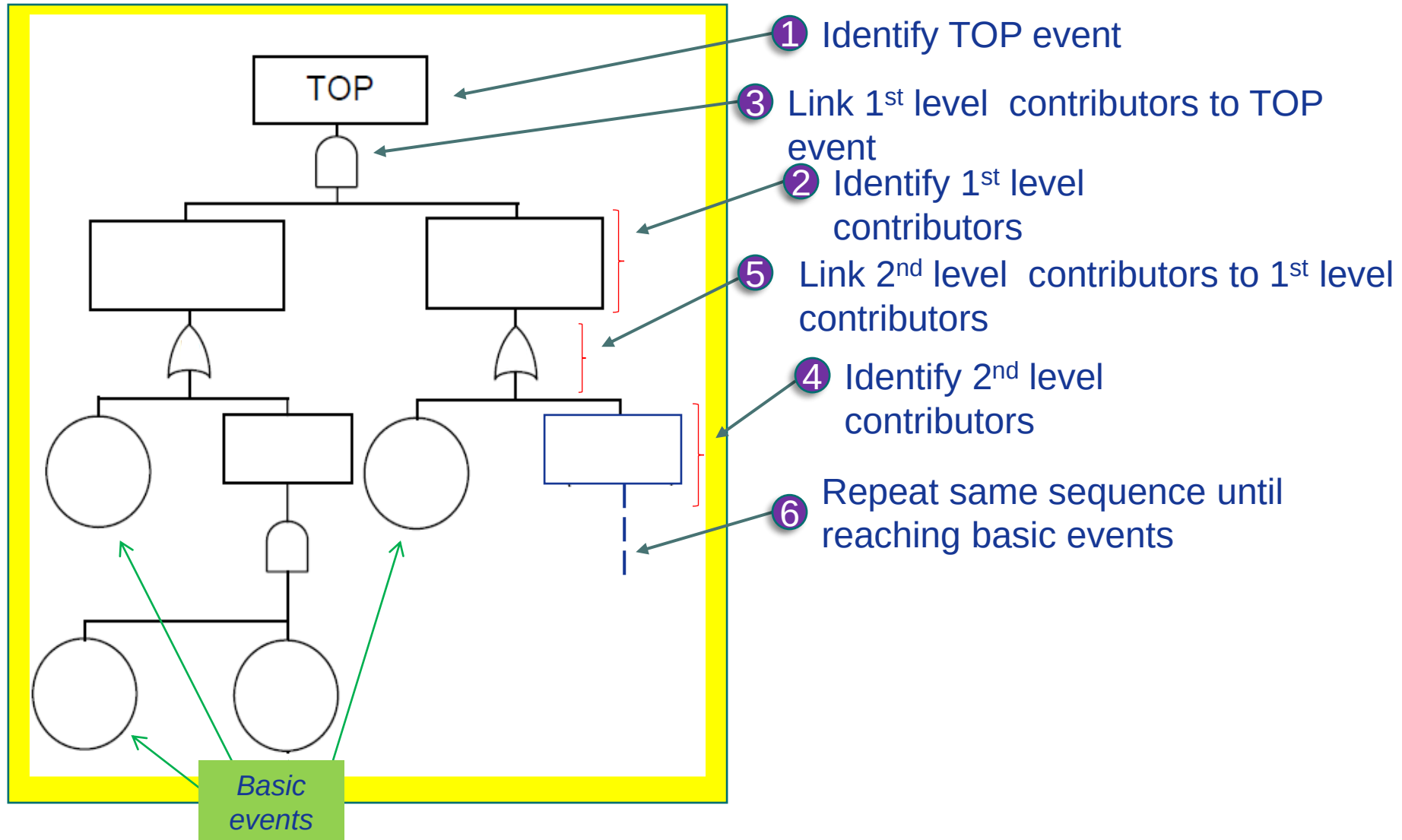
1. Define TOP event
2. Determine the immediate, necessary and sufficient (INS) causes for the TOP event to occur (1st level contributors).
3. Contributors = Independent Fault or Failure condition
4. Link 1st level contributors to TOP event via logic gates (AND or OR gate)
5. Identify 2nd level contributors
6. Link 2nd level contributor to 1st level contributor via logic gates
7. Proceed in this way until we reach appropriate level (basic events)

Appropriate level:

Independent basic events

Events for which we have failure data

Fault tree construction - Illustration



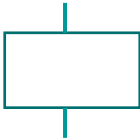
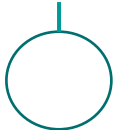
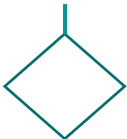
Immediate, necessary and sufficient (INS) causes

- Read the top / intermediate event wording
- Identify all INS events to cause the top / intermediate event
 - **Immediate** – do not skip past events
 - **Necessary** – include only what is actually necessary
 - **Sufficient** - do not include more than the minimum necessary
- Test mentally the event and logic until satisfied

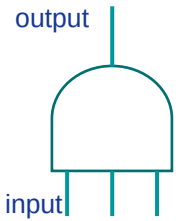
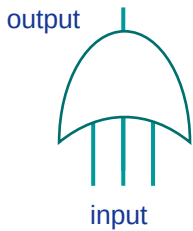
Failure vs. Fault

- It is very important to define clearly each event either as a failure or fault, so it can be identified as a basic cause or be further resolved
- **Failure event**
 - the occurrence of a basic component failure
 - the result of an internal inherent failure mechanism, hence requires no further breakdown
- **Fault event**
 - the occurrence or existence of an *undesired state* for a component, subsystem or system
 - it can be further breakdown

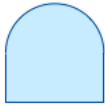
Fault tree symbols

Category	Symbol	Description
Event description		Description of fault event that results from logical combination of fault events through the input of logic gate
Input events (states)		The basic fault event represents a basic equipment failure that requires no further development of failure causes
		The basic fault event whose causes have not been fully developed either because of lack of information or because its consequences are insignificant
Transfer	 transfer in  transfer out	The triangle signifies a transfer of a fault tree branch to another location within the tree, to avoid repeating segments of the fault tree

Fault tree symbols (cont'd)

Category	Symbol	Description
Logical gates	 output input AND - Gate	The AND-gate indicates that the output fault event occurs only if all the input fault events occur simultaneously
	 output input OR - Gate	The OR-gate indicates that the output fault event occurs if at least one of the input fault events occurs

Fault tree symbols (cont'd)



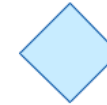
AND gate

The output event occurs if all input events occur.



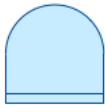
OR gate

The output event occurs if at least one of the input events occurs.



Undeveloped event

An event which is no further developed. It is a basic event that does not need further resolution.



Priority AND gate

The output event occurs if all input events occur in a specific sequence.



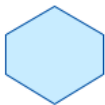
XOR gate

The output event occurs if exactly one input event occurs.



House event

An event that is normally expected to occur. In general, these events can be set to occur or not occur, i.e. they have a fixed probability of 0 or 1.



Inhibit gate

The input event occurs if all input events occur and an additional conditional event occurs.



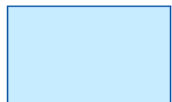
Voting gate

The output event occurs if k or more of the input events occur.



Conditional event

A specific condition or restriction that can apply to any gate.



Event



Basic event

A basic initiating fault (or failure event)

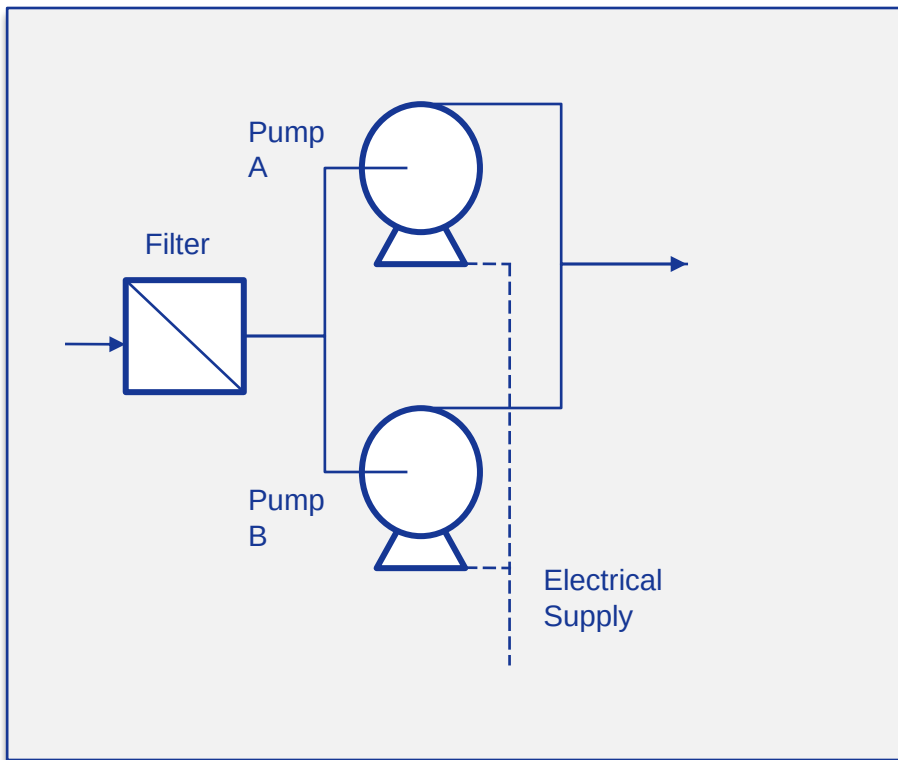


Transfer symbol

Indicates a transfer continuation to a sub tree.

Example 1 : Fault Tree construction

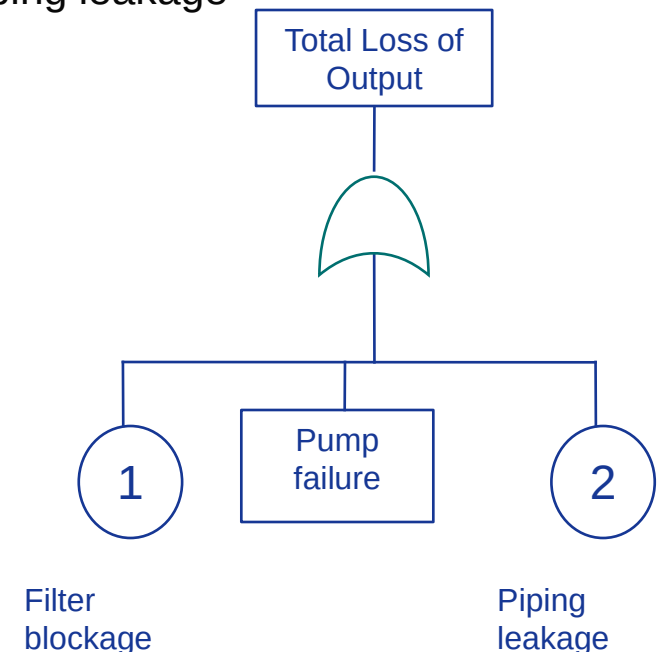
Consider a simple pumping sub-system below. There is a need to find out causes of the "Total loss of output"



STEPS

1. Identify TOP event = Total loss of output
2. Identify immediate, necessary and sufficient (INS) causes of TOP event. This can be caused either by:

- a. Filter blockage
- b. Pump failure
- c. Piping leakage



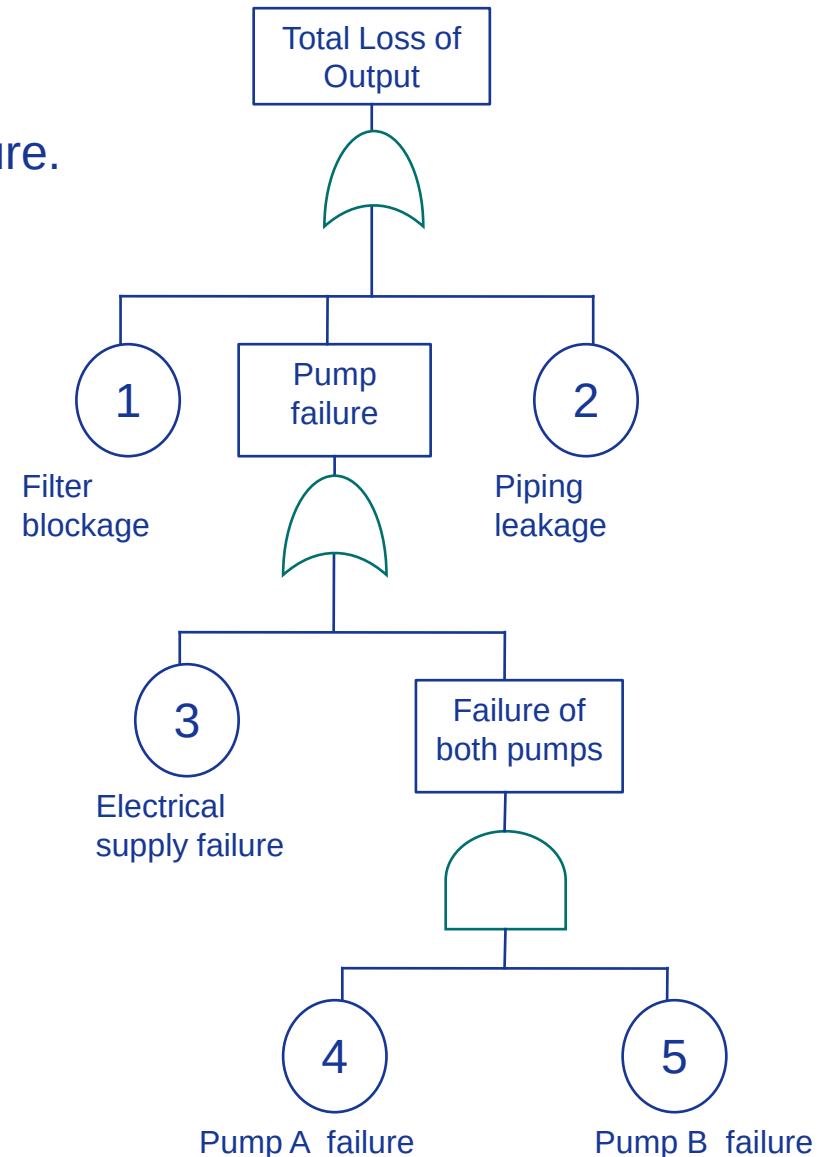
Example 1 : Fault Tree construction (cont'd)

STEPS (cont'd)

3. Identify the INS causes for pump failure.
This can be caused either by:

- a. Failure of electrical supply
- b. Failure of both pump

The corresponding Fault tree can be drawn as this

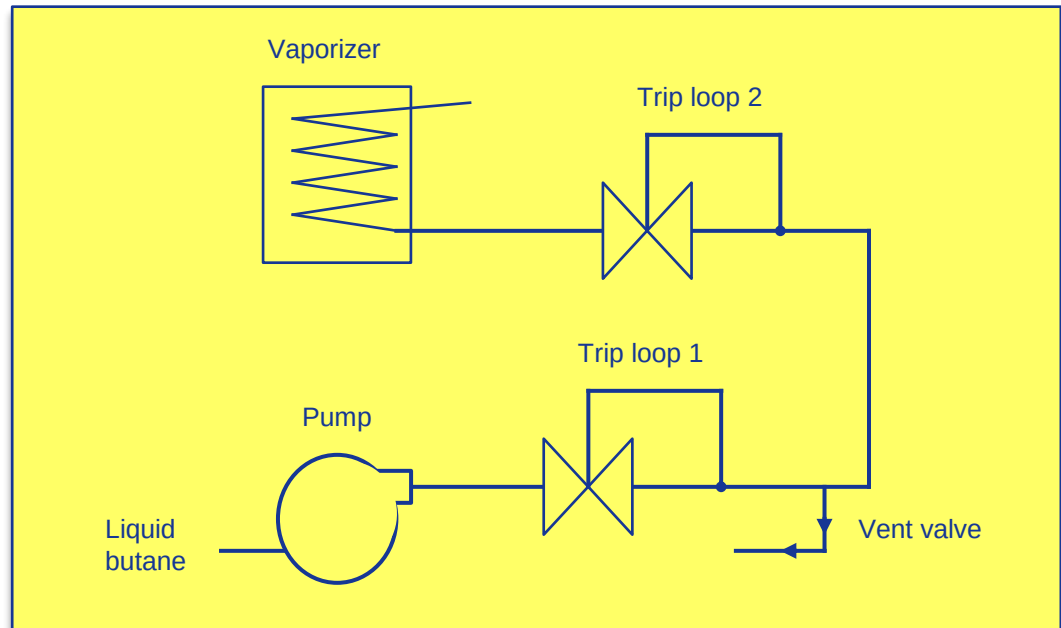


Exercise 1

In the system below, Liquid butane is pumped from a tank into a vaporizer where it is heated to form a gas. In the event of a pump surge the pressure in the vaporizer exceeds the rating of the vaporizer tubes. To prevent the tubes from rupturing, three safety systems have been placed in the system, which will shut down the process in case of pump surge. They include two trip loops which close a valve halting the butane flow, and a vent valve which opens allowing the butane to return to tank if the pressure exceeds the preset limit.

Q: Develop a Fault tree, where the top event is “Vaporizer coil ruptures under high pressure”

Note: Assume the occurrence of a high pressure will definitely rupture the tank





Introduction to FTA



Steps in FTA



Minimal Cut Sets



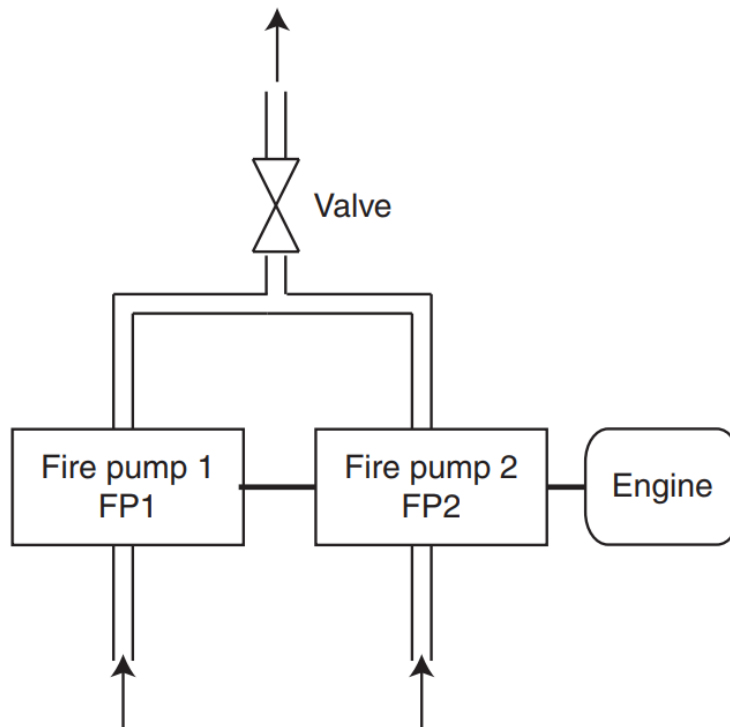
FTA analysis



FTA and RBD

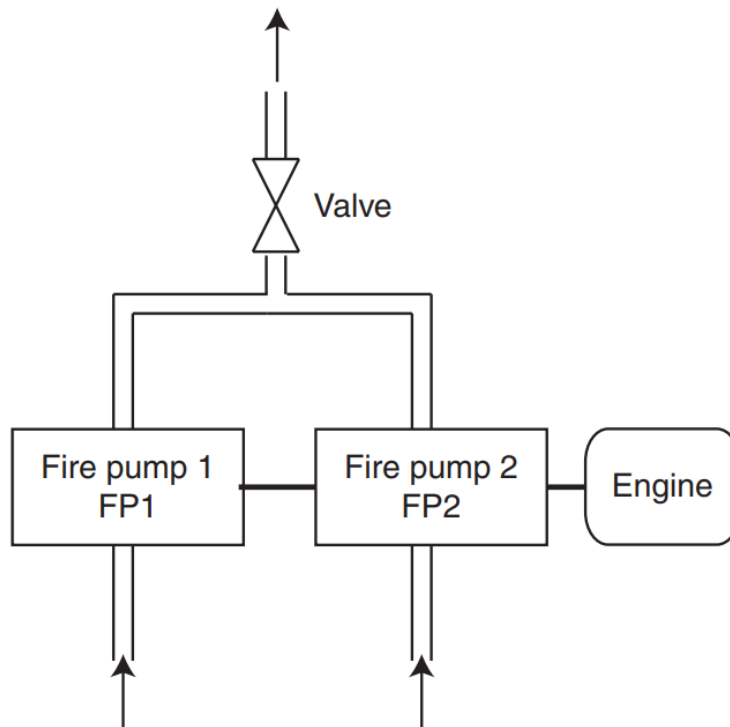
Cut sets

Consider the Redundant Fire water system.
TOP event is “No water from fire water system”



Construct the Fault Tree

Cut sets



Causes for TOP event:

VF = Valve failure

G1 = No output from any of the fire pumps

G2 = No water from FP1 G3 = No water from FP2

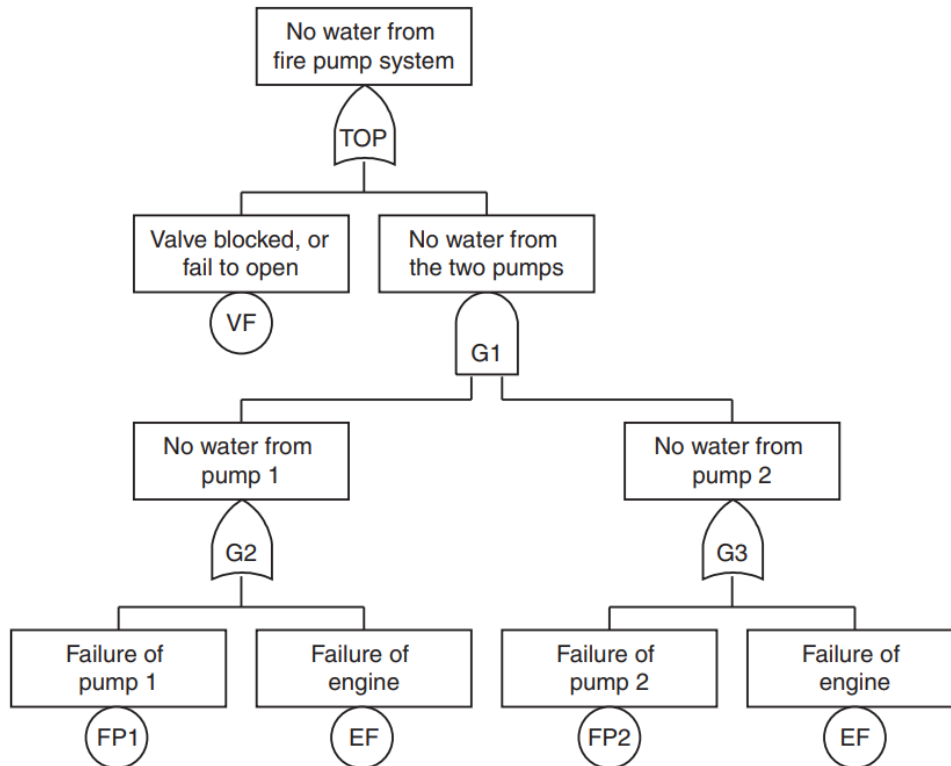
FP1 = failure of FP1

EF = Failure of engine

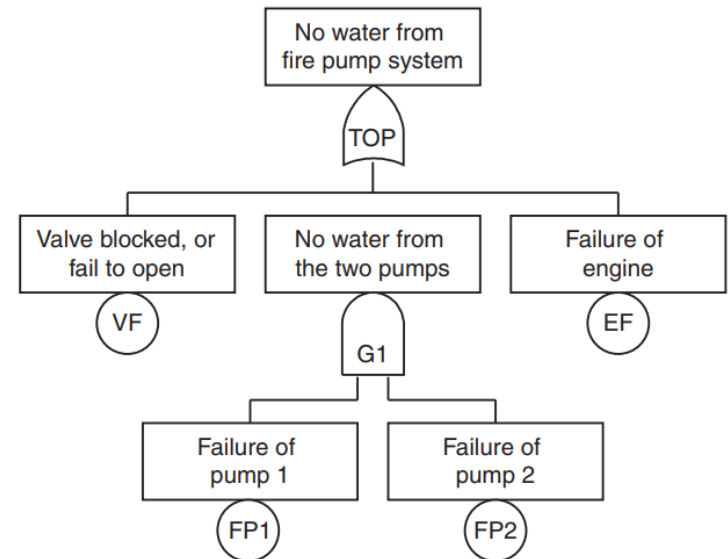
FP2 = Failure of FP2

Cut sets

FTA (a)



FTA (b)



Are they the same?

Both are logically identical, giving the same info

Cut sets

- **CUT SET** - any set of basic events which, if all occur, will cause the TOP event to occur.
- **MINIMAL CUT SET** - a least set of basic events which, if all occur, will cause the TOP event to occur.
- A cut set is said to be minimal if the set cannot be reduced without losing its status as a cut set
- The minimal cut set analysis provides a new fault tree, logically equivalent to the original, with an OR gate beneath the top event, whose inputs (bottom) are minimal cut sets.
- Each minimal cut set is an AND gate with a set of basic event inputs necessary and sufficient to cause the top event.
- Once the minimal cut sets are identified the quantification of the fault tree can be carried out

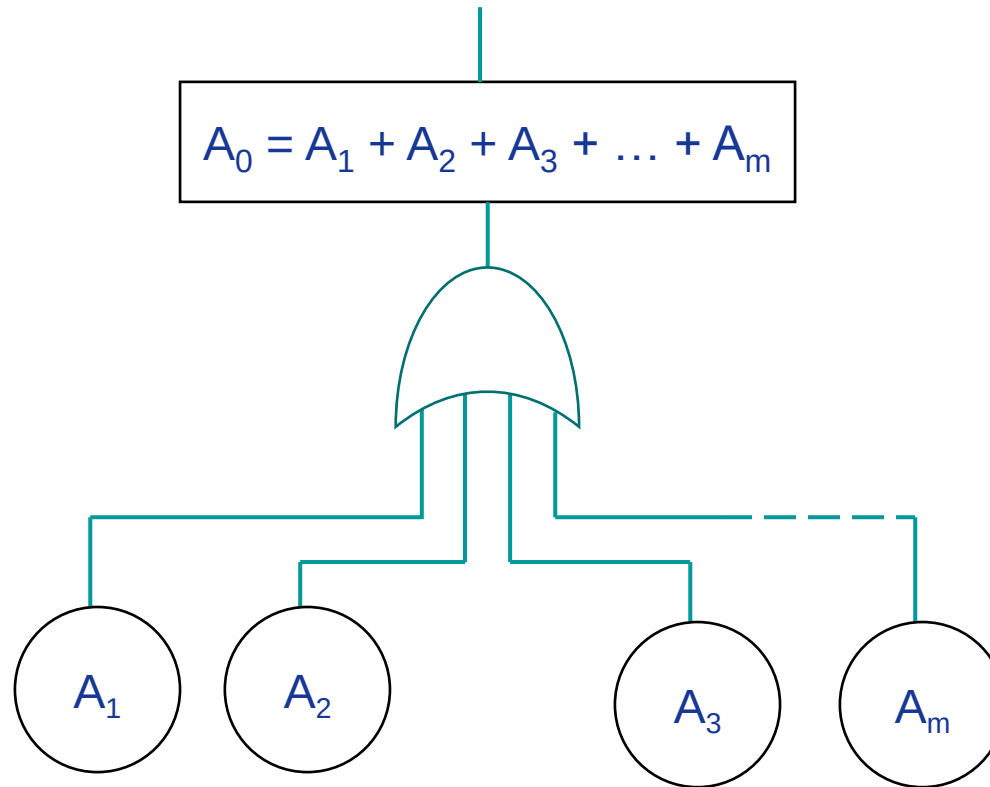
Finding minimal cut sets

- **Boolean expression reduction approach**
 - A fault tree can be translated into an equivalent set of Boolean Expression
 - Simplification of this expression is required to find the minimal cut sets
 - Simplification is performed based on Boolean algebra rules
 - Two commonly used approaches :
 - **Top-down**
 - **Bottom-up**

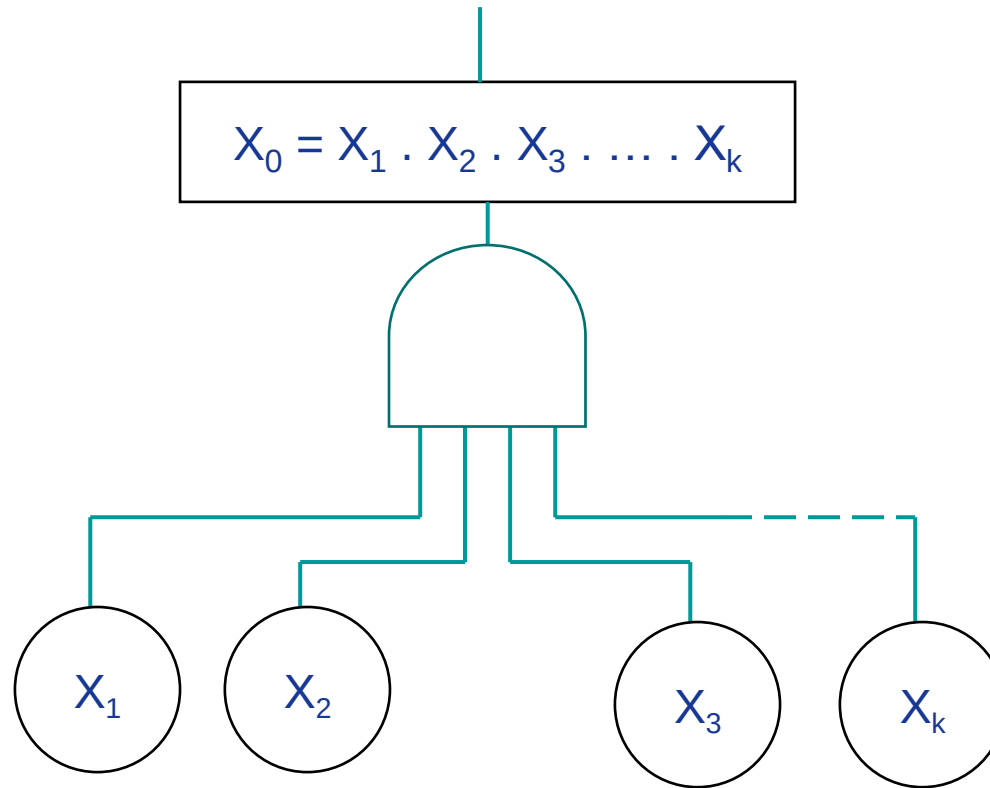
Boolean Algebra rules

Rules	Expressions
1. Associative Law	$(A + B) + C = A + (B + C) = A + B + C$ $(AB)C = A(BC) = ABC$
2. Distributive Law	$X(Y + Z) = XY + XZ$ $X + YZ = (X + Y)(X + Z)$
3. Commutative Law	$AB = BA$ $A + B = B + A$
4. Absorption Law	$X + XY = X$ $X(X + Y) = X$
5. Idempotent Law	$A \cdot A = A$ $A + A = A$

Boolean Expressions – OR Gate

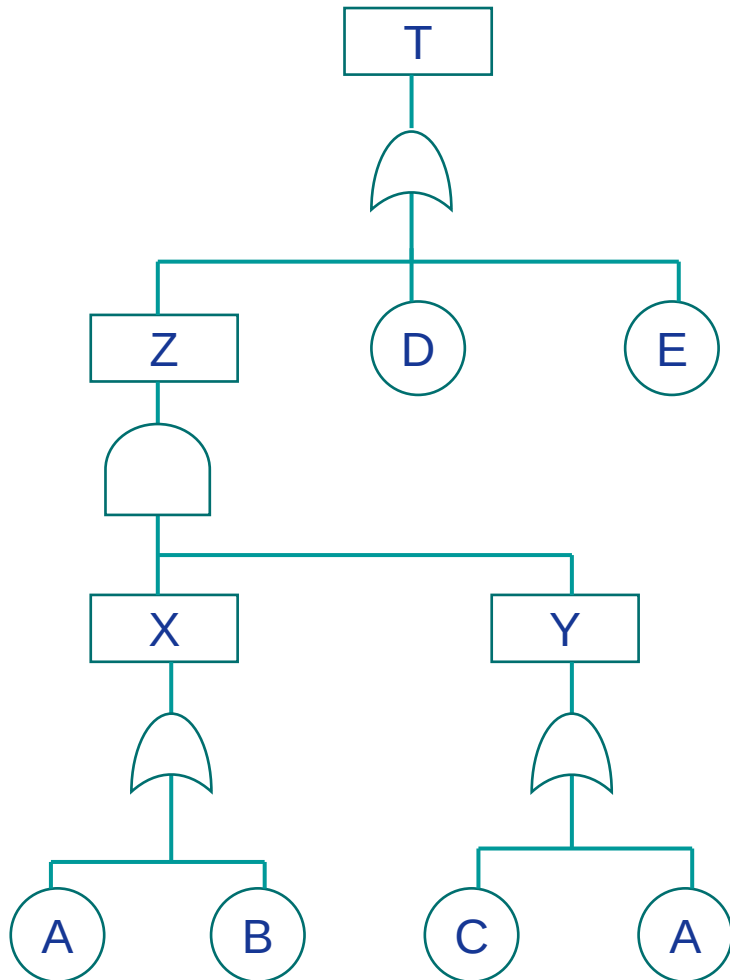


Boolean Expressions – AND Gate



Example 2: Top-down approach

Q: Find the minimal cut sets



Solution

$$T = Z + D + E$$

substitute $Z = XY$

$$T = XY + D + E$$

substitute $X = A + B, Y = A + C$

$$\begin{aligned} T &= (A + B)(A + C) + D + E \\ &= AA + AC + BA + BC + D + E \end{aligned}$$

*apply **Idempotent law** ($AA = A$)*

$$T = A + AC + AB + BC + D + E$$

*apply **Absorption law** ($A + AB = A$)*

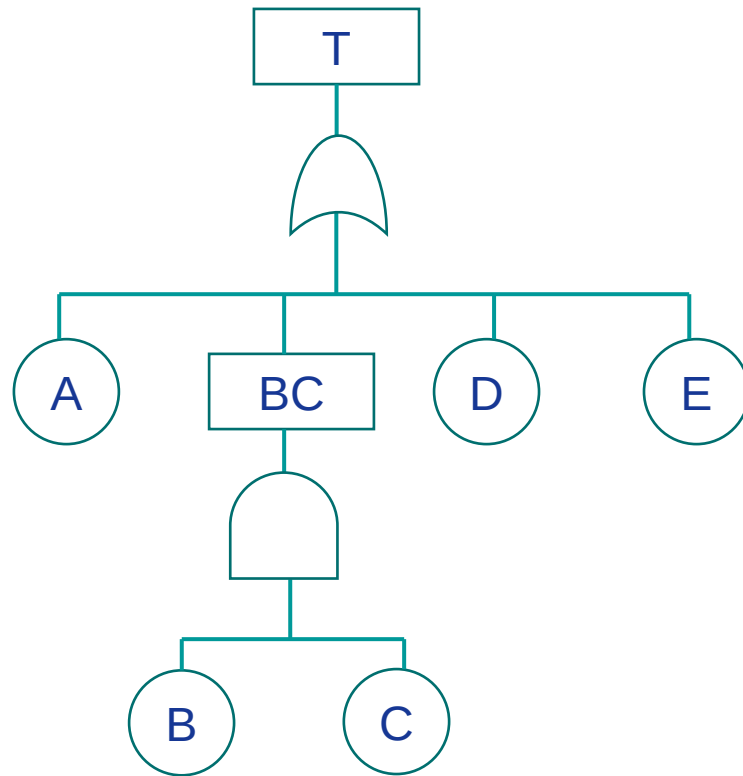
$$\underline{T = A + BC + D + E}$$

Minimal
cut sets

Note: alternatively use Distributive law $(A + B)(A + C) = A + BC$

Example 2: Top-down approach (cont'd)

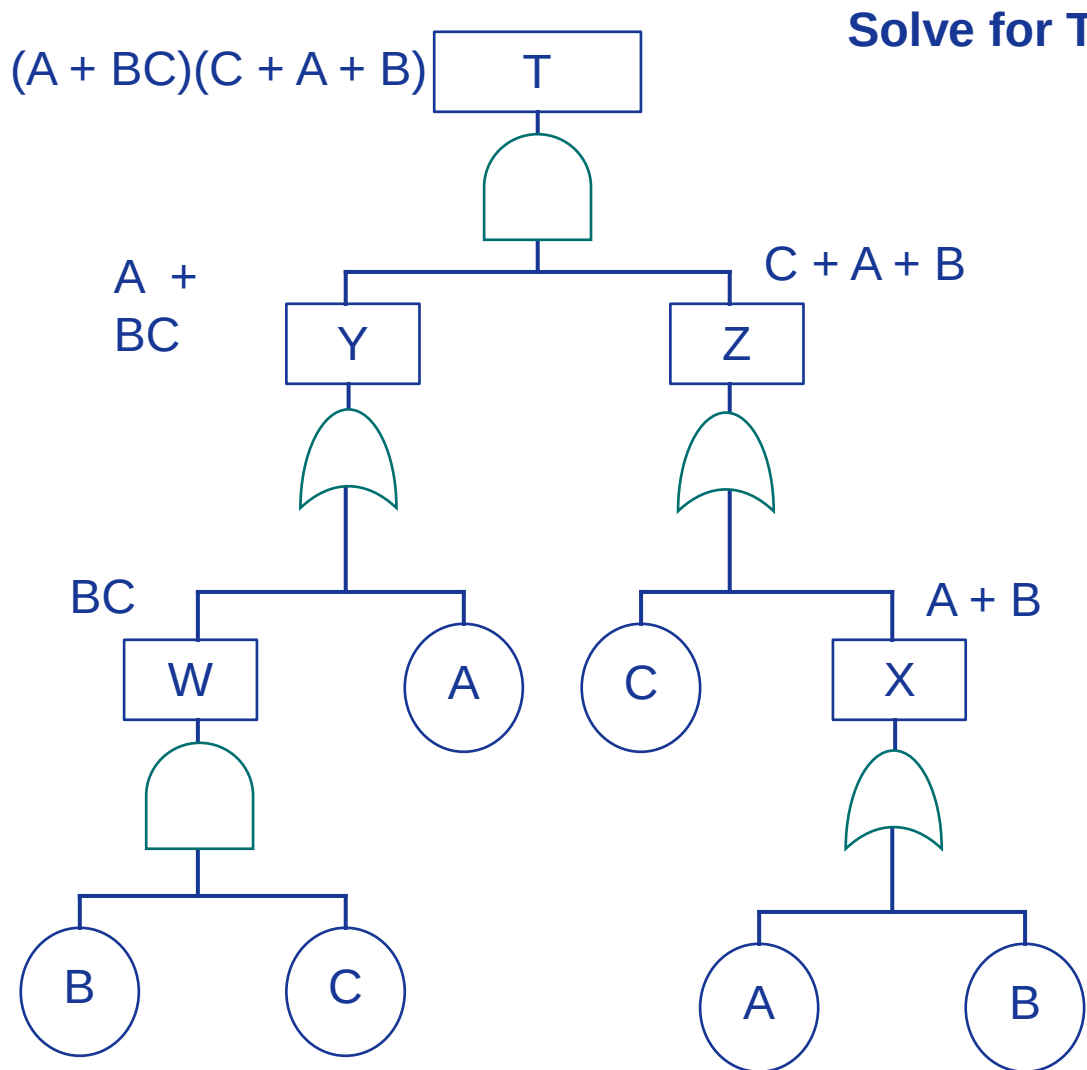
The minimal cut sets expression for the TOP event can be represented by the fault tree below



$$T = A + BC + D + E$$

This “**equivalent**” fault tree is a repeated event free fault tree

Example 3: Bottom-up approach



Find Boolean expression at each gate starting from the bottom i.e. W, X, Y, Z

$$T = (A + BC)(C + A + B)$$

$$= AA + AB + AC + ABC + BBC + BCC$$

$$(AA = A)$$

$$= A + AB + AC + AC + ABC + BC + BC$$

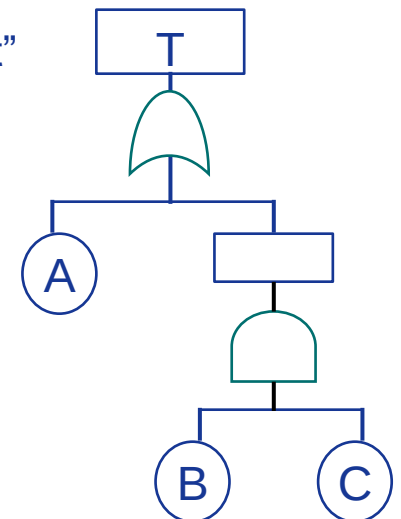
$$(A + A = A)$$

$$= A + AB + AC + ABC + BC$$

$$(A + AB = A)$$

$$= \underline{A + BC}$$

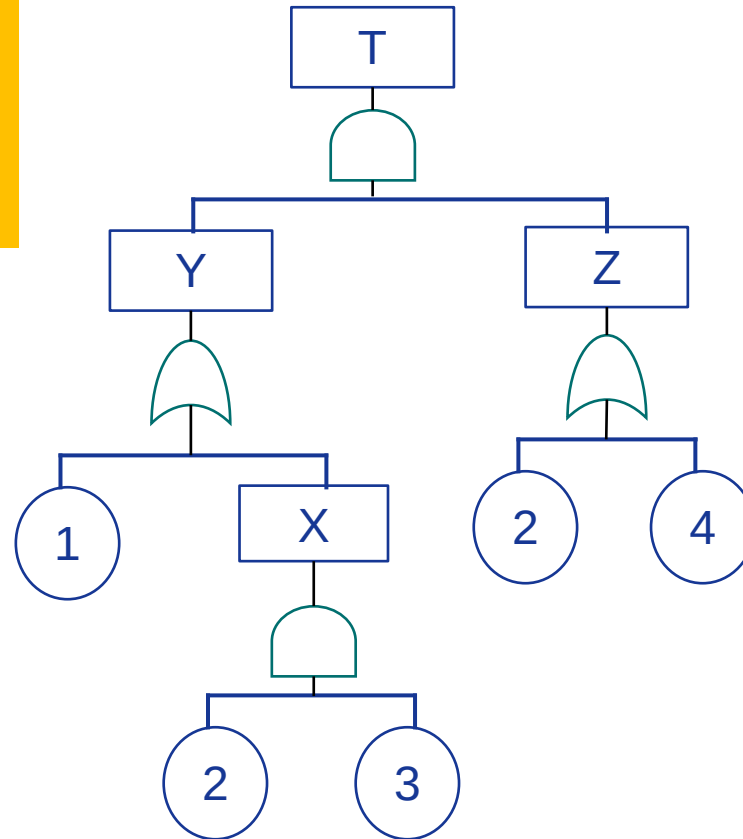
“Equivalent”
Fault tree



Exercise 2

Q:

- i. Find the minimal cut sets for the given Fault Tree
- ii. Its new “equivalent” Fault tree





Introduction to FTA



Steps in FTA



Minimal Cut Sets



FTA analysis



FTA and RBD

Qualitative Analysis

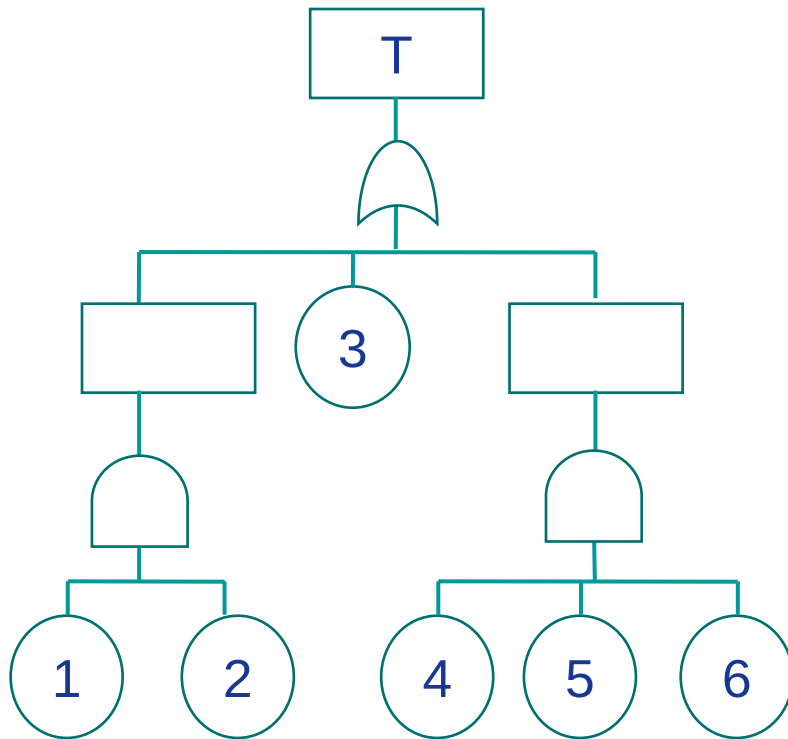
Qualitative assessment by analyzing the minimal cut sets:

- Ranking the order of minimal cut sets based on the criticality importance
- Ranking based on the type of basic events involved
 - **Human error** (most critical)
 - Failure of active equipment
 - Failure of passive equipment

Rank	Basic event 1	Basic event 2
1	Human error	Human error
2	Human error	Failure of active unit
3	Human error	Failure of passive unit
4	Failure of active unit	Failure of active unit
5	Failure of active unit	Failure of passive unit
6	Failure of passive unit	Failure of passive unit

Qualitative Analysis - Importance measures

“equivalent” fault tree



Qualitative ranking of contributions to system failure by analyzing structural importance

- Minimal cut sets important by order number

Minimal cut sets

1	2	
3		
4	5	6

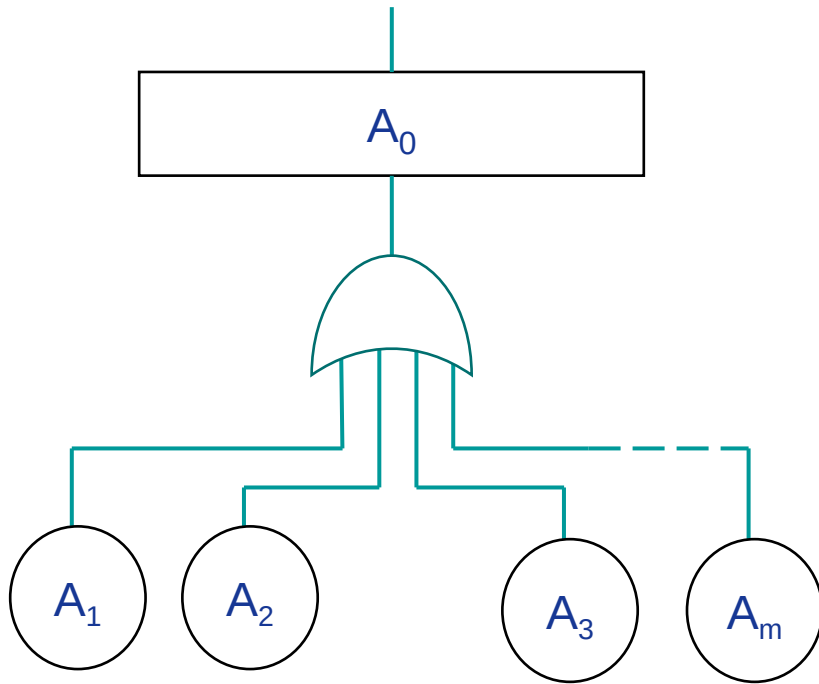
- singlet cut set indicates a potential Single-Point Failure
- Long cut set indicates low vulnerability
- Short cut set indicates higher vulnerability

- Component important by number of times it occurs in different cut sets
- Evaluate each cut set / minimal cut set for design weak point and common cause problems

Quantitative Analysis

- Based on the minimal cut sets, we can calculate the probability of the occurrence of the TOP event
- This probability is obtained by estimating the probability of occurrence of the output events of lower and intermediate logic gates, i.e. AND gate and OR gate

Probability of occurrence – single OR Gate

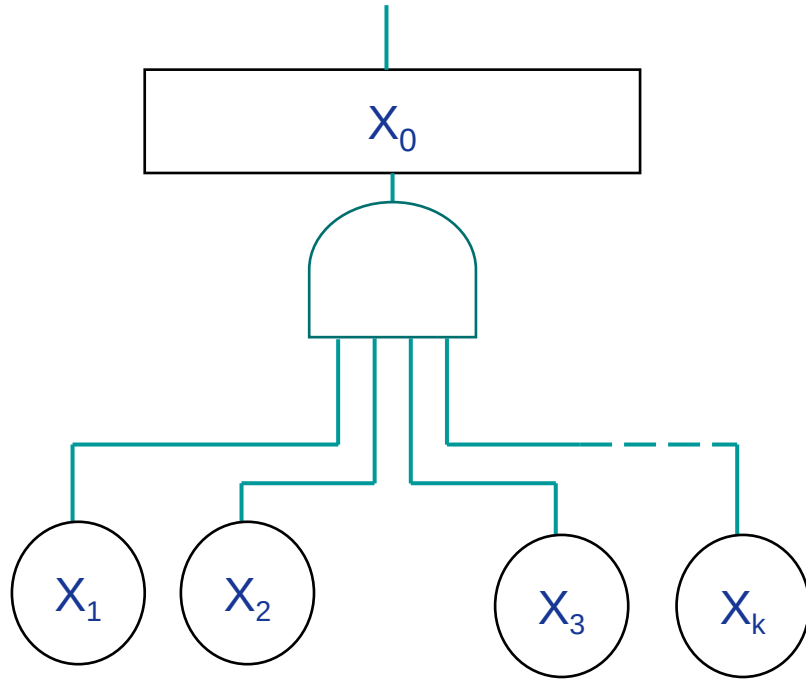


$$P(A_0) = 1 - \prod_{i=1}^m \{1 - P(A_i)\}$$

Example, for $m = 3$

$$\begin{aligned} P(A_0) &= 1 - \prod_{i=1}^3 [1 - P(A_i)] \\ &= 1 - [1 - P(A_1)][1 - P(A_2)][1 - P(A_3)] \end{aligned}$$

Probability of occurrence – single AND Gate



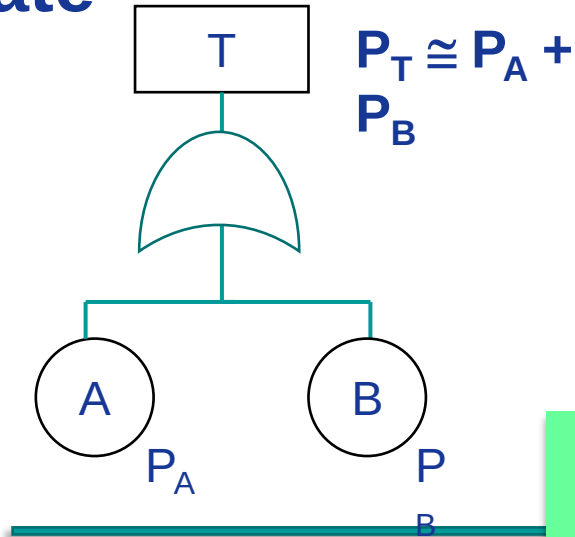
$$P(X_0) = \prod_{i=1}^k P(X_i)$$

Example, for $k = 4$

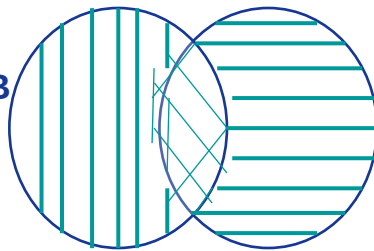
$$P(X_0) = P(X_1)P(X_2)P(X_3)P(X_4)$$

Probability of occurrence - comparison

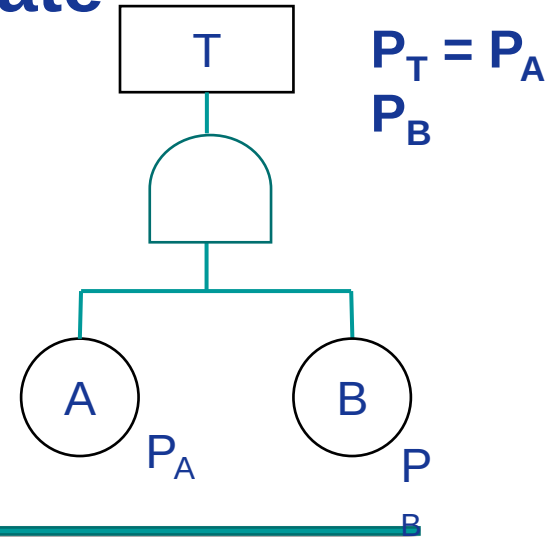
OR Gate



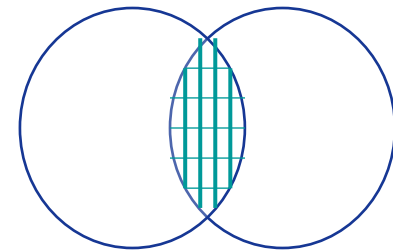
Union ($\cup$)



AND Gate



Intersection ($\cap$)



A and B are
INDEPENDENT
Events

$$P_T = P_A P_B$$

$$P_T = P_A + P_B - P_A P_B$$

Normally ignored – insignificant, when $P_i < 0.1$ (Rare event approximation), (P_T is accurate to within about ten percent of the true probability)

Example 4: Quantitative Analysis

Q: Assume that the probability of occurrence of basic events are as follows :

E_1	E_2	E_3	E_4	E_5	E_6
0.12	0.13	0.11	0.07	0.08	0.05

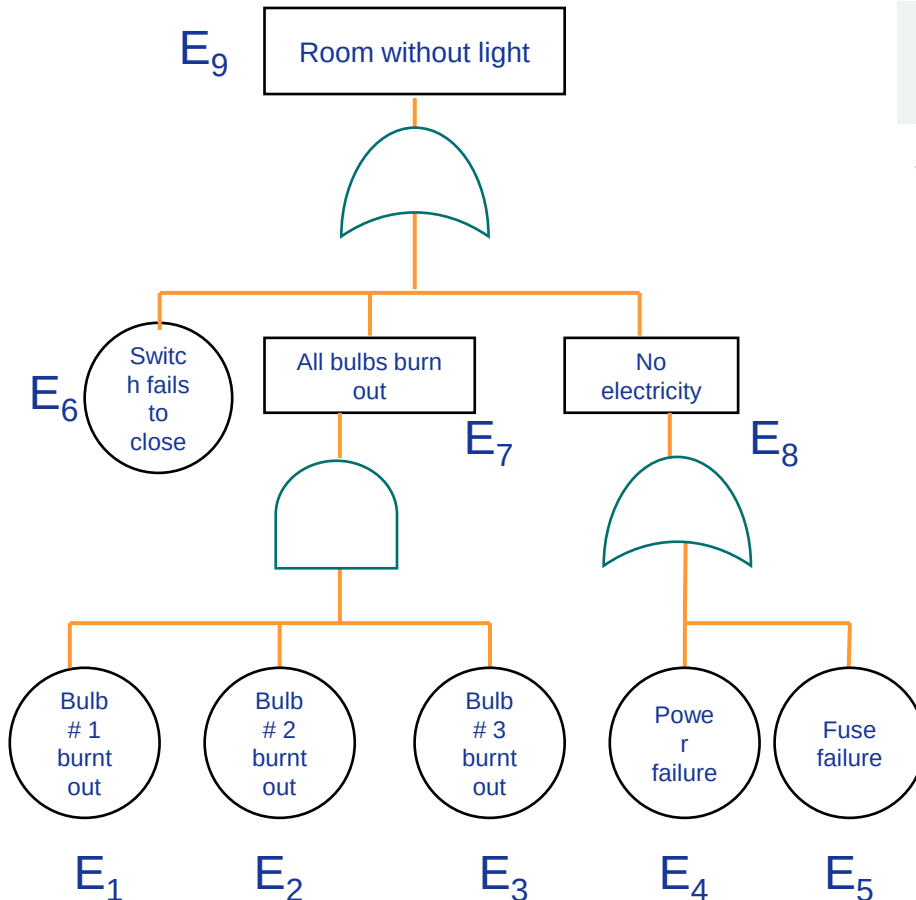
Compute the probability of occurrence of the room without light, $P(E_9)$.

Steps

$$\begin{aligned}
 1. \text{ Calculate } P(E_7) &= P(E_1) P(E_2) P(E_3) \\
 &= (0.12)(0.13)(0.11) \\
 &= 0.00172
 \end{aligned}$$

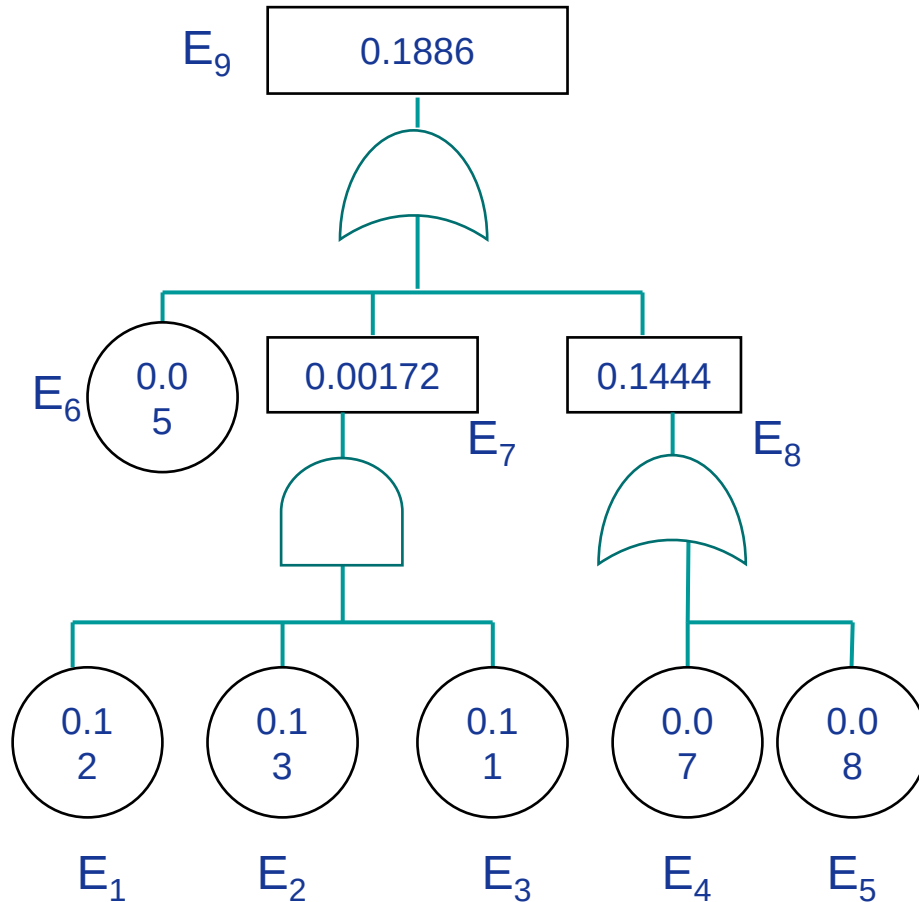
$$\begin{aligned}
 2. \text{ Compute } P(E_8) &= P(E_4) + P(E_5) - P(E_4) P(E_5) \\
 &= (0.07) + (0.08) - (0.07)(0.08) \\
 &= 0.1444
 \end{aligned}$$

$$\begin{aligned}
 3. \text{ Calculate } P(E_9) &= 1 - [1 - P(E_6)][1 - P(E_7)][1 - P(E_8)] \\
 &= 1 - [1 - 0.05][1 - 0.00172][1 - 0.1444] \\
 &= \mathbf{0.1886}
 \end{aligned}$$



Example 4: Quantitative Analysis (cont'd)

In Summary



$$P(E_7) = (0.12)(0.13)(0.11) \\ = 0.00172$$

$$P(E_8) = (0.07) + (0.08) - (0.07)(0.08) \\ = 0.1444$$

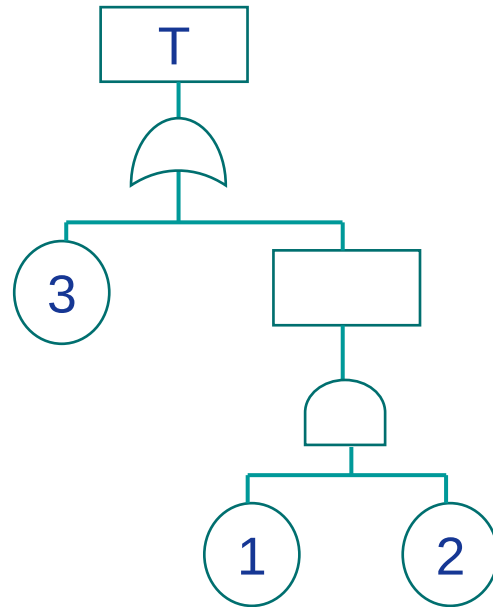
$$P(E_9) = 1 - [1 - 0.05][1 - 0.00172][1 - 0.1444] \\ = \mathbf{0.1886}$$

Example 5 : Quantitative Analysis

The minimal cut sets is

3	
1	2

“Equivalent” Fault Tree



Note: Calculating P_T based on Boolean Indicated cut sets, instead of minimal cut sets will result in inaccurately high P_T

$$P_T \cong P_1 \times P_3 + P_2 \times P_3 + P_3 + P_1 \times P_2 + P_1 \times P_2 + P_1 \times P_2 \times P_3$$

$$P_T \cong \mathbf{0.05596 !}$$

Q: Calculate P_T , given the event probability (P_e), $P_1 = 0.03$, $P_2 = 0.04$ and $P_3 = 0.05$

Solution

Let P_k = cut set probability

$$P_k = \prod P_e = P_1 \times P_2 \times \dots \times P_n$$

$$P_T \cong \sum P_k$$

$$P_T \cong P_3 + P_1 \times P_2$$

3	
1	2

$$P_T \cong 0.03 + (0.04 \times 0.05) = \mathbf{0.032}$$

1	3	
2	3	
3		
1	2	
1	2	
1	2	3

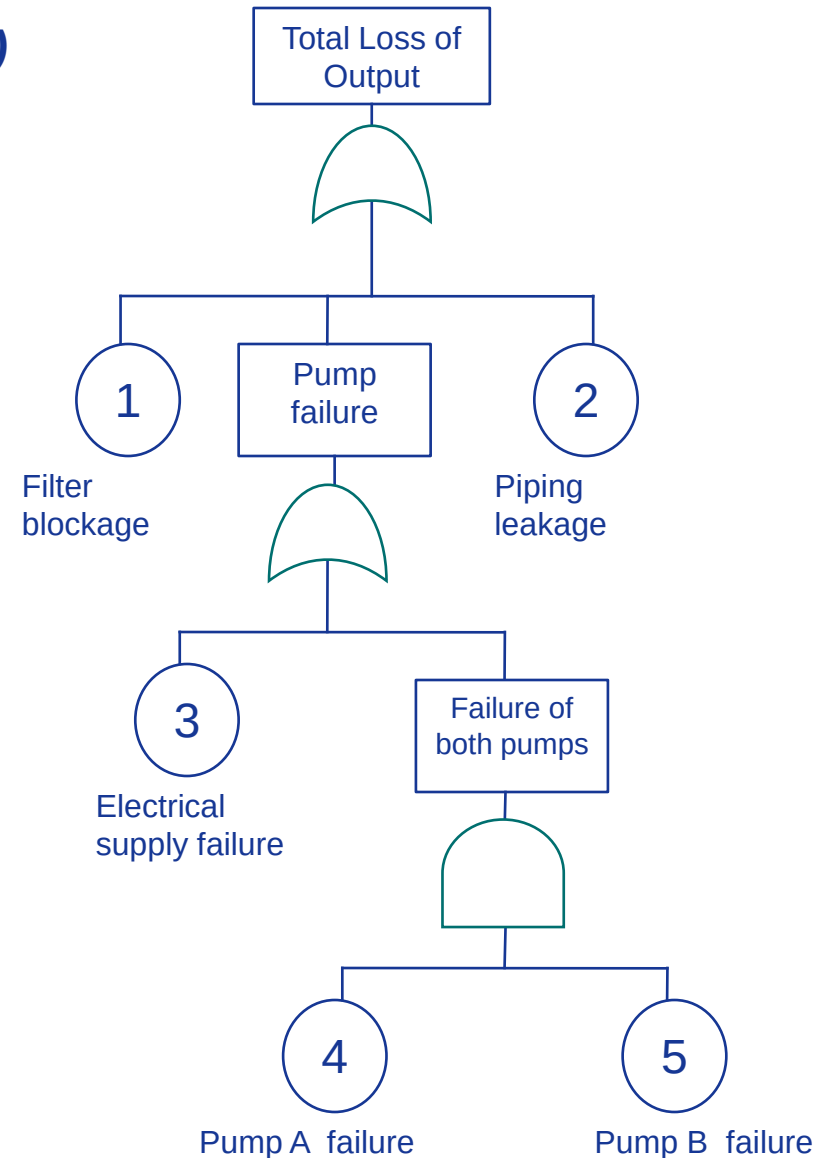
Exercise 3

Refer to Example 1 (Pumping sub-system)

Q: Assume that the probability of occurrence of basic events are as follows :

Event	1	2	3	4	5
Prob.	0.01	0.05	0.05	0.1	0.05

Estimate the probability of occurrence of total loss of output for the pumping sub-system



Quantitative Analysis - Importance measures

- Another valuable information that can be derived from a Fault tree analysis is the importance measure for each component or minimal cut set
- An important measure is a sensitivity analysis which identifies weak areas of the system and can be valuable particularly at the design stage
- The importance of each component is signified by the role it plays in either causing or contributing to the occurrence of the top event
- For analysis of importance measure, a numerical value is assigned to each of the component or minimal cut set, which allows them to be ranked according to the extent of their contribution to the top event incidence

Basic Fault Tree Importance Measures

- ❖ FV Importance = Relative contribution to the system failure probability from a component failure
- ❖ RAW = Factor increase in the system failure probability when a component is assumed to be failed
- ❖ RRW = Factor decrease in the system failure probability when a component is assumed to succeed

FV Importance = "Fussell-Vesely Importance"

RAW = "Risk Achievement Worth"

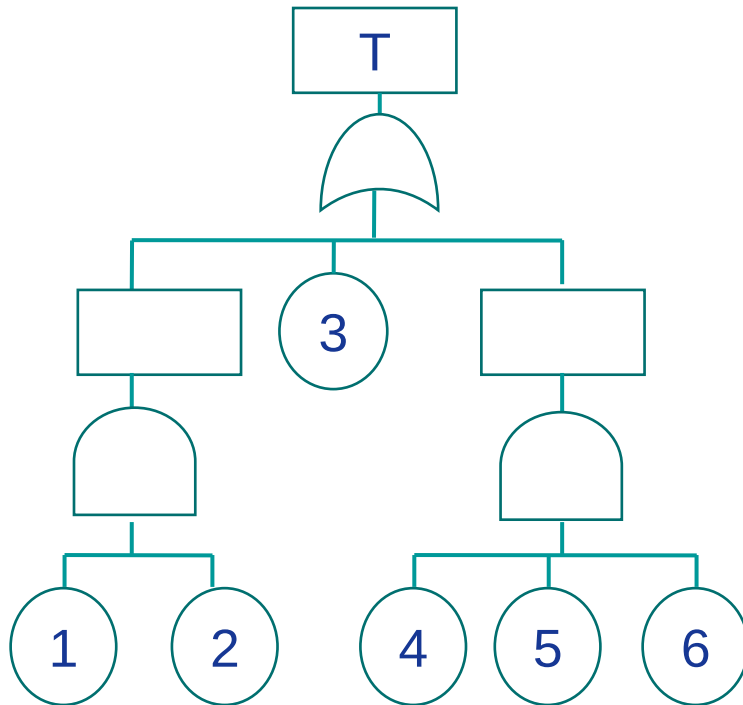
RRW = "Risk Reduction Worth"

Uses of the Importance Measures

- ❖ Focus system safety on the top contributors (FV)
- ❖ Review possible relaxations for the lowest contributors (FV, RAW)
- ❖ Focus on upgrades having the greatest improvements (RRW)
- ❖ Define contingency measures to be consistent with the failure impact (RAW)
- ❖ Establish assurance requirements to be consistent with their importance (FV, RAW)

Quantitative Analysis - Importance measures

“equivalent” fault tree



Fussell-Vesely Measure

The quantitative measure of importance of a cut set, I_k , is defined as the probability of occurrence of cut set k given that the system has failed

Minimal cut sets

1	2	
3		
4	5	6

$$I_k = \frac{P_k}{P_T}$$

where

$$P_k = \prod P_e = P_1 \times P_2 \times \dots \times P_n$$

→ $P_k = P_4 \times P_5 \times P_6$

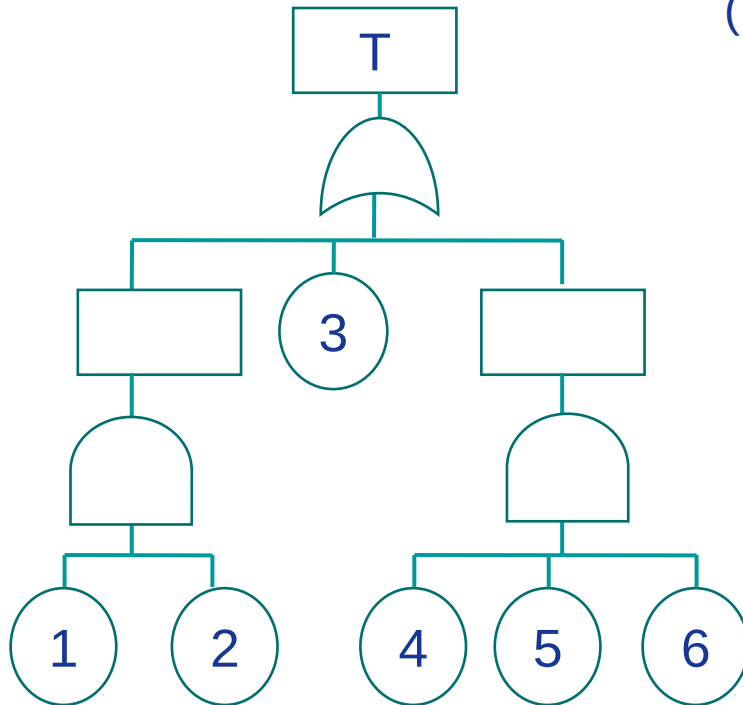
To reduce system vulnerability, focus on cut sets having greater importance

Example 6: Importance measures

“equivalent” fault tree

Minimal cut sets
(MCS)

1	2	
3		
4	5	6



Assuming the probability of the event is as follows:

$$P(1) = 0.05$$

$$P(2) = 0.05$$

$$P(3) = 0.01$$

$$P(4) = 0.1$$

$$P(5) = 0.1$$

$$P(6) = 0.1$$

Example 6: Importance measures (cont'd)

$$\begin{aligned} P_T &= P(1.2 + 3 + 4.5.6) \\ &= (0.05)(0.05) + (0.01) + (0.1)(0.1)(0.1) = 0.0135 \end{aligned}$$

$$\begin{aligned} \text{MCS (12)} \quad I_{(12)} &= \frac{P(1.2)}{P_T} = \frac{0.0025}{0.0135} = 0.185 \end{aligned}$$

$$\begin{aligned} \text{MCS (3)} \quad I_{(3)} &= \frac{P(3)}{P_T} = \frac{0.01}{0.0135} = 0.741 \end{aligned}$$

Highest
importance



$$\begin{aligned} \text{MCS (456)} \quad I_{(456)} &= \frac{P(4.5.6)}{P_T} = \frac{0.001}{0.0135} = 0.0741 \end{aligned}$$



Introduction to FTA



Steps in FTA



Minimal Cut Sets



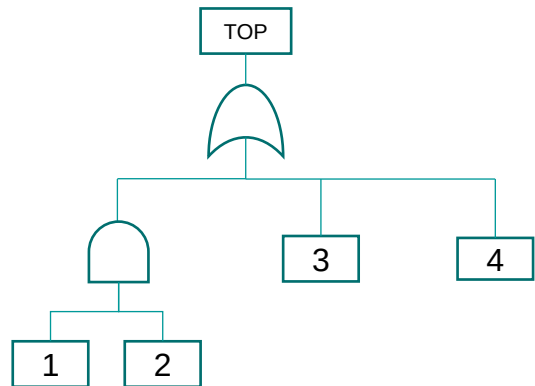
FTA analysis



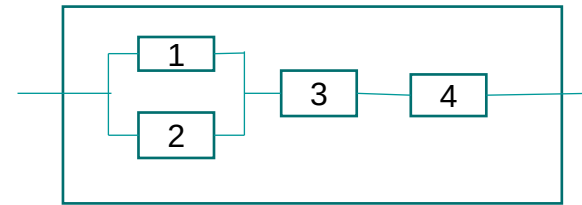
FTA and RBD

FTA and Reliability Block Diagram (RBD)

- FTA can be constructed by using RBD
- The fundamental different between FTA and RBD is that analysis of **FTA** is done in “**failure space**”, where as **RBD** is in “**success space**”



FTA : “failure space” analysis

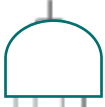
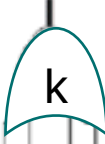


RBD : “success space” analysis

- FTA traditionally uses fixed (instantaneous) probability for calculation of probability of top event failure. RBD may include probability distribution for determining the reliability of the block.

FTA and RBD

Most of the logical constructions in a fault tree diagram can also be modeled with an RBD. In general, a fault tree can be easily converted to an RBD. It is generally more difficult to convert an RBD into a fault tree, especially if one allows for highly complex configuration

Table 1: Classic Fault Tree Gates and their Traditional RBD Equivalents			
Name of Gate	Classic FTA Symbol	Description	RBD Equivalent
AND		The output event occurs if all input events occur.	Simple Parallel Configuration
OR		The output event occurs if at least one of the input events occurs.	Series Configuration
Voting OR (k-out-of-n)		The output event occurs if k or more of the input events occur.	k-out-of-n Parallel Configuration
Inhibit		The input event occurs if all input events occur and an additional conditional event occurs.	Simple Parallel Configuration of all the events plus the condition
Priority AND		The output event occurs if all input events occur in a specific sequence.	Standby Configuration: Special case with a switch.
XOR		The output event occurs if exactly one input event occurs.	Cannot be represented. See discussion in article.
Sequence Enforcing	Various representations	The output event occurs if all input events occur in a specified sequence.	Cold Standby Configuration
Transfer		Used to indicate a transfer/continuation to another tree.	Subdiagram Block

FTA and RBD

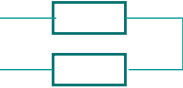
Table 2: Classic Fault Tree Event Symbols and their RBD Equivalents

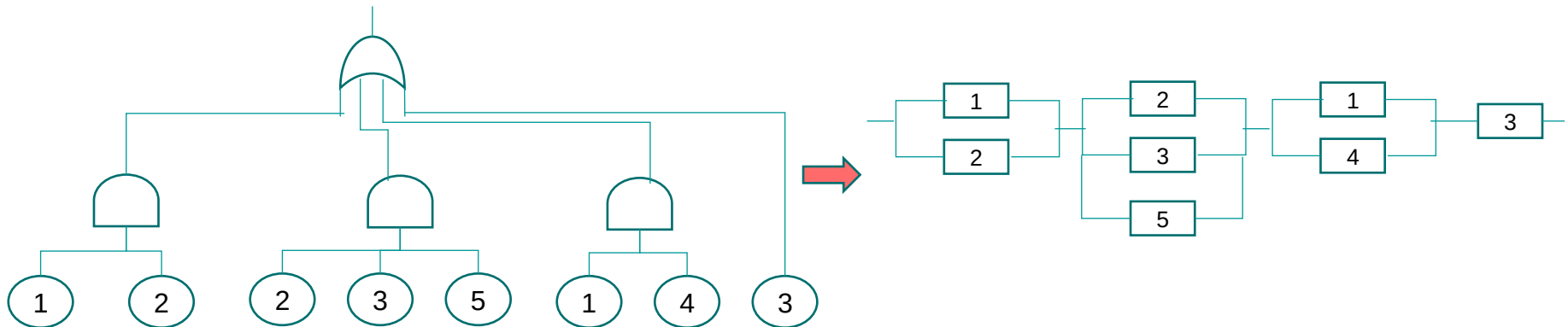
Primary Event Block	Classic FTA Symbol	Description	RBD Equivalent
Basic Event		A basic initiating fault (or failure event).	Block
External Event (House Event)		An event that is normally expected to occur. In general, these events can be set to occur or not occur, i.e. they have a fixed probability of 0 or 1.	Block that cannot fail or that is in a failed state.
Undeveloped Event		An event which is no further developed. It is a basic event that does not need further resolution.	Block
Conditioning Event		A specific condition or restriction that can apply to any gate.	Block: Placement of the block will vary depending on the gate applied.

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Conversion FTA:RBD

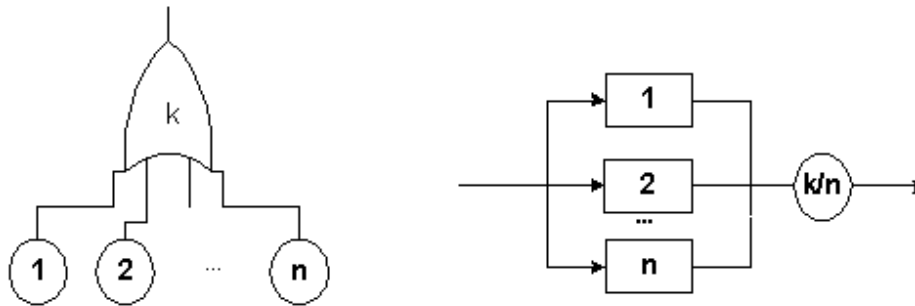
- Start from TOP event and replace gates successively

FTA	RBD
Logical AND gate 	Parallel structure 
Logical OR gate 	Serial structure 



Conversion FTA:RBD

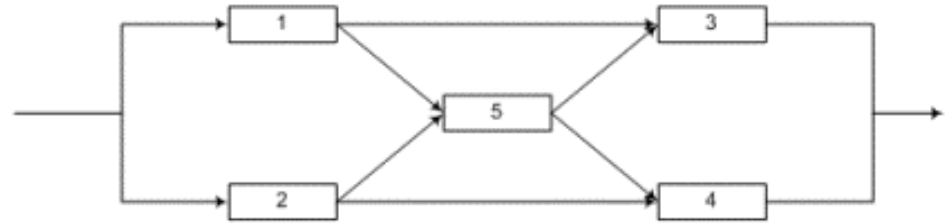
FTA diagram with a **voting gate** and the RBD equivalent.



Conversion FTA:RBD

Fault trees and complex

RBDs: Example of a complex reliability block diagram is the so-called "bridge." The following RBD represents such a bridge.



Representation of this bridge as a fault tree diagram requires the utilization of duplicate events, since gates can only represent components in series and parallel. An inspection of this system reveals that any of the following failures will cause the system to fail:

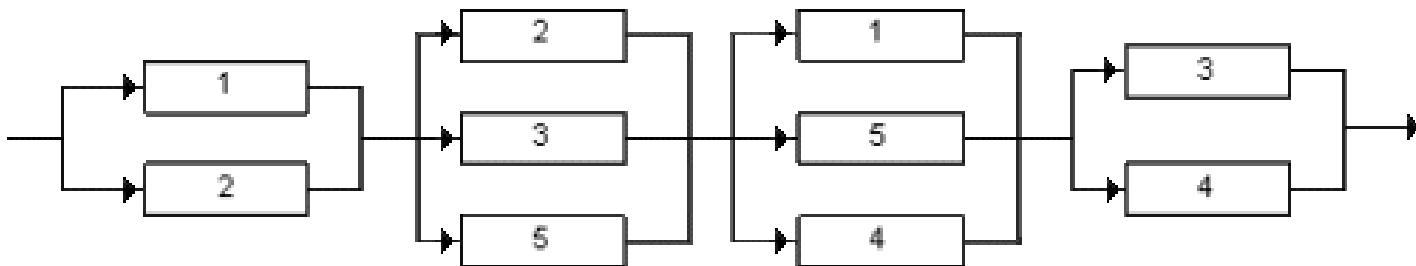
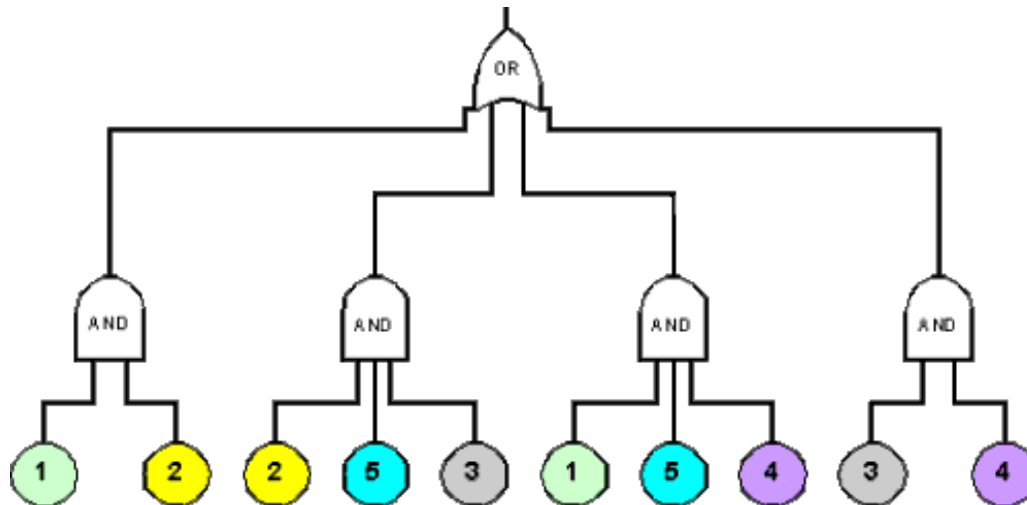
- Failure of components 1 and 2.
- Failure of components 3 and 4.
- Failure of components 1 and 5 and 4.
- Failure of components 2 and 5 and 3.

In probability terminology, we have:

- $(1 \text{ And } 2) \text{ Or } (3 \text{ And } 4) \text{ Or } (1 \text{ And } 5 \text{ And } 4) \text{ Or } (2 \text{ And } 5 \text{ And } 3).$

These sets of events are also called *minimal cut sets*. It can now be seen how the fault tree can be created by representing the above set of events in the following fault tree.

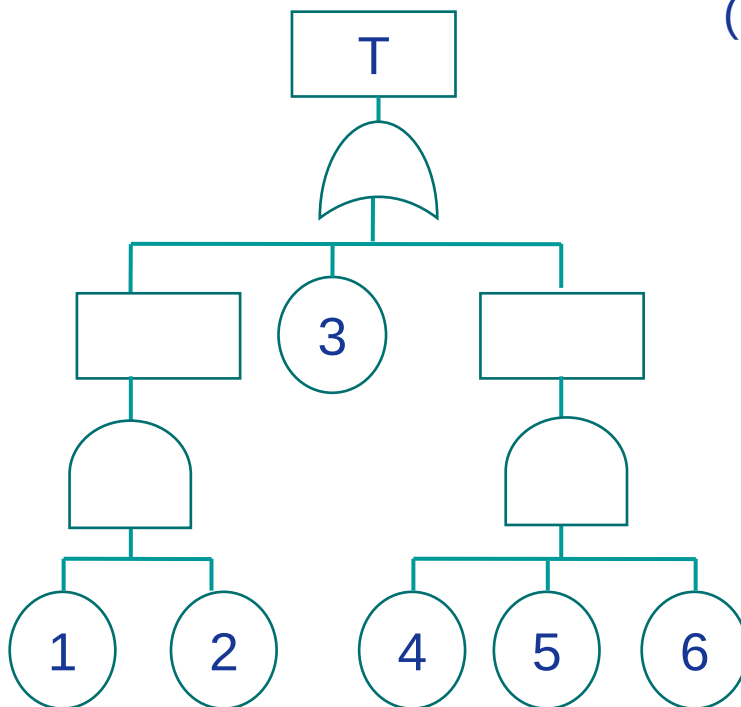
Conversion FTA:RBD



Exercise 4

From Example 6: Construct an equivalent RBD and calculate the reliability of the system with respect to this Top event (Use Blocksim)

“equivalent” fault tree



Minimal cut sets
(MCS)

1	2	
3		
4	5	6

Assuming the probability of the event is as follows:

$$P(1) = 0.05$$

$$P(2) = 0.05$$

$$P(3) = 0.01$$

$$P(4) = 0.1$$

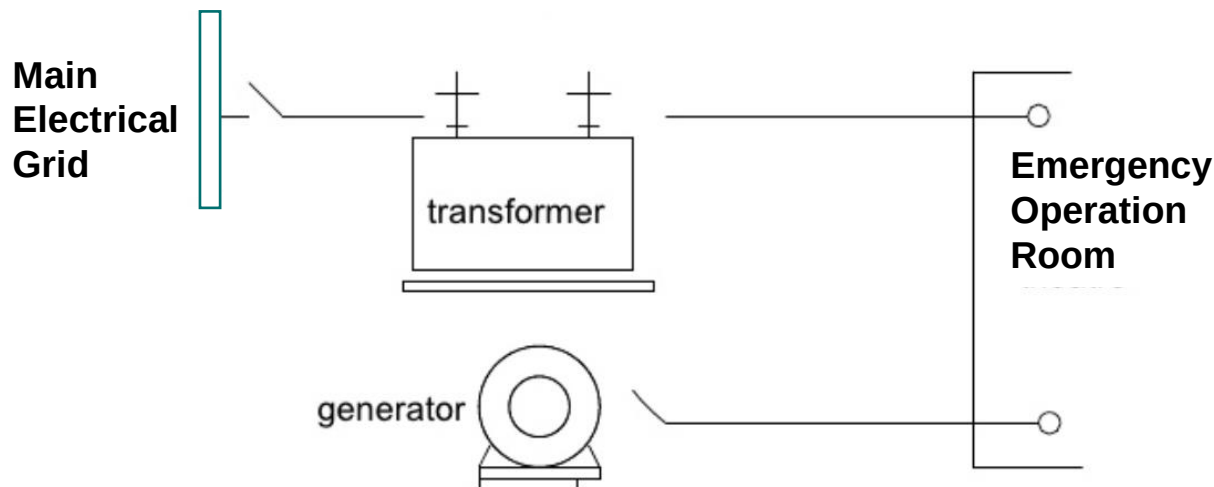
$$P(5) = 0.1$$

$$P(6) = 0.1$$

Exercise 5

An Emergency operation room obtains power from the main Electric grid through a Transformer connected in series. To ensure uninterrupted supply, an auxiliary generator is also used with a suitable switch-over. Probability of failure of the main grid is 0.01. Reliability of the Transformer and Generator are 0.996 and 0.99, respectively.

1. Draw RBD for the system
2. Calculate Reliability of the system
3. Construct FT
4. Calculate probability of failure for the system





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Thank You !

References

Books / Handbooks

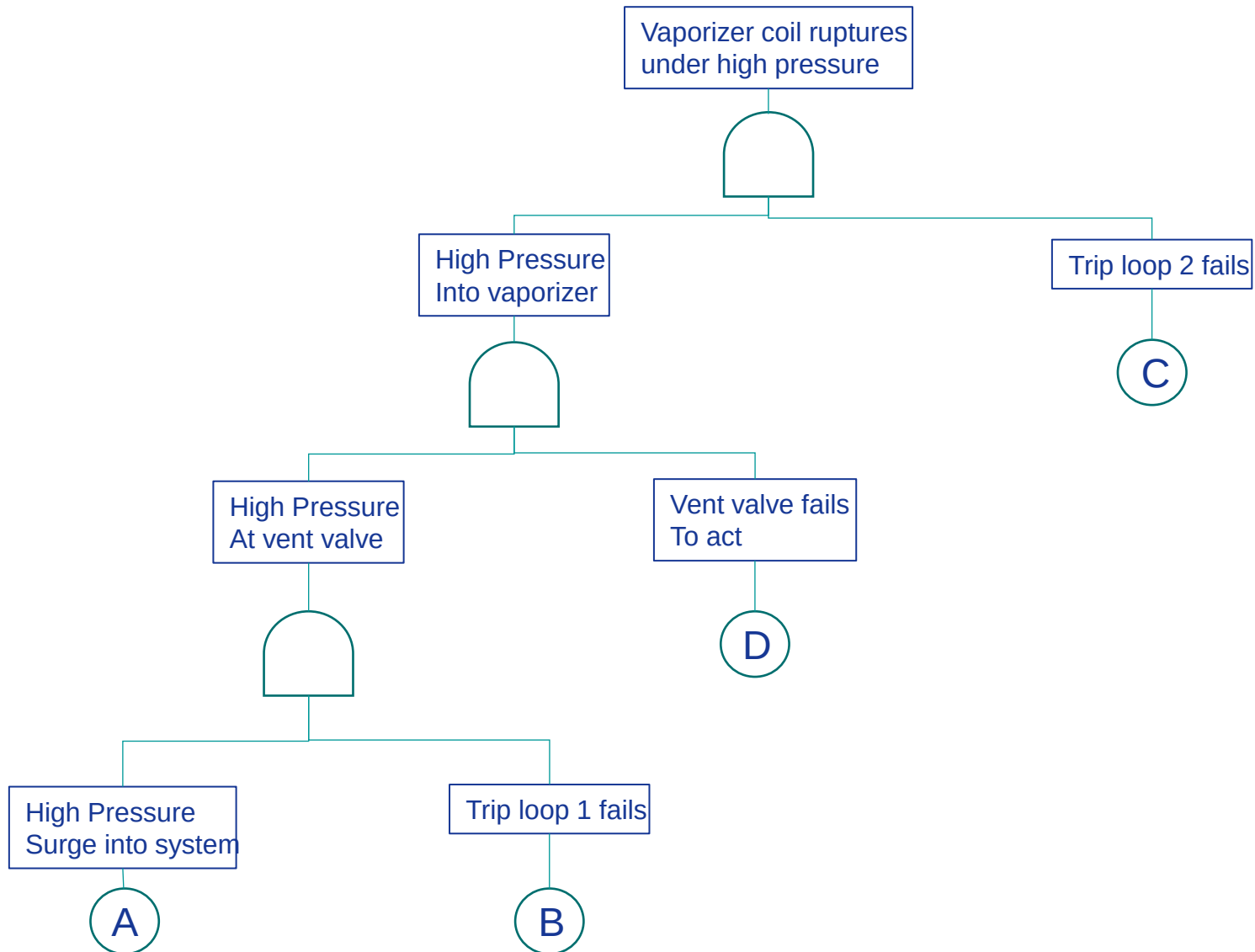
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- Andrew, J. (1998) Tutorial : Fault Tree Analysis, Proceeding of the 16th International System Safety Conference-1998 [PowerPoint slides]. Retrieved from <http://www.fault-tree.net/papers/andrews-fta-tutor.pdf>
- Ericson, C.A. (2000) Fault Tree Analysis [PowerPoint slides]. Retrieved from <http://www.fault-tree.net/papers/ericson-fta-tutorial.pdf>
- Rausand, M. (2005) Chapter 3 System Analysis Fault Tree Analysis [PowerPoint slides]. Retrieved from <http://www.ntnu.no/ross/srt/slides/fta.pdf>

Exercise 1: Solution



Exercise 2: Solution

$$T = Y.Z$$

$$T = (1 + X) (2 + 4)$$

$$T = (1 + 2.3) (2 + 4)$$

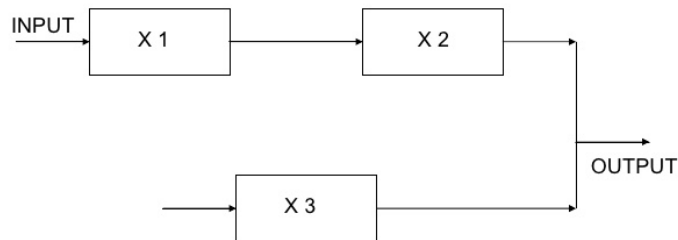
$$= 1.2 + 1.4 + 2.2.3 + 2.3.4$$

$$= 1.2 + 1.4 + 2.3 + 2.3.4$$

$$= 1.2 + 1.4 + 2.3$$

Exercise 5: Solution

BLOCK DIAGRAM



FAILURE OF THE SYSTEM

$$F(S) = (P(X1) \text{ or } P(X2)) \text{ and } P(X3)$$

$$P(X1) = 0.01$$

$$P(X2) = 1 - 0.996 = 0.004$$

$$P(X3) = 1 - 0.99 = 0.001$$

$$F(S) = (P(X1) + P(X2) - (P(X1) * P(X2))) * P(X3)$$

$$= (0.01 + 0.004 - (0.01 * 0.004)) * (0.001)$$

$$F(S) = 0.0001396$$

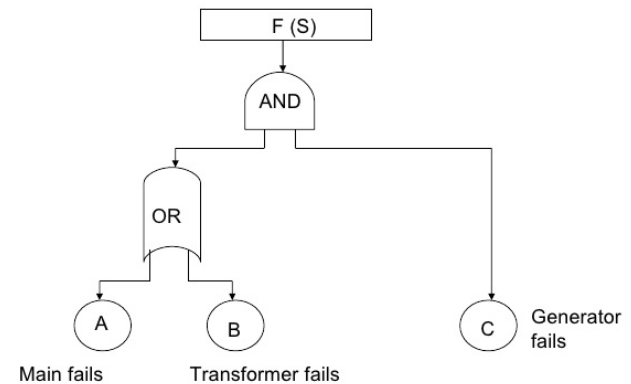
FOR RELIABILITY

$$R(S) = 1 - F(S)$$

$$= 1 - 0.0001396$$

$$R(S) = 0.99986$$

Fault tree for problem



Introduction to Markov Process in Reliability

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Session Objectives

To understand the application of Markov theory in reliability.

To be able to construct Markov-chain model.

To be able to perform a basic analysis of a Markov chain model.

Covered	Not Covered
Brief explanation on stochastic modeling Basic property of Markov Discrete-state Markov models: - Discrete-time - Continuous-time Examples of Markov chains Simulation of discrete-time Markov model using spreadsheet	Continuous -state Markov models: - Discrete-time - Markov chain Monte Carlo methodology - Continuous-time Methods to solve discrete-state, continuous -time Markov model

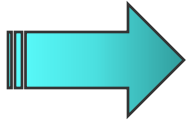
Introduction to Markov model

Basic property of Markov

Types of Markov model

International Standards

Contents



Introduction to Markov model

Basic property of Markov

Types of Markov model

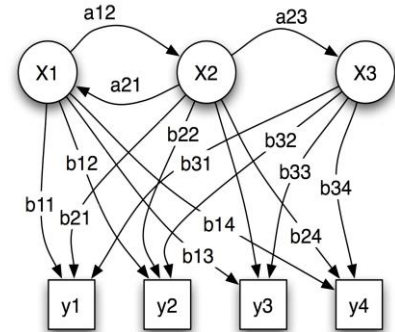
International Standards

Introduction

A Markov chain is a mathematical system that undergoes transitions from one state to another in a chainlike manner.

It is named for Andrey Markov, a Russian mathematician

- Born in June 14, 1856 and died in July 20, 1922.
- Graduated from Saint Petersburg University in 1878.
- He is best known for his work on theory of stochastic processes.
- His research later became known as [Markov chains](#).



Stochastic Process

Stochastic means random; specifically, it involves a random variable (i.e. chance or probability).

A stochastic process

- Sometimes random process.
- There is some indeterminacy in its future evolution described by probability distributions. This means that even if the initial condition is known, there are many possibilities the process might go to, but some paths may be more probable and others less so.

Example of stochastic process: stock market and exchange rate fluctuations, signals such as speech, audio and video, medical data such as a patient's blood pressure or temperature.



Stochastic Modelling

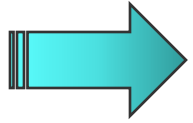
Stochastic modeling can be defined as

- a mathematical model that describes a natural phenomenon as stochastic process; the model predicts the outcomes using probabilities.

In reliability, availability and maintainability, stochastic modeling is used to describe a system's operation with respect to time.

- The component failure and repair times become the random variables.





Introduction to Markov model

Basic property of Markov

Types of Markov model

International Standards

Markov Process

A Markov process is a time-varying random phenomenon for which a Markov property holds.

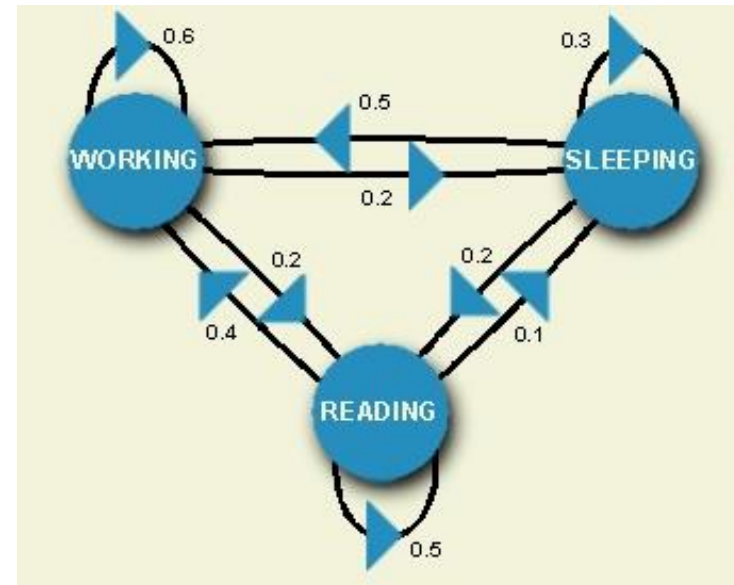
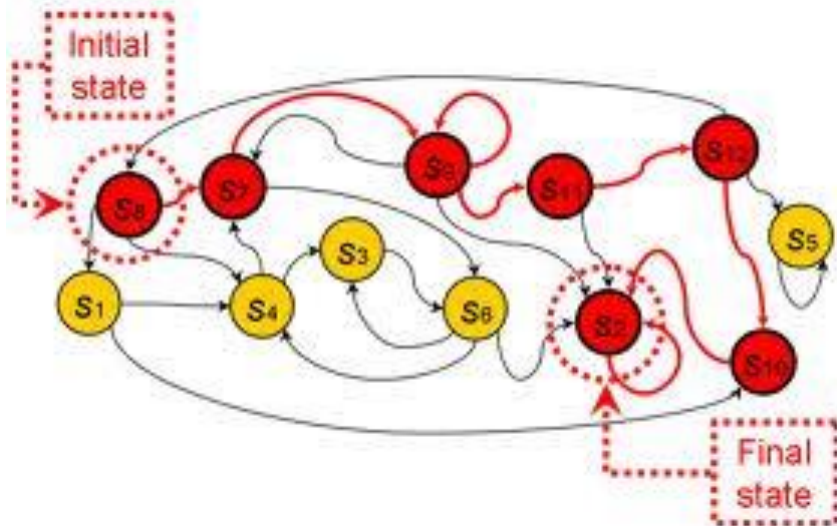
Markov property refers to the memoryless property that is the conditional probability distribution of future states of the process, given the present state and the past states, depends only upon the present state.

$$P\left(\underbrace{X_{n+1} = x}_{\text{Future state}} \mid \underbrace{X_1 = x_1, X_2 = x_2, \dots, X_n = x_n}_{\text{Past states}} \mid \underbrace{X_n = x_n}_{\text{Present state}}\right) = P\left(X_{n+1} = x \mid \underbrace{X_n = x_n}_{\text{Present state}}\right)$$

Main Assumptions

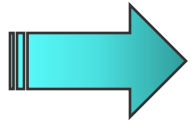
The Markov model assumes that the future is independent of the past given the present.

When using Markov, the random variable is indexed in time, which can be either discrete or continuous.



Introduction to Markov model

Basic property of Markov



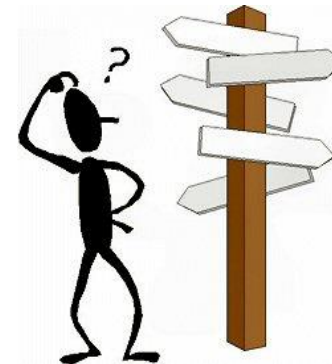
Types of Markov model

International Standards

Types of Markov Model

1. Discrete-state, discrete-time
2. Discrete-state, continuous-time
3. Continuous-state, discrete-time
 - Markov chain Monte Carlo methodology
4. Continuous-state, continuous-time

Will not
covered
in this
module



Discrete-Time Markov Model

Typically known as “Markov chain”.

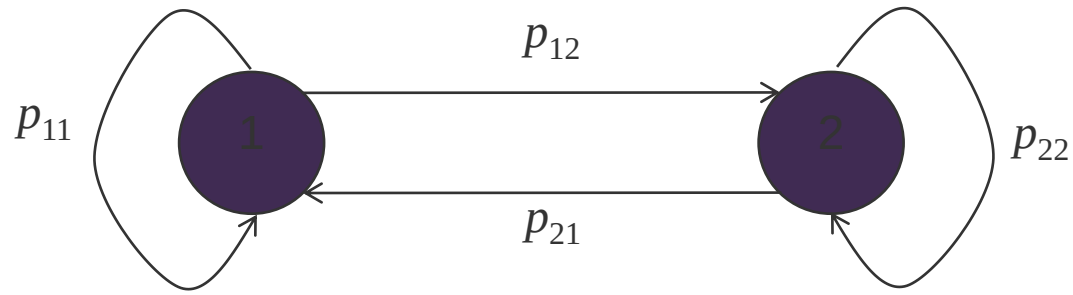
Means a system whose state changing randomly between steps.

The steps are often thought of as time, but they can equally well refer to physical distance or any other discrete measurement

The steps are just the integers.

Discrete-Time Markov Model

The probability of going from state i to state j in single step is



The n -step transition is

$$p_{ij} = \Pr \left(X_1 = j \mid X_0 = i \right)$$

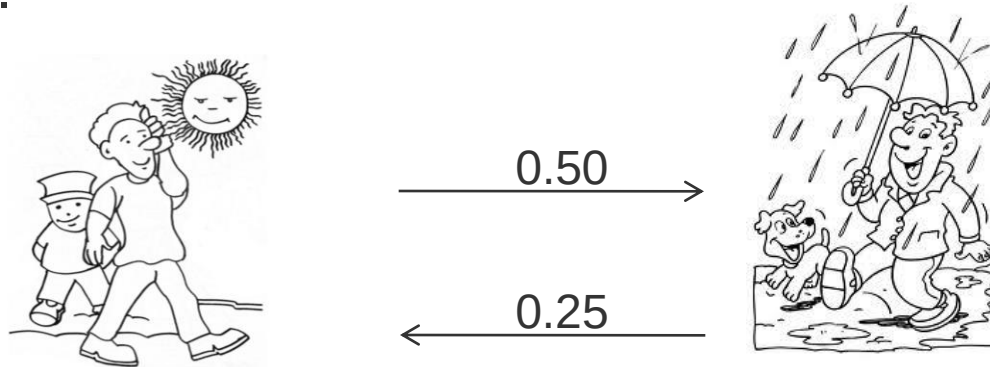
$$p_{ij}^{(n)} = \Pr \left(X_n = j \mid X_0 = i \right)$$

Example 1: Discrete Time

Today's weather does have an influence on what tomorrow's weather will be. After observing a long sequence of rainy and sunny days, a Markov model could be used to analyze the probability that one kind of weather is followed by another.

Assume that 25% of the time, a sunny day follows a rainy day and 75% of the time, rain was followed by more rain.

Assume that sunny days were followed 50% of the time by rain and 50% by sun.



Example 1: Discrete Time (cont'd)

Using this analysis, a new sequence of statistically similar weather can be generated by following these steps:

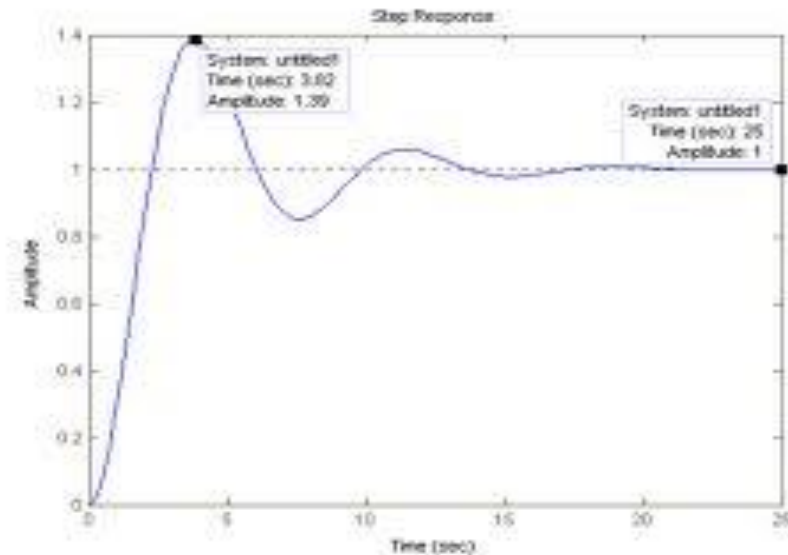
1. Start with today's weather.
2. Given today's weather, choose a random number to pick tomorrow's weather.
3. Make tomorrow's weather as "today's weather" and go back to step 2.

Results

Sunny, Sunny, Rainy, Rainy, Rainy, Rainy, Rainy, Rainy, Rainy, Sunny, Sunny...

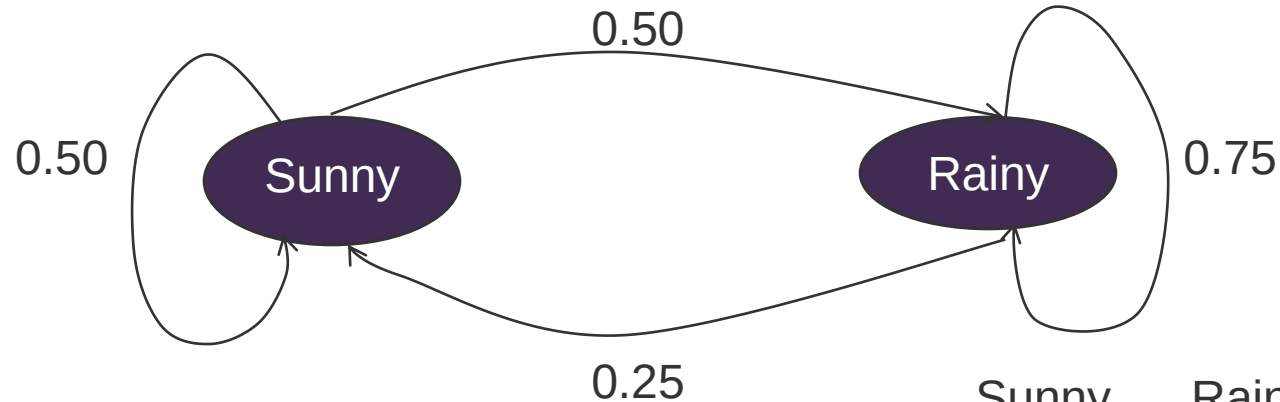
Example 1: Discrete Time (cont'd)

1. Formulate a Markov chain that describes the state of the day and draw the state-space diagram.
2. Assuming it is sunny at day 0,
 - a) What is the probability that it will be raining in day 3 and day 10?
 - b) Determine the steady-state probabilities of the Markov chain.



Example 1: Discrete Time (cont'd)

1. Formulate a Markov chain that describes the state of the day and determine the transition probability matrix.



Transition probability matrix



$$Q = \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix} \begin{matrix} \text{Sunny} \\ \text{Rainy} \end{matrix}$$

Example 1: Discrete Time (cont'd)

2. Assuming it is sunny at day 0,
a) What is the probability that it will be raining in day 3?

$$P_0 = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$P_1 = P_0 Q = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 1(0.5) + 0(0.25) & 1(0.5) + 0(0.75) \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \end{bmatrix}$$

$$P_2 = P_1 Q = \begin{bmatrix} 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.375 & 0.625 \end{bmatrix}$$

$$P_3 = P_2 Q = \begin{bmatrix} 0.375 & 0.625 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.344 & 0.656 \end{bmatrix} \longrightarrow \text{Day 3}$$

Example 1: Discrete Time (cont'd)

2. Assuming it is sunny at day 0,
 b) What is the probability that it will be raining in day 7 and day 10?

$$P_3 = P_2Q = \begin{bmatrix} 0.375 & 0.625 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.344 & 0.656 \end{bmatrix}$$

$$P_4 = P_3Q = \begin{bmatrix} 0.344 & 0.656 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.336 & 0.664 \end{bmatrix}$$

$$P_5 = P_4Q = \begin{bmatrix} 0.336 & 0.664 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.334 & 0.666 \end{bmatrix}$$

$$P_6 = P_5Q = \begin{bmatrix} 0.334 & 0.666 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.333 & 0.667 \end{bmatrix}$$

$$P_7 = P_6Q = \begin{bmatrix} 0.333 & 0.667 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.333 & 0.667 \end{bmatrix} \longrightarrow \text{Day 7}$$

⋮

$$P_{10} = P_9Q = \begin{bmatrix} 0.333 & 0.667 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.333 & 0.667 \end{bmatrix} \longrightarrow \text{Day 10}$$

Example 1: Discrete Time(cont'd)

2. Assuming it is sunny at day 0,
 - c) Determine the steady-state probabilities of the Markov chain.

$$P_5 = P_4 Q = \begin{bmatrix} 0.336 & 0.664 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.334 & 0.666 \end{bmatrix}$$

$$P_6 = P_5 Q = \begin{bmatrix} 0.334 & 0.666 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.333 & 0.667 \end{bmatrix}$$

$$P_7 = P_6 Q = \begin{bmatrix} 0.333 & 0.667 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.333 & 0.667 \end{bmatrix}$$

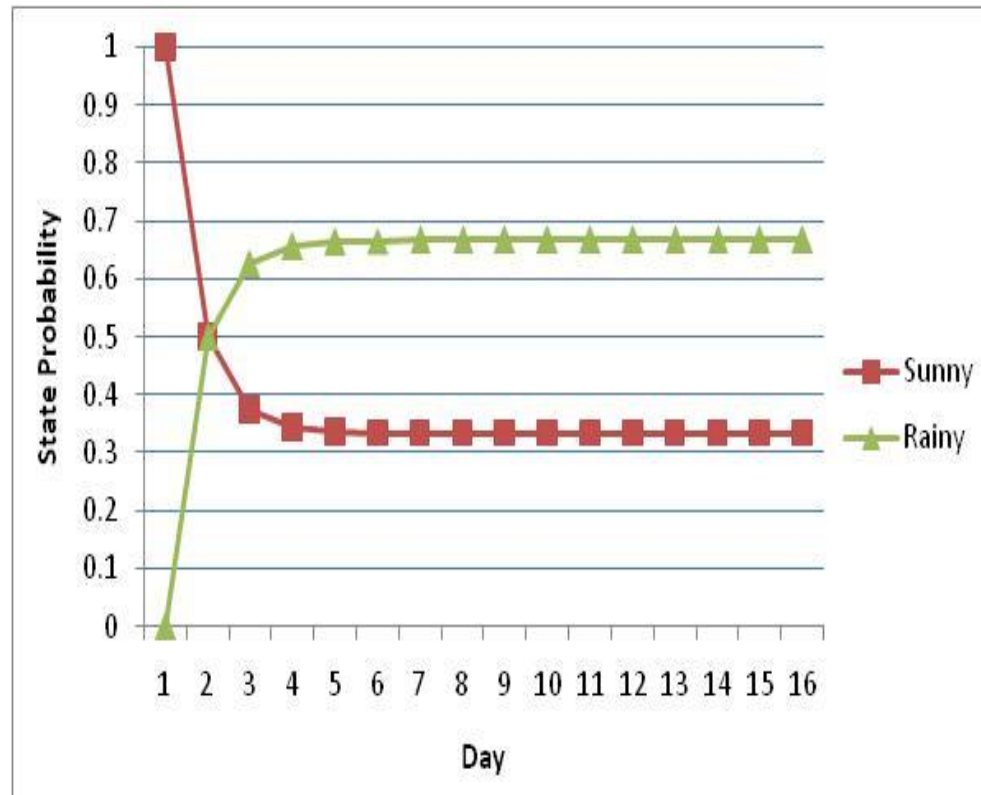
⋮

$$P_{10} = P_9 Q = \begin{bmatrix} 0.333 & 0.667 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.333 & 0.667 \end{bmatrix}$$

**Steady State
reaches at day 6**

Example 1: Discrete Time(cont'd)

2. Assuming it is sunny at day 0,
 - c) Determine the steady-state probabilities of the Markov chain.



Example 1: Discrete Time (cont'd)

Using Excel to solve the weather problem

B20		fx {=MMULT(B19:C19,\$B\$11:\$C\$12)}					
	A	B	C	D	E	F	G
1	Weather Problem						
2							
3	Input Data						
4		Sunny	Rainy				
5	Sunny	0.5	0.5				
6	Rainy	0.25	0.75				
7							
8		Sunny	Rainy				
9	Initial state	1	0				
10							
11	Transition probability	0.5	0.5				
12		0.25	0.75				
13							
14	Day	Sunny	Rainy				
15	0	1	0				
16	1	0.5	0.5				
17	2	0.375	0.625				
18	3	0.344	0.656				
19	4	0.338	0.664				
20	5	0.334	0.666				
21	6	0.333	0.667				
22	7	0.333	0.667				
23	8	0.333	0.667				
24	9	0.333	0.667				
25	10	0.333	0.667				
26	11	0.333	0.667				
27	12	0.333	0.667				
28	13	0.333	0.667				
29	14	0.333	0.667				
30	15	0.333	0.667				
31							

MMULT: Returns the matrix product of two arrays. The result is an array with the same number of rows as array1 and the same number of columns as array2.

Syntax
MMULT(array1,array2)

Input transition probability

Input the initial state

**Steady State
reaches at day 6**

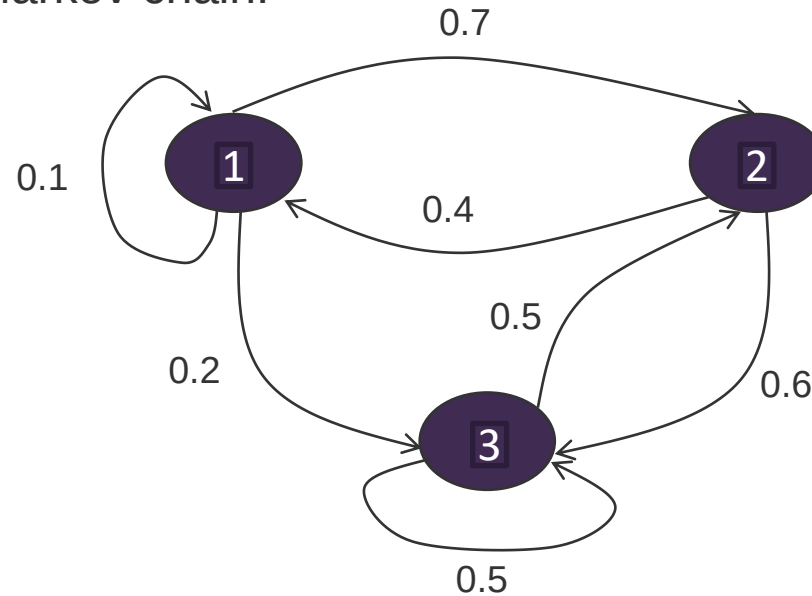
Monte Carlo for Markov

How to use Monte Carlo in Markov?

Excel file: [Weather example_2-state Markov model.xlsm](#)

In-Class Exercise

Consider the following Markov chain.



1. Determine the transition probability matrix.
2. Determine the probability at each state for $t = 3$.
3. When the steady state can be reached? What is the steady state probability?

Continuous-Time Markov Model

A Markov chain in discrete time, $\{X_n: n > 0\}$, remains in any state for exactly one unit of time before making a transition (change of state).

Now, we proceed to relax this restriction by allowing a chain to spend a continuous amount of time in any state, but in such a way as to retain the Markov property.

The time spent in any state i is called the holding time in state i .

When the holding time ends, the process then makes a transition into state j according to transition probability P_{ij} , independent of the past, and so on.

The holding times will have to be exponentially distributed.

Continuous-Time Markov Model

A continuous-time Markov chain (CTMC) makes transitions from state to state, independent of the past, according to a discrete-time Markov chain, but once entering a state remains in that state, independent of the past, for an exponentially distributed amount of time before changing state again.

Example 3: CTMC

Consider a two-unit parallel system with a standby redundancy, each of having failure rate of λ . Let us define some possible states.

State 1: Unit 1 in operation, Unit 2 fails

State 2: Unit 2 in operation, Unit 1 fails

State 3: Both Unit 1 and 2 fail

The probability of passing from State 1 to State 2 in any given time interval $[t, t + \delta t]$, given that the current state is State 1 is $\lambda \delta t$.

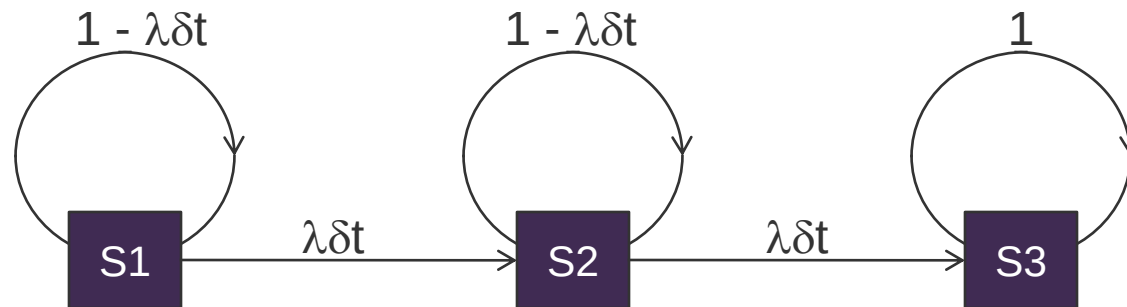
Similarly, conditional on being in state 2 at time t , the probability of passing to State 3 in the next δt is $\lambda \delta t$.

The probabilities of not moving from State 1 and State 2 follow and since there is no repair in this situation, once arrived at State 3, the system remains in State 3.

Example 3: CTMC (cont'd)

Markov analysis requires:

1. A matrix of transition probabilities describing the probability of moving from state i to state j , given that a move is being made.
2. The 'waiting time' distribution for each state, until the move is made.



Markov diagram for a two-unit parallel system with standby redundancy, each unit with exponentially distributed lifetime.

Example 3: CTMC (cont'd)

The transition matrix shows the probabilities of passing from one state to another during the time interval $[t, t + \delta t]$

		Current state		
		1	2	3
Next state	1	$1 - \lambda \delta t$	0	0
	2	$\lambda \delta t$	$1 - \lambda \delta t$	0
	3	0	$\lambda \delta t$	1

The probability of being in state i at time $[t, t + \delta t]$ is $P_i[t, t + \delta t]$

$$\begin{bmatrix} 1 - \lambda \delta t & 0 & 0 \\ \lambda \delta t & 1 - \lambda \delta t & 0 \\ 0 & \lambda \delta t & 1 \end{bmatrix} \begin{bmatrix} P_1(t) \\ P_2(t) \\ P_3(t) \end{bmatrix} = \begin{bmatrix} P_1(t + \delta t) \\ P_2(t + \delta t) \\ P_3(t + \delta t) \end{bmatrix}$$

Example 3: CTMC (cont'd)

Solving the differential equations, yield the probability of each state

$$P_1(t) = e^{-\lambda t}$$

$$P_2(t) = \lambda t e^{-\lambda t}$$

$$P_3(t) = 1 - P_1(t) - P_2(t)$$

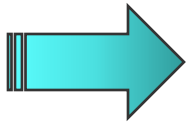
So the reliability of the system is

$$P_1(t) + P_2(t) = e^{-\lambda t} (1 + \lambda t)$$

Introduction to Markov model

Basic property of Markov

Different types of Markov model



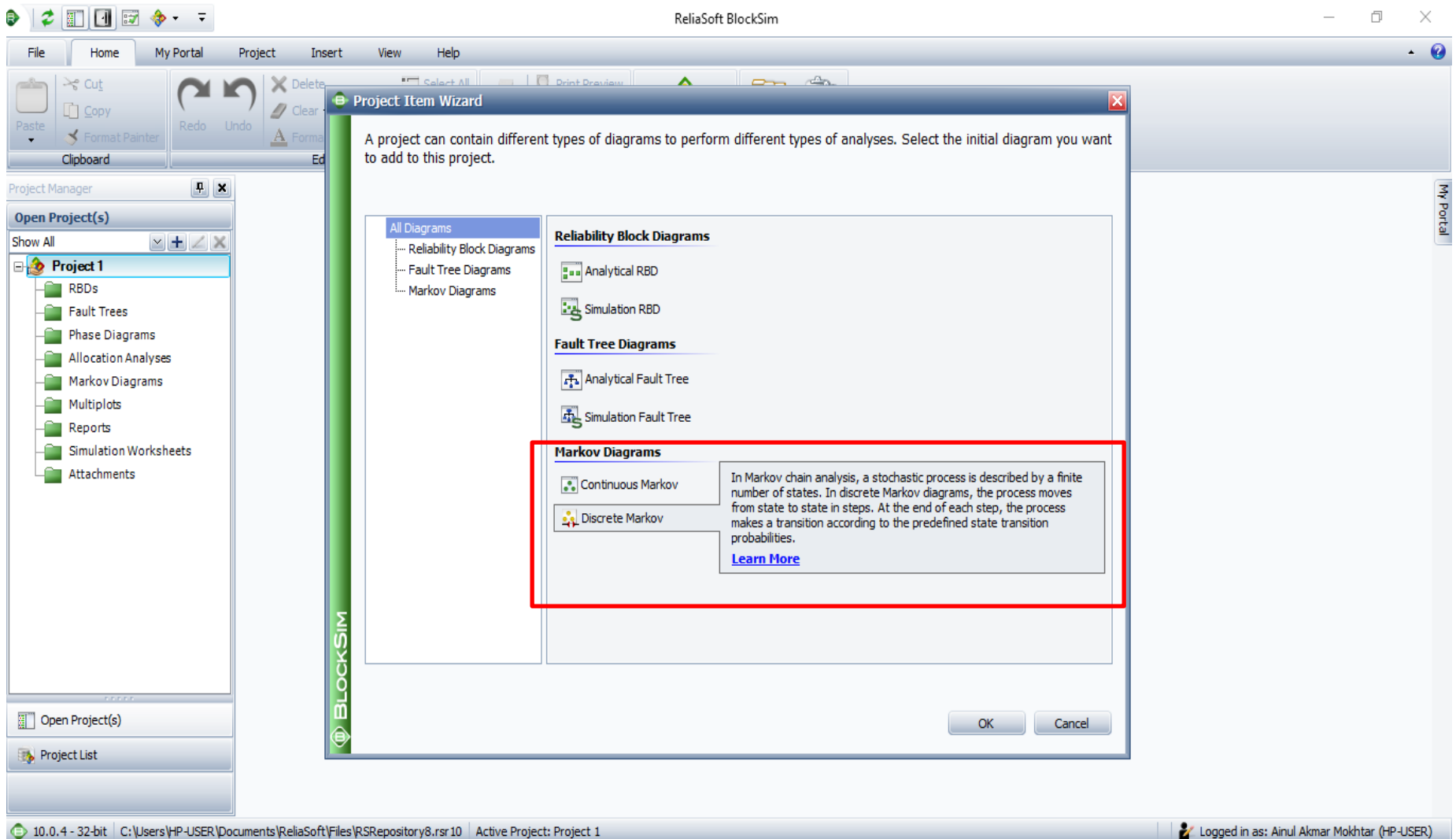
International Standards

International Standards

1. IEC 61165: Application of Markov techniques
 - Provides guidance on the application of Markov techniques to dependability analysis.

2. IEC 61508: Functional safety of electrical/electronic/programmable electronic safety-related systems
 - Has revitalized the need for Markov analysis due to the standard's requirement to analyze disparate failure modes.
 - Has 7 parts.
 - Main objective is to estimate the probability of failure on demand for critical system being examined.

Discrete Markov Model in BlockSim



ReliaSoft BlockSim

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Clipboard Redo Undo Delete Clear Format

Project Manager

Open Project(s)

Show All

Project 1

- RBDs
- Fault Trees
- Phase Diagrams
- Allocation Analyses
- Markov Diagrams
- Multiplots
- Reports
- Simulation Worksheets
- Attachments

Open Project(s)

Project List

Project Item Wizard

A project can contain different types of diagrams to perform different types of analyses. Select the initial diagram you want to add to this project.

All Diagrams

- Reliability Block Diagrams
- Fault Tree Diagrams
- Markov Diagrams

Reliability Block Diagrams

- Analytical RBD
- Simulation RBD

Fault Tree Diagrams

- Analytical Fault Tree
- Simulation Fault Tree

Markov Diagrams

- Continuous Markov
- Discrete Markov

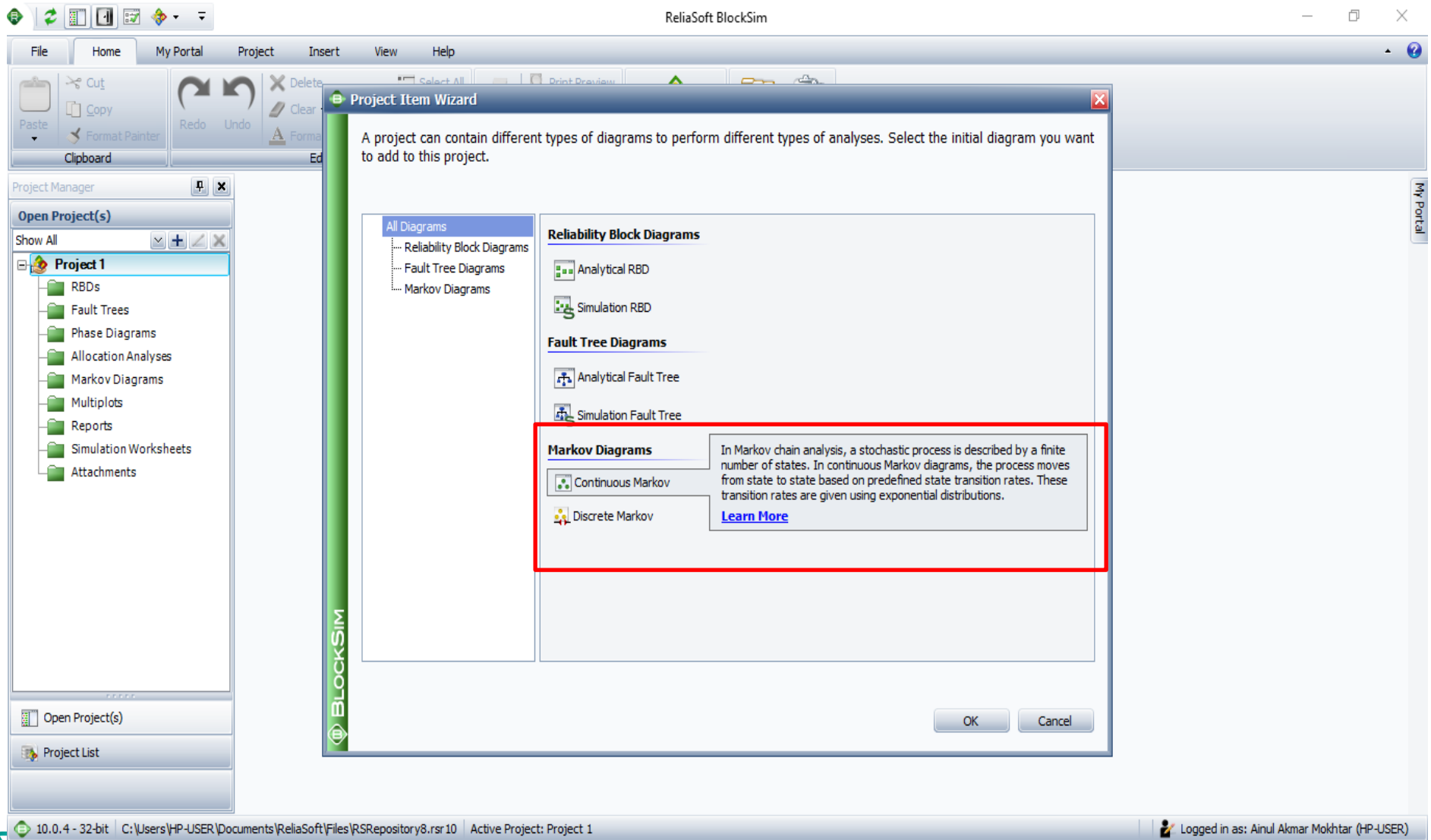
In Markov chain analysis, a stochastic process is described by a finite number of states. In discrete Markov diagrams, the process moves from state to state in steps. At the end of each step, the process makes a transition according to the predefined state transition probabilities.

[Learn More](#)

OK Cancel

10.0.4 - 32-bit | C:\Users\HP-USER\Documents\ReliaSoft\Files\RSRepository8.rsr10 | Active Project: Project 1 | Logged in as: Ainul Akmar Mokhtar (HP-USER)

Continuous Markov Model in BlockSim



ReliaSoft BlockSim

File Home My Portal Project Insert View Help

Clipboard: Cut, Copy, Paste, Format Painter, Redo, Undo, Delete, Clear, Format

Project Manager: Open Project(s), Show All, Project 1, RBDs, Fault Trees, Phase Diagrams, Allocation Analyses, Markov Diagrams, Multiplots, Reports, Simulation Worksheets, Attachments

Project Item Wizard

A project can contain different types of diagrams to perform different types of analyses. Select the initial diagram you want to add to this project.

All Diagrams

- Reliability Block Diagrams
- Fault Tree Diagrams
- Markov Diagrams

Reliability Block Diagrams

- Analytical RBD
- Simulation RBD

Fault Tree Diagrams

- Analytical Fault Tree
- Simulation Fault Tree

Markov Diagrams

- Continuous Markov
- Discrete Markov

In Markov chain analysis, a stochastic process is described by a finite number of states. In continuous Markov diagrams, the process moves from state to state based on predefined state transition rates. These transition rates are given using exponential distributions.

[Learn More](#)

OK Cancel

10.0.4 - 32-bit C:\Users\HP-USER\Documents\ReliaSoft\Files\RSRepository8.rsr10 Active Project: Project 1 Logged in as: Ainul Akmar Mokhtar (HP-USER)

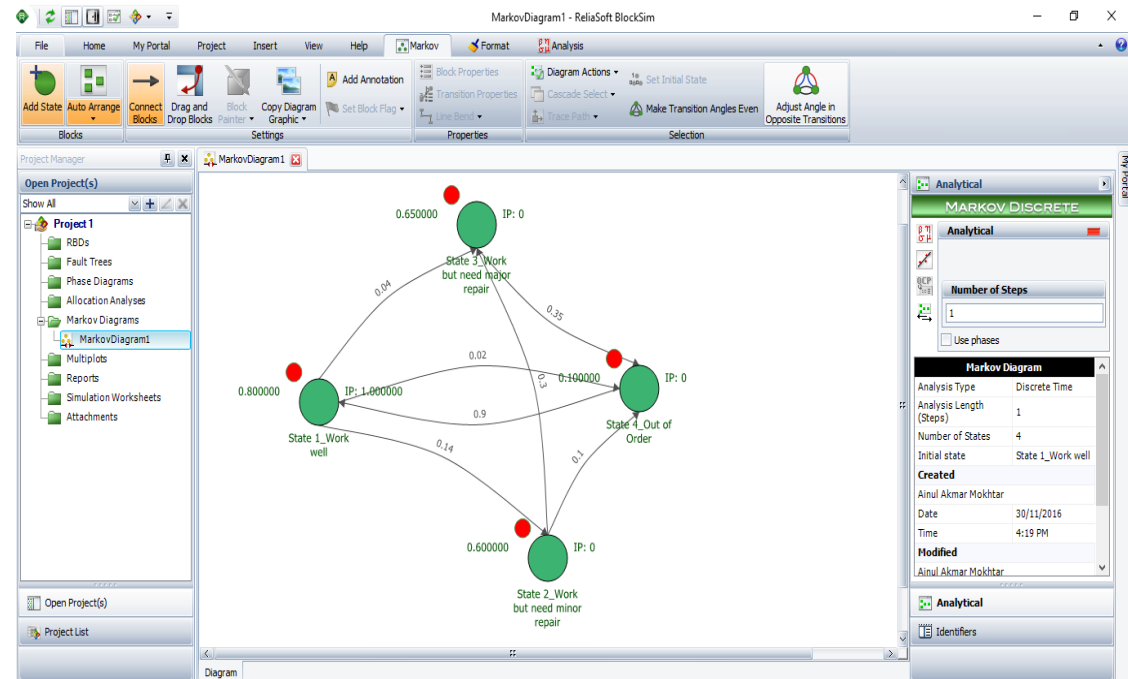
Example: Discrete Markov Model in BlockSim

A centrifugal pump is in one of four possible states. The corresponding transition probability matrix is

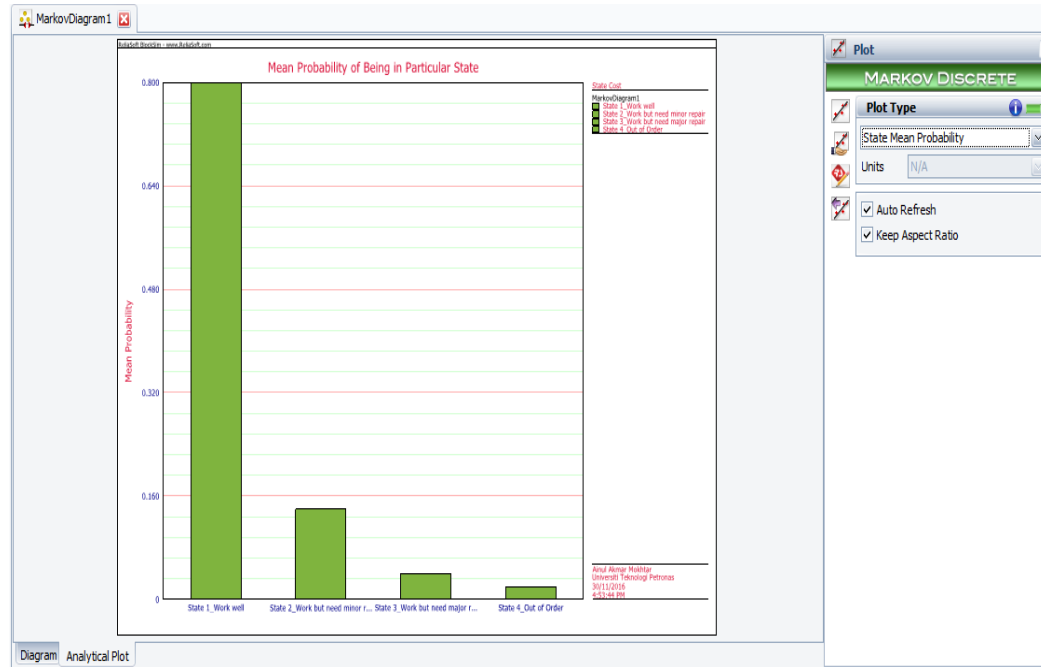
$$\begin{bmatrix} 0.80 & 0.14 & 0.04 & 0.02 \\ 0 & 0.6 & 0.3 & 0.10 \\ 0 & 0 & 0.65 & 0.35 \\ 0.90 & 0 & 0 & 0.10 \end{bmatrix}$$

Find the steady-state distribution.

State	Description
1	working without a problem
2	working but in need of minor repair
3	working but in need of major repair
4	out-of-order



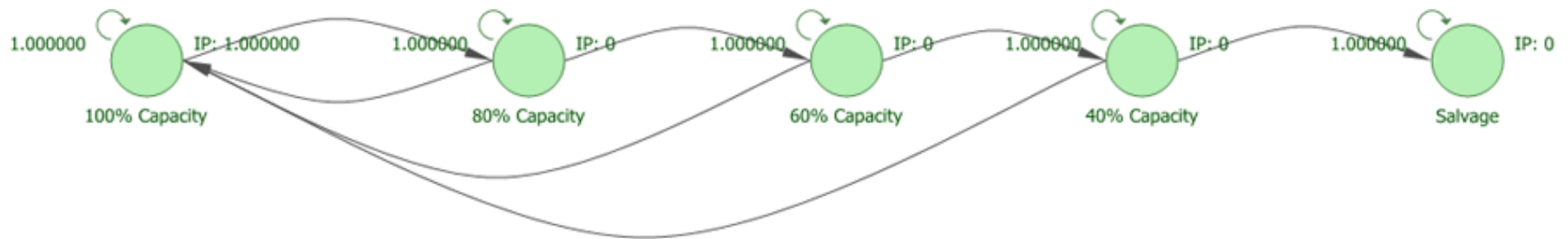
Example: Discrete Markov Model in BlockSim



Home				
Paste	Cut	Quick Print	Print	Print Preview
Paste Special	Copy	Set Colors	Transfer Report Sheet	Full Report
Analysis				
Analysis Summary				
Result Summary				
Transition Probability Matrix				
State Point Results				
Results After 1 Steps				
State Name	Initial Probability	Mean Probability	Point Probability	Steps Spent in State
State 1_ Work well	1	0.8	0.8	0.8
State 2_ Work but need minor repair	0	0.14	0.14	0.14
State 3_ Work but need major repair	0	0.04	0.04	0.04
State 4_ Out of Order	0	0.02	0.02	0.02

In-Class Exercise

An initial estimation analysis on the life cycle of a complex drilling system that starts off as brand new (100% initial probability in the full capacity state) has to be conducted. The system has a probability to degrade into various states of capacity with time and can eventually enter a salvage state. There is also a probability of being returned to the as-good-as-new condition from each degraded state, except from the salvage state. The salvage state is considered to be a "sink," a state from which there are no transitions to any other state and therefore we have zero probability of leaving. The objective is to determine, on average, what percent of the time will be spent in each state over a 10-year period. To perform the analysis, a discrete Markov chain diagram is developed. The initial setup looks like this:



In-Class Exercise

It is estimated that the probabilities per month to move between states are as follows:

1% chance to degrade from 100% to 80% capacity.

10% chance to be restored from 80% to 100% capacity.

3% chance to degrade from 80% to 60% capacity.

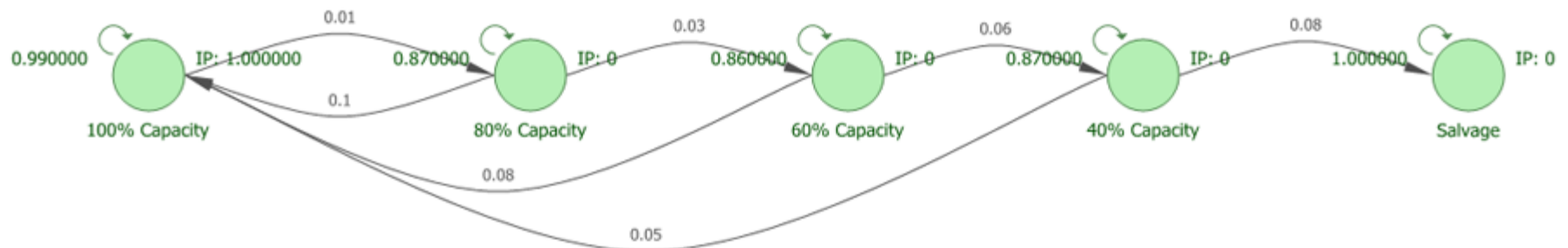
8% chance to be restored from 60% to 100% capacity.

6% chance to degrade from 60% to 40% capacity.

8% chance to be restored from 40% to 100% capacity.

8% chance to degrade from 40% capacity to salvage.

Based on these percentages, the final diagram that is ready for analysis looks like this:



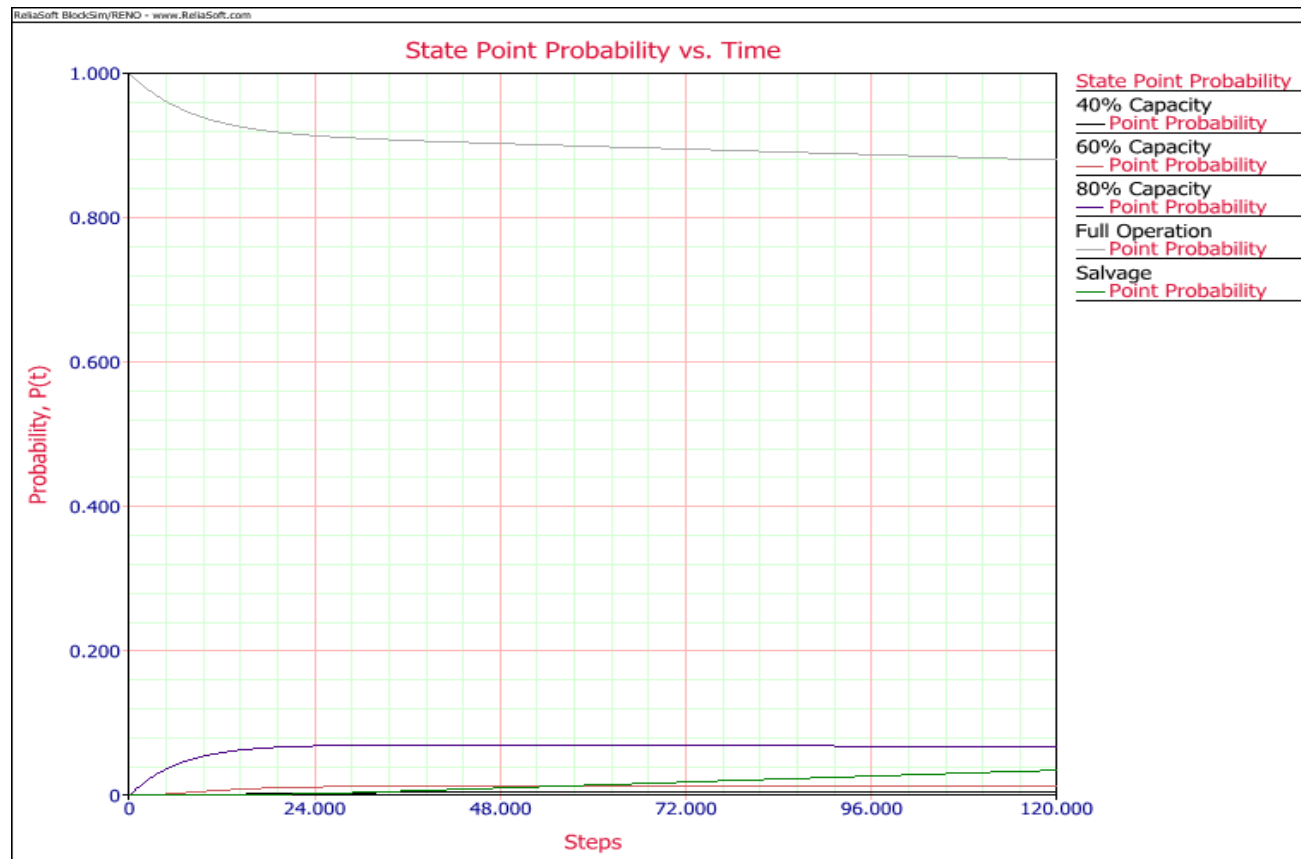
In-Class Exercise

Since the estimated probabilities are on a month scale, each step of the analysis will be taken to be the equivalent of one month (i.e. run the calculation for 120 steps). Below is the transition probability matrix between the states:

Full Diagram					
FROM - >TO	100% Capacity	80% Capacity	60% Capacity	40% Capacity	Salvage
100% Capacity	0.99	0.01	0	0	0
80% Capacity	0.1	0.87	0.03	0	0
60% Capacity	0.08	0	0.86	0.06	0
40% Capacity	0.05	0	0	0.87	0.08
Salvage	0	0	0	0	1

In-Class Exercise

Figure below shows the state point probability plot to see if the system has reached steady state within the time frame.



In-Class Exercise

Conclusions

From the results, we can conclude that the majority of the time (89.4%) the system should be running at 100% capacity and that after the 10-year period there is about a 5.3% chance that the system will degrade to a point from which it cannot be restored (the salvage state).

Results After 120 Steps			
State Name	Initial Probability	Mean Probability	Point Probability
100% Capacity	1	0.89354	0.859252
80% Capacity	0	0.065122	0.066382
60% Capacity	0	0.013105	0.014282
40% Capacity	0	0.005624	0.00662
Salvage	0	0.022608	0.053464

Case Study 2: Continuous Markov

A system with two pumps, A and B.

Define the state:

- 1: B is active, A is in standby. Both are in working condition. This is the state in which you want analysis to start.
- 2: A is active, B is in standby. Both are in working condition.
- 3: B fails. A is active and B is undergoing repair.
- 4: A fails. B is active while A is undergoing repair.
- 5: Both pumps are undergoing repair.

Exercise 2: Continuous Markov

Transition rates

- A running pump will fail according to 1-P Exponential distribution with mean of 100 Hrs
- A standby pump will fail according to 1-P Exponential distribution with mean of 200 Hrs
- Repair times follow an exponential distribution with MTTR of 20 hours

Calculate system availability after running for 1000 hrs

Use the data in RBD, Any difference in the results?

Application of Markov in IPF

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PFD calculation for 1oo2 configuration

➤ Since the transitions are given as rates (not probabilities), the probabilities at each states should be solved using Continuous time Markov chain.

➤ The 1oo2 diagram as shown in Figure 1, can be described by the following Kolmogorov equations

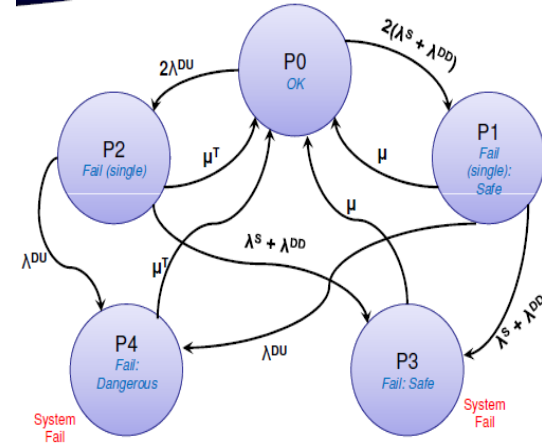
$$\frac{dP_0(t)}{dt} = -2(\lambda^{du} + \lambda^s + \lambda^{dd})P_0(t) + \mu P_1(t) + \mu^T P_2(t) + \mu P_3(t) + \mu^T P_4(t)$$

$$\frac{dP_1(t)}{dt} = -(\lambda^s + \lambda^{dd} + \lambda^{du} + \mu)P_1(t) + 2(\lambda^s + \lambda^{dd})P_0(t)$$

$$\frac{dP_2(t)}{dt} = -(\lambda^s + \lambda^{dd} + \lambda^{du} + \mu^T)P_2(t) + 2\lambda^{du}P_0(t)$$

$$\frac{dP_3(t)}{dt} = -\mu P_3(t) + (\lambda^s + \lambda^{dd})P_1(t) + (\lambda^s + \lambda^{dd})P_2(t)$$

$$\frac{dP_4(t)}{dt} = -\mu^T P_4(t) + \lambda^{du}P_1(t) + \lambda^{du}P_2(t)$$



$$\text{Safe ratio (Sr)} = \frac{\sum \lambda^s \text{ (for all safe failure modes)}}{\sum \lambda \text{ (for all failure modes)}}$$

Diagnostic Coverage Factor for Safe Failures (C^{SD}) :

$$C^{SD} = \frac{\sum \lambda^{SD} \text{ (for all safe detected failure modes)}}{\sum \lambda^s \text{ (for all safe failure modes)}}$$

Diagnostic Coverage Factor for Dangerous Failures (C^{DD}) :

$$C^{DD} = \frac{\sum \lambda^{DD} \text{ (for all dang. detected failure modes)}}{\sum \lambda^D \text{ (for all dang. failure modes)}}$$

Therefore :

$$\lambda^{SD} = \text{Sr} C^{SD} \lambda$$

$$\lambda^{SU} = \text{Sr} (1 - C^{SD}) \lambda$$

$$\lambda^{DD} = (1 - \text{Sr}) C^{DD} \lambda$$

$$\lambda^{DU} = (1 - \text{Sr}) (1 - C^{DD}) \lambda$$

Reference:
 Document No.: IC-EWB-IPF-003
 Revision: 1
 February 2011

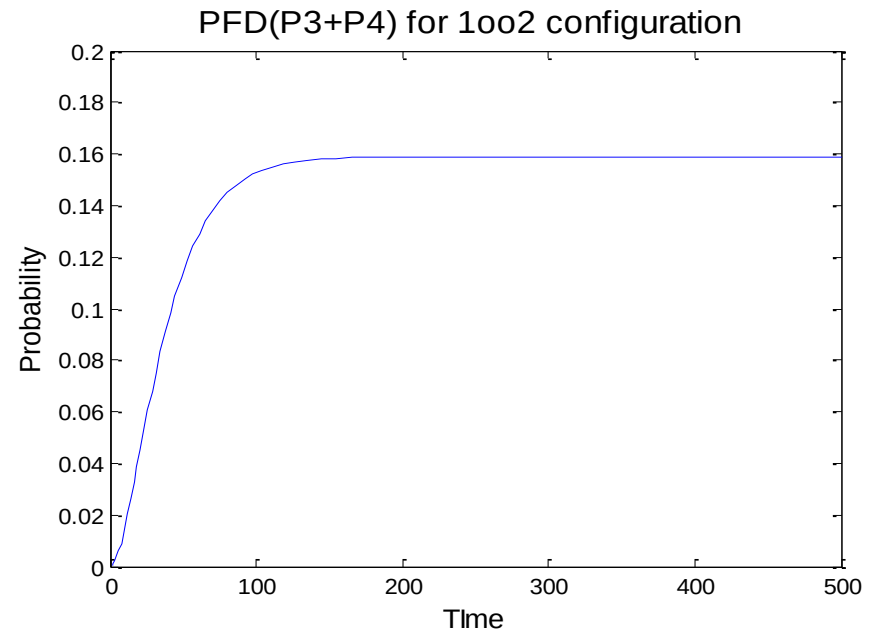
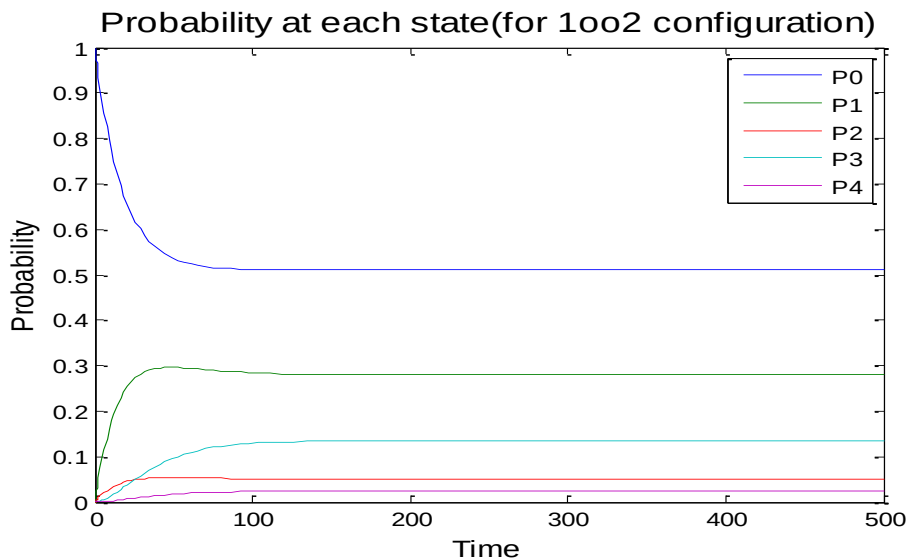


Solution based on solving the differential equations

The transition rates

λ_S	0.01
λ_{DD} and λ_{DU}	0.0022
μ and μ^T	0.03

μ : Repair rate for detected failure
 μ^T : Repair rate for undetected failure



$$\text{PFD} = P3 + P4 = 0.1589$$

Solution based on Matrix

Assuming steady state, thus the differential equations are set to equal zero (no change in probability with respect to time)

In matrix form

$$\begin{bmatrix} 1-2(\lambda^S + \lambda^{DD} + \lambda^{DU}) & 1+\mu & 1+\mu^T & 1+\mu & 1+\mu^T \\ 2(\lambda^S + \lambda^{DD}) & -(\mu + \lambda^S + \lambda^{DD} + \lambda^{DU}) & 0 & 0 & 0 \\ 2\lambda^{DU} & 0 & -(\mu + \lambda^S + \lambda^{DD} + \lambda^{DU}) & 0 & 0 \\ 0 & \lambda^S + \lambda^{DD} & \lambda^S + \lambda^{DD} & -\mu & 0 \\ 0 & \lambda^{DU} & \lambda^{DU} & 0 & -\mu^T \end{bmatrix} \times \begin{bmatrix} P0 \\ P1 \\ P2 \\ P3 \\ P4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Inversing the matrix and multiply by the initial conditions, the probability of each state is given as

P0	0.510204
P1	0.280382
P2	0.050561
P3	0.134584
P4	0.024269



PFD=P3+P4=0.1587

The solution is the same as solving the differential equation

Solution based on Matrix

Represent in matrix form, and solve:

$$\begin{bmatrix}
 1 - 2(\lambda^S + \lambda^{DD} + \lambda^{DU}) & 1 + \mu & 1 + \mu^T & 1 + \mu & 1 + \mu^T \\
 2(\lambda^S + \lambda^{DD}) & -(\mu + \lambda^S + \lambda^{DD} + \lambda^{DU}) & 0 & 0 & 0 \\
 2\lambda^{DU} & 0 & -(\mu + \lambda^S + \lambda^{DD} + \lambda^{DU}) & 0 & 0 \\
 0 & \lambda^S + \lambda^{DD} & \lambda^S + \lambda^{DD} & -\mu & 0 \\
 0 & \lambda^{DU} & \lambda^{DU} & 0 & -\mu^T
 \end{bmatrix}
 \times
 \begin{bmatrix}
 P0 \\
 P1 \\
 P2 \\
 P3 \\
 P4
 \end{bmatrix}
 =
 \begin{bmatrix}
 1 \\
 0 \\
 0 \\
 0 \\
 0
 \end{bmatrix}$$

Assumptions

λ_S	0.01
λ_{DD} and λ_{DU}	0.0022
μ and μ^T	0.03

Matrix				
0.9712	1.03	1.03	1.03	1.03
0.0244	-0.0444	0	0	0
0.0044	0	-0.0444	0	0
0	0.0122	0.0122	-0.03	0
0	0.0022	0.0022	0	-0.03

P0
P1
P2
P3
P4

=

Initial Condition
1
0
0
0
0

P0
P1
P2
P3
P4

=

Inverse Matrix				
0.5102	17.5170	17.5170	17.5170	17.5170
0.2804	-12.8961	9.6265	9.6265	9.6265
0.0506	1.7359	-20.7866	1.7359	1.7359
0.1346	-4.5385	-4.5385	-28.7126	4.6207
0.0243	-0.8184	-0.8184	0.8332	-32.5001

Initial Condition
1
0
0
0
0

=

0.510204
0.280382
0.050561
0.134584
0.024269

Thank you

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